# Data Mining and Knowledge Discovery Technologies



DAVID TANIAR

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David Taniar
Monash University, Australia




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*David Taniar, Monash University, Australia*

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# Preface

This is the second volume of the Advances in Data Warehousing and Mining (ADWM) book series. ADWM publishes books in the areas of data warehousing and mining. The topic of this volume is data mining and knowledge discovery. This volume consists of 14 chapters in four section, contributed by authors and editorial board members from the International Journal of Data Warehousing and Mining, as well as invited authors who are experts in the data mining field.

Section I, *Association Rules*, consists of four chapters covering association rule techniques for multidimensional data, XML data, Web data, as well as rule interestingness measures.

Chapter I, *OLEMAR: An Online Environment for Mining Association Rules in Multidimensional Data*, by Riadh Ben Messaoud (University of Lyon 2), Sabine Loudcher Rabaséda (University of Lyon 2, France), Rokia Missaoui (University of Québec in Outaouais, Canada), and Omar Boussaid (University of Lyon 2, France), proposes to extend OLAP with data mining focusing on mining association rules in data cubes. OLEMAR (online environment for mining association rules) extracts associations from multidimensional data and allows extraction of inter-dimensional association rules, as well.

Chapter II, *Current Interestingness Measures for Association Rules: What do they Really Measure?*, by Yun Sing Koh (Auckland University of Technology, New Zealand), Richard O'Keefe (University of Otago, New Zealand), and Nathan Rountree (University of Otago, New Zealand), focuses on interestingness measurements for association rules. Rule interestingness measure is important as most of the association rule mining techniques, such as Apiori, commonly extract a very large number of rules, which might be difficult for decision makers to digest. It therefore makes sense to have these rules presented in a certain order or in groups or rules. This chapter studies the inter-relationship among variables in order to study the behaviour of the interestingness measures. It also introduces a classification of the current interestingness measures.

Chapter III, *Mining Association Rules from XML Data*, by Qin Ding and Gnanasekaran Sundarraj (The Pennsylvania State University at Harrisburg, USA), focuses on XML data.

XML data is growingly popular—used for data exchange as well as to represent semi-structured data. This chapter proposes a framework for association rule mining on XML data, and presents a Java-based implementation of the Apriori and FP-Growth algorithms for mining XML data.

Chapter IV, *A Lattice-Based Framework for Interactively and Incrementally Mining Web Traversal Patterns*, by Yue-Shi Lee and Show-Jane Yen (Ming Chuan University, Taiwan), concentrates on Web mining in order to improve Web services. It particularly focuses on Web traversal pattern mining, which discovers user access patterns from Web logs. This information is important, as it may be able to give Web users navigation suggestions. This chapter discusses efficient incremental and interactive mining algorithms to discover Web traversal patterns and make the mining results to satisfy the users' requirements.

Section II, *Clustering and Classification*, consists of four chapters covering clustering using genetic algorithm (GA) and rbf-Kernel, as well as classification of sequence data. This part also includes a chapter on the privacy issue.

Chapter V, *Determination of Optimal Clusters Using a Genetic Algorithm*, by Tushar, Shibendu Shekhar Roy, and Dilip Kumar Pratihar (IIT, Kharagpur), discusses the importance of clustering techniques. Besides association rules, clustering is an important data mining technique. A clustering method analyzes the pattern of a dataset and groups the data into several clusters based on the similarity among them. This chapter discusses clustering techniques using fuzzy c-means (*FCM*) and entropy-based fuzzy clustering (*EFC*) algorithms.

Chapter VI, *K-Means Clustering Adopting rbf-Kernel*, by ABM Shawkat Ali (Central Queensland University, Australia), focuses on the k-means clustering technique. This chapter presents an extension of the k-means algorithm by adding the radial basis function (rbf) kernel in order to achieve a better performance compared with the classical k-means algorithm.

Chapter VII, *Advances in Classification of Sequence Data*, by Pradeep Kumar (University of Hyderabad, Gachibowli, India), P. Radha Krishna (University of Hyderabad, Gachibowli, India), Raju S. Bapi (University of Hyderabad, Gachibowli, India), and T. M. Padmaja (University of Hyderabad, Gachibowli, India), focuses on sequence data. It reviews the state of the art for sequence data classification, including *kNN*, *SVM*, and Bayes classification. It describes the use of $S^3M$ similarity metric. The chapter closes by pointing out various application areas of sequence data and describes open issues in sequence data classification.

Chapter VIII, *Using Cryptography for Privacy-Preserving Data Mining*, by Justin Zhan (Carnegie Mellon University, USA), focuses on privacy issues in *kNN* classification. Privacy concerns may prevent the parties from directly sharing the data and some types of information about the data. Therefore, the main issue is how multiple parties could share data in the collaborative data mining without breaching data privacy. The other issue is how to obtain accurate data mining results while preserving data privacy.

Section III on *Domain Driven and Model Free*, consists of two chapters covering domain driven and model free data mining.

Chapter IX, *Domain Driven Data Mining*, by Longbing Cao and Chengqi Zhang (University of Technology Sydney, Australia), proposes a practical data mining methodology called domain-driven data mining, whereby it meta-synthesizes quantitative intelligence and qualitative intelligence in mining complex applications. It targets actionable knowledge discovery in a constrained environment for satisfying user preference.

Chapter X, *Model Free Data Mining*, by Can Yang, Jun Meng, Shanan Zhu, and Mingwei Dai (Zhejiang University, Hangzhou, P. R. China and Xi'an Jiao Tong University, Xi'an, P. R. China), presents a model free data mining. This chapter shows the underlying relationship between sensitivity analysis and consistency analysis for input selection, and then derives an efficient model free method using common sense. It utilizes a fuzzy logic called fuzzy consistency analysis (*FCA*), which is a model free method and can be implemented efficiently as a classical model free method.

The final section, Section IV, *Issues and Applications*, consists of four chapters, discussing the minus sides of data mining, as well as presenting applications in bioinformatics and social sciences.

Chapter XI, *Minimizing the Minus Sides of Mining Data*, by John Wang (Montclair State University, USA), Xiaohua Hu (Drexel University, USA), and Dan Zhu (Iowa State University, USA), explores the effectiveness of data mining from a commercial perspective. It discusses several issues including the statistical issues, technical issues, and organizational issues.

Chapter XII, *Study of Protein-Protein Interactions from Multiple Data Sources*, by Tu Bao Ho, Thanh Phuong Nguyen, and Tuan Nam Tran (Japan Advanced Institute of Science and Technology, Japan), focuses on an application of data mining in the bioinformatics domain. This chapter gives a survey of computational methods for protein-protein interaction (PPI). It describes the use of inductive logical programming to learn prediction rules for protein-protein and domain-domain interactions.

Chapter XIII, *Data Mining in the Social Sciences and Iterative Attribute Elimination*, by Anthony Scime (SUNY Brockport, USA), Gregg R. Murray (SUNY Brockport, USA), Wan Huang (SUNY Brockport, USA), and Carol Brownstein-Evans (Nazareth College), presents an application in the social sciences domain. This domain is still underrepresented in the data mining area. With the large collection of social data, it gives potential opportunities to find society's pressing problems.

Finally, Chapter XIV, *A Machine Learning Approach for One-Stop Learning*, by Marco A. Alvarez and SeungJin Lim (Utah State University, USA), presents an application in the learning and education area. As the Web is nowadays an important source of learning, having an efficient tool and method for effective learning is critical. This chapter describes the use of *SVM*, AdaBoost, Naïve Bayes, and neural network in one-stop learning.

Overall, this volume covers important foundations to researches and applications in data mining, covering association rules, clustering, and classification, as well as new directions in domain driven and model free data mining. Issues and applications, particularly in bioinformatics, social and political sciences, and learning and education, show a full spectrum of the coverage of important and emerging topics in data mining.

*David Taniar, Editor-in-Chief*

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# *Section I*

# Association Rules

## Chapter I

# OLEMAR: An Online Environment for Mining Association Rules in Multidimensional Data

Riadh Ben Messaoud, University of Lyon 2, France
Sabine Loudcher Rabaséda, University of Lyon 2, France
Rokia Missaoui, University of Québec, Canada
Omar Boussaid, University of Lyon 2, France

## Abstract

*Data warehouses and OLAP (online analytical processing) provide tools to explore and navigate through data cubes in order to extract interesting information under different perspectives and levels of granularity. Nevertheless, OLAP techniques do not allow the identification of relationships, groupings, or exceptions that could hold in a data cube. To that end, we propose to enrich OLAP techniques with data mining facilities to benefit from the capabilities they offer. In this chapter, we propose an online environment for mining association rules in data cubes. Our environment called OLEMAR (online environment for mining association rules), is designed to extract associations from multidimensional data. It allows the extraction of inter-dimensional association rules from data cubes according to a sum-based aggregate measure, a more general indicator than aggregate values provided by the traditional COUNT measure. In our approach, OLAP users are able to drive a mining process guided by a meta-rule, which meets their analysis objectives. In*

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*addition, the environment is based on a formalization, which exploits aggregate measures to revisit the definition of the support and the confidence of discovered rules. This formalization also helps evaluate the interestingness of association rules according to two additional quality measures: lift and loevinger. Furthermore, in order to focus on the discovered associations and validate them, we provide a visual representation based on the graphic semiology principles. Such a representation consists in a graphic encoding of frequent patterns and association rules in the same multidimensional space as the one associated with the mined data cube. We have developed our approach as a component in a general online analysis platform called Miningcubes according to an Apriori-like algorithm, which helps extract inter-dimensional association rules directly from materialized multidimensional structures of data. In order to illustrate the effectiveness and the efficiency of our proposal, we analyze a real-life case study about breast cancer data and conduct performance experimentation of the mining process.*

# Introduction

Data warehousing and OLAP (online analytical processing) technologies have gained a widespread acceptance since the 90's as a support for decision-making. A data warehouse is a collection of subject-oriented, integrated, consolidated, time-varying, and non-volatile data (Kimball, 1996; Inmon, 1996). It is manipulated through OLAP tools, which offer visualization and navigation mechanisms of multidimensional data views commonly called *data cubes*.

A data cube is a multidimensional representation used to view data in a warehouse (Chaudhuri & Dayal, 1997). The data cube contains *facts* or *cells* that have *measures*, which are values based on a set of dimensions where each dimension usually consists of a set of categorical descriptors called *attributes* or *members*. Consider for example a *sales* application where the dimensions of interest may include, *costumer*, *product*, *location*, and *time*. If the measure of interest in this application is the *sales amount*, then an OLAP fact represents the sales measure corresponding to a single member in the considered dimensions. A dimension may be organized into a hierarchy. For instance, the location dimension may form the hierarchy *city* → *state* → *region*. Such dimension hierarchies allow different levels of granularity in the data warehouse. For example, a *region* corresponds to a high level of granularity whereas a *city* corresponds to a lower level. Classical aggregation in OLAP considers the process of summarizing data values by moving from a hierarchical level of a dimension to a higher one. Typically, additive data are suitable for simple computation according to aggregation functions (SUM, AVERAGE, MAX, MIN, and COUNT). For example, according to such a computation, a user may observe the sum of sales of products according to year and region.

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Furthermore, with efficient techniques developed for computing data cubes, users have become widely able to explore multidimensional data. Nevertheless, the OLAP technology is quite limited to an exploratory task and does not provide automatic tools to identify and visualize patterns (e.g., clusters, associations) of huge multidimensional data.

In order to enhance its analysis capabilities, we propose to couple OLAP with data mining mechanisms. The two fields are complementary, and associating them can be a solution to cope with their respective limitations. OLAP technology has the ability to query and analyze multidimensional data through exploration, while data mining is known for its ability to discover knowledge from data. The general issue of coupling database systems with data mining was already discussed and motivated by Imieliński and Mannila (1996). The authors state that data mining leads to new challenges in the database area, and to a second generation of database systems for managing KDD (knowledge discovery in databases) applications just as classical ones manage business ones. More generally, the association of OLAP and data mining allows elaborated analysis tasks exceeding the simple exploration of data. Our idea is to exploit the benefits of OLAP and data mining techniques and to integrate them in the same analysis framework. In spite of the fact that both OLAP and data mining were considered two separate fields for a while, several recent studies showed the benefits of coupling them.

In our previous studies, we have shown the potential of coupling OLAP and data mining techniques through two main approaches. Our first approach deals with the reorganization of data cubes for a better representation and exploration of multidimensional data (Ben Messaoud, Boussaid, & Loudcher, 2006a). The approach is based on multiple correspondence analysis (MCA), which allows the construction of new arrangements of modalities in each dimension of a data cube. Such a reorganization aims at bringing together cells in a reduced part of the multidimensional space, and hence giving a better view of the cube. Our second approach constructs a new OLAP operator for data clustering called *OpAC* (Ben Messaoud, Boussaid, & Loudcher, 2006b), which is based on the agglomerative hierarchical clustering (AHC).

In this chapter, we present a third approach which also follows the general issue of coupling OLAP with data mining techniques but concerns the mining of association rules in multidimensional data. In Ben Messaoud, Loudcher, Boussaid, and Missaoui (2006), we have proposed a guided-mining process of association rules in data cubes. Here, we enrich this proposal and establish a complete online environment for mining association rules (*OLEMAR*). In fact, it consists of a mining and visualization package for the extraction and the representation of associations from data cubes. Traditionally, with OLAP analysis, we used to observe summarized facts by aggregating their measures according to groups of descriptors (members) from analysis dimensions. Here, with *OLEMAR*, we propose to use association rules in order to better understand these summarized facts according to their descriptors. For

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instance, we can note from a given data cube that sales of *sleeping bags* are particularly high in a given city. Current OLAP tools do not provide explanations of such particular fact. Users are generally supposed to explore the data cube according to its dimensions in order to manually find an explanation for a given phenomenon. For instance, one possible interpretation of the previous example consists in associating sales of *sleeping bags* with the *summer season* and *young tourist costumers*.

In the recent years, many studies addressed the issue of performing data mining tasks on data warehouses. Some of them were specifically interested in mining patterns and association rules in data cubes. For instance, Kamber, Han, and Chiang (1997) state that it is important to explore data cubes by using association rule algorithms. Further, Imieliński, Khachiyan, and Abdulghani (2002) believe that OLAP is closely interlinked with association rules and shares with them the goal of finding patterns in the data. Goil and Choudhary (1998) argue that automated techniques of data mining can make OLAP more useful and easier to apply in the overall scheme of decision support systems. Moreover, cell frequencies can facilitate the computation of the support and the confidence, while dimension hierarchies can be used to generate multilevel association rules.

*OLEMAR* is mainly based on a mining process, which explains possible relationships of data by extracting *inter-dimensional* association rules from data cubes (i.e., rules mined from multiple dimensions without repetition of predicates in each dimension). This process is guided by the notion of *inter-dimensional meta-rule*, which is designed by users according to their analysis needs. Therefore, the search of association rules can focus on particular regions of the mined cube in order to meet specific analysis objectives. Traditionally, the COUNT measure corresponds to the frequency of facts. Nevertheless, in an analysis process, users are usually interested in observing multidimensional data and their associations according to measures more elaborated than simple frequencies. In our approach, we propose a redefinition of the support and the confidence to evaluate the interestingness of mined association rules when *SUM-based* measures are used. Therefore, the support and the confidence according to the COUNT measure become particular cases of our general definition. In addition to support and confidence, we use two other descriptive criteria (*lift* and *loevinger*) in order to evaluate the interestingness of mined associations. These criteria are also computed for *sum-based aggregate measures* in the data cube and reflect interestingness of associations in a more relevant way than what is offered by support and confidence.

The mining algorithm works in a *bottom-up* manner and is an adaptation of the *Apriori* algorithm (Agrawal, Imieliński, & Swami, 1993) to multidimensional data. It is also guided by user's needs expressed through the meta-rule, takes into account a user selected measure in the computation of the support and the confidence, and provides further evaluation of extracted association rules by using *lift* and *loevinger* criteria.

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In addition to the mining process, the environment also integrates a visual tool, which aims at representing the mined frequent patterns and the extracted association rules according to an appropriate graphical encoding based on the *graphic semiology* principles of Bertin (1981). The peculiarity of our visualization component lies on the fact that association rules are represented in a multidimensional space in a similar way as facts (cells).

This chapter is organized as follows. In the second section, we define the formal background and notions that will be used in the sequel. The third section presents the key concepts of our approach for mining inter-dimensional association rules: the concept of inter-dimensional meta-rule; the general computation of support and confidence based on OLAP measures; and criteria for the advanced evaluation of mined association rules. The fourth section deals with the visualization of the mined inter-dimensional association rules while the fifth section provides the implementation of the online mining environment and describes our algorithm for mining inter-dimensional association rules. In the sixth section, we use a case study about mammographies to illustrate our findings while the seventh section concerns the experimental analysis of the developed algorithm. In the eighth section, we present a state of the art about mining association rules in multidimensional data. We also provide a comparative study of existing work and our own proposal. Finally, we conclude this chapter and address future research directions.

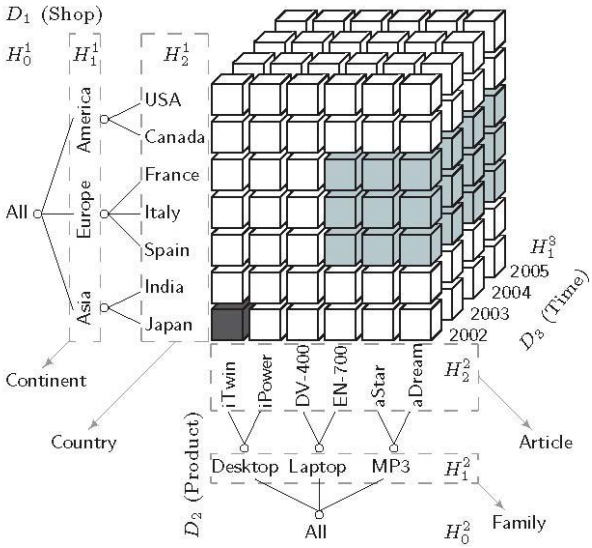
## Formal Background and Notations

In this section, we define preliminary formal concepts and notations we will use to describe our mining process. Let *C* be a data cube with a non empty set of *d* dimensions $\mathbf{D} = \{D_1, \ldots, D_i, \ldots, D_d\}$ and a non empty set of measures **M**. We consider the following notations:

- Each dimension $D_i \in \mathbf{D}$ has a non empty set of hierarchical levels. *C*;
- $H_j^i$ is the $j^{th}$ $(j \geq 0)$ level hierarchical level in $D_i$. The coarse level of $D_i$, denoted $H_0^i$, corresponds to its total aggregation level *All*. For example, in Figure 1, dimension *Shop* ($D_1$) has three levels: *All*, *Continent*, and *Country*. The *All* level is denoted $H_0^1$, the *Continent* level is denoted $H_1^1$, and the *Country* level is denoted $H_2^1$;

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*Figure 1. Example of sales data cube*




- $\mathbf{H}_i$ is the set of hierarchical levels of dimension $D_i$, where each level $H_j^i \in \mathbf{H}_i$ consists of a non empty set of members denoted $A_{ij}$. For example, in Figure 1, the set of hierarchical levels of $D_2$ is $\mathbf{H}_2 = \{H_0^2, H_1^2, H_2^2\}$= {*All*, *Family*, *Article*}, and the set of members of the *Article* level of $D_2$ is $A_{22}$ ={*iTwin*, *iPower*, *DV-400*, *EN-700*, *aStar*, *aDream*}.

**Definition 1. (Sub-cube)**

Let $\mathbf{D} \subseteq \mathbf{D}$ be a non empty set of $p$ dimensions {$D_1$, …, $D_p$} from the data cube $C$ ($p \leq d$). The $p$-tuple $(\Theta_1,\ldots,\Theta_p)$ is called a sub-cube on $C$ according to $\mathbf{D}$ iff $\forall i \in \{1,\ldots,p\}$, $\Theta_i \neq \emptyset$ and there exists a unique $j$ such that $\Theta_i \subseteq A_j$.

As previously defined, a sub-cube according to a set of dimensions $\mathbf{D}$ corresponds to a portion from the initial data cube $C$. It consists in setting for each dimension from $\mathbf{D}$ a non-empty subset of member values from a single hierarchical level of that dimension. For example, consider $\mathbf{D} = \{D_1, D_2\}$ a subset of dimensions from the cube of Figure 1. $(\Theta_1,\Theta_2)$ = (*Europe*, {*EN-700*, *aStar*, *aDream*}) is therefore a pos-

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sible sub-cube on *C* according to $\mathbf{D}$, which is displayed by the grayed portion of the cube in the figure. Note that the same portion of the cube can be defined differently by considering the sub-cube $(\Theta_1, \Theta_2, \Theta_3)$ = (*Europe*, {*EN-700*, *aStar*, *aDream*}, *All*) according to $\mathbf{D} = \{D_1, D_2, D_3\}$.

One particular case of the sub-cube definition is when it is defined on *C* according to $\mathbf{D} = \{D_1, \ldots, D_d\}$ and $\forall i \in \{1, \ldots, d\}$, $\Theta_i$ is a single member from the finest hierarchical level of $D_i$. In this case, the sub-cube corresponds to a cube cell in *C*. For example, the black cell in Figure 1 can be considered as the sub-cube (*Japan*, *iTwin*, *2002*) on *C* according to $\mathbf{D} = \{D_1, D_2, D_3\}$. Each cell from the data cube *C* represents an OLAP fact which is evaluated in $\Re$ according to one measure from **M**. In our proposal, we evaluate a sub-cube according to its *sum-based aggregate measure* which is defined as follows:

**Definition 2. (Sum-based aggregate measure)**

Let $(\Theta_1, \ldots, \Theta_p)$ be a sub-cube on *C* according to $\mathbf{D} \subseteq \mathbf{D}$. The sum-based aggregate measure of sub-cube $(\Theta_1, \ldots, \Theta_p)$ according to a measure $M \in \mathbf{M}$, noted $M(\Theta_1, \ldots, \Theta_p)$, is the SUM of measure *M* of all facts in the sub-cube.

For instance, the *sales turnover* of the grayed sub-cube in Figure 1 can be evaluated by its sum-based aggregate measure according to the expression *Turnover*(*Europe*, {*EN-700*, *aStar*, *aDream*}), which represents the SUM of the *sales turnover* values contained in grayed cells in the *Sales* cube.

**Definition 3. (Dimension predicate)**

Let $D_i$ be a dimension of a data cube. A dimension predicate $\alpha_i$ in $D_i$ is a predicate of the form $\langle a \in A_{ij} \rangle$.

A dimension predicate is a predicate which takes a dimension member as a value. For example, one dimension predicate in $D_1$ of Figure 1 can be of the form $\mathsf{a}_1 = \langle a \in A_{ij} \rangle = \langle a \in \{America, Europe, Asia\} \rangle$.

Definition 4. (Inter-dimensional predicate)

Let $\mathbf{D} \subseteq \mathbf{D}$ be a non empty set of *p* dimensions $\{D_1, \ldots, D_p\}$ from the data cube *C* $(2 \leq p \leq d)$. $(\mathsf{a}_1 \wedge \ldots \wedge \mathsf{a}_p)$ is called an inter-dimensional predicate in **D** iff $\forall i \in \{1, \ldots, p\}$, $\alpha_i$ is a dimension predicate in $D_i$.

For instance, let consider $\mathbf{D} = \{D_1, D_2\}$ a set of dimensions from the cube of Figure 1.

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An inter-dimensional predicate can be of the form: $(\langle a_1 \in A_1 \rangle, \langle a_2 \in A_2 \rangle)$. An inter-dimensional predicate defines a conjunction of non-repetitive predicates (i.e., each dimension has a distinct predicate in the expression).

# The Proposed Mining Process

As previously mentioned, our mining process consists in (i) exploiting meta-rule templates to mine rules from a limited subset of a data cube, (ii) revisiting the definition of support and confidence based on the measure values, (iii) using advanced criteria to evaluate interestingness of mined associations, and (iv) proposing an *Apriori*-based algorithm for mining multidimensional data.

## Inter-Dimensional Meta-Rules

We consider two distinct subsets of dimensions in the data cube *C*: (i) $D_C \subset \mathbf{D}$ is a subset of *p context dimensions*. A sub-cube on *C* according to $D_C$ defines the context of the mining process; and (ii) $D_A \subset \mathbf{D}$ is a subset of *analysis dimensions* from which predicates of an inter-dimensional meta-rule are selected. An inter-dimensional meta-rule is an association rule template of the following form:

In the context $(\Theta_1,\ldots,\Theta_p)$

$$(\alpha_1 \wedge \ldots \wedge \alpha_s) \Rightarrow (\beta_1 \wedge \ldots \wedge \beta_r) \quad (1)$$

where $(\Theta_1,\ldots,\Theta_p)$ is a sub-cube on *C* according to $D_C$. It defines the portion of cube *C* to be mined. Unlike the meta-rule proposed in Kamber et al. (1997), our proposal allows the user to target a mining context by identifying the sub-cube $(\Theta_1,\ldots,\Theta_p)$ to be explored. Note that in the case when $D_C = \emptyset$, no particular analysis context is selected. Therefore, the mining process covers the whole cube *C*.

We note that $\forall k \in \{1,\ldots,s\}$ (respectively $\forall k \in \{1,\ldots,r\}$), $\alpha_k$ (respectively $\beta_k$) is a dimension predicate in a distinct dimension from $D_A$.

Therefore, the conjunction $(\alpha_1 \wedge \ldots \wedge \alpha_s) \wedge (\beta_1 \wedge \ldots \wedge \beta_r)$ is an inter-dimensional predicate in $D_A$, where the number of predicates $(s+r)$ in the meta-rule is equal to the number of dimensions in $D_A$. We also note that our meta-rule defines a non-repetitive predicate association rules since each analysis dimension is associated with a distinct

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predicate. For instance, suppose that in addition to the three dimensions displayed in Figure 1, the *Sales* cube contains four other dimensions: *Profile* ($D_4$), *Profession* ($D_5$), *Gender* ($D_6$), and *Promotion* ($D_7$). Let consider the following subsets from the *Sales* data cube: $D_C = \{D_5, D_6\} = \{Profession, Gender\}$, and $D_A = \{D_1, D_2, D_3\} = \{Shop, Product, Time\}$. One possible inter-dimensional meta-rule scheme is:

In the context (*Student*, *Female*)

$$\langle a_1 \in Continent \rangle \wedge \langle a_3 \in Year \rangle \Rightarrow \langle a_2 \in Article \rangle \quad (2)$$

According to the previous inter-dimensional meta-rule, association rules are mined in the sub-cube (*Student*, *Female*) which covers the population of sales concerning female students. The dimensions *profile* and *promotion* do not interfere in the mining process. Dimension predicates in $D_1$ and $D_3$ are set in the body of the rule whereas the dimension predicate in $D_2$ is set in the head of the rule. The first dimension predicate is set to the *continent* level of $D_1$, the second one is set to the *Year* level of $D_3$, and the third dimension predicate is set to the *article* level of $D_2$.

## Measure-Based Support and Confidence

Traditionally, as it was introduced in Agrawal et al. (1993), the support (SUPP) of an association rule $X \Rightarrow Y$ in a database of transactions *T*, is the probability that the population of transactions contains both *X* and *Y*. The confidence (CONF) of $X \Rightarrow Y$ is the conditional probability that a transaction contains *Y* given that it already contains *X*. Rules that do not satisfy user provided minimum support (*minsupp*) and minimum confidence (*minconf*) thresholds are considered uninteresting. A rule is said *large*, or *frequent*, if its support is no less than *minsupp*. In addition, a rule is said *strong* if it satisfies both *minsupp* and *minconf*.

In the case of a data cube *C*, the structure of data facilitates the mining of multidimensional association rules. The aggregate values needed for discovering association rules are already computed and stored in *C*, which facilitates calculus of the support and the confidence and therefore reduces the testing and the filtering time. In fact, a data cube stores the particular COUNT measure which represents pre-computed frequencies of OLAP facts. With this structure, it is straightforward to calculate support and confidence of associations in a data cube based on this summary information. For instance, suppose that a user needs to discover association rules according to meta-rule (2). In this case one association rule can be $R_1 : America \wedge 2004 \Rightarrow Laptop$. The support and confidence of $R_1$ are computed as follows:

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$$\text{SUPP}(R_1) = \frac{COUNT(America, Laptop, 2004, All, Student, Female, All)}{COUNT(All, All, All, All, Student, Female, All)}$$

$$\text{CONF}(R_1) = \frac{COUNT(America, Laptop, 2004, All, Student, Female, All)}{COUNT(America, All, 2004, All, Student, Female, All)}$$

Note that in the previous expressions, the support (respectively the confidence) is computed according to the frequency of units of facts based on the COUNT measure. In other words, only the number of facts is taken into account to decide whether a rule is *large* (respectively *strong*) or not. However, in the OLAP context, users are usually interested to observe facts according to summarized values of measures more expressive than their simple number of occurrences. It seems naturally significant to compute the support and the confidence of multidimensional association rules according to the sum of these measures. For example, consider a fragment from the previous *sales* sub-cube (*student, female*) by taking once the COUNT measure and then the SUM of the *sales turnover* measure. Table 4(a) and Table 4(b) sum-up views of these sub-cube fragments. In this example, for a selected *minsupp*, some itemsets are *large* according to the COUNT measure in Table 4(a), whereas they are not frequent according to the SUM of the *sales turnover* measure in Table 4(b), and *vice versa*. For instance, with a *minsupp* = 0.2, the itemsets (<America>, <MP3>, <2004>) and (<America>, < MP3>, <2005>) are *large* according to the COUNT measure (grayed cells in Table 4(a)); whereas, these itemsets are not *large* in Table 4 (b). The *large* itemsets according to the SUM of the profit measure are rather (<Europe>, <Laptop>, <2004>) and (<Europe>, <Laptop>, <2005>).

In the OLAP context, the rule mining process needs to handle any measure from the data cube in order to evaluate its interestingness. Therefore, a rule is not merely

*Table 5. Fragment of the sales cube according to the (a) COUNT measure and the (b) SUM of the sales turnover measure*

| | 2004 | | 2005 | |
|---|---|---|---|---|
| | America | Europe | America | Europe |
| Desktop | 1,200 | 800 | 950 | 500 |
| Laptop | 2,500 | 2,700 | 2,800 | 3,200 |
| MP3 | 10,600 | 5,900 | 11,400 | 9,100 |

(a)

| | 2004 | | 2005 | |
|---|---|---|---|---|
| | America | Europe | America | Europe |
| Desktop | $ 60,000 | $ 33,000 | $ 28,000 | $ 10,000 |
| Laptop | $ 500,000 | $ 567,000 | $ 420,000 | $ 544,000 |
| MP3 | $ 116,000 | $ 118,000 | $ 57,000 | $ 91,000 |

(b)

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evaluated according to probabilities based on frequencies of facts, but needs to be evaluated according to quantity measures of its corresponding facts. In other words, studied associations do not concern the population of facts, but they rather concern the population of units of measures of these facts. The choice of the measure closely depends on the analysis context according to which a user needs to discover associations within data. For instance, if a firm manager needs to see strong associations of sales covered by achieved profits, it is more suitable to compute the support and the confidence of these associations based on units of profits rather than on units of sales themselves. Therefore, we define a general computation of support and confidence of inter-dimensional association rules according to a user defined (sum-based) measure *M* from the mined data cube. Consider a general rule *R*, which complies with the defined inter-dimensional meta-rule (1):

In the context $(\Theta_1,\ldots,\Theta_p)$

$(x_1 \wedge \ldots \wedge x_s) \Rightarrow (y_1 \wedge \ldots \wedge y_r)$

The support and the confidence of this rule are therefore computed according to the following general expressions:

$$\text{SUPP}(R) = \frac{M(x_1,\ldots,x_s,y_1,\ldots,y_r,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}{M(All,\ldots,All,\Theta_1,\ldots,\Theta_p,All,\ldots,All)} \quad (3)$$

$$\text{CONF}(R) = \frac{M(x_1,\ldots,x_s,y_1,\ldots,y_r,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}{M(x_1,\ldots,x_s,All,\ldots,All,\Theta_1,\ldots,\Theta_p,All,\ldots,All)} \quad (4)$$

where $M(x_1,\ldots,x_s,y_1,\ldots,y_r,\Theta_1,\ldots,\Theta_p,All,\ldots,All)$ is the sum-based aggregate measure of a sub-cube. From a statistical point of view, the collection of facts is not studied according to frequencies but rather with respect to the units of mass evaluated by the OLAP measure *M* of the given facts. Therefore, an association rule $X \Rightarrow Y$ is considered *large* if both *X* and *Y* are supported by a sufficient number of the units of measure *M*. It is important to note that we provide a definition of support and confidence which generalizes the traditional computation of probabilities. In fact, traditional support and confidence are particular cases of the above expressions which can be obtained by the COUNT measure. In the above expressions, in order to insure the validity of our new definition of support and confidence, we suppose that the measure *M* is additive and has positive values.

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## Advanced Evaluation of Association Rules

Support and confidence are the mostly known measures for the evaluation of association rule interestingness. These measures are key elements of all *Apriori*-like algorithms (Agrawal et al., 1993) which mine association rules such that their support and confidence are greater than user defined thresholds. However, they usually produce a large number of rules which may not be interesting. Various properties of interestingness criteria of association rules have been investigated. For a large list of criteria, the reader can refer to Lallich, Vaillant, and Lenca (2005) and Lenca, Vaillant, and Lallich (2006).

Let's consider again the association rule $R: X \Rightarrow Y$, which complies with the inter-dimensional meta-rule (1), where $X = (x_1 \wedge \ldots \wedge x_s)$ and $Y = (y_1 \wedge \ldots \wedge y_r)$ are conjunctions of dimension predicates. We also consider a user-defined measure *M* from data cube *C*. We denote by $P_X$ (respectively, $P_Y$, $P_{XY}$) the relative measure *M* of facts matching *X* (respectively *Y*, *X* and *Y*) in the sub-cube defined by the instance ($\Theta_1,\ldots,\Theta_p$) in the context dimensions $D_C$. We also denote by $P_{\overline{X}} = 1 - P_X$ (respectively, $P_{\overline{Y}} = 1 - P_Y$) the relative measure *M* of facts not matching *X* (respectively *Y*), i.e., the probability of not having *X* (respectively *Y*). The support of *R* is equal to $P_{XY}$ and its confidence is defined by the ratio:

$$\frac{P_{X\overline{Y}}}{P_X}$$

which is a conditional probability, denoted $P_{\overline{X/Y}}$, of matching *Y* given that *X* is already matched.

$$P_X = \frac{M(x_1,\ldots,x_s,All,\ldots,All,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}{M(All,\ldots,All,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}$$

$$P_Y = \frac{M(All,\ldots,All,y_1,\ldots,y_r,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}{M(All,\ldots,All,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}$$

$$P_{XY} = \text{SUPP}(R) = \frac{M(x_1,\ldots,x_s,y_1,\ldots,y_r,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}{M(All,\ldots,All,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}$$

$$P_{\overline{Y/X}} = \text{CONF}(R) = \frac{M(x_1,\ldots,x_s,y_1,\ldots,y_r,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}{M(x_1,\ldots,x_s,All,\ldots,All,\Theta_1,\ldots,\Theta_p,All,\ldots,All)}$$

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There are two categories of frequently used evaluation criteria to capture the interestingness of association rules: *descriptive* criteria and *statistical* criteria. In general, one of the most important drawbacks of a statistical criterion is that it depends on the size of the mined population (Lallich et al., 2005). In fact, when the number of examples in the mined population becomes large, such a criterion loses its discriminating power and tends to take a value close to one. In addition, a statistical criterion requires a *probabilistic* approach to model the mined population of examples. This approach is quite heavy to undertake and assumes advanced statistical knowledge of users, which is not particularly true for OLAP users.

On the other hand, descriptive criteria are easy to use and express interestingness of association rules in a more natural manner. In our approach, in addition to support and confidence, we add two descriptive criteria for the evaluation of mined association rules: the lift criterion (LIFT) (Brin, Motwani, & Silverstein, 1997) and the loevinger criterion (LOEV) (Loevinger, 1947). These two criteria take the independence of itemsets $X$ and $Y$ as a reference, and are defined on rule $R$ as follows:

$$\text{LIFT}(R) = \frac{P_{XY}}{P_X P_Y} = \frac{\text{SUPP}(R)}{P_X P_Y}$$

$$\text{LOEV}(R) = \frac{P_{Y/X} - P_Y}{P_{\bar{Y}}} = \frac{\text{CONF}(R) - P_Y}{P_{\bar{Y}}}$$

The lift of a rule can be interpreted as the deviation of the support of the rule from the expected support under the independence hypothesis between the body $X$ and the head $Y$ (Brin et al., 1997). For the rule $R$, the lift captures the deviation from the independence of $X$ and $Y$. This also means that the lift criterion represents the probability scale coefficient of having Y when $X$ occurs. For example, LIFT($R$) = 2 means that facts matching with X have twice more chances to match with Y. As opposed to the confidence, which considers directional implication, the lift directly captures correlation between body $X$ and its head $Y$. In general, greater Lift values indicate stronger associations.

In addition to support and confidence, the loevinger criterion is one of the oldest used interestingness evaluations for association rules (Loevinger, 1947). It consists in a linear transformation of the confidence in order to enhance it. This transformation is achieved by centering the confidence on $P_Y$ and dividing it by the scale coefficient $P_{\bar{Y}}$. In other words, the loevinger criterion normalizes the centered confidence of a rule according to the probability of not satisfying its head.

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# The Visualization of Inter-Dimensional Association Rules

In addition to the previous mining process, our online mining environment includes facilities for a graphic representation of the mined inter-dimensional association rules. This representation offers an easier access to the knowledge expressed by a huge number of mined associations. Users can therefore get more insight about rules and easily focus on interesting ones. A particular feature of our visualization solution consists in representing association rules in a multidimensional way so that they can be explored like any part of the data cube.

Traditionally, a user observes the measures associated with facts (cells) in a data cube according to a set of dimensions in a multidimensional space. In our visualization approach, we embed in this space representation, a graphic encoding of inter-dimensional association rules. This encoding refers to the principles of *graphic semiology* of Bertin (1981). Such principles consist to organize the visual and perceptual components of graphics according to features and relations between data. They mainly use the visual variables of *position*, *size*, *luminosity*, *texture*, *color*, *orientation,* and *form*. The position variable has a particular impact on human retention since it concerns dominant visual information from a perceptual point of view. The other variables have rather a retinal property since it is quite possible to see their variations independently from their positions. The size variable generally concerns surfaces rather than lengths. According to Bertin, the variation of surfaces is a sensible stimulus for the variation of size and more relevant to human cognition than variation of length.

We note that the position of each cell in the space representation of a data cube is important since it represents a conjunction of predicate instances. For instance, let *c* be a cell in the space representation of the data cube *C*. The position of *c* corresponds to the intersection of row *X* with column *Y*. *X* and *Y* are conjunctions of modalities where each modality comes from a distinct dimension. In other words, *X* and *Y* are inter-dimensional instance predicates in the analysis dimensions retained for the visualization. Therefore, cell *c* corresponds to the itemset $\{X, Y\}$. According to the properties of the itemset $\{X, Y\}$, we propose to represent the appropriate graphic encoding as follows (see Figure 2):

- If $\{X, Y\}$. is not frequent, only the value of the measure *M*, if it exists, is represented in cell *c*.
- If $\{X, Y\}$. is frequent and it does not generate association rules, a white square is represented in cell *c*.

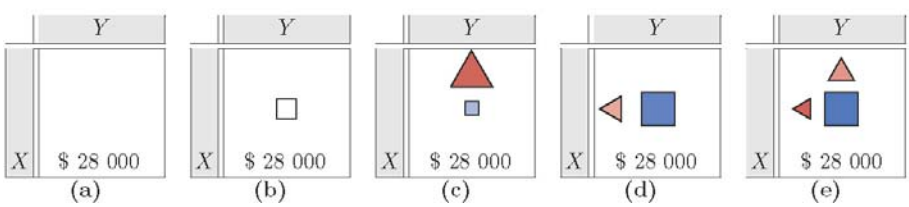
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- If $\{X, Y\}$. is frequent and generates the association rule $X \Rightarrow Y$, a blue square and a red triangle are displayed in cell *c*. The triangle points to *Y* according to the implication of the rule.
- If $\{X, Y\}$. is frequent and generates the association rule $Y \Rightarrow X$, a blue square and a red triangle are displayed in cell *c*. The triangle points to *X* according to the implication of the rule.
- If $\{X, Y\}$ is frequent and generates the association rules $X \Rightarrow Y$ and $Y \Rightarrow X$, a blue square and two red triangles are displayed in cell *c*. The first triangle points to *Y* according to the implication of the rule $X \Rightarrow Y$, and the second triangle points to X according to the implication of the rule $Y \Rightarrow X$.

For a given association rule, we use two different forms and colors to distinguish between the itemset of the rule and its implication. In fact, the itemset $\{X, Y\}$ is graphically represented by a blue square and the implication $X \Rightarrow Y$ is represented by a red equilateral triangle. We also use the surface of the previous forms in order to encode the importance of the support and the confidence. The support of the itemset $\{X, Y\}$is represented by the surface of the square and the confidence of the rule $X \Rightarrow Y$ is represented by the surface of the triangle. Since the surface is one of the most relevant variables to human perception, we use it to encode most used criteria to evaluate the importance of an association rule. For high values of the support (respectively, the confidence), the blue square (respectively, the red triangle) has a large surface, while low values correspond to small surfaces of the form. Therefore, the surfaces are proportionally equal to the values of the support and the confidence.

The lift and the loevinger criteria are highlighted with the luminosity of their respective forms. We represent high values of the lift (respectively, the loevinger criterion) by a low luminosity of the blue square (respectively, the red triangle). We note that a high luminosity of a form corresponds to a pale color, whereas, a low luminosity of a form corresponds to a dark color.

*Figure 2. Examples of association rule representations in a cube cell*




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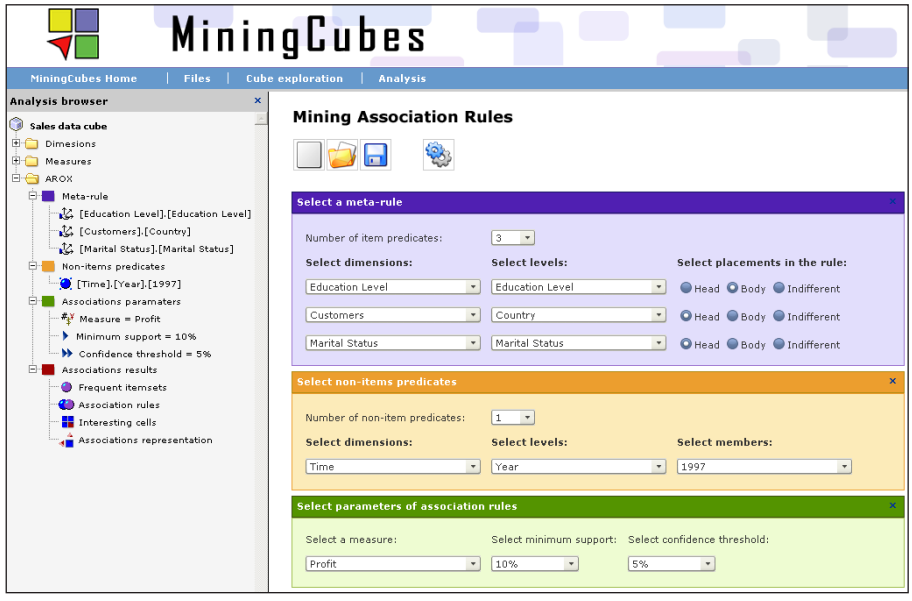
# Implementation and Algorithms

We have developed *OLEMAR* as a module of on a client/server analysis platform called *MiningCubes*, which already includes our previous proposals dealing with coupling OLAP and data mining (Ben Messaoud et al., 2006a, 2006b). *MiningCubes* is equipped with a *data loader component* that enables connection to multidimensional data cubes stored in *analysis services* of *MS SQL server 2000*. The *OLEMAR module* allows the definition of required parameters to run an association rule mining process. In fact, as shown in the interface of Figure 3, a user is able to define analysis dimensions $D_A$, context dimensions $D_C$, a meta-rule with its context sub-cube $(\Theta_1,\ldots,\Theta_p)$ and its inter-dimensional predicates scheme $(\alpha_1 \wedge \ldots \wedge \alpha_s) \Rightarrow (\beta_1 \wedge \ldots \wedge \beta_r)$, the measure *M* used to compute quality criteria of association rules, and the thresholds *minsupp* and *minconf*.

The generation of association rules from a data cube closely depends on the search for *large* (frequent) itemsets. Traditionally, frequent itemsets can be mined according to two different approaches:

- The *top-down* approach, which starts with *k*-itemsets and steps down to 1-itemsets. The decision whether an itemset is frequent or not is directly based

*Figure 3. Interface of the OLEMAR module in MiningCubes*




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on the *minsupp* value. In addition, it assumes that if a *k*-itemset is frequent, then all sub-itemsets are frequent too.

- The *bottom-up* approach which goes from 1-itemsets to larger itemsets. It complies with the *Apriori* property of *anti-monotony* (Agrawal et al., 1993) which states that *for each non-frequent itemset, all its super-itemsets are definitely not frequent*.

The previous property enables the reduction of the search space, especially when it deals with large and sparse data sets, which is particularly the case of OLAP data cubes. We implemented the mining process by defining an algorithm based on the *Apriori* property according to a *bottom-up* approach for searching large itemsets. As summarized in Algorithm 1, we proceed by an increasing level wise search for large *i*-itemsets, where *i* is the number of items in the itemset. We denote by $C(i)$ the sets of *i-candidates* (i.e., *i*-itemsets that are potentially frequent), and $F(i)$ the sets of *i-frequents* (i.e., frequent *i*-itemsets).

At **the initialization step**, our algorithm captures the 1-candidates from user defined analysis dimensions $D_A$ over the data cube *C*. These 1-candidates correspond to members of $D_A$, where each member complies with one dimension predicate $\alpha_k$ or $\beta_k$ in the meta-rule *R*. In other words, for each dimension $D_i$ of $D_A$, we capture 1-candidates from $A_{ij}$, which is the set of members of the $j^{\text{th}}$ hierarchical level of $D_i$ selected in its corresponding dimensional predicate in the meta-rule scheme. For example, let consider the data cube of Figure 4. We assume that, according to a user meta-rule, mined association rules need to comply with the meta-rule scheme:

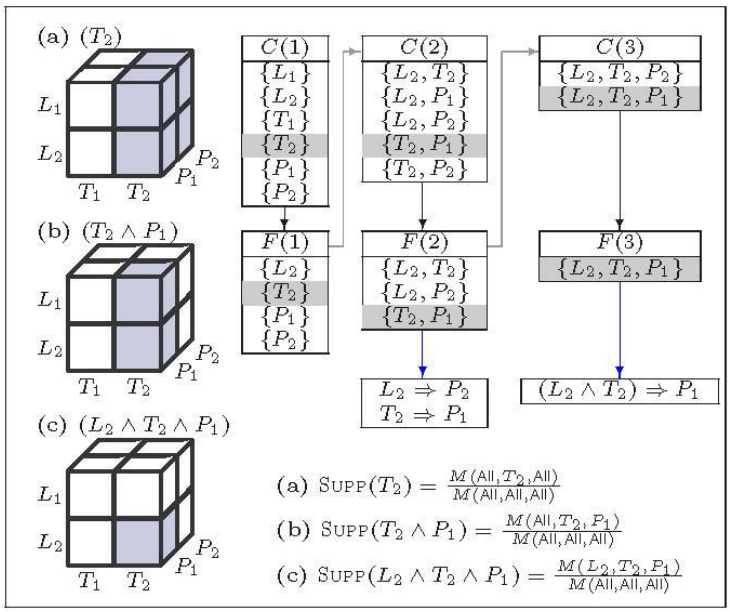
$$\langle a_1 \in \{L_1, L_2\} \rangle \wedge \langle a_2 \in \{T_1, T_2\} \rangle \Rightarrow \langle a_3 \in \{P_1, P_2\} \rangle.$$

Therefore, the set of 1-candidates is: $C(1) = \{\{L_1\}, \{L_2\}, \{T_1\}, \{T_2\}, \{P_1\}, \{P_2\}\}$. For each level *i*, if the set $F(i)$ is not empty and *i* is less than $s + r$, **the first step** of the algorithm derives frequent itemsets $F(i)$ from $C(i)$ according to two conditions: (i) an itemset $A \in C(i)$ should be an instance of an inter-dimensional predicates in $D_A$, i.e., *A* must be a conjunction of members from *i* distinct dimensions from $D_A$; and (ii) in addition to the previous condition, to be included in $F(i)$, an itemset $A \in C(i)$ must have a support greater than the minimum support threshold *minsupp*. As shown in Figure 4, SUPP(*A*) is a measure-based support computed according to a user selected measure *M* from the cube.

From each $A \in F(i)$, **the second step** extracts association rules based on two conditions: (i) an association rule $X \Rightarrow Y$ must comply with the user defined meta-rule *R*, i.e., items of *X* (respectively, items of *Y*) must be instances of dimension predicates defined in the body (respectively, in the head) of the meta-rule scheme of *R*. For example, in Figure 4, $P_2 \Rightarrow L_2$ can not be derived from $F(2)$ because, according to the

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*Figure 4. Example of a bottom-up generation of association rules from a data cube*



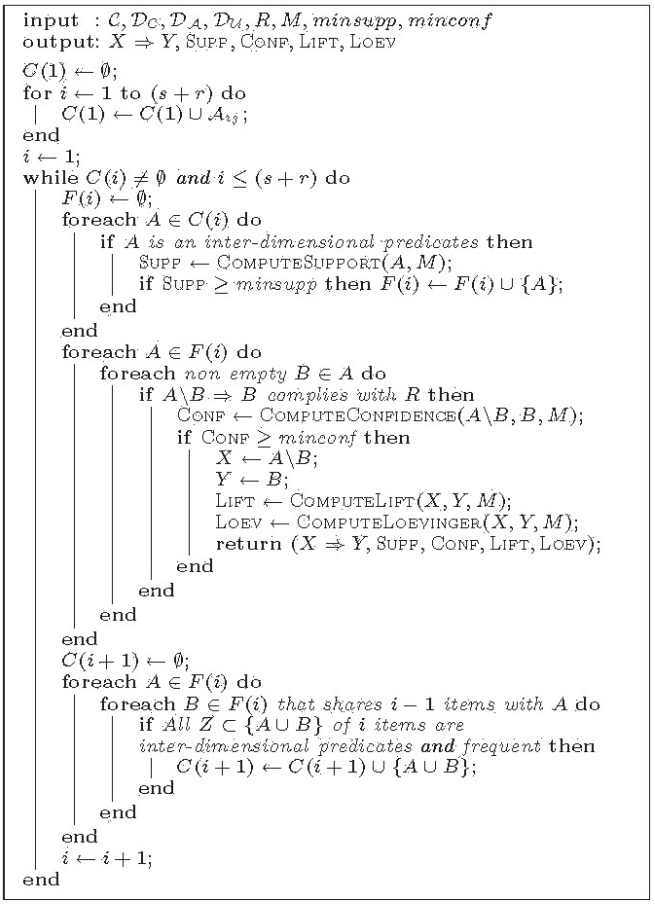

previous meta-rule scheme, instances of $\langle a_1 \in \{L_1, L_2\} \rangle$ must be in the body of mined rules and not in their head; and (ii) an association rule must have a confidence greater than the minimum confidence threshold *minconf*. The computation of confidence is also based on the user defined measure *M*. When an association rule satisfies the two previous conditions, the algorithm computes its Lift and Loevinger criteria according to the formulae we gave earlier. Finally, the rule, its support, confidence, Lift and Loevinger criteria are returned by the algorithm.

Based on the *Apriori* property, **the third step** uses the set *F*(*i*) of large *i*-itemsets to derive a new set *C*(*i* + 1) of (*i* + 1) $_{(i+1)}$-candidates. A given (*i* + 1)-candidate is the union of two *i*-itemsets *A* and *B* from *F*(*i*) that verifies three conditions: (i) *A* and *B* must have *i* – 1 common items; (ii) all non empty sub-itemsets from $A \cup B$ must be instances of inter-dimensional predicates in $D_A$; and (iii) all non empty sub-itemsets from $A \cup B$ must be frequent itemsets. For example in Figure 4, itemsets $A = \{L_2, T_2\}$ and $B = \{L_2, P_2\}$ from *F*(2) have $\{L_2\}$ as a common 1-itemset, all non empty sub-itemsets from $A \cup B = \{L_2, T_2, P_2\}$ are frequents and represent instances of inter-dimensional predicates. Therefore, $\{L_2, T_2, P_2\}$ is a 3-candidate included in *C*(3).

Note that the computation of support, confidence, Lift, and Loevinger criteria are performed respectively by the functions: COMPUTESUPPORT, COMPUTECONFIDENCE,

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*Algorithm 1. The algorithm for mining inter-dimensional association rules from data cubes*




COMPUTELIFT and COMPUTELOEVINGER. These functions take the measure *M* into account and are implemented using MDX (Multi-Dimensional eXpression language in *MS SQL Server 2000*) that provides required pre-computed aggregates from the data cube. For instance, reconsider the *Sales* data cube of Figure 1, the meta-rule (2), and the rule $R_1$: $America \wedge 2004 \Rightarrow Laptop$. According to formula (3) and considering the *sales turnover* measure, the support of $R_1$ is written as follows:

$$\text{SUPP}(R_1) = \frac{Sales_turnover(America, Laptop, 2004, All, Student, Female, All)}{Sales_turnover(All, All, All, All, Student, Female, All)}$$

The numerator value of SUPP($R_1$) is therefore returned by the following MDX query:

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```
SELECT
        NON EMPTY {[Shop].[Continent].[America]} ON AXIS(0),
        NON EMPTY {[Time].[Year].[2004]} ON AXIS(1),
        NON EMPTY {[Product].[Family].[Laptop]} ON AXIS(2)
FROM Sales
WHERE  ([Measures].[Sales_turnover],
                [Profession].[Profession category].[Student],
                [Gender].[Gender].[Female])
```

# A Case Study

In order to validate our approach, this section presents the results of a case study conducted on clinical data dealing with the breast cancer research domain. More precisely, data refer to suspicious regions extracted from the *digital database for screening mammography* (DDSM). In the following, we present the DDSM and the generated data cube.

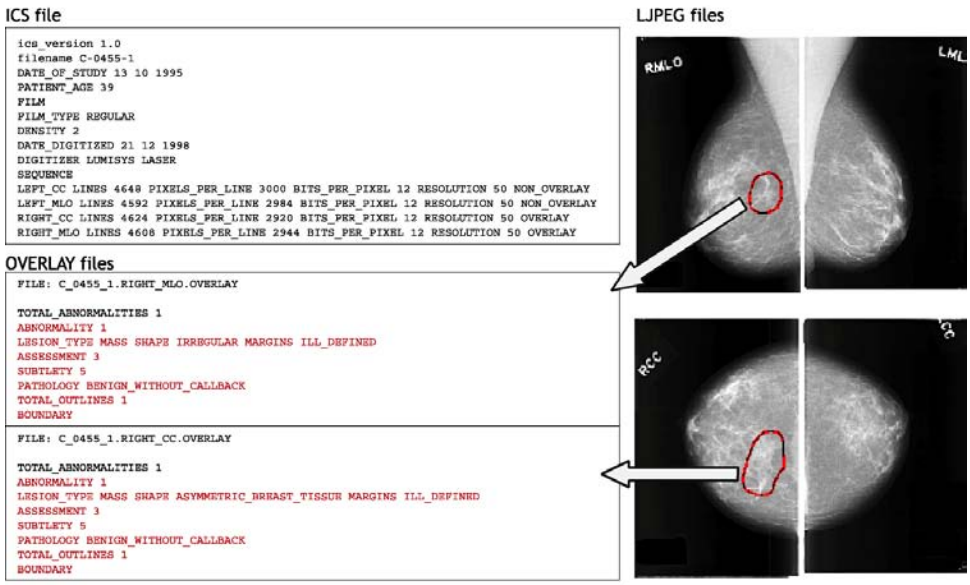
## The Digital Database for Screening Mammography (DDSM)

The DDSM is basically a resource used by the mammography image analysis research community in order to facilitate sound research in the development of analysis and learning algorithms (Heath, Bowyer, Kopans, Moore, & Jr, 2000). The database contains approximately 2,600 studies, where each study corresponds to a patient case.

As shown in Figure 5, a patient case is a collection of images and text files containing medical information collected along exams of screening mammography. The DDSM contains four types of patient cases: *normal*, *benign without callback*, *benign*, and *cancer*. *Normal* type covers mammograms from screening exams that were read as normal and had a normal screening exam. *Benign without callback* cases are exams that had an abnormality that was noteworthy but did not require the patient to be recalled for any additional checkup. In *benign* cases, something suspicious was found and the patient was recalled for some additional checkup that resulted in a benign finding. *Cancer* type corresponds to cases in which a proven cancer was found.

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*Figure 5. An example of a patient case study taken from the DDSM*



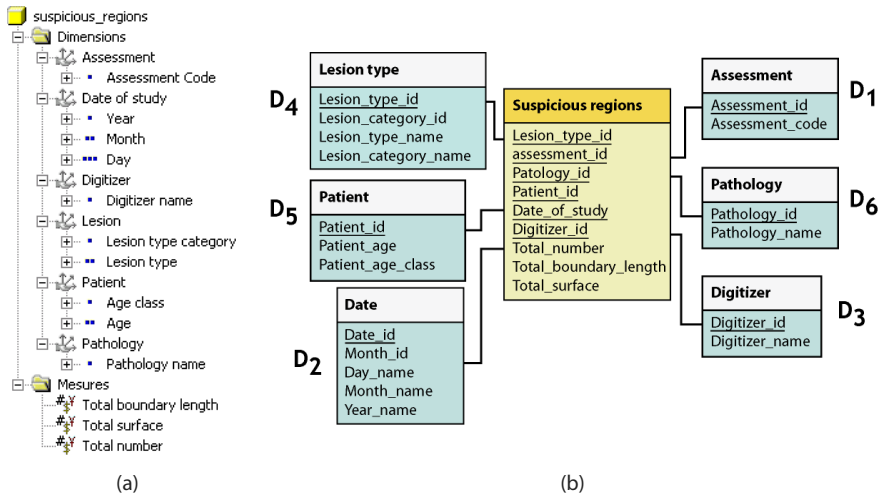

## The Suspicious Regions Data Cube

A patient file refers to different data formats and encloses several subjects that may be studied according to various points of view. In our case study, we focus on studying the screening mammography data by considering suspicious regions (abnormalities) detected by an expert as an OLAP fact.

Under *analysis services* of *MS SQL Server 2000*, we have constructed the *suspicious regions data cube* from the DDSM data. Our data cube contains 4 686 OLAP facts. Figure 6(a) and Figure 6(b) illustrate, respectively, the physical structure and the conceptual model of the constructed cube as they are presented in the cube editor of *analysis services*. According to this data cube, a set of suspicious regions can be analyzed according to several axes: the *lesion*, the *assessment*, the *subtlety*, the *pathology*, the *date of study*, the *digitizer*, the *patient*, etc. The fact is measured by the *total number* of regions, the *total boundary length*, and the *total surface* of the suspicious regions. We note that, in this cube, the set of concerned facts deals only with *benign*, *benign without callback*, and *cancer* patient cases. *Normal* cases are not concerned since they do not contain suspicious regions.

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*Figure 6. (a) the physical structure, and (b) the conceptual model of the suspicious regions data cube*




## Application on the Suspicious Regions Data Cube

We have applied our online mining environment on the suspicious regions data cube *C*. To illustrate this mining process, we suppose that an expert radiologist looks for associations that could explain the reasons of cancer tumors. We assume that the expert restricts his study to suspicious regions found on scanners of mammograms digitized thanks to a *Lumisis Laser* machine. This means that the subset of context dimensions $\mathbf{D}_C$ contains the dimension *Digitizer* ($D_3$) and the selected context corresponds to the sub-cube (*Lumisis Laser*) according to $\mathbf{D}_C$. We also suppose that the expert needs to explain the different types of pathologies in these mammograms. In order to do so, he chooses to explain the modalities of the *pathology name* level ($H_1^6$), included in the dimension *pathology* ($D_6$), by both those of the *assessment code* level ($H_1^1$), from dimension *assessment* ($D_1$), and those of the *lesion type category* level ($H_1^4$), from dimension *lesion* ($D_4$). In other words, the subset of analysis dimensions $\mathbf{D}_A$ consists of the dimensions *assessment* ($D_1$), *lesion* ($D_4$), and *pathology* ($D_6$). Thus, according to our formalization:

- The subset of context dimensions is $\mathbf{D}_C = \{D_3\} = \{Digitizer\}$;
- The subset of analysis dimension is $\mathbf{D}_A = \{D_1, D_4, D_6\} = \{Assessment, Lesion, Pathology\}$.

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*Table 1.*

| | **Association rule *R*** | **SUPP** | **CONF** | **LIFT** | **LOEV** |
|---|---|---|---|---|---|
| 1 | {*All, Calcification type pleomorphic*} ⇒ {*Benign*} | 5.03% | 24.42% | 0.73 | -0.14 |
| 2 | {***3***, *All*} ⇒ {***Cancer***} | 5.15% | 8.50% | 0.60 | -0.62 |
| 3 | {*0, All*} ⇒ {*Benign*} | 5.60% | 66.72% | 1.99 | 0.50 |
| 4 | {***4***, *Calcification type pleomorphic*} ⇒ {***Cancer***} | 6.10% | 61.05% | 1.01 | 0.01 |
| 5 | {*All, Mass shape lobulated*} ⇒ {*Cancer*} | 6.14% | 48.54% | 0.80 | -0.31 |
| 6 | {*All, Mass shape lobulated*} ⇒ {*Benign*} | 6.21% | 49.03% | 1.47 | 0.23 |
| 7 | {*3, All*} ⇒ {*Benign*} | 7.09% | 49.99% | 1.99 | 0.09 |
| 8 | {*All, Mass shape oval*} ⇒ {*Benign*} | 8.59% | 65.82% | 1.97 | 0.49 |
| 9 | {***5***, *Calcification type pleomorphic*} ⇒ {***Cancer***} | 8.60% | 98.92% | 1.63 | 0.97 |
| 10 | {***5***, *Mass shape irregular*} ⇒ {***Cancer***} | 14.01% | 96.64% | 1.60 | 0.91 |
| 11 | {*All, Calcification type pleomorphic*} ⇒ {*Cancer*} | 15.43% | 74.97% | 1.24 | 0.36 |
| 12 | {***4***, *All*} ⇒ {***Cancer***} | 16.43% | 46.06% | 0.76 | -0.37 |
| 13 | {*4, All*} ⇒ {*Benign*} | 18.64% | 52.29% | 1.56 | 0.28 |
| 14 | {*All, Mass shape irregular*} ⇒ {*Cancer*} | 20.38% | 87.09% | 1.44 | 0.67 |
| 15 | {***5***, *All*} ⇒ {***Cancer***} | 36.18% | 98.25% | 1.62 | 0.96 |

Therefore, with respect to the previous subset of dimensions, to guide the mining process of association rules, the expert needs to express the following inter-dimensional meta-rule:

In the context (*Lumisis Laser*)

$$\langle a_1 \in \textit{Assessment code} \rangle \wedge \langle a_4 \in \textit{Lesion type category} \rangle \Rightarrow \langle a_6 \in \textit{Patho}\log \textit{y name} \rangle$$

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Note that, in order to explain the pathologies of suspicious regions, the dimension predicate in $D_6$ ($\langle a_6 \in Pathology\ name \rangle$) is set to the head of the meta-rule (conclusion) whereas the other dimension predicates ($\langle a_4 \in Lesion\ type\ category \rangle$ and $\langle a_1 \in Assessment\ code \rangle$) are rather set to its body (consequence).

Assume that *minsupp* and *minconf* are set to 5%, and *Surface* of suspicious regions is the measure on which the computation of the support, the confidence, the Lift, and the Loevinger criteria will be based. The guided mining process provides the association rules that we summarize in Table 1.

Note that the previous association rules comply with the designed inter-dimensional meta-rule, which aims at explaining pathologies according to assessments and lesions. From these associations, an expert radiologist can easily note that cancer cases of suspicious regions are mainly caused by high values of assessment codes. For example, rule $R_{15}$:$\{5, All\} \Rightarrow \{Cancer\}$ is supported by 36.18% of surface units of suspicious regions. In addition, its confidence is equal to 98.25%. In other words, knowing that a suspicious region has an assessment code of 5, the region has 98.25% chances to be a cancer tumor. Rule $R_{15}$ has also a Lift equal to 1.62, which means that the total surface of cancer suspicious regions having an assessment code equal to 5 is 1.62 times greater than the expected total surface under the independence situation between the assessment and the pathology type.

The lesion type can also explain pathologies. From the previous results, we note that the *mass shape irregular* and the *calcification type pleomorphic* are the major lesions leading to cancers. In fact, rules $R_{11}$:$\{All, Calcification\ type\ pleomorphic\} \Rightarrow \{Cancer\}$ and $R_{14}$:$\{All, Mass\ shape\ irregular\} \Rightarrow \{Cancer\}$ confirm this observation with supports respectively equal to 15.43% and 20.38%, and confidences respectively equal to 74.97% and 87.09%.

Recall that our online mining environment is also able to provide an interactive visualization of its extracted inter-dimensional association rules. Figure 7 shows a part of the data cube where association rules $R_4$, $R_9$, and $R_{10}$ are displayed in the visualization interface.

## Performance Evaluation

We have evaluated the performance of our mining process algorithm for the suspicious regions data cube. We conducted a set of experiments to measure time processing for different situations of input data and parameters of the *OLEMAR module* supported by *miningcubes*. These experiments are achieved under Windows XP on a 1.60GHz PC with 480MB of main memory, and an Intel Pentium 4 processor. We also used *Analysis Services* of *MS SQL Server 2000* as a *local-host* OLAP server.

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*Figure 7. Visualization of extracted association rules in MiningCubes*

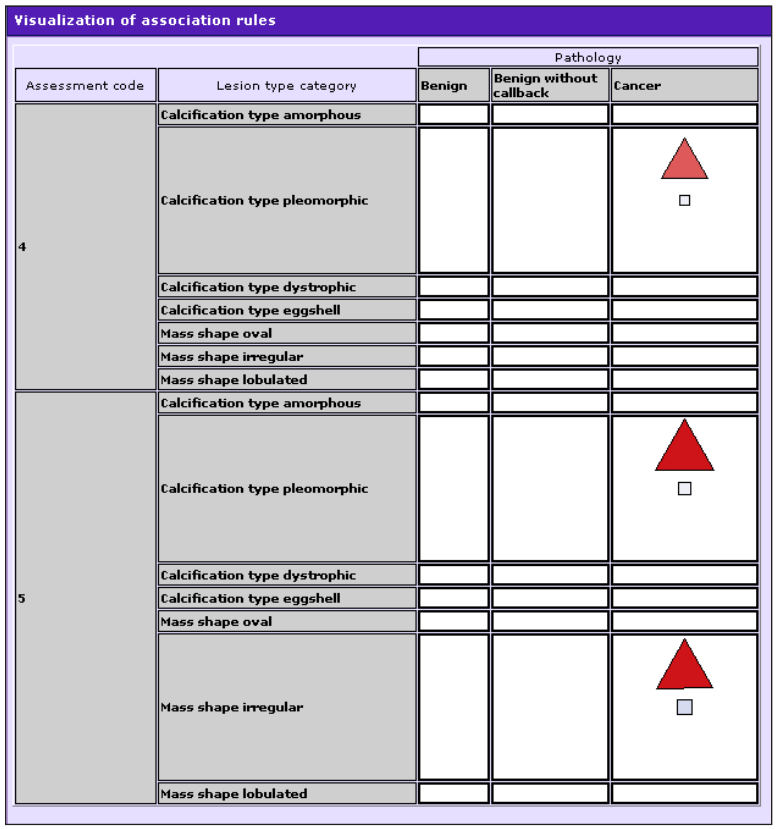



Figure 8 shows the relationship between the runtime of our mining process and the support of mined association rules according to several confidence thresholds. In general, the mining of association rules needs less time when it deals with increasing values of the support.

Figure 9 presents a test of our algorithm for several numbers of facts. For small support values, the running time considerably increases with the number of mined facts. However, for large supports, the algorithm has already equal response times independently from the number of mined facts. Another point of view of this phenomenon can be illustrated by Figure 10, which indicates that for a support and a confidence threshold equal to 5%, the efficiency of the algorithm closely depends on the number of extracted frequent itemsets and association rules. The running time obviously increases according to the number of extracted frequent itemsets and association rules. Nevertheless, the generation of association rules from frequent itemsets is more time consuming than the extraction of frequent itemsets themselves.

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*Figure 8. The running times of the mining process according to support with different confidences*

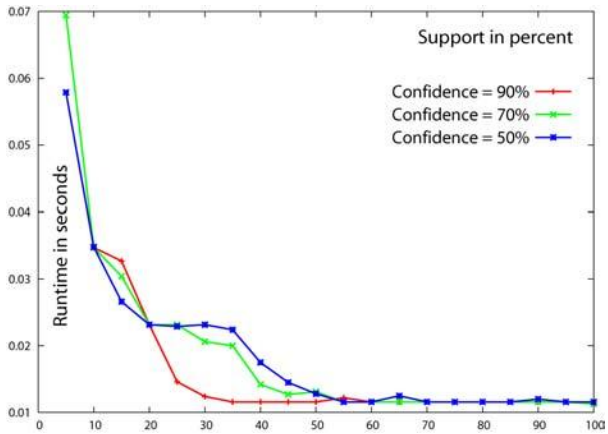
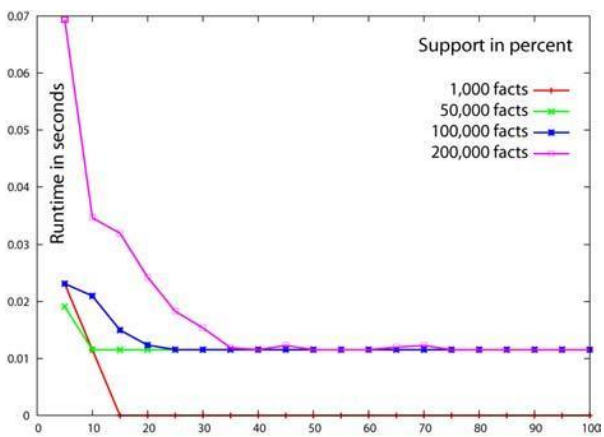


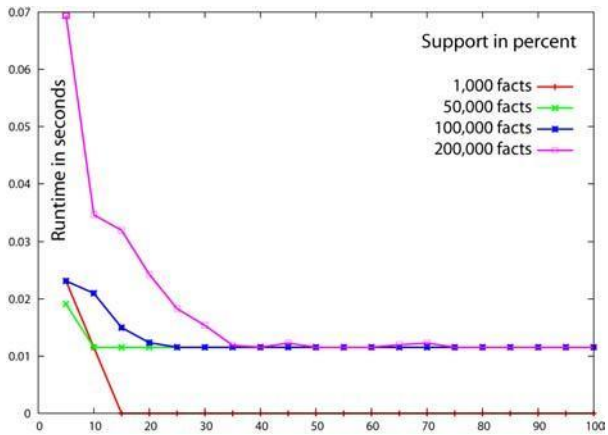

*Figure 9. The running times of the mining process according to support with different numbers of facts*




An *apriori*-based algorithm is efficient for searching frequent itemsets and has a low complexity level especially in the case of sparse data. Nevertheless, the *apriori* property does not reduce the running time of extracting association rules from a frequent itemset. For each frequent itemset, the algorithm must generate all possible association rules that comply with the meta-rule scheme and search those having a confidence greater than *minconf*.

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*Figure 10. The running times the mining process according to the number of frequent itemsets and the number of association rules*




In general, these experiments highlight acceptable runtime processing. The efficiency of our algorithm is due to: (i) the use of inter-dimensional meta-rules which reduce the search space of association rules and therefore, considerably decreases the runtime of the mining process; (ii) the use of pre-computed aggregates of the multidimensional cube which helps compute the support and the confidence via MDX queries; and (iii) the use of the anti-monotony property of *apriori*, which is definitely suited to sparse data cubes and considerably reduces the complexity of large itemsets search.

# Related Work

## Association Rule Mining in Multidimensional Data

Association rule mining was first introduced by Agrawal et al. (1993) who were motivated by *market basket analysis* and designed a framework for extracting rules from a set of transactions related to items bought by customers. They also proposed the *apriori* algorithm that discovers *large* (*frequent*) itemsets satisfying given minimal support and confidence. Since then, many developments have been performed in order to handle various types and data structures.

To the best of our knowledge, Kamber et al. (1997) were the first researchers who addressed the issue of mining *association rules from multidimensional data*. They

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introduced the concept of *meta-rule-guided mining* which consists in using rule templates defined by users in order to guide the mining process. They provide two kinds of algorithms for extracting association rules from data cubes: (1) algorithms for materialized MOLAP (*multidimensional OLAP*) data cubes and (2) algorithms for non-materialized ROLAP (*relational OLAP*) data cubes. These algorithms can mine *inter-dimensional* association rules with distinct predicates from single levels of dimensions. An *inter-dimensional* association rule is mined from multiple dimensions without repetition of predicates in each dimension, while an *intra-dimensional* association rule cover repetitive predicates from a single dimension. The support and the confidence of mined associations are computed according to the COUNT measure.

Zhu considers the problem of mining three types of associations: *inter-dimensional*, *intra-dimensional*, and *hybrid* rules (Zhu, 1998). The latter type consists in combining intra and inter-dimensional association rules. Unlike Kamber et al. (1997)—where associations are directly mined from multidimensional data—Zhu (1998) generates a *task-relevant working cube* with desired dimensions, flattens it into a tabular form, extracts frequent itemsets, and finally mines association rules. Therefore, this approach does not profit from hierarchical levels of dimensions since it flattens data cubes in a pre-processing step. In other words, it adapts multidimensional data and prepares them to be handled by classical iterative association mining process. Further, the proposal uses the COUNT measure and does not take into account further aggregate measures to evaluate discovered rules. We also note the lack of a general formalization for the proposed approach.

*Cubegrades*, proposed in Imieliński et al. (2002), are a generalization of association rules. They focus on significant changes that affect measures when a cube is modified through specialization (*drill-down*), generalization (*roll-up*), or mutation (*switch*). The authors argue that traditional association rules are restricted to the COUNT aggregate and can only express relative changes from body of the rule to body and head. In a similar way, Dong, Han, Lam, Pei, and Wang (2001) proposed an interesting and efficient version of the *cubegrade* problem called *multidimensional constrained gradients*, which also seeks significant changes in measures when cells are modified through generalization, specialization or mutation. To capture significant changes only and prune the search space, three types of constraints are considered. The concept of *cubegrades* and *constrained gradients* is quite different from classical mining of association rules. It discovers modifications on OLAP aggregates when moving from a *source-cube* to a *target-cube*, but it is not capable of searching patterns and association rules included in the cube itself. We consider a *cubegrade* as an inter-dimensional association rule with repetitive predicates, which implicitly takes into account hierarchical levels of dimensions.

Chen, Dayal, and Hsu (2000) propose a *distributed OLAP based infrastructure* which combines data warehousing, data mining, and OLAP-based engine for Web access analysis. In the data mining engine, the authors mine intra-dimensional as-

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sociation rules from a single level of a dimension, called *base dimension*, by adding *features* from other dimensions. They also propose to consider the used features at multiple levels of granularity. In addition, the generated association rules can also be materialized by particular cubes, called *volume cubes*. However, in this approach, the use of association rules closely depends on the specific domain of Web access analysis for a *sale* application. Furthermore, it lacks a formal description that enables its generalization to other application domains.

*Extended association rules* were proposed by Nestorov and Jukić (2003) as an output of a cube mining process. An *extended association rule* is a repetitive predicate rule which involves attributes of *non-item* dimensions (i.e., dimensions not related to items/products). Their proposal deals with an extension of classical association rules since it provides additional information about the precise context of each rule. However, the authors focus on mining associations from transaction databases and do not take dimension hierarchy and data cube measures into account when computing support and confidence.

Tjioe and Taniar (2005) propose a method for mining association rules in data warehouses. Based on the multidimensional data organization, their method is able to extract associations from multiple dimensions at multiple levels of abstraction by focusing on summarized data according to the COUNT measure. In order to do so, they prepare multidimensional data for the mining process according to four algorithms: VAvg, HAvg, WMAvg, and ModusFilter. These algorithms prune all rows in the fact table which have less than the average quantity and provide an *initialized table*. This table is next used for mining both non-repetitive predicate and repetitive predicate association rules.

## Discussion and the Position of our Proposal

The previous work on mining association rules in multidimensional data can be studied and compared according to various aspects.

As shown in Table 1, most of the proposals are designed and validated for sales data cubes. Their applications are therefore inspired by the well-known *basket market analysis* problem (*BMA*) driven on transactional databases. Nevertheless, we believe that most of the proposals (except for the proposals of Chen et al. (2000) and Nestorov et al. (2003)) can easily be extended to other application domains. Our approach covers a wide spectrum of application domains. It depends neither on a specific domain nor on a special context of data.

Almost all the previous proposals are based on the frequency of data, by using the COUNT measure, in order to compute the support and the confidence of the discovered association rules. As indicated earlier, Imieliński et al. (2002) can exploit any measure to detect *cubegrades*. Nevertheless, the authors do not compute the support

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and the confidence of the produced *cubegrades*. Tjioe et al. (2005) use the average (AVG) of measures in order to prune uninteresting itemsets in a pre-processing step. However, in the mining step, they only exploit the COUNT measure to compute the support and the confidence of association rules. Our approach revisits the support and the confidence of association rules when SUM-based aggregates are used.

According to Table 2, some of the proposals mine inter-dimensional association rules, whereas others deal with intra-dimensional rules. In general, an inter-dimensional association rule relies on more than one dimension from the mined data cube and consists of non-repetitive predicates, where the instance of each predicate comes from a distinct dimension. An intra-dimensional rule relies rather on a single dimension. It is constructed within repetitive predicates where their instances represent modalities from the considered dimension. Nevertheless, a *cubegrade* (Imieliński et al., 2002), or a *constrained gradient* (Dong et al., 2001), can be viewed as an inter-dimensional association rule which has repetitive predicates. The instances of these predicates can be redundant in both the head and the body of the implication. Furthermore, the proposal of Tjioe et al. (2005) is mostly the only one which allows the mining of inter-dimensional association rules with either repetitive or non-repetitive predicates. In our proposal, we focus on the mining of inter-dimensional association rules with non-repetitive predicates.

We note that, except for (Kamber et al., 1997; Zhu, 1998), most of the previous proposals try to exploit the hierarchical aspect of multidimensional data by expressing associations according to multiple levels of abstractions. For example, a *cubegrade* is an association which can be expressed within multiple levels of granularity. Association rules in Chen et al. (2000) also exploit dimension hierarchies. In our case, the definition of the context in the meta-rule can be set to a given level of granularity.

*Table 1. Comparison of association rule mining proposals from multidimensional data across application domain, data representation, and measure*

| Proposal | Application domain | | Mined data structure | | Measure | |
|---|---|---|---|---|---|---|
| | General | BMA | Tabular | MD | COUNT | All measures |
| Kamber, Han and Chiang (1997) | • | | | • | • | |
| Zhu (1998) | • | | • | | • | |
| Imieliński, Khachiyan and Abdulghani (2002) | • | | • | | | • |
| Dong, Han, Lam, Pei and Wang (2001) | • | | • | | | • |
| Chen, Dayal and Hsu (2000) | | • | • | | • | |
| Nestorov and Jukić (2003) | | • | • | | • | |
| Tjioe and Taniar (2005) | • | | • | | • | |
| Our proposal | • | | | • | | • |

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*Table 2. Comparison of association rule mining proposals from multidimensional data across dimension, level, and predicate*

| Proposal | Dimension | | Hierarchy | | Predicate | |
|---|---|---|---|---|---|---|
| | Intra | Inter | Single | Multiple | Repetitive | Non-repetitive |
| Kamber, Han and Chiang (1997) | | • | • | | | • |
| Zhu (1998) | • | • | • | | • | • |
| Imieliński, Khachiyan and Abdulghani (2002) | | • | | • | • | |
| Dong, Han, Lam, Pei and Wang (2001) | | • | | • | • | |
| Chen, Dayal and Hsu (2000) | • | | | • | • | |
| Nestorov and Jukić (2003) | • | | • | | • | |
| Tjioe and Taniar (2005) | • | • | | • | • | • |
| Our proposal | | • | | • | | • |

*Table 3. Comparison of association rule mining proposals from multidimensional data across user interaction, formalization, and association exploitation*

| Proposal | User interaction | | Formalization | | Association exploitation | |
|---|---|---|---|---|---|---|
| | Yes | No | Yes | No | Yes | No |
| Kamber, Han and Chiang (1997) | • | | | • | | • |
| Zhu (1998) | • | | | • | • | |
| Imieliński, Khachiyan and Abdulghani (2002) | • | | • | | | • |
| Dong, Han, Lam, Pei and Wang (2001) | • | | • | | | • |
| Chen, Dayal and Hsu (2000) | | • | | • | | • |
| Nestorov and Jukić (2003) | • | | | • | | • |
| Tjioe and Taniar (2005) | • | | | • | | • |
| Our proposal | • | | • | | • | |

According to Table 3, we note that the proposal of Chen et al. (2000) does not consider any interaction between users and the mining process. In fact, in the proposed *Web infrastructure*, analysis objectives are already predefined over transactional data and therefore users can not interfere with these objectives. In Kamber et al. (1997), user's needs are expressed through the definition of a meta-rule.

Except for *cubegrades* (Imieliński et al., 2002) and *constrained gradients* (Dong et al., 2001), almost all proposals miss a theoretical framework which establishes a general formalization of the mining process of association rules in multidimensional data.

In addition, in all these proposals, Zhu (1997) is mostly the only one who proposes association rule visualization. Nevertheless, the proposed graphical representation is similar to the ones commonly used in traditional association rules mining, and hence does not take into account multidimensionality.

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*OLEMAR* is entirely driven by user's needs. It uses meta-rules to meet the analysis objectives. It is also based on a general formalization of the mining process of inter-dimensional association rules. Moreover, we include a visual representation of rules based on the graphic semiology principles.

## Conclusion, Discussion, and Perspectives

In this chapter, we design an online environment for mining inter-dimensional association rules from data cubes as a part of a platform called *CubeMining*. We use a guided rule mining facility, which allows users to limit the mining process to a specific context defined by a particular portion in the mined data cube. We also provide a computation of the support and the confidence of association rules when a SUM-based measure is used. This issue is quite interesting since it expresses associations which do not restrict users' analysis to associations driven only by the traditional COUNT measure. The support and the confidence may lead to the generation of large number of association rules. Therefore, we propose to evaluate interestingness of mined rules according to two additional descriptive criteria (lift and loevinger). These criteria can express the relevance of rules in a more precise way than what is offered by the support and the confidence. Our association rule mining procedure is an adaptation of the traditional level-wise *apriori* algorithm to multidimensional data. In order to make extracted knowledge easier to interpret and exploit, we provide a graphical representation for the visualization of inter-dimensional association rules in the multidimensional space of the mined data cube. Empirical analysis showed the efficiency of our proposal and the acceptable runtime of our algorithm.

In the current development of our mining solution, we integrate SUM-based measures in the computation of interestingness criteria of extracted association rules. However, this choice assumes that the selected measure is additive and has only positive values. In the suspicious regions data cube, the *surface of regions* is an appropriate measure for the computation of the revisited criteria. Nevertheless, the *total boundary length* of regions can not be used for that computation since the SUM of boundary lengths does not make concrete sense. In some cases, an OLAP context may be expressed by facts with non-additive or negative measures. For instance, in the traditional example of a sales data cube, the *average of sales* is typically a non-additive measure. Furthermore, the *profit of sales* is also an OLAP measure that can have negative values. In such situations, we obviously need a more appropriate interestingness estimation of association rule to handle a wider spectrum of measure types and aggregate functions (e.g., AVG, MAX).

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Our proposal provides inter-dimensional association rules with non-repetitive predicates. Such rules consist of a set of predicate instances where each one represents a modality coming from a distinct dimension. This kind of association rules helps explain a value of a dimension by other values drawn from other dimensions. Nevertheless, an inter-dimensional association rule does not explain a modality by other ones from the same dimension. For instance, the latter type of rules is not able to explain the sales of a *product* by those of other *products* or even other *product categories*. In order to cope with this issue, we also need to extend our proposal in order to cover the mining of inter-dimensional association rules *with repetitive predicates* as well as *intra-dimensional* association rules. In addition, these new kinds of associations should profit from dimension hierarchies and allow modalities from multiple granularity levels.

The association rule mining process in our environment is based on an adaptation of the traditional level-wise *apriori* algorithm to multidimensional data. The *anti-monotony* property (Agrawal et al., 1993) allows a fast search of frequent itemsets, and the guided mining of association rules we express as a meta-rule limits the search space according to the analysis objectives of users. However, some recent studies have shown the limitations of *Apriori* and privileged the notion of frequent closed itemsets like in *close* (Pasquier, Bastide, Taouil, & Lakhal, 1999), *pascal* (Bastide, Taouil, Pasquier, Stumme, & Lakhal, 2000), *closet* (Pei, Han, & Mao, 2000), *Charm* (Zaki & Hsiao, 2002), and *galicia* (Valtchev, Missaoui & Godin, 2004).

Finally, measures are used in our environment for computing interestingness criteria. We plan to study the semantics of association rules when measures appear in the expression of rules.

# References

Agrawal, R., Imieliński, T., & Swami, A. (1993, May). Mining association rules between sets of items in large databases. In *Proceedings of the ACM SIGMOD International Conference on Management of Data (SIGMOD 1993)* (pp 207-216). Washington, DC.

Bastide, Y., Taouil, R., Pasquier, N., Stumme, G., & Lakhal, L. (2000). Mining frequent patterns with counting inference. *SIGKDD Explor. Newsl., 2,* 66-75.

Ben Messaoud, R., Boussaid, O., & Loudcher, R. S. (2006a). Efficient multidimensional data representation based on multiple correspondence analysis. In *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2006)*, Philadelphia (pp. 662-667).

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Ben Messaoud, R., Boussaid, O., & Loudcher, R. S. (2006b). A data mining-based OLAP aggregation of complex data: Application on XML documents. *International Journal of Data Warehousing and Mining*, *2*(4), 1-26.

Ben Messaoud, R., Loudcher, R. S., Boussaid, O., & Missaoui, R. (2006). Enhanced mining of association rules from data cubes. In *Proceedings of the 9th ACM International Workshop on Data Warehousing and OLAP (DOLAP 2006)* (pp. 11-18). Arlington, VA.

Bertin, J. (19981). *Graphics and graphic information processing*. de Gruyter.

Brin, S., Motwani, R., & Silverstein, C. (1997). Beyond market baskets: Generalizing association rules to correlations. In *Proceedings of the ACM SIGMOD International Conference on Management of Data (SIGMOD 1997)* (pp 265-276).

Chaudhuri. S., & Dayal, U. (1997). An overview of data warehousing and OLAP technology. *SIGMOD Record*, *26*(1), 65-74.

Chen, Q., Dayal. U., & Hsu, M. (2000). An OLAP-based scalable Web access analysis engine. In *Proceedings of the 2nd International Conference on Data Warehousing and Knowledge Discovery (DaWaK 2000)* (pp. 210-223). London.

Chen, Q., Dayal. U., & Hsu. M. (1999). A distributed OLAP infrastructure for e-commerce. In *Proceedings of the 4th IECIS International Conference on Cooperative Information Systems (COOPIS 1999)* (pp. 209-220). Edinburgh, Scotland.

Dong, G., Han, H., Lam, J. M. W., Pei, J., & Wang, K. (2001). Mining multi-dimensional constrained gradients in data cubes. In *Proceedings of the 27th International Conference on Very Large Data Bases (VLDB 2001)* (pp. 321-330). Rome, Italy.

Goil, S., & Choudhary, A. (1998). High performance multidimensional analysis and data mining. In *Proceedings of the 1st International Workshop on Data Warehousing and OLAP (DOLAP 1998)* (pp. 34-39). Bethesda, Maryland.

Heath, M., Bowyer, K., Kopans, D., Moore, R., & Jr, P.K. (2000). The digital database for screening mammography. In *Proceedings of the 5th International Workshop on Digital Mammography*, Toronto, Canada.

Imieliński, T., Khachiyan, L., & Abdulghani, A. (2002). Cubegrades: Generalizing association rules. *Data Mining and Knowledge Discovery*, *6*(3), 219-258.

Inmon, W. H. (1996). *Building the data warehouse*. John Wiley & Sons.

Kamber, M., Han, J., & Chiang, J. (1997). Multi-dimensional association rules using data cubes. In *Proceedings of the 3rd International Conference on Knowledge Discovery and Data Mining (KDD 1997)* (pp. 207-210). Newport Beach, CA.

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Kimball, R. (1996). *The data warehouse toolkit*. John Wiley & Sons.

Lallich, S., Vaillant, B., & Lenca, P. (2005). Parametrised measures for the evaluation of association rules interestingness. In *Proceedings of the 6th International Symposium on Applied Stochastic Models and Data Analysis (ASMDA 2005)* (pp. 220-229). Brest, France.

Lenca, P., Vaillant, B., & Lallich, S. (2006). On the robustness of association rules. In *Proceedings of the 2006 IEEE International Conference on Cybernetics and Intelligent Systems (CIS 2006)* (pp. 596-601). Bangkok, Thailand.

Loevinger, J. (1974). A systemic approach to the construction and evaluation of tests of ability. *Psychological Monographs*, *61*(4).

Nestorov, S., & Jukić, N. (2003). Ad-hoc association-rule mining within the data warehouse. In *Proceedings of the 36th Hawaii International Conference on System Sciences (HICSS 2003)* (pp. 232-242).

Pasquier, N., Bastide, Y., Taouil, R., & Lakhal, L. (1999). Efficient mining of association rules using closed itemset lattices. *Information Systems*, *24*(1), 25-46.

Pei, J., Han, J., & Mao, R. (2000). Closet: An efficient algorithm for mining frequent closed itemsets. In *Proceedings of the ACM SIGMOD International Workshop on Data Mining and Knowledge Discovery (DMKD 2000)* (pp. 21-30). Dallas, Texas.

Tjioe, H. C., & Taniar, D. (2005). Mining association rules in data warehouses. *International Journal of Data Warehousing and Mining*, *1*(3), 28-62.

Valtchev, P., Missaoui, R., & Godin, R. (2004). Formal concept analysis for knowledge and data discovery: New challenges. In *Proceedings of the 2nd International Conference on Formal Concept Analysis (ICFCA 2004)* (pp. 352-371).

Zaki, M. J., & Hsiao, C. J. (2002). CHARM: An efficient algorithm for closed itemset mining. In *Proceeding of the 2nd SIAM International Conference on Data Mining (SDM'02)*, Arlington, VA.

Zhu, H. (1998). *Online analytical mining of association rules*. Master's thesis, Simon Faster University, Burnaby, British Columbia, Canada, December.

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## Chapter II

# Current Interestingness Measures for Association Rules:
## What Do They Really Measure?

Yun Sing Koh, Auckland University of Technology, New Zealand
Richard O'Keefe, University of Otago, New Zealand
Nathan Rountree, University of Otago, New Zealand

## Abstract

*Association rules are patterns that offer useful information on dependencies that exist between the sets of items. Current association rule mining techniques such as apriori often extract a very large number of rules. To make sense of these rules we need to order or group the rules in some fashion such that the useful patterns are highlighted. The study of this process involves the investigation of an "interestingness" in the rules. To date, various measures have been proposed but unfortunately, these measures present inconsistent information about the interestingness of a rule. In this chapter, we show that different metrics try to capture different dependencies among variables. Each measure has its own selection bias that justifies the rationale for preferring it compared to other measures. We present an experimental study of the behaviour of the interestingness measures such as lift, rule interest, Laplace, and information gain. Our experimental results verify that many of these measures*

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*are very similar in nature. From the findings, we introduce a classification of the current interestingness measures.*

## Introduction

Interestingness measures are divided into two types: objective and subjective measures. Objective measures are based on probability, statistics, or information theory. They use a data driven approach to assess the interestingness of a rule. They are domain independent and require minimal user participation. Objective measures emphasise conciseness, generality, reliability, peculiarity, or diversity of the rules found. Some objective measures are symmetric with respect to the permutation of items, while others are not. From an association rule mining perspective, symmetric measures are often used for itemsets whereas asymmetric measures are applied to rules. Using these measures each association rule is treated as an isolated rule and they are not compared against each other. Subjective measures take into account both the data and the user of these data. Hence, subjective measures require access to domain knowledge on the data. These measures determine whether a rule is novel, actionable, and surprising. A rule is interesting if it is both surprising and actionable. However, this is a highly subjective view as actionable is determined by both the problem domain and the user's goals (Silberschatz & Tuzhilin, 1995).

In this chapter, we only concentrate on objective measures, as they do not need expert domain knowledge. A large number of rules may be extracted as we lower the minimum support threshold or increase the number of items in the database. For this reason, the number of possible association rules grows exponentially with the number of items and the complexity of the rules being considered. For this reason, objective measures are used to rank, order, and prune the rules for presentation.

Currently there are more than 50 objective measures proposed and at present, there are a number of reviews conducted to make sense of the interestingness measures for association rules (Geng & Hamilton, 2006; McGarry, 2005; Tan & Kumar, 2000). Here we make two major contributions. We present a new visualisation technique to visualise and evaluate the current objective measures and also discuss the suitability of these measures in detecting meaningful rules.

Most objective measures are probability based. They are normally functions of a 2×2 contingency table. Table 1 shows the contingency table for $A \rightarrow B$ in dataset $D$. Here $n(AB)$ denotes the number of transactions containing both $A$ and $B$ in dataset $D$ or count($AB$,$D$). $N$ denotes the total number of transactions or $|D|$. For this pur-

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*Table 1. 2 × 2 contingency for rule A → B*

| Item | $A$ | $\neg A$ | Total |
|---|---|---|---|
| $B$ | $n(AB)$ | $n(\neg AB)$ | $n(B)$ |
| $\neg B$ | $n(A\neg B)$ | $n(\neg A\neg B)$ | $n(\neg B)$ |
| Total | $n(A)$ | n($\neg A$) | $N$ |

pose we use $\Pr(A) = \frac{n(A)}{N}$ to denote the probability of A. $\Pr(B|A)$ is the conditional probability of $B$ given A, $\frac{n(AB)}{n(A)}$.

Considering that there are now more than 50 objective measures proposed to find useful patterns with association mining, it is justifiable that some research be conducted to understand and analyse the functions of these measures. We note that all these measures find different properties within a rule interesting.

# Evaluation of Other Objective Measures

In this section, we will discuss some previous experimental studies. We then propose a visual approach to evaluating these measures. Currently there is a very large number of objective measures. For comparison purposes we have limited the set of measures to consist of the commonest objective measures discussed in previous interestingness measure literature (Huynh, Guillet, & Briand, 2005; Lenca, Meyer, Vaillant, & Lallich, 2004; Tan, Kumar, & Srivastava, 2004; Vaillant, Lenca, & Lallich, 2004).

## Related Work

Many experimental studies have been conducted to analyse the usage of these measures. Here we discuss the analyses conducted by previous research carried out in this area of study.

A comparative study of 21 objective measures was carried out by Tan et al. (2004). They suggest that the rankings of the measures become highly correlated when support based pruning is used. Tan et al. (2004) proposed five properties to evaluate an objective measure, M, based on operations for 2×2 contingency tables. They suggest that a good measure should have the following five properties.

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- **Property 1:** $M$ is symmetric under variable permutations. This property states that the rules $A \rightarrow B$ and $B \rightarrow A$ should have the same interestingness value. This is not true for many existing measures. For example, confidence violates this property. Confidence is an asymmetrical measure. It is the conditional probability of $B$ occurring, given $A$. To cater for this property, Tan et al. (2004) transformed every asymmetrical measure $M$ to a symmetrical measure by taking the maximum value produced by $M$ on both $A \rightarrow B$ and $B \rightarrow A$. For example the symmetrical confidence value is calculated as $\max(\Pr(A|B), \Pr(B|A))$.
- **Property 2:** $M$ is invariant under row and column scaling. This property states that the rule $A \rightarrow B$ should have the same $M$ value when we scale any row or column by a positive factor. Odds ratio, Yule's Y, and Yule's Q are examples of measures that follow this property (Tan et al., 2004).
- **Property 3:** A normalised measure $M$ is antisymmetric when the rows or columns are permuted. A normalised measure has values ranging between −1 and +1. In this property, swapping within the rows or columns in the contingency table makes interestingness values change their signs. For example, $M$ should become $-M$ if the rows or columns are permuted.
- **Property 4:** $M$ is invariant under the inversion operation. Inversion is a case of row and column permutation. In this process both rows and columns are swapped simultaneously. In this property $M$ should remain the same if both the rows and columns are permuted. It states that $M(A \rightarrow B) = M(\neg A \rightarrow \neg B)$.
- **Property 5:** A binary measure is null invariant when adding transactions that do not contain $A$ and $B$ does not change $M$. In this property, $M$ has no relationship with the number of transactions that do not contain $A$ and $B$.

We note that these properties are:

- Plausible for associations, but highly implausible for rules
- Impossible for k-valued variables

They proposed a method to rank measures based on a specific dataset. In this method an expert is first required to rank a set of mined patterns. Then the measures with similar ranking are selected. However this is only suitable for a small set of rules, and is not directly applicable if the set of rules is large. In this case, the method attempts to capture rules with the highest conflicts in the rankings done by the selected measures. It means the rules which are ranked in a different order by different measures are presented to the experts for ranking. The method then selects the measure that gives a ranking closest to the manual ranking. The disadvantage of this method is that it requires expert domain knowledge and also expert involvement.

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Lenca et al. (2004) introduced the use of a multicriteria decision aid method for objective measure selection. They propose an initial array of 20 objective measures evaluated based on 8 different evaluation criteria. Using the list of evaluation criteria, they analyse the objective measures.

In this approach, weights are assigned to each property that the user considers to be of importance. Then a decision matrix is created, wherein each row represents a measure and a column represents a property. An entry in the matrix represents the weight for the measure according to the property. For example if an asymmetric property is needed, the measure is assigned 1 if it is asymmetric, and 0 if it is symmetric. Then applying the multicriteria decision process would generate a ranking of results. Unlike the method proposed by Tan et al. (2004), this method does not require the mined patterns to be ranked. Instead the user must be able to identify the desired properties of a rule.

Lenca et al. (2004) use eight properties to evaluate an objective measure, *M*, given a rule $A \rightarrow B$.

- **Property 1:** *M* should be asymmetric. In this property, it is desirable to make a distinction between measures that evaluate $A \rightarrow B$ differently from $B \rightarrow A$, and those which do not.
- **Property 2:** *M* decreases as the number of transactions containing *B* but not *A* increases.
- **Property 3:** *M* is constant if *A* and *B* are statistically independent. The value at independence should be constant and independent of the marginal frequencies.
- **Property 4:** *M* is constant if there is no counterexample. This property states a rule with a confidence value of 1 should have the same interestingness value regardless of its support.
- **Property 5:** *M* decreases with $\Pr(A\neg B)$ in a linear, concave, or convex fashion and with $\Pr(A\neg B)$ around 0+. A concave decrease with $\Pr(A\neg B)$, reflects the ability to tolerate a few counterexamples without a significant loss of interest. A convex decrease around $\Pr(A\neg B)$ increases the sensitivity to a false positive.
- **Property 6:** *M* increases as the total number of transactions increases. This property describes the changes assuming that $\Pr(A)$, $\Pr(B)$, and $\Pr(AB)$ are held constant. Measures that are sensitive to the number of records are called statistical measures while those that are not affected by the number of records are called descriptive measures.
- **Property 7:** The threshold used to identify interesting from uninteresting rules is easy to fix. This property states that when a threshold is used to identify interesting from uninteresting rules, it should be easy to locate.

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- **Property 8:** The semantics of the measure are easy to express. This property denotes the ability of *M* to express a comprehensible idea of the interestingness of a rule.

Despite all this, the effectiveness of this method is dependent on the list of evaluation criteria which is clearly not exhaustive. New criteria may lead to a better distinction between measures, which are similar.

Vaillant et al. (2004) proposed a method for analysing measures by clustering the measures into groups. Like the previous approaches, the clustering methods are based on either the ruleset generated by the experiments or the properties of the measure. There are two types of clustering: property based clustering and experiment based clustering. Property based clustering groups measures based on the similarity of the measure. In experiment based clustering, measures are considered similar if they produce similar interestingness values on the same set of association rules. They compared the behaviour of 20 measures on 10 datasets. They used a pre-order agreement coefficient based on Kendall's $\tau$ for synthetic comparison of the rankings of the rules by two given measures. Based on the datasets they generated 10 pre-ordered comparison matrices. They identified four main groups of measures. But there are some differences in the results depending on which database is considered.

Huynh et al. (2005) introduced ARQUAT, a new tool to study the specific behaviour of 34 objective measures using a specific dataset and from an exploratory data analysis perspective. ARQUAT has five task oriented groups: rule analysis, correlation, and clustering analysis, sensitivity analysis and comparative analysis. In rule analysis, the tool summarises some simple statistics in the rule set structure, whereas in the correlation analysis, the correlations between the objective measures were computed in the pre-processing stages by using Pearson's correlation function. To make better sense, the rules are clustered together based on the best rules given for each measure.

Despite the fact that there have been many exploratory analyses carried out on these measures, a way to compare the behaviour of the objective measures effectively has still not been found. Our main objective is to help users select an appropriate objective measure with regards to goals, preferences, and properties of the measure. In the next section, we suggest a visualisation model based on the characteristics of the rules analysed.

## Framework for Visualising Objective Measures

To make sense of these measures we took a visualisation approach. We introduced a new visualisation framework, which was used on the results from each measure. An experimental study of the behaviour of the objective measures was conducted.

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*Table 2. List of objective measures*

| Objective Measure | Formula |
|---|---|
| Added Value | $\max(\Pr(B\|A) - \Pr(B), \Pr(A\|B) - \Pr(A))$ |
| Certainty Factor | $\begin{cases} \frac{\Pr(B\|A) - \Pr(B)}{1 - \Pr(B)}, \text{if } \Pr(B\|A) > \Pr(B); \\ \frac{\Pr(B\|A) - \Pr(B)}{\Pr(B)}, \text{if } \Pr(B\|A) < \Pr(B); \text{and} \\ 0 \text{ otherwise.} \end{cases}$ |
| Collective Strength | $\frac{(\Pr(AB) + \Pr(\neg A\neg B))(1 - \Pr(A)\Pr(B) - \Pr(\neg A)\Pr(\neg B))}{\Pr(A)\Pr(B) + \Pr(\neg A)\Pr(\neg B)(1 - \Pr(AB) - \Pr(\neg A\neg B))}$ |
| Confidence | $\Pr(B\|A)$ |
| Conviction | $\frac{\Pr(A)\Pr(\neg B)}{\Pr(A\neg B)}$ |
| Cosine | $\frac{\Pr(AB)}{\sqrt{\Pr(A)\Pr(B)}}$ |
| Descriptive Confirmed-Conf | $1 - 2\frac{\Pr(A\neg B)}{\Pr(A)}$ |
| Example & Contra-Example | $1 - \frac{\Pr(A\neg B)}{\Pr(A) - \Pr(A\neg B)}$ |
| Ganascia | $2\Pr(B\|A) - 1 = 2(\text{Confidence}) - 1$ |
| Gini | $\Pr(A)(\Pr(A\|B)^2 + \Pr(A\|\neg B)^2) + \Pr(\neg A)(\Pr(B\|\neg A)^2 + \Pr(\neg B\|\neg A)^2) - \Pr(B)^2 - \Pr(\neg B)^2$ |
| Information Gain | $\log\frac{\Pr(AB)}{\Pr(A)\Pr(B)} = \log(\text{lift})$ |

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*Table 2. continued*

| Objective Measure | Formula |
|---|---|
| Jaccard | $\frac{Pr(AB)}{Pr(A) + Pr(B) - Pr(AB)}$ |
| J-measure | $Pr(AB)\log\frac{Pr(AB)}{Pr(A)Pr(B)} + Pr(A\neg B)\log\frac{Pr(A\neg B)}{Pr(A)Pr(\neg B)}$ |
| Cohen's Kappa | $\frac{Pr(AB) + Pr(\neg A\neg B) - Pr(A)Pr(B) - Pr(\neg A)Pr(\neg B)}{1 - Pr(A)Pr(B) - Pr(\neg A)Pr(\neg B)}$ |
| Klösgen | $\sqrt{Pr(AB)}\max(Pr(B \mid A) - Pr(B), Pr(A \mid B) - Pr(A))$ $= \sqrt{Pr(AB)}.\text{Added Value}$ |
| Loevinger | $1 - \frac{Pr(A\neg B)}{Pr(A)Pr(\neg B)} = 1 - \frac{1}{\text{Conviction}}$ |
| Lift | $\frac{Pr(AB)}{Pr(A)Pr(B)}$ |
| Mutual Information | $Pr(AB)\log\frac{Pr(AB)}{Pr(A)Pr(B)} + Pr(A\neg B)\log\frac{Pr(A\neg B)}{Pr(A)Pr(\neg B)}$ $+ Pr(\neg AB)\log\frac{Pr(\neg AB)}{Pr(\neg A)Pr(B)} + Pr(\neg A\neg B)\log\frac{Pr(\neg A\neg B)}{Pr(\neg A)Pr(\neg B)}$ |
| Odds Multiplier | $\frac{Pr(A) - Pr(A\neg B)Pr(\neg B)}{Pr(B)Pr(A\neg B)}$ |
| Odds Ratio | $\frac{Pr(AB)Pr(\neg A\neg B)}{Pr(A\neg B)Pr(\neg AB)}$ |
| Pavillion | $Pr(\neg B) - \frac{Pr(A\neg B)}{Pr(A)}$ |
| $\phi$ - Coefficient | $\frac{Pr(AB) - Pr(A)Pr(B)}{\sqrt{Pr(AB)Pr(\neg A)Pr(\neg B)}}$ |
| Rule Interest (RI) | $Pr(AB) - Pr(A)Pr(B)$ |
| Yule's Q | $\frac{Pr(AB)Pr(\neg A\neg B) - Pr(A\neg B)Pr(\neg AB)}{Pr(AB)Pr(\neg A\neg B) + Pr(A\neg B)Pr(\neg AB)}$ |
| Yule's Y | $\frac{\sqrt{Pr(AB)Pr(\neg A\neg B)} - \sqrt{Pr(A\neg B)Pr(\neg AB)}}{\sqrt{Pr(AB)Pr(\neg A\neg B)} + \sqrt{Pr(A\neg B)Pr(\neg AB)}}$ |

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*Table 3. Contingency table for A and B*

| | sup($A$,$D$) | sup($\neg A$,$D$) |
|---|---|---|
| sup($B$,$D$) | $T$ | $R$ |
| sup($\neg B$,$D$) | $L$ | $N$ |

Due to the large number of measures, we have limited the research to the measures shown in Table 2. Some of the measures proposed are direct monotone functions of $M$. We suggest that if $M$ is explored as $\beta_0 + \beta_1 M$, any monotone function of $M$ need not be explored, as it will produce the same ranking as $M$. For example. Ganascia is a monotone function of confidence (Ganascia = 2(conf) – 1), and as such will rank rules in the same way as confidence. However for completeness, we discuss all the measures in Table 2 as they have been published in previous literature.

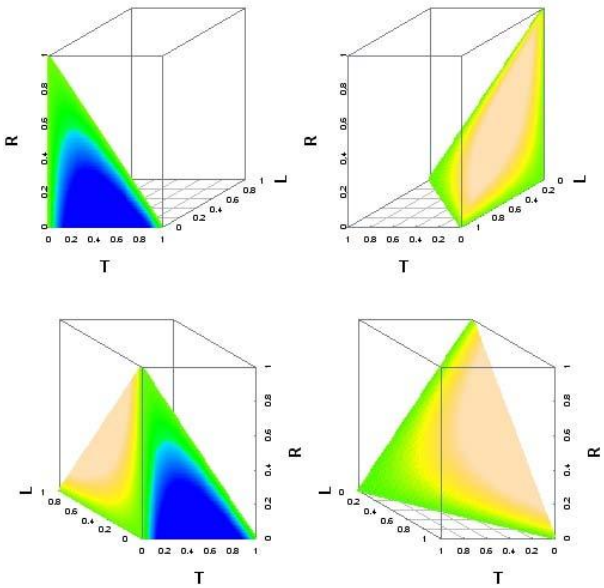
In general, the relationship between two binary variables $A$ and $B$ in $A \rightarrow B$, can be tabulated in Table 3. We generated results given the range of Pr($A$), Pr($B$), and Pr($AB$) from 0 to 1. From the results produced from each measure, we then created a three dimensional plot of the results of the objective measures. The x-axis contains values of $T$, y-axis contains values of $R$, and z-axis contains values of $L$. Here $T$ = Pr($AB$), $L$ = Pr($A\neg B$), $R$ = Pr($\neg AB$) and $N$ = Pr($\neg A\neg B$). In this experiment we generated every combination possible for $A \rightarrow B$ in dataset $D$. Here sup($A$,$D$) ranged from 0 to 1 with an increment of 0.01, for each of these values sup($B$,$D$) ranged from 0 to 1 with an increment of 0.01, and sup($AB$,$D$) ranged from 0 to min(sup($A$,$D$), sup($B$,$D$)). We then calculate the values produced by the objective measures for each of the cases.

All the results produced by the objective measures have to be normalised between 0 and 1. As the results from some of the objective measures may tend towards $-\infty$ or $+\infty$, we had to normalise the results using the tenth percentile and ninetieth percentile. Any result below the tenth percentile is considered "not interesting" and results above ninetieth percentile is considered "interesting." In the plots, the darker shade represents the most interesting rules; it gradually changes to a lighter shade to represent uninteresting rules.

From the visualisation carried out some of the plots generated have similar shaded regions. They show that some rules with similar characteristics are more interesting when compared to the rest. We suggest that these measures should be categorised together as they find the same rules interesting, and thus they work in the same manner. The plots generated were categorised into different types based on the area of the plot, which they considered interesting. In order to make certain that they belonged within the same category we ran a Spearman's rank correlation test between the objective measures using the R statistical package.

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*Figure 1. Three dimensional plot for Rule Interest*



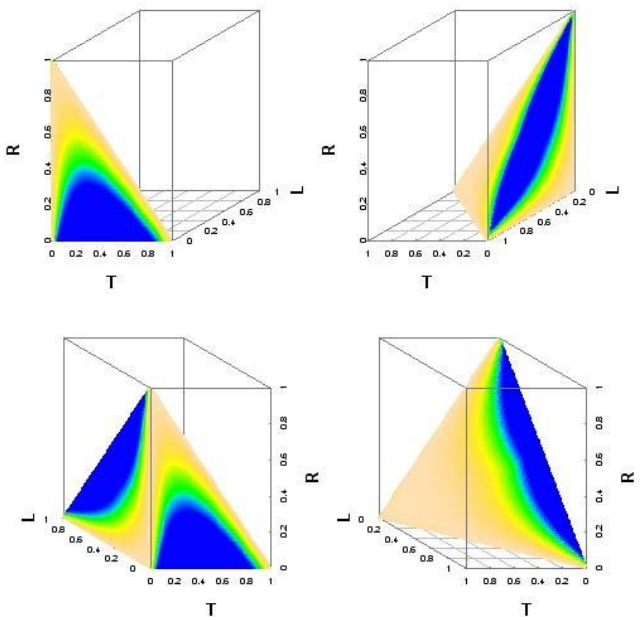

Spearman's rank correlation coefficient is a nonparametric (distribution free) rank statistic. It was proposed by Spearman in 1904 as a measure of the strength of the association between two variables, without making any assumptions about the frequency distribution of the variables (Lehmann, 1998). Like all other correlation coefficients, Spearman's rank correlation coefficient produces a value between −1.00 and +1.00. A positive correlation is one in which the ranks of both variables increase together. A negative correlation is one in which the rank of one variable increases as the rank of the other variable decreases. A correlation coefficient close to 0 means there is no linear relationship between the ranks. A correlation coefficient of −1.00 or +1.00 will arise if the relationship between the two variables is exactly linear. We ran Spearman's rank correlation on the results produced by two different objective

*Table 4. Correlation between objective measures (Type 1)*

| Measure | Collective Strength | Cohen's Kappa | Klösgen | Rule Interest |
|---|---|---|---|---|
| Collective Strength | 1.00 | 0 .99 | 0.95 | 0 .99 |
| Cohen's Kappa | 0.99 | 1.00 0 | .95 | 0.99 |
| Klösgen | 0.95 | 0 .95 | 1.00 | 0 .95 |
| Rule Interest | 0.99 | 0 .99 | 0.95 | 1 .00 |

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*Figure 2. Three dimensional plot for J-measure*




measures. Here we say that values close to +1.00 have a strong correlation because the objective measures rank the results in a similar fashion.

## Types of Objective Measures

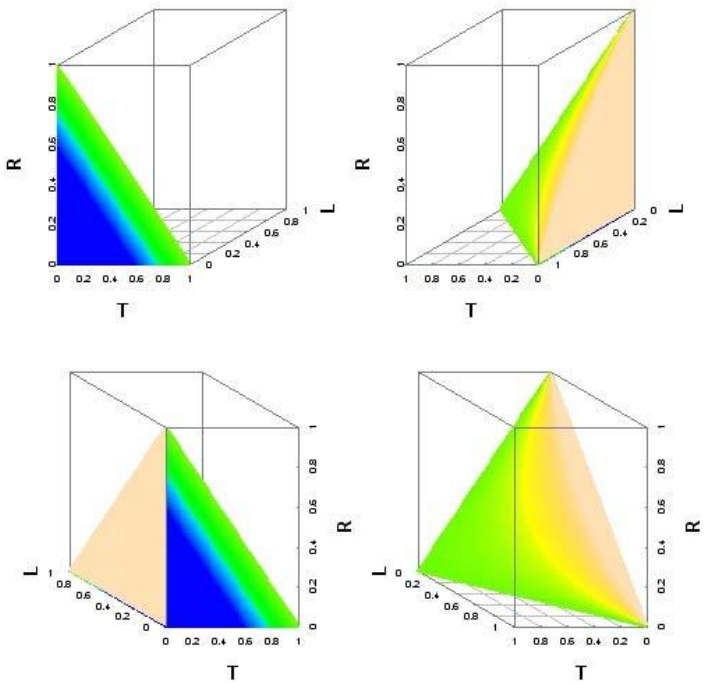
Each of the plots fitted into a particular category depending on the areas which were rated strongly by it. We found seven different categories, and in the following section we will take a closer look at each type of measure. The measures in each particular type produce plots which have similar areas shaded.

*Table 5. Correlation between objective measures (Type 2)*

| Measure | Gini | J-measure | Mutual Information |
|---|---|---|---|
| Gini | 1.00 | 0.97 | 1.00 |
| J-measure | 0.97 | 1.00 | 0.98 |
| Mutual Information | 1.00 | 0.98 | 1.00 |

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*Figure 3. Three dimensional plot for Lift*




## Type 1

The first type consist of Piateskty-Shapiro's rule interest (Freitas, 1999; Piatetsky-Shapiro, 1991), collective strength (Aggarwal & Yu, 2001), Cohen's Kappa (Tan, Kumar, & Srivastava, 2002), and Klösgen (1996).

This type of measure considers a particular point to be of specific interest. Figure 1 shows the plot produced by Piateskty-Shapiro's rule interest. Notice that the graph produced reflects that the area which is considered interesting (darker shade) seems to disperse from the middle of the *T*-axis. This type of objective measure considers a rule interesting if both its antecedent and consequent almost always occur together, and appear nearly half the time in the dataset (*T* is close to 0.5, *L* is close to 0, and *R* close to 0). This type of measure suggests that rules with support ≈ 0.5 with confidence ≈ 1 are more interesting than those that lie at either extreme end of the graph where $T \approx 0$ and $T \approx 1$. Table 4 details the correlation values between objective measures in Type 1. The correlation values ranged from 0.95 to 1.00.

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*Table 6. Correlation between objective measures (Type 3)*

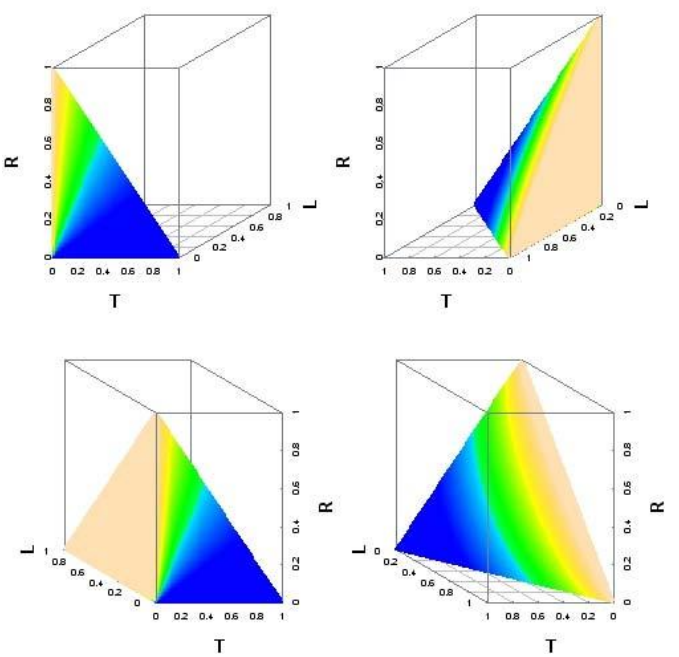
| Measure | Added Value | Information Gain | Lift |
|---|---|---|---|
| Added Value | 1.00 | 0.97 | 0.97 |
| Information Gain | 0.97 | 1.00 | 1.00 |
| Lift | 0.97 | 1.00 | 1.00 |

## Type 2

The second type of measure consists of J-measure (Smyth & Goodman, 1992), Gini, and mutual information. These measures consider two different points within the plot to be of specific interest. Figure 2 is the plot produced by J-measure.

This type of objective measure considers rules in two areas of the graph interesting. The first area corresponds to the previous interesting rules that Type 1 measures detect. The second area which this type of measure finds interesting is when both the antecedent and consequent individually have support ≈ 0.5 but they almost

*Figure 4. Three dimensional plot for cosine*



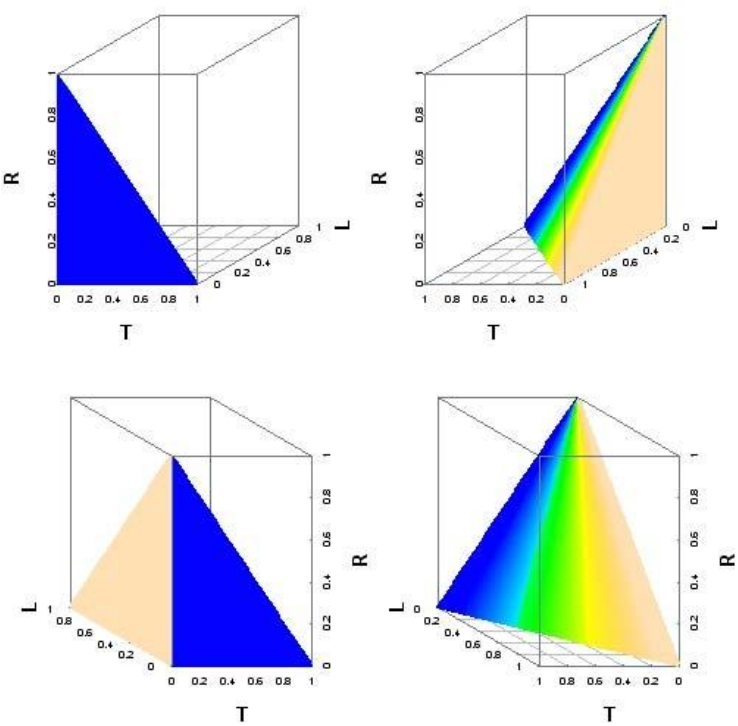

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*Table 7. Correlation between objective measures (Type 4)*

| Measure | Cosine | Jaccard |
|---|---|---|
| Cosine | 1.00 | 1.00 |
| Jaccard | 1.00 | 1.00 |

always do not occur together (*T* is close to 0, *L* is close to 0.5, and *R* is close to 0.5). This measure takes into consideration positive association as well as negative association. The first area corresponds to interesting positive frequent association rule mining whereas the second area shows negative association rules. Negative association rules identify itemsets that conflict with each other. An example of such rule is $\neg A \rightarrow \neg B$. The correlation results calculated using Spearman's rank correlation for Type 2 objective measures are shown in Table 5. In the results the correlation values are at least 0.97.

*Figure 5. Three dimensional plot for confidence*




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*Table 8. Correlation between objective measures (Type 5)*

| Measure | Confidence | Descriptive Confirm Confidence | Example & Contra Example | Ganascia |
|---|---|---|---|---|
| Confidence | 1.00 | 1.00 | 1.00 | 1.00 |
| Descriptive Confirm Confidence | 1.00 | 1 .00 | 1.00 | 1.00 |
| Example & Contra Example | 1.00 | 1 .00 | 1.00 | 1.00 |
| Ganascia | 1 .00 | 1.00 | 1.00 | 1.00 |

## Type 3

The third type of measure consists of added value (Tan et al., 2002), information gain (Vaillant et al., 2004), and lift.

Figure 3 shows the plot produced by lift. Note that when $T = 0$, the lift value would be 0, thus the side of the plot where $T = 0$ and $L = 0$ is shaded beige. For these measures, the rules with low support and whose constituent items almost always appear together ($T$ is closer to 0) seem more interesting. Theoretically these measures would be able to detect interesting infrequent rules but in reality they are not able to differentiate interesting low support rules from noise. Table 6 shows the correlations between measures in Type 6. All three measures are strongly correlated with the lowest correlation as 0.97.

## Type 4

Cosine (Tan et al., 2000) and Jaccard (Tan et al., 2002) fall into the fourth type of measure. Figure 4 shows the plot produced by cosine.

Here it considers rules that have a higher support more interesting. In this type of objective measure we concentrate on antecedents and consequents that appear together almost all the time. The rule is considered slightly more interesting when $T$ moves closer to 1. The area that is interesting is determined by both $R$ and $L$; as the values for a rule on the axis $R$ and $L$ increases, a rule is considered more interesting. This measure is suitable for finding interesting frequent rules. Table 7 shows the correlations between measures in Type 7. The correlation between Cosine and Jaccard is 1.00.

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## Type 5

The fifth type of measure consists of confidence, Ganascia (Ganascia, 1988; Lallich, Vaillant, & Lenca, 2005), descriptive confirmed-confidence (Huynh et al., 2005), and example and contra example (Lallich et al., 2005). Figure 5 shows the results produced by confidence. The plots produced by other measures discussed here are similar. This type of objective measure concentrates on antecedents and consequents that appear together almost all the time. A rule is considered slightly more interesting when *T* moves closer to 1. This type of interest measure considers rules with high support more interesting. But notice from the graph, that the interesting area is not affected by the *R* value. The area that is interesting is determined by *T* and *L*, when the values for the rules on the axis *T* and *L* increases the rules are considered more interesting.

We calculated the correlation value for these four measures shown in Table 8. We note that the correlation value for all these is 1.00.

## Type 6

The sixth type of measure consists of conviction (Brin, Motwani, Ullman, & Tsur, 1997), Loevinger (Lallich et al., 2005; Loevinger, 1947), odds multiplier (Lallich et al., 2005), and Pavilion (Huynh et al., 2005).

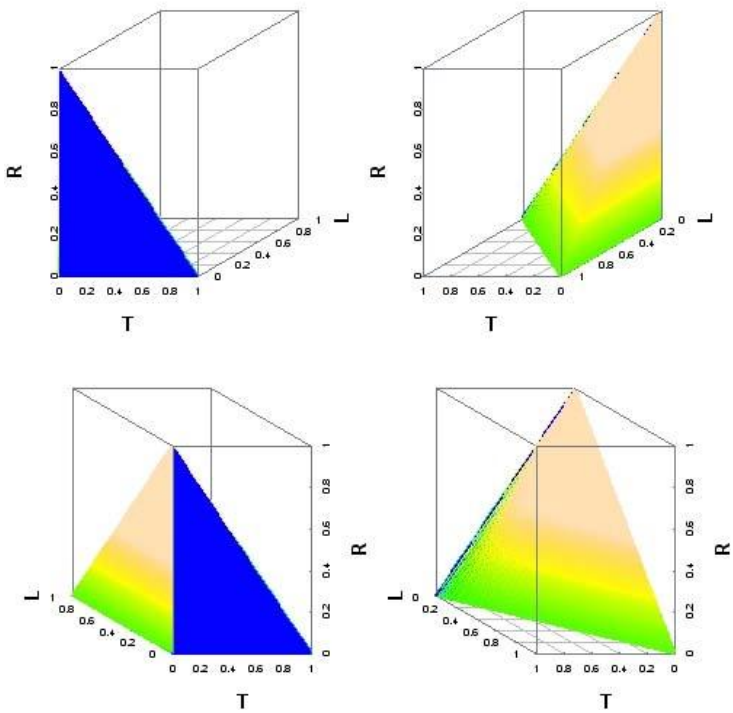
Figure 6 shows the results produced by Loevinger. The plots produced by other measures discussed here are similar. From the graph we notice that this type of measure finds rules that have both their antecedent and consequent occurring together interesting. But they also consider seeing the antecedent and not the consequent most of the time together interesting as well (noted by the green area when *R* is closer to 0). Like the previous measures, we calculated the correlation for these four measures shown in Table 9. We note that the correlations range between 0.96 and 1.00.

*Table 9. Correlation between objective measures (Type 6)*

| Measure | Conviction | Loevinger | Odds Multipler | Pavillion |
|---|---|---|---|---|
| Conviction | 1.00 | 1.00 | 0.96 | 0.96 |
| Loevinger | 1.00 | 1.00 | 0.96 | 0.96 |
| Odds Multipler | 0.96 | 0.96 | 1.00 | 0.97 |
| Pavillion | 0.96 | 0.96 | 0.97 | 1.00 |

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*Figure 6. Three dimensional plot for Loevinger*




## Type 7

The seventh type of measure consists of certainty factor (Berzal, Blanco, Sánchez, & Vila, 2001; Mahoney & Mooney, 1994), ϕ-coefficient (Gyenesei, 2001; Tan et al., 2000), Yule's Y (Reynolds, 1977), Yule's Q (Reynolds, 1977), and odds ratio (Reynolds, 1977). Figure 7 shows the results produced by certainty factor. The plots produced by other measures discussed here are similar. The entire area when either $L = 0$ or $R = 0$ was considered most interesting.

In this category of objective measure, we notice the measures consider two particular situations interesting. They consider it more interesting when either *L* or *R* is close to 0. This means that a rule is interesting when either the antecedent almost always appears with the consequent (but the consequent does not have to always appear with the antecedent) or when the consequent almost always appears with the antecedent. Here we calculated the correlation value for these five measures shown in Table 10. We note that the correlations range between 0.94 and 1.00.

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*Figure 7. Three dimensional plot for certainty factor*

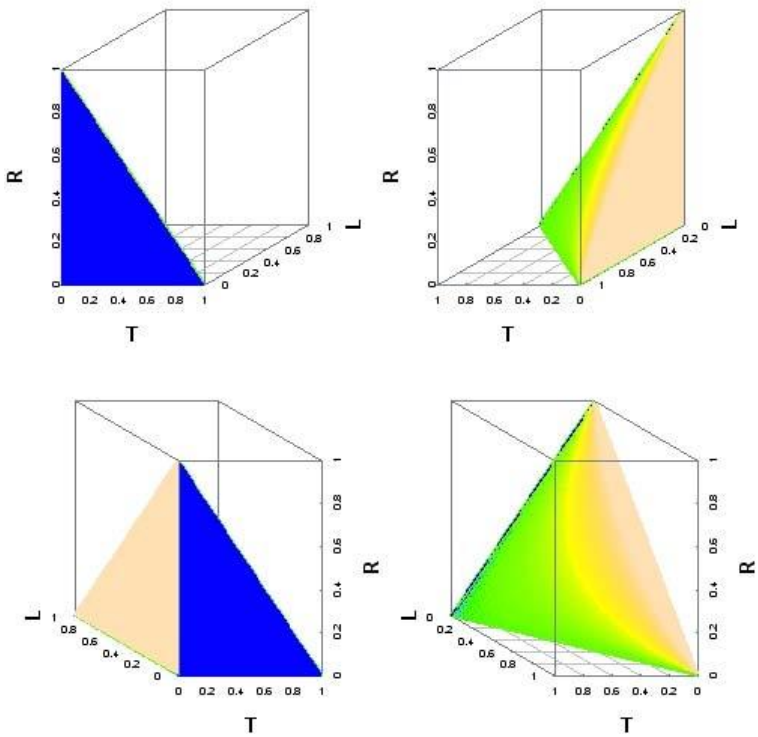
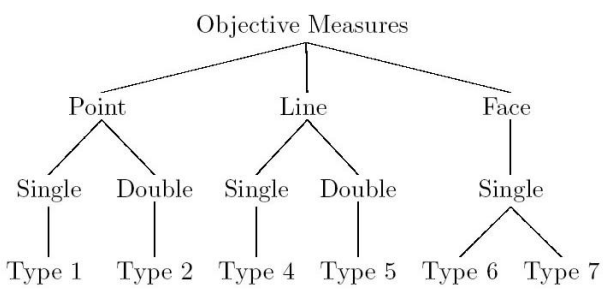



*Figure 8. Classification of objective measures*




**Type 1** Collective Strength, Cohen's Kappa, Klösgen, $\phi$-coefficient, Rule Interest

**Type 2** Gini, J-Measure, Mutual Information

**Type 3** Added Value, Information Gain, Lift

**Type 4** Cosine, Jaccard

**Type 5** Confidence, Descriptive Confirm-Confidence, Example & Contra Example, Ganascia

**Type 6** Conviction, Loevinger, Odds Multiplier, Pavilion

**Type 7** Certainty Factor, Odds Ratio, Yule's Q, Yule's Y

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*Table 10. Correlation between objective measures (Type 7)*

| Measure | Certainty Factor | Odds Ratio | f -coefficient | Yule's Q | Yule's Y |
|---|---|---|---|---|---|
| Certainty Factor | 1.00 | 0.94 | 0.94 | 0.94 | 0.94 |
| Odds Ratio | 0.94 | 1.00 | 0.95 | 1.00 | 1.00 |
| f -coefficient | 0.94 | 0.95 | 1.00 | 0.95 | 0.95 |
| Yule's Q | 0.94 | 1.00 | 0.95 | 1.00 | 1.00 |
| Yule's Y | 0.94 | 1.00 | 0.95 | 1.00 | 1.00 |

## Summary of Types of Objective Measures

Note that strong correlations exist between results from the measures within the same category. This means that all these measures rank the interestingness of particular rules in a similar order. Thus they work in the same manner.

Based on the results from the visualisation the objective measures were categorised into seven types. Each type defines a certain set of rules interesting. Figure 8 shows the classification of the objective measures. Our visualisation technique defines a "pyramid" of space in which an interestingness measure can exist at some value. We therefore classify the measures according to what parts of the pyramid register as "high" interest. "Points" refers to that group of measures that cause (one or two) points on the pyramid to have high interest; "Lines" refers to (one or two) edges. "Face" refers to the group of measures that causes an entire face of the pyramid to have high interest.

Here we show the different correlation value between the types of measures. We chose a measure from each type. We analyse confidence, conviction, cosine, Gini, Klösgen, Lift, and Odds Ratio. We calculated the correlation of the results produced by each measure given the combination of the $L$, $R$, and $T$ in the range from 0 to 1 with the increment of 0.01.

Table 11 shows the correlation between these measures. Note that in this table, most of the measures have a low correlation value. However, we notice that there are some measures that have slightly higher correlation between them. This means that they rank the rules in a similar fashion. Despite this it does not mean that they should belong in the same type. This is because the measures may have a different tolerance level for counterexamples. For example a measure may have a concave decrease with $\Pr(A \neg B)$, which reflects the ability to tolerate a few counterexamples without a loss of interest. Hence, the boundary between interesting and uninteresting rules may vary between measures.

Even though there are numerous objective measures defined to detect the most interesting rules, we were able to classify the objective measures into seven types.

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*Table 11. Correlation between the seven types of measures*

| Measure | Confidence | Conviction | Cosine | Gini | Klösgen | Lift | Odds Ratio |
|---|---|---|---|---|---|---|---|
| Confidence | 1.00 | 0.66 | 0.84 | 0.00 | 0.58 | 0.66 | 0.67 |
| Conviction | 0.66 | 1.00 | 0.72 | 0.02 | 0.90 | 0.88 | 0.92 |
| Cosine | 0.84 | 0.72 | 1.00 | 0.06 | 0.65 | 0.73 | 0.74 |
| Gini | 0.00 | 0.02 | 0.06 | 1.00 | 0.06 | -0.02 | 0.00 |
| Klösgen | 0.58 | 0.90 | 0.65 | 0.06 | 1.00 | 0.88 | 0.90 |
| Lift | 0.66 | 0.88 | 0.73 | -0.02 | 0.88 | 1.00 | 0.92 |
| Odds Ratio | 0.67 | 0.92 | 0.74 | 0.00 | 0.90 | 0.92 | 1.00 |

There are other objective measures we have not discussed here including Gray and Orlowska's weighting dependency (1998), causal support (Huynh et al., 2005; Kodratoff, 2001), causal confirm (Huynh et al., 2005; Kodratoff, 2001), dependence (Huynh et al., 2005; Kodratoff, 2001), Bayes factor (Jeffreys, 1935), gain (Fukuda, Morimoto, Morishita, & Tokuyama, 1996), Hellinger's divergence (Lee & Shin, 1999), Goodman-Kruskal (Goodman & Kruskal, 1954; Kim & Chan, 2004), mutual confidence, implication index (Vaillant et al., 2004), Laplace (Clark & Boswell, 1991; Roberto Bayardo & Agrawal, 1999), Sebag and Schoenaur (Sebag & Schoenauer, 1988; Vaillant et al., 2004), similarity index (Huynh et al., 2005), and Zhang (Vaillant et al., 2004; Zhang, 2000). Each measure in any particular group actually proposes a region as interesting similar to that proposed by other measures in the same group. Hence, when a user decides to select an interestingness measure for post-processing they must first take into consideration the properties of the rules they are looking at.

## Summary

In this chapter, we analysed some of the current objective measures used in association rule mining. To date there are various objective measures but each measure has its own selection bias that justifies the rational for preferring it compared to other measures. We analysed the properties of the current measures. We noted problems with current methods of evaluating, mixing, and weighting existing interestingness measure. In order to get a better understanding, we also developed a framework to evaluate the particular rules, which these measures rank most interesting.

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# References

Aggarwal, C. C., & Yu, P. S. (2001). Mining associations with the collective strength approach. *IEEE Transactions on Knowledge and Data Engineering, 13*(6), 863-873.

Berzal, F., Blanco, I., Sánchez, D., & Vila, M. A. (2001). A new framework to assess association rules. *Lecture Notes in Computer Science, in Advances in Intelligent Data Analysis* (Vol. 2189, pp. 95-104).

Brin, S., Motwani, R., Ullman, J. D., & Tsur, S. (1997). Dynamic itemset counting and implication rules for market basket data. In *SIGMOD'97: Proceedings of the 1997 ACM SIGMOD International Conference on Management of Data* (pp. 255-264). New York: ACM Press.

Clark, P., & Boswell, R. (1991). Rule induction with CN2: Some recent improvements. In *Proceedings of the 5th European Working Session On Learning* (pp. 151-163). Berlin: Springer.

Freitas, A. (1999, October). On rule interestingness measures. *Knowledge-Based Systems, 12*(5-6), 309-315.

Fukuda, T., Morimoto, Y., Morishita, S., & Tokuyama, T. (1996). Data mining using two-dimensional optimized association rules: Scheme, algorithms, and visualization. In *SIGMOD '96: Proceedings of the 1996 ACM SIGMOD International Conference on Management of Data* (pp. 13-23). New York: ACM Press.

Ganascia, J. G. (1988). Improvement and refinement of the learning bias semantic. In *ECAI* (pp. 384-389).

Geng, L., & Hamilton, H. J. (2006). Interestingness measures for data mining: A survey. *ACM Computer Surveys, 38*(3), 9(1-32).

Goodman, L. A., & Kruskal, W. H. (1954). Measures of association for cross classification. *Journal of the American Statistical Association, 49*, 732-764.

Gray, B., & Orlowska, M. (1998). CCAIIA: Clustering categorical attributes into interesting association rules. In *Proceedings of PAKDD'98* (pp. 132-143).

Gyenesei, A. (2001). Interestingness measures for fuzzy association rules. In *PKDD '01: Proceedings of the 5th European Conference on Principles of Data Mining and Knowledge Discovery* (pp. 152-164). London: Springer-Verlag.

Huynh, X. H., Guillet, F., & Briand, H. (2005). ARQAT: An exploratory analysis tool for interestingness measures. In *ASMDA 2005 Conference International Symposium on Applied Stochastic Models and Data Analysis* (pp. 334-344).

Jeffreys, H. (1935). Some tests of significance, treated by the theory of probability. *Proceedings of the Cambridge Philosophical Society* (Vol. 31, pp. 203-222).

Kim, H. rae, & Chan, P. K. (2004). Identifying variable-length meaningful phrases

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with correlation functions. In *ICTAI '04: Proceedings of the 16th IEEE International Conference on Tools with Artificial Intelligence* (pp. 30-38). Washington, DC: IEEE Computer Society.

Klösgen, W. (1996). EXPLORA: A multipattern and multistrategy discovery assistant. In *Advances in Knowledge Discovery and Data Mining* (pp. 249-271). AAAI Press.

Kodratoff, Y. (2001). Comparing machine learning and knowledge discovery in databases: An application to knowledge discovery in texts. Machine Learning and Its Applications: Advanced Lectures, 1-21.

Lallich, S., Vaillant, B., & Lenca, P. (2005). Parametrised measures for the evaluation of association rules interestingness. In *ASMDA 2005 Conference International Symposium on Applied Stochastic Models and Data Analysis* (pp. 220-229).

Lee, C. H., & Shin, D. G. (1999). A multistrategy approach to classification learning in databases. *Data Knowledge Engineering, 31*(1), 67-93.

Lehmann, E. L. (1998). *Nonparametrics: Statistical methods based on ranks* (Revised ed.). Pearson Education.

Lenca, P., Meyer, P., Vaillant, B., & Lallich, S. (2004). *Multicriteria decision aid for interestingness measure selection* (Tech. Rep. No. LUSSI-TR- 2004-01-EN). LUSSI Department, GET / ENST Bretagne.

Loevinger, J. (1947). A systematic approach to the construction and evaluation of tests of ability. *Psychological Monographs, 61*(4), 1-49.

Mahoney, J. J., & Mooney, R. J. (1994). Comparing methods for refining certainty-factor rule-bases. In *International Conference on Machine Learning* (pp. 173-180).

McGarry, K. (2005). A survey of interestingness measures for knowledge discovery. Knowledge *Engineering Review, 20*(1), 39-61.

Piatetsky-Shapiro, G. (1991). Discovery, analysis, and presentation of strong rules. In Knowledge discovery in databases (p. 229-248). AAAI/MIT Press.

Reynolds, H. T. (1977). The analysis of cross-classifications. Free Press.

Roberto Bayardo, J., & Agrawal, R. (1999). Mining the most interesting rules. In *KDD '99: Proceedings of the Fifth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 145-154). New York: ACM Press.

Sebag, M., & Schoenauer, M. (1988). Generation of rules with certainty and confidence factors from incomplete and incoherent learning bases. In J. Boose, B. Gaines, & M. Linster (Eds.), *Proceedings of the European Knowledge Acquisition Workshop, (ekaw'88)* (pp. 28-1-28-20). Gesellschaft f¨ur Mathematik und Datenverarbeitung mbH.

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Silberschatz, A., & Tuzhilin, A. (1995). On subjective measures of interestingness in knowledge discovery. In Knowledge Discovery and Data Mining (pp. 275-281).

Smyth, P., & Goodman, R. M. (1992). An information theoretic approach to rule induction from databases. *IEEE Transactions on Knowledge and Data Engineering, 4*(4), 301-316.

Tan, P. N., & Kumar, V. (2000). *Interestingness measures for association patterns: A perspective* (Tech. Rep. No. TR 00-036). Department of Computer Science and Engineering, University of Minnesota.

Tan, P. N., Kumar, V., & Srivastava, J. (2004). Selecting the right objective measure for association analysis. *Inf. Syst., 29*(4), 293-313.

Tan, P. N., Kumar, V., & Srivastava, J. (2002). Selecting the right interestingness measure for association patterns. In *KDD '02: Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 32-41). New York: ACM Press.

Vaillant, B., Lenca, P., & Lallich, S. (2004). A clustering of interestingness measures. In *Lecture Notes in Computer Science* (Vol. 3245, pp. 290-297).

Zhang, T. (2000). Association rules. In T. Terano, H. Liu, & A. L. P. Chen (Eds.), *Proceedings of the 4th Pacific-Asia Conference Knowledge Discovery and Data Mining, Current Issues and New Applications, PAKDD '00* (Vol. 1805, pp. 245-256). Lecture Notes in Computer Science. Springer.

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## Chapter III

# Mining Association Rules from XML Data

Qin Ding, East Carolina University, USA

Gnanasekaran Sundarraj,
The Pennsylvania State University at Harrisburg, USA

## Abstract

*With the growing usage of XML in the World Wide Web and elsewhere as a standard for the exchange of data and to represent semi-structured data, there is an imminent need for tools and techniques to perform data mining on XML documents and XML repositories. In this chapter, we propose a framework for association rule mining on XML data. We present a java-based implementation of the apriori and the FP-growth algorithms for this task and compare their performances. We also compare the performance of our implementation with an XQuery-based implementation.*

## Introduction

Advances in data collection and storage technologies have led organizations to store vast amounts of data pertaining to their business activities. Extracting "use-

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ful" information from such huge data collections is of importance in many business decision-making processes. Such an activity is referred to as *data mining* or *knowledge discovery in databases* (KDD) (Han & Kamber, 2006). The term data mining refers to tasks such as *classification*, *clustering*, *association rule mining*, *sequential pattern mining*, and so forth (Han et al., 2006).

The task of association rule mining is to find correlation relationships among different data attributes in a large set of data items, and this has gained a lot of attention since its introduction (Agrawal, Imieliński, & Swami, 1993). Such relationships observed between data attributes are called *association rules* (Agrawal et al., 1993). A typical example of association rule mining is the *market basket analysis*. Consider a retail store that has a large collection of items to sell. Often, business decisions regarding discount, cross-selling, grouping of items in different aisles, and so on need to be made in order to increase the sales and hence the profit. This inevitably requires knowledge about past transaction data that gives the buying habits of customers. The association rules in this case will be of the form "*customers who bought item A also bought item B*," and association rule mining is to extract such rules from the given historical transaction data.

Explosive use of World Wide Web to buy and sell items over the Internet has led to similar data mining requirements from online transaction data. In an attempt to standardize the format of data exchanged over the Web and to achieve interoperability between the different technologies and tools involved, World Wide Web consortium (W3C) introduced *Extensible Markup Language* (XML) (Goldfarb, 2003). XML is a simple but very flexible text format derived from *Standard Generalized Markup Language* (SGML) (Goldfarb, 2003), and has been playing an increasingly important role in the exchange of wide variety of data over the Web. Even though it is a markup language much like the *HyperText Markup Language* (HTML) (Goldfarb, 2003), XML was designed to describe data and to focus on what the data is, whereas HTML was designed to display data and to focus on how the data looks on the Web browser. A data object described in XML is called an *XML document*.

XML also plays the role of a meta-language, and allows document authors to create customized markup languages for limitless different types of documents, making it a standard data format for online data exchange. This growing usage of XML has naturally resulted in increasing amount of available XML data, which raises the pressing need for more suitable tools and techniques to perform data mining on XML documents and XML repositories.

In this chapter, we study the various approaches that have been proposed for association rule mining from XML data, and present a java-based implementation for the two well-known algorithms for association rule mining: *apriori* (Agrawal & Srikant, 1994) and FP-growth (Han, Pei, Yin, & Mao 2004). The rest of this chapter is organized as follows. In the second section, we describe the basic concepts and definitions for association rule mining. In this section, we also explain the above two

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algorithms briefly. In the third section, we detail the various approaches to association rule mining on XML data, and in the fourth section we present our Java-based implementation for this task. Finally, we give the experimental results in the fifth section before concluding this chapter.

# Association Rule Mining

The first step in association rule mining is to identify *frequent sets*, the sets of items that occur together often enough to be investigated further. Because of the exponential scale of the search space, this step is undoubtedly the most demanding in terms of computational power and in the need for the use of efficient algorithms and data structures. These factors become really important when dealing with real time data.

Next we give the basic concepts and definitions for association rule mining, and then briefly explain the apriori and FP-growth algorithms. Note that when describing the data, we use the terms "transaction" and "item" in the rest of the chapter just to be consistent with our examples.

## Basic Concepts and Definitions

Let $I = \{i_1, i_2, i_3, \ldots, i_m\}$ be a set of items. Let $D$ be the set of transactions where each transaction $T \in D$ is a set of items such that $T \subseteq I$. An association rule is of the form $A \Rightarrow B$ where $A \subseteq I$, $B \subseteq I$, and $A \cap B = \varnothing$. The set of items $A$ is called *antecedent* and the set $B$ the *consequent*. Such rules are considered to be interesting if they satisfy some additional properties, and the following two properties have been mainly used in association rule mining: *Support* and *Confidence*. Though other measures have been proposed for this task in the literature, we will consider only the previous two (Brin, Motwani, Ullman, & Tsur, 1997; Silverstein, Brin & Motwani, 1998). Support $s$ for a rule $A \Rightarrow B$, denoted by $s(A \Rightarrow B)$, is the ratio of the number of transactions in $D$ that contain all the items in the set $A \cup B$ to the total number of transactions in $D$. That is:

$$s(A \Rightarrow B) = \frac{\sigma(A \cup B)}{|D|} \quad (1)$$

where the function $\sigma(X)$ of a set of items $X$ denotes the number of transactions in $D$ that contain all the items in $X$. $\sigma(X)$ is also called the *support count* of $X$. Confidence

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$c$ for a rule $A \Rightarrow B$, denoted by $c(A \Rightarrow B)$, is the ratio of the support count of $A \cup B$ to that of the antecedent $A$. That is:

$$c(A \Rightarrow B) = \frac{\sigma(A \cup B)}{\sigma(A)} \quad (2)$$

For a user-specified minimum support $s_{min}$ and minimum confidence $c_{min}$, the task of association rule mining is to extract, from the given data set $D$, the association rules that have support and confidence greater than or equal to the user-specified values. Formal definition of this problem is given below.

**Input:** A non-empty set of transaction data $D$ where each transaction $T \in D$ is a non-empty subset of item set $I = \{i_1, i_2, i_3, \ldots, i_m\}$, minimum support $s_{min}$, and minimum confidence $c_{min}$.

**Output:** Association rules of the form "$A \Rightarrow B$ with support $s$ and confidence $c$" where $A \subseteq I$, $B \subseteq I$, $A \cap B = \varnothing$, $s >= s_{min}$, and $c >= c_{min}$.

A set of items is referred to as an *itemset*. An itemset that contains $k$ items is called a *k-itemset*. A $k$-itemset $L_k$ is frequent if $\sigma(L_k) >= s_{min} \times |D|$. Such a $k$-itemset is also referred to as a *frequent k-itemset*. A frequent 1-itemset is simply called a *frequent item*.

Consider the sample transaction data given in Table 1. Let us assume that $s_{min} = 0.4$ and $c_{min} = 0.6$. It can be seen that the rule $\{i_2, i_4\} \Rightarrow \{i_3\}$ has support 0.4 and confidence 0.66. This is a valid association rule satisfying the given $s_{min}$ and $c_{min}$ values.

The task of mining association rules from a given large collection of data is a two-step process:

1. Find all frequent itemsets satisfying $s_{min}$
2. Generate association rules from the frequent itemsets satisfying $s_{min}$ and $c_{min}$

*Table 1. Sample transaction data*

| Transaction | Items |
|---|---|
| $T_1$ | $\{i_1, i_2\}$ |
| $T_2$ | $\{i_1, i_3, i_4, i_5\}$ |
| $T_3$ | $\{i_2, i_3, i_4, i_6\}$ |
| $T_4$ | $\{i_1, i_2, i_3, i_4\}$ |
| $T_5$ | $\{i_1, i_2, i_4, i_6\}$ |

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The second step is straightforward and in this chapter we will be concentrating only on the first step.

## Apriori Algorithm

As the name implies, this algorithm uses prior knowledge about frequent itemset properties. It employs an iterative approach where *k*-itemsets are used to explore (*k* +1)-itemsets. To improve the efficiency of generating frequent itemsets, it uses an important property called the "a priori property," which states that all nonempty subsets of a frequent itemset must also be frequent. In other words, if an itemset *A* does not satisfy the minimum support, then for any item $i_i \in I$, the set $A \cup i_i$ cannot satisfy the minimum support either.

The apriori algorithm first computes the frequent 1-itemsets, $L_1$. To find frequent 2-itemsets $L_2$, a set of candidate 2-itemsets, $C_2$, is generated by joining $L_1$ with itself, i.e., $C_2 = L_1 |\times| L_1$. The join is performed in such a way that for $L_k |\times| L_k$, the *k*-itemsets $l_1$ and $l_2$, where $l_1 \in L_k$ and $l_2 \in L_k$, must have $k - 1$ items in common. Once $C_2$ is computed, for every 2-itemset $c_2$ in $C_2$, all possible 1-subsets of $c_2$ are checked to make sure that all of them are frequent. If any one of them is not a frequent itemset, then $c_2$ is removed from $C_2$. Once all the 2-itemsets in $C_2$ are checked, the set now becomes $L_2$ from which $L_3$ can be computed. This process is continued until, for some value *k*, $L_{k+1}$ becomes an empty set. The algorithm is shown in Figure 1.

In order to generate the frequent *k*-itemset $L_k$, this algorithm scans the input dataset *k* times. Also during the beginning stages the number of candidate itemsets generated

*Figure 1. Apriori algorithm*

```
APRIORI(D, s_min)
 1  C_1 ← {{i} | i ∈ I}
 2  k ← 1
 3  while C_k ≠ ∅ do
 4     for each transaction T ∈ D do
 5        for each candidate itemset c_k ∈ C_k do
 6           if c_k ⊆ T then
 7              s(c_k) ← s(c_k) + 1
 8     L_k ← {c_k | c_k ∈ C_k, s(c_k) ≥ s_min}
 9     C_{k+1} ← ∅
10     for each l_1, l_2 ∈ L_k, |l_1 ∩ l_2| = k − 1 do
11        I ← l_1 ∪ l_2
12        if ∀J ⊂ I, |J| = k and J ∈ L_k then
13           C_{k+1} ← C_{k+1} ∪ I
14     k ← k + 1
15  return L_k
```

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could be very large. These factors greatly affect the running time of the algorithm. In the next subsection, we describe the FP-growth algorithm, which is normally faster than the apriori algorithm.

## FP-Growth Algorithm

The FP-growth algorithm adopts *divide-and-conquer* strategy. First it computes the frequent items and represents the frequent items as a compressed database in the form of a tree called *frequent-pattern tree*, or FP-tree. The rule mining is then performed on this tree. This means the dataset $D$ needs to be scanned only once. Also this algorithm does not require the candidate itemset generation. So it is normally many times faster than the apriori algorithm.

The frequent items are computed as in the apriori algorithm and represented in a table called *header table*. Each record in the header table will contain the frequent item and a link to a node in the FP-tree that has the same item name. Following this link from the header table, one can reach all nodes in the tree having the same item name. Each node in the FP-tree, other than the root node, will contain the item name, support count, and a pointer to link to another node in the tree that has the same item name. The steps for creating the FP-tree are given next:

- Scan the transaction data $D$ once and create $L_1$ along with the support count for each frequent item in $L_1$. Sort $L_1$ in the descending order of support count and create the header table $L$.
- Create the FP-tree with an empty root node $M$. For each transaction $T \in D$, perform the following:
    - Select and sort the frequent items in $T$ to the order of $L$. Let the sorted frequent items in $T$ be $p|P$, where $p$ is the first element and $P$ is the remaining list. Let INSERT_TREE be the function that is called recursively to construct the tree. Call INSERT_TREE($p|P, M$), which does the following. If $M$ has a child $N$ such that *N.itemname = p.item-name*, then increment $N$'s support count by 1; else create a new node $N$ with support count 1, let $M$ be its parent, and link $N$ to other node in $M$ with the same item-name. If $P$ is not empty, call INSERT_TREE($P$, $N$) recursively.

Once the header table and the FP-tree are constructed, then for each frequent item in the header table, the *conditional pattern base*, which is a list of nodes that link the frequent item's node in the FP-tree to the root node, is formed. Each pattern base is assigned a support count, which is the minimum of the support counts for the items in the pattern base. If the support count of a pattern base is less than $s_{min}$, then

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it is ignored. So if a frequent item appears *n* times in the FP-tree, then it can have at most *n* conditional pattern bases. For each included pattern base of a frequent item, an FP-tree called *conditional FP-tree* is constructed, and the mining process is repeated until the conditional FP-tree is empty or there is only one conditional pattern base. The set of items in such single pattern bases form the frequent itemsets. Finally, association rules are extracted from these frequent itemsets.

In the apriori and the FP-growth algorithms previously described, once the frequent itemsets are computed, association rules can be extracted using that information. The algorithm to perform this is the same for both apriori and FP-growth, and as mentioned before it is outside the scope of this chapter.

## Various Approaches to XML Rule Mining

Association rule mining from XML data has gained momentum over the last few years and is still in its nascent stage. Several techniques have been proposed to solve this problem. The straightforward approach is to map the XML documents to relational data model and to store them in a relational database. This allows us to apply the standard tools that are in use to perform rule mining from relational databases. Even though it makes use of the existing technology, this approach is often time consuming and involves manual intervention because of the mapping process. Due to these factors, it is not quite suitable for XML data streams.

Recently, World Wide Web consortium introduced an XML query language called *XQuery* (Brundage, 2004). This query language addresses the need for the ability to intelligently query XML data sources. It is also flexible enough to query data from different types of XML information sources, including XML databases and XML documents. Naturally, this led to the use of XQuery to perform the association rule mining directly from XML documents. Since XQuery is designed to be a general purpose XML query language, it is often very difficult to implement complicated algorithms. So far only the apriori algorithm has been implemented using XQuery (Wan & Dobbie, 2003). It has been raised as an open question in Wan et al. (2003) whether or not the FP-growth algorithm can be implemented using XQuery, and there is no such implementation available at this point.

The other approach is to use programs written in a high level programming language for this task. Most of such implementations require the input to be in a custom text format and do not work with XML documents directly. In order to adopt this approach to XML rule mining, it requires an additional step to convert the XML documents into the custom text files and apply these tools. This step often affects the overall performance of this approach.

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Our approach for XML rule mining is to use programs written in Java to work directly with XML documents. This offers more flexibility and performs well compared to other techniques. The implementation details and experimental results of our approach are given in the following sections.

## Implementation Details

Java provides excellent support for handling XML documents. Programs written in Java can access XML documents in one of the following two ways:

1. **Document Object Model (DOM):** This allows programs to randomly access any node in the XML document, and requires that the entire document is loaded into memory.
2. **Simple API for XML (SAX):** This approach follows event-driven model and allows programs to perform only sequential access on the XML document. This does not load the entire document into memory.

Since the apriori algorithm needs to scan the input data many times, we used DOM for implementing this algorithm. Similarly, SAX is the natural choice for FP-growth, since it needs to scan the input data only once and works with the FP-tree constructed in memory for further processing.

An XML document contains one root level element with the corresponding opening and closing tags. These tags surround all other data content within the XML document. The format of a sample XML data used to test our algorithm is shown in Figure 2. The *transactions* tag is the root element that contains zero to many *transaction* elements. Each *transaction* element is uniquely identified by its *id* attribute. Each *transaction* element contains one *items* element, which in turn contains zero to many *item* elements. An *item* element has the name of the particular item in the given transaction.

Note that the input XML document can have a very complicated structure containing the transaction data at different depths. We assume in this case that the input document is preprocessed by using an XML style sheet language, like XSLT, to convert it into a simply structured format as shown in Figure 2 (Gardner & Rendon, 2001). This preprocessing can be done quickly and easily, and is outside the scope of this chapter.

The configuration settings for our implementation are given in Figure 3. These configurations are stored in a Java property file as property name-value pairs. The first four properties are self-explanatory. In order to make our implementation more

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*Figure 2. XML input file format*

```
<transactions>
    <transaction id="1">
        <items>
            <item>Bread</item>
            <item>Milk</item>
        </items>
    </transaction>
    <transaction id="2">
        <items>
            .
            .
            .
        </items>
    </transaction>
    .
    .
    .
</transactions>
```

*Figure 3. XML input file format*

```
INPUT_FILE_NAME=data.xml
OUTPUT_FILE_NAME=rules.xml
MINIMUM_SUPPORT=0.6
MINIMUM_CONFIDENCE=1.0
FIRST_LEVEL_ELEMENT_NAME=transactions
SECOND_LEVEL_ELEMENT_NAME=transaction
THIRD_LEVEL_ELEMENT_NAME=items
FOURTH_LEVEL_ELEMENT_NAME=item
SECOND_LEVEL_ELEMENT_UNIQUE_
    ATTRIBUTE_NAME=id
```

generic in being able to work with any XML tag names, we allow the user to pass the name of these tags through the properties in lines 5 through 9.

Our implementation outputs the association rules in XML format as shown in Figure 4. The root level element name is *rules* which may contain zero or more *rule* elements. Each *rule* element has one *antecedent* and one *consequent* element, and each rule has two attributes: *support* and *confidence*.

Our implementation includes several optimization strategies outlined in the previous literatures (Park, Chen, & Yu, 1995; Savasere, Omiecinski, & Navathe, 1995; Silverstein et al., 1998). Also we used custom-built data structures to improve the performance instead of using the ones provided in *Java Software Development Kit* (JSDK) library. In our FP-growth implementation, we stored the FP-tree in the form of an XML document. This allowed us to use XPath (Gardner et al., 2001) expressions to quickly query any node in the FP-tree.

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*Figure 4. XML output file format*

```
<rules>
    <rule support="0.6" confidence="1.0">
        <antecedent>
            <item>Bread</item>
        </antecedent>
        <consequent>
            <item>Milk</item>
        </consequent>
    </rule>
    .
    .
    .
</rules>
```

# Experimental Results

We studied the performance of our implementation on three transaction datasets created randomly. The details of these datasets are given in Table 2. We used 30 distinct items and a maximum of 20 items in each transaction for all the datasets. We created our own datasets due to the fact that there are no benchmark data available for this problem. The experiments were performed on a Pentium 4, 3.2 GHz system running Windows XP Professional with 1 GB of main memory. Figure 5 shows the running time comparison between the apriori algorithm and the FP-growth algorithm for the Dataset 1. It can be seen that the FP-growth always outperforms the apriori for all values of minimum support. This was the case in all the three datasets tested.

Figure 6 shows the running time comparison between the Java-based apriori and the XQuery-based apriori implementations. We used the XQuery implementation from (Wan et al., 2003) for this comparison. We observed that the Java-based apriori outperforms the XQuery implementation on all three datasets. But the gap between the two narrows as the number of transactions increases. All these graphs were obtained for a minimum confidence of 0.6.

It can be observed that the performance of these algorithms largely depends on the number of frequent itemsets. For lower values of minimum support, it is expected to have many frequent itemsets, and this number will decrease as the minimum support increases. So the running time decreases as the minimum support increases. The large gap between the apriori and the FP-growth at lower values of minimum support was caused by the large number of candidate itemsets created in apriori. It is our opinion that the data structure overhead in the XQuery implementation is what led to the performance difference between the Java-based apriori and XQuery-based apriori. The performance graphs for the remaining two datasets resembled the ones shown here except for the numerical values on the time axis.

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*Table 2. Test datasets*

| Datasets | Number of Transactions |
|---|---|
| Dataset 1 | 100 |
| Dataset 2 | 500 |
| Dataset 3 | 1000 |

*Figure 5. Apriori vs. FP-growth on dataset 1*

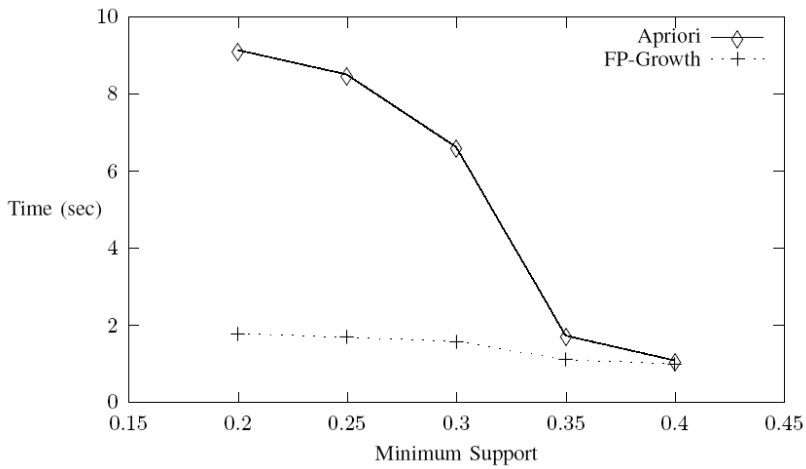
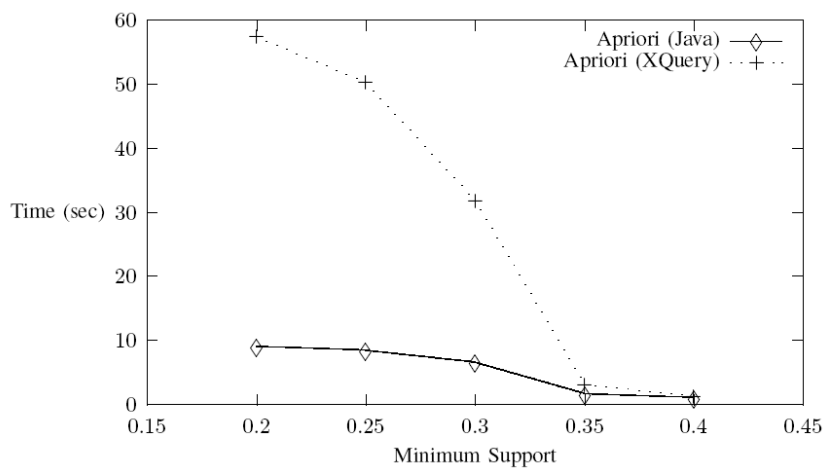



*Figure 6. Java-based apriori vs. XQuery-based apriori on dataset 1*




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# Conclusion

In this chapter, we studied the association rule mining problem and various approaches for mining association rules from XML documents and XML repositories. We presented a java-based approach to this problem and compared ours with an XQuery-based implementation.

Our approach performed very well against the one that we compared. There are several modifications that have been proposed to both apriori and FP-growth algorithms, which include modifications to the data structures used in the implementation and modifications to the algorithm itself (Park et al., 1995; Savasere et al., 1995; Silverstein et al., 1998). Though our implementation includes many such techniques, more analysis can be done on this front.

Though FP-growth algorithm is normally faster than the apriori algorithm, it is harder to implement the first one. One future direction is to use XQuery to implement the FP-growth algorithm and compare its results with our current java-based implementation.

# References

Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules. In *Proceedings of the International Conference on Very Large Data Bases* (pp. 487-499). Santiago, Chile.

Agrawal, R., Imielinski, T., & Swami, A. (1993). Mining association rules between sets of items in large databases. In *Proceedings of the ACM SIGMOD International Conference on Management of Data* (pp. 207-216). Washington, DC.

Brin, S., Motwani, R., Ullman, J. D., & Tsur, S. (1997). Dynamic itemset counting and implication rules for market basket data. In *Proceedings of the ACM SIGMOD International Conference on Management of Data* (pp. 255-264). Tucson, AZ.

Brundage, M. (2004). *XQuery: The XML query language*. Addison-Wesley Professional.

Gardner, J. R., & Rendon, Z. L. (2001). *XSLT and XPATH: A guide to XML transformations*. Prentice Hall.

Goldfarb, G. F. (2003). *XML handbook*. Prentice Hall.

Han, J., & Kamber, M. (2006). *Data mining: Concepts and techniques*. Morgan Kaufmann.

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Han, J., Pei, J., Yin, Y., & Mao, R. (2004). Mining frequent patterns without candidate generation: A frequent-pattern tree approach. *Data Mining and Knowledge Discovery, 8*(1), 53-87.

Park, J. S., Chen, M. S., & Yu, P. S. (1995). An effective hash-based algorithm for mining association rules. In *Proceedings of the ACM SIGMOD International Conference on Management of Data* (pp. 175-186). San Jose, CA.

Savasere, A., Omiecinski, E., & Navathe, S. (1995). An efficient algorithm for mining association rules in large databases. In *Proceedings of the International Conference on Very Large Databases* (pp. 432-444). Zurich, Switzerland.

Silverstein, C., Brin, S., & Motwani, R. (1998). Beyond market baskets: Generalizing association rules to dependence rules. *Data Mining and Knowledge Discovery, 2*(1), 39-68.

Wan, J. W. W., & Dobbie, G. (2003). Extracting association rules from XML documents using XQuery. In *Proceedings of the 5th ACM International Workshop on Web Information and Data Management* (pp. 94-97). New Orleans, LA.

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## Chapter IV

# A Lattice-Based Framework for Interactively and Incrementally Mining Web Traversal Patterns

Yue-Shi Lee, Ming Chuan University, Taiwan, R.O.C.
Show-Jane Yen, Ming Chuan University, Taiwan, R.O.C.

## Abstract

*Web mining is one of the mining technologies, which applies data mining techniques in large amounts of Web data to improve the Web services. Web traversal pattern mining discovers most of the users' access patterns from Web logs. This information can provide the navigation suggestions for Web users such that appropriate actions can be adopted. However, the Web data will grow rapidly in the short time, and some of the Web data may be antiquated. The user behaviors may be changed when the new Web data is inserted into and the old Web data is deleted from Web logs. Besides, it is considerably difficult to select a perfect minimum support threshold during the mining process to find the interesting rules. Even the experienced experts also cannot determine the appropriate minimum support. Thus, we must constantly*

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*adjust the minimum support until the satisfactory mining results can be found. The essences of incremental or interactive data mining are that we can use the previous mining results to reduce the unnecessary processes when the minimum support is changed or Web logs are updated. In this chapter, we propose efficient incremental and interactive data mining algorithms to discover Web traversal patterns and make the mining results to satisfy the users' requirements. The experimental results show that our algorithms are more efficient than the other approaches.*

# Introduction

With the trend of the information technology, huge amounts of data would be easily produced and collected from the *electronic commerce* environment every day. It causes the Web data in the database to grow up at amazing speed. Hence, how should we obtain the useful information and knowledge efficiently based on the huge amounts of Web data has already been the important issue at present.

*Web mining* (Chen, Park, & Yu, 1998; Chen, Huang, & Lin, 1999; Cooley, Mobasher, & Srivastava, 1997; EL-Sayed, Ruiz, & Rundensteiner, 2004; Lee, Yen, Tu, & Hsieh, 2003, 2004; Pei, Han, Mortazavi-Asl, & Zhu, 2000; Yen, 2003; Yen & Lee, 2006) refers to extracting useful information and knowledge from Web data, which applies data mining techniques (Chen, 2005; Ngan, 2005; Xiao, 2005) in large amount of Web data to improve the Web services. Mining *Web traversal patterns* (Lee et al., 2003, 2004; Yen, 2003) is to discover most of users' access patterns from Web logs. These patterns can not only be used to improve the Web site design (e.g., provide efficient access between highly correlated objects, and better authoring design for Web pages, etc.), but also be able to lead to better marketing decisions (e.g., putting advertisements in proper places, better customer classification, and behavior analysis, etc.)

In the following, we describe the definitions about Web traversal patterns: Let $I = \{x_1, x_2, \ldots, x_n\}$ be a set of all Web pages in a Web site. A *traversal sequence* $S = \langle w_1, w_2, \ldots, w_m \rangle$ $(w_i \in I, 1 \le i \le m)$ is a list of Web pages, which is ordered by traversal time, and each Web page can repeatedly appear in a traversal sequence, that is, backward references are also included in a traversal sequence. For example, if there is a path which visits Web page <B>, and then go to Web page <G> and <A> sequentially, and come back to Web page <B>, and then visit Web page <C>. The sequence <BGABC> is a traversal sequence. The *length* of a traversal sequence *S* is the total number of Web pages in *S*. A traversal sequence with length *l* is called an *l-traversal sequence*. For example, if there is a traversal sequence $\alpha$ = <AGDFAB>, the length of $\alpha$ is 6 and we call $\alpha$ a 6-traversal sequence. Suppose that there are two traversal sequences $\alpha = \langle a_1, a_2, \ldots, a_m \rangle$ and $\beta = \langle b_1, b_2, \ldots, b_n \rangle$ $(m \le n)$, if there

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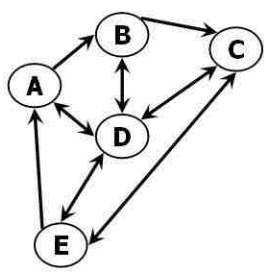
*Table 1. Traversal sequence database*

| TID | User sequence |
|---|---|
| 1 | ABCED |
| 2 | ABCD |
| 3 | CDEAD |
| 4 | CDEAB |
| 5 | CDAB |
| 6 | ABDC |

exists $i_1 < i_2 < \ldots < i_m$, such that $b_{i1} = a_1$, $b_{i2} = a_2$, $\ldots b_{im} = a_m$, then β *contains* α, α is a *sub-sequence* of β, and β is a *super-sequence* of α. For instance, if there are two traversal sequences α = <*BEA*> and β = <*ABCEA*>, then α is a sub-sequence of β and β is a super-sequence of α.

A *traversal sequence database D*, as shown in Table 1, contains a set of records. Each record includes *traversal identifier (TID)* and a *user sequence.* A user sequence is a traversal sequence, which stands for a complete browsing behavior by a user. The *support* for a traversal sequence α is the ratio of user sequences, which contains α to the total number of user sequences in *D*. It is usually denoted as *Support (α)*. The *support count* of α is the number of user sequences which contain α. For a traversal sequence $<x_1, x_2, \ldots, x_l>$, if there is a link from $x_i$ to $x_{i+1}$ (for all $i$, $1 \le i \le l\text{-}1$) in the Web site structure, then the traversal sequence is a *qualified traversal sequence*. A traversal sequence α is a *Web traversal pattern* if α is a qualified traversal sequence and *Support (α)* ≥ *min_sup*, in which the *min_sup* is the user specified *minimum support* threshold. For instance, in Table 1, if we set *min_sup* to 80%, then *Support (<AB>)* = 4/5 = 80% ≥ *min_sup* = 80%, and there is a link from "A" to "B" in the Web site structure shown in Figure 1. Hence, <*AB*> is a Web traversal pattern. If the length of a Web traversal pattern is *l*, then it can be called an *l-Web traversal pattern*.

*Figure 1. Web site structure*




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However, the user sequences will grow rapidly and some of the user sequences may be antiquated. The Web traversal patterns may be changed when the new user sequences are inserted into and the old user sequences are deleted from traversal sequence database. Therefore, we must re-discover the Web traversal patterns from the updated database. For example, if a new movie "Star Wars" is coming, in a DVD movies selling Web site, the users may rent or buy the new movie from the Web site. Hence, the users may change their interests to the science-fiction movie. That is, the user behaviors may be changed. Therefore, if we do not re-discover the Web traversal patterns from updated databases, some of the new information (about science-fiction movie) will be lost. However, it is very time-consuming to re-find the Web traversal patterns. For this reason, an incrementally mining method is needed to avoid re-mining the entire database.

Besides, based on the *min_sup*, all the Web traversal patterns can be found. Thus, it is very important to set an appropriate *min_sup*. If the *min_sup* is set too high, it will not find enough information for us. On the other hand, if the *min_sup* is set too low, unimportant information may be found and we will waste a lot of time finding all the information. However, it is very difficult to select a perfect minimum support threshold in the mining procedure to find the interesting rules. Even though they are experienced experts, they also cannot determine the appropriate minimum support threshold. Therefore, we must constantly adjust the minimum support until the satisfactory results can be found. It is very time consuming on these repeated mining processes. In order to find appropriate minimum support threshold, an interactive scheme is needed.

In this chapter, we use a uniform framework and propose two novel *incremental Web traversal pattern mining* algorithm *IncWTP* and *interactive Web traversal pattern mining* algorithm *IntWTP* to find all the Web traversal patterns when the database is updated and the *min_sup* is changed, respectively. If the database is updated and the minimum support is changed simultaneously, then the two algorithms can be executed successively. These two algorithms utilize the previous mining results to find new Web traversal patterns such that the mining time can be reduced. Therefore, how to choose a storage structure to store previous mining results becomes very important. In this chapter, lattice structure is selected as our storage structure. Not only utilizes the previous mining results, we also use the Web site structure to reduce mining time and storage space.

The rest of this chapter is organized as follows. Section 2 introduces the most recent researches related to this work. Section 3 describes the data structure for mining Web traversal patterns incrementally and interactively. Section 4 proposes our incremental Web traversal pattern mining algorithms. The interactive Web traversal pattern mining algorithm is presented in section 5. Because our approach is the first work on the maintenance of Web traversal patterns, we evaluate our algorithm by comparing with Web traversal pattern mining algorithm *MFTP* (Yen, 2003) in section 6. Finally, we conclude our work and present some future research in section 7.

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# Related Work

*Path traversal pattern* mining (Chen et al., 1998, 1999; Pei et al., 2000; Yen et al., 2006) is the technique that finds navigation behaviors for most of the users in the Web environment. The Web site designer can use this information to improve the Web site design (Sato, Ohtaguro, Nakashima, & Ito, 2005; Velásquez, Ríos, Bassi, Yasuda, & Aoki, 2005), and to increase the Web site performance. Many researches focused on this field (e.g., *FS (full scan)* algorithm, *SS (selective scan)* algorithm (Chen et al., 1998), and *MAFTP (maintenance of frequent traversal patterns)* algorithm (Yen et al., 2006), etc. Nevertheless, these algorithms have the limitations that they can only discover the *simple path traversal pattern*, which there is no repeated page in the pattern, that is, there is no backward reference in the pattern and the support for the pattern is no less than the minimum support threshold. These algorithms just consider the forward references in the traversal sequence database. Hence, the simple path traversal patterns discovered by the above algorithms are not fit in the real Web environment. Besides, *FS* and *SS* algorithms must rediscover simple path traversal patterns from entire database when the minimum support is changed or the database is updated.

The *MAFTP* algorithm (Yen et al., 2006) is an incremental updating technique to maintain the discovered path traversal patterns when the user sequences are inserted into or deleted from the database. The *MAFTP* algorithm partitions the database into some segments and scans the database segment by segment. For each segment scan, the candidate traversal sequences that cannot be frequent traversal sequences can be pruned and the frequent traversal sequences can be found out earlier. However, the *MAFTP* algorithm cannot deal with the backward references. Besides, *MAFTP* needs to re-mine the simple path traversal patterns from the original database when the minimum support is changed. Our approach can discover the non-simple path traversal patterns, that is, both forward references and backward references are considered. Besides, only a small number of the candidate traversal sequences need to be counted from the original traversal sequence database when the database is updated or the minimum support is changed for our algorithms.

The *non-simple path traversal pattern* (i.e., Web traversal pattern) contains not only forward references but also backward references. This information can present user navigation behaviors completely and correctly. The related researches are *MFTP (mining frequent traversal patterns)* algorithm (Yen, 2003), *IPA (integrating path traversal patterns and association rules)* algorithm (Lee et al., 2003, 2004), and *FS-miner* algorithm (EL-Sayed, 2004). *MFTP* algorithm can discover Web traversal patterns from traversal sequence database. This algorithm considers not only forward references, but also backward references. Unfortunately, *MFTP* algorithm must rediscover Web traversal patterns from entire database when the minimum support is changed or the database is updated. Our approach can discover the Web traversal

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patterns and both database insertion and deletion are also considered. Besides, our approach can use the discovered information to avoid re-mining entire database when the minimum support is changed.

The *IPA* algorithm can not only discover Web traversal patterns, but also user purchase behavior. It also considers the Web site structure to avoid generating un-qualified traversal sequences. Nevertheless, *IPA* algorithm does not consider incremental and interactive situations. It must rediscover Web traversal patterns from entire database when the minimum support is changed or the database is updated.

The *FS-miner* algorithm can discover Web traversal patterns from the traversal sequence database. *FS-miner* algorithm scans database twice to build a *FS-tree* (frequent sequences tree structure), and then it discovers Web traversal patterns from the FS-tree. However, the FS-tree may be too large to fit into memory. Besides, *FS-miner* finds the consecutive reference sequences traversed by a sufficient number of users, that is, they just consider the consecutive reference sub-sequences of the user sequences. However, there may be some noises, which exist in a user sequence, that is, some pages in a user sequence may be not the pages that the user really wants to visit. If all sub-sequences for a user sequence are considered, then *FS-miner* cannot work. Hence, some important Web traversal patterns may be lost for the *FS-miner* algorithm.

Besides, the *FS-miner* algorithm needs to set a system-defined minimum support, and then the FS-tree is constructed according to the system-defined minimum support. For the interactive mining, the user specified minimum support must be no less than the system-defined minimum support. Otherwise *FS-miner* cannot work. If the system-defined minimum support is too small, then the constructed FS-tree will be very large such that the FS-tree is hard to maintain. If the system-defined minimum support is too large, then users cannot set smaller minimum support than the large system-defined minimum support and the range for setting the user-specified minimum support is rather restricted. Hence, it is difficult to apply FS-miner on the incremental and interactive mining. For our approach, all the sub-sequences for a user sequence are considered, that is, the noises which exist in a user sequence can be ignored. Besides, there is no restriction on setting the user-specified minimum support, that is, users can set any value as the minimum support threshold for our algorithm. Furthermore, because our algorithm discovers the Web traversal patterns level-by-level in the lattice structure, it will not cause the memory be broken when we just load one level of lattice structure into memory.

*Sequential pattern mining* (Cheng, Yan, & Han, 2004; Lin, & Lee, 2002; Parthasarathy, Zaki, Ogihara, & Dwarkadas, 1999; Pei et al., 2001; Pei et al., 2004) is also similar to Web traversal pattern mining; they discover sequential patterns from *customer sequence database*. The biggest difference between Web traversal pattern and sequential pattern is that Web traversal pattern considers the link between two Web pages in the Web structure, that is, there must be a link from each page

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to the next page in a Web traversal pattern. Parthasarathy et al. (1999) proposed an incremental sequential pattern mining algorithm *ISL (incremental sequence lattice) algorithm*. *ISL* algorithm updates the lattice structure when the database is updated. The lattice structure keeps all the sequential patterns, and candidate sequences and their support counts, such that just new generated candidate sequences need to be counted from the original database and the mining efficiency can be improved. The candidate sequences whose support count is 0 are also kept in the lattice. It will cause the lattice structure too huge to fit into memory.

The other incremental sequential pattern mining algorithm is *IncSpan (incremental mining in sequential pattern)*, which was proposed by Cheng et al. (2004). This algorithm is based on the *PrefixSpan (prefix-projected sequential pattern mining)* algorithm (Pei et al., 2001; Pei et al., 2004). *IncSpan* uses the concept of *projected-database* to recursively mine the sequential patterns. However, *ISL* and *IncSpan* algorithms cannot deal with the situation that when the new user sequences are inserted into customer sequence database. They just considered inserting the transactions into the original user sequences. Because the user sequences will grow up at any time in the Web environment, our work focuses on mining Web traversal patterns when the user sequences are inserted into and deleted from the traversal sequence database. Besides, *ISL* and *IncSpan* algorithms are applied on mining sequential patterns and must re-mine the sequential patterns from the entire database when the minimum support is changed. Our work need to consider the Web site structure to avoid finding unqualified traversal sequences. For this reason, we cannot apply these two algorithms on mining Web traversal patterns.

For interactive data mining, *KISP (knowledge base assisted incremental sequential pattern)* algorithm (Lin et al., 2002) has been proposed for interactively finding sequential patterns. The *KISP* algorithm constructs a *KB* (*knowledge base)* structure in hard disk to minimize the response time for iterative mining. Before discovering the sequential patterns, all the sequences are stored in *KB* structures ordered by the sequence length. For every sequence length, *KB* stores the sequences ordered by their supports. *KISP* algorithm uses the previous information in *KB* and extends the content of *KB* for further mining. Based on the *KB*, *KISP* algorithm can mine the sequential patterns on different minimum support thresholds without re-mining the sequential patterns from the original database.

However, the *KB* structure simply stores the sequences ordered by sequence lengths and supports. There is no relationship among sequences about super-sequence or sub-sequence in *KB* structure. Hence, some information cannot be obtained from *KB* structure directly. For example, we may want to find sequential patterns related to certain items or find the longest sequential patterns, which are not sub-sequence of any other sequential patterns. For our algorithm, we use lattice structure to keep the previous mining results. The information mentioned above about Web traversal patterns can be obtained easily by traversing the lattice structure. Besides, *KISP* algorithm must re-mine the sequential patterns from the entire database when the

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database is updated. For our algorithm, we can mine Web traversal patterns interactively and incrementally in one lattice-based framework.

## Data Structure for Mining Web Traversal Patterns

In order to mine Web traversal patterns incrementally and interactively, we use previous mining results to discover new patterns such that mining time can be reduced. In this chapter, we use a lattice structure to keep previous mining results. Figure 2 shows the simple lattice structure *O* for the database described in Table 1, when *min_sup* is set to 50%. In the lattice structure *O*, only Web traversal patterns are stored in this structure. To incrementally and interactively mine the Web traversal patterns and speed up mining processes, we extend lattice structure *O* to

*Figure 2. Simple lattice structure*

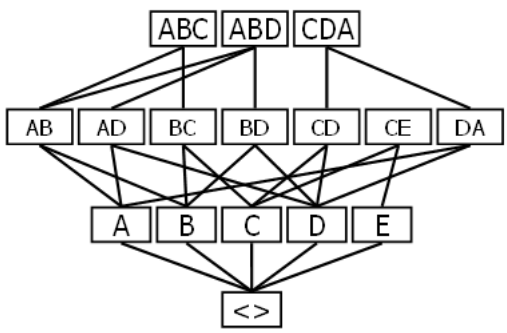
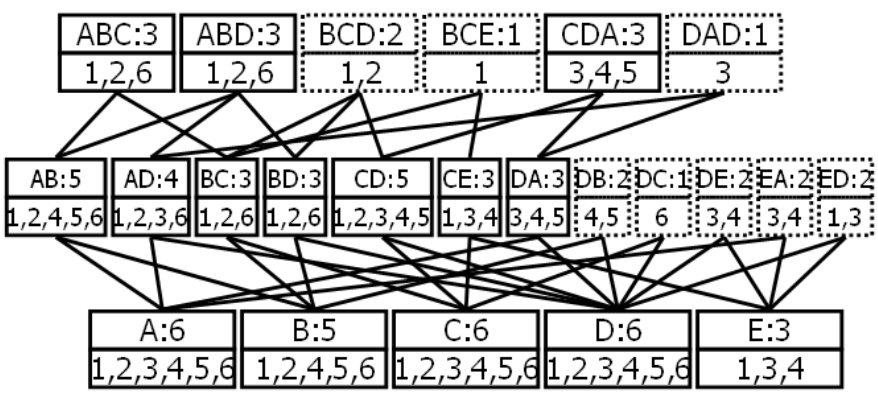



*Figure 3. Extended lattice structure*




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record more information. The extended lattice structure *E* is shown in Figure 3. In Figure 3, each node contains a traversal sequence whose support count is more or equal to 1. We append support information into the upper part of each node. We use this information to calculate and accumulate supports when the incremental and interactive mining is proceeding.

Moreover, we also append TID information in which the traversal sequence occurs into the lower part of each node. We can use this information to reduce unnecessary database scans. Different from simple lattice structure, we put all candidate traversal sequences, whose support counts are greater than or equal to one, into the lattice structure, and the lattice is stored on a disk by levels.

We can use lattice structure to quickly find the relationships between patterns. For example, if we want to search for the patterns related to Web page "A", we can just traverse the lattice structure from the node "A". Moreover, if we want to find *maximal Web traversal patterns* which are not sub-sequences of the other Web traversal patterns, we just need to traverse the lattice structure once and return the patterns in top nodes, whose supports are greater than or equal to *min_sup*. For example, in Figure 3, Web traversal patterns <CE>, <ABC>, <ABD> and <CDA> are maximal Web traversal patterns.

We utilize Web site structure, which is shown in Figure 1 to mine Web traversal patterns from the traversal sequence database shown in Table 1. The final results are shown in Figure 3 when the *min_sup* set to 50%. The reason for using Web site structure is that we want to avoid unqualified Web traversal sequences to be generated in the mining process. For example, assume that our Web site has 300 Web pages and all of them are all 1-Web traversal patterns. If we do not refer to Web site structure, then 299×300=89,700 candidate 2-sequences can be generated. However, in most situations, most of them are unqualified. Assume that the average out-degree for a node is 10. If we refer to the Web site structure, then just about 300×10=3,000 candidate 2-sequences are generated.

The candidate generation method is like the join method proposed in (Cheng et al., 2004). For any two distinct Web traversal patterns , say $<s_1, \ldots, s_{k-1}>$ and $<u_1, \ldots, u_{k-1}>$, we join them together to form a *k*-traversal sequence only if either $<s_2, \ldots, s_{k-1}>$ exactly is the same with $<u_1, \ldots, u_{k-2}>$ or $<u_2, \ldots, u_{k-1}>$ exactly the same with $<s_1, \ldots, s_{k-2}>$ (i.e., after dropping the fist page in one Web traversal pattern and the last page in the other Web traversal pattern, the remaining two (*k*-2)-traversal sequence are identical). For example, candidate sequence <ABCDE> can be generated by joining two Web traversal patterns <ABCD> and <BCDE>. For a candidate *l*-traversal sequence $\alpha$, if a qualified length (*l*-1) sub-sequence of $\alpha$ is not a Web traversal pattern, then $\alpha$ must not be Web traversal pattern and $\alpha$ can be pruned. Hence, we also check all of the qualified Web traversal sub-sequences with length *l*-1 for a candidate *l*-traversal sequence to reduce some unnecessary candidates. In this example, we need to check if <ABDE> and <ABCE> are Web traversal patterns.

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If one of them is not a Web traversal pattern, <ABCDE> is also not a Web traversal pattern. We do not need to check <ACDE>, because <ACDE> is an unqualified Web traversal sequence (i.e., no link from A to C).

## Algorithm for Incremental Web Traversal Pattern Mining

In this section, we propose an algorithm *IncWTP* for the maintenance of Web traversal patterns when the database is updated. Our algorithm *IncWTP* mines the Web traversal patterns from the first level to the last level in the lattice structure. For each level $k$ ($k \geq 1$), the $k$-Web traversal patterns are generated. There are three main steps in each level $k$: In the first step, the deleted user sequences' TIDs are deleted from each node of the $k$th level and the support count of the node is decreased if the node contains the TID of the deleted user sequence. It is very easy to obtain the support count of each node in this step, because our lattice structure keeps not only TID information but also the support count for each node.

In the second step, we deal with the inserted user sequences. For each inserted user sequence $u$, we decompose $u$ into several traversal sequences with length $k$, that is, all the length $k$ sub-sequences of the user sequence $u$ are generated. According to the Web site structure, the unqualified traversal sequences can be pruned. This pruning can avoid searching for the unqualified traversal sequences from the candidate sequences for counting their supports. For each qualified $k$-traversal sequence $s$, if $s$ has been contained in a node of the lattice structure, then we just increase the support count of this node and add TID of user sequence $u$ to the node. Otherwise, if all the qualified length ($k$-1) sub-sequences of $s$ are Web traversal patterns, then a new node $n_s$ which contains traversal sequence $s$ and the TID of user sequence $u$ is created in the $k$th level. The links between the nodes which contain the qualified length ($k$-1) sub-sequences of $s$ in the ($k$-1)th level and the new node $n_s$ are created in the lattice structure. Because our lattice structure always maintains the qualified candidate $k$-traversal sequences and the links between a traversal sequence s and all the length ($k$-1) sub-sequences of $s$, the relationships between super-sequences and sub-sequences can easily obtained by traversing the lattice structure.

After processing inserted and deleted user sequences, all the $k$-Web traversal patterns can be generated. If the support count of a node is equal to 0, then the node and all the links related to the node can be deleted from the lattice structure. If the support of a node is less than *min_sup*, then all the links between the node and the nodes $N$ in the ($k$+1)th level are deleted, and the nodes in $N$ are marked, because the traversal sequences in nodes $N$ turn out to be not candidate traversal sequences. Hence, in

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the *k*th level, if a node has been marked, then this node and the links between this node and the nodes in the (*k*+1)th level are also deleted.

In the last step, the candidate (*k*+1)-traversal sequences will be generated. The new Web traversal patterns in level *k* can be joined by themselves to generate new candidate (*k*+1)-traversal sequences. Besides, the original Web traversal patterns in level *k* are also joined with the new Web traversal patterns to generate the other new candidate (*k*+1)-traversal sequences. The original *k*-Web traversal patterns need not be joined each other, because they are joined before. The candidate generation method can avoid generating redundant candidate traversal sequences such that the number of the candidate traversal sequences can be reduced.

After generating the new candidate *(k*+1*)*-traversal sequences, the original database needs to be scanned to obtain the original support count and the TID information for each new candidate (*k*+1)-traversal sequence *c*. The new node $n_c$ which contains *c* is created and inserted into the lattice structure. The links between the nodes which contain the qualified length *k* sub-sequences of *c* in the *k*th level and the new node $n_c$ are created in the lattice structure. If there is no Web traversal patterns generated, then the mining process terminates.

Our incremental mining algorithm *IncWTP* is shown in algorithm 1, which is the c++ like algorithm. Algorithm 2 shows the function *CandidateGen*, which generates and processes the candidate traversal sequences. In algorithm 1, *D* denotes the traversal sequence database, *W* denotes the Web site structure, *L* denotes the lattice structure, *s* denotes the *min_sup*, *NewWTP* denotes new Web traversal patterns, *OriWTP* denotes original Web traversal patterns, *InsTID* denotes the inserted user sequences' TIDs, *DelTID* denotes the deleted user sequences' TIDs, *k* denotes current process level in *L*, and the *maximum level* of the original lattice structure is *m*. For instance, the maximum level of the lattice structure in Figure 3 is 3. All the Web traversal patterns will be outputted as the results.

For example in Table 1, we insert one user sequence (7, ABCEA) and delete two user sequences (1, ABCED) and (2, ABCD) as shown in Table 2. The *min_sup* also

*Table 2. Traversal sequence database after inserting and deleting user sequences from Table 1*

| TID | User sequence |
|---|---|
| 3 | CDEAD |
| 4 | CDEAB |
| 5 | CDAB |
| 6 | ABDC |
| 7 | ABCEA |

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*Figure 4. Updated lattice structure after processing level 1*

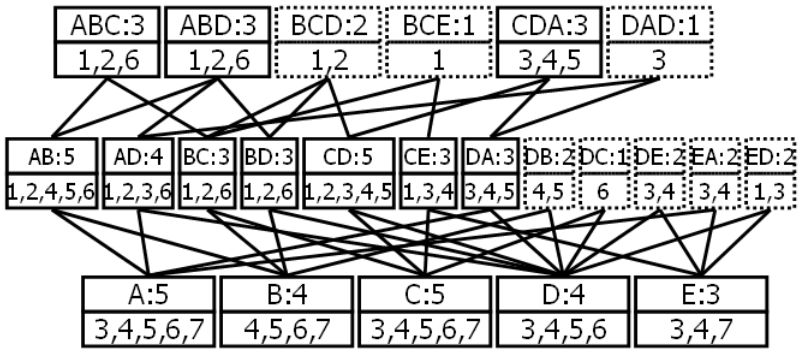



*Figure 5. Updated lattice structure after processing level 2*

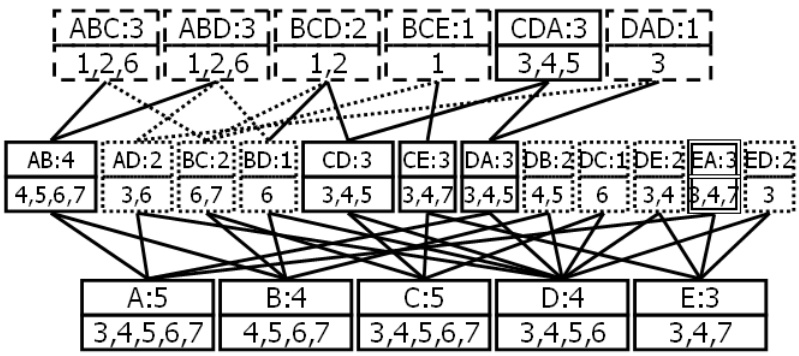
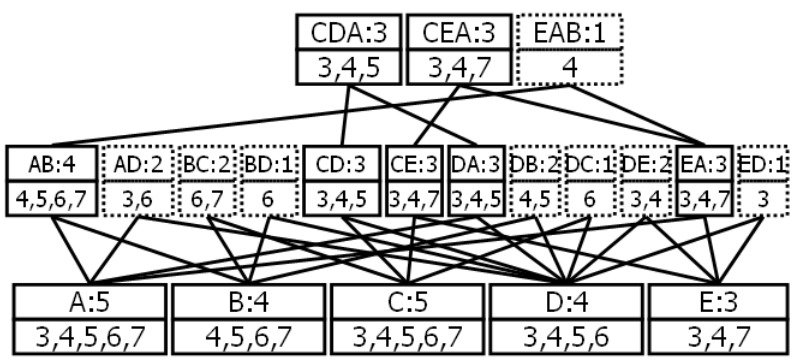



*Figure 6. Updated lattice structure after processing level 3*




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sets to 50%. At the first level of the lattice structure in Figure 3, TID 1 and TID 2 are deleted and the support count is decreased from each node which contains TID 1 or TID 2 for level 1. Then, the inserted user sequence TID 7 is decomposed into length 1 traversal sequences. The TID 7 is added and support count is increased to each node which contains one of the decomposed 1-traversal sequences. Because there is no new Web traversal patterns generated in level 1, we continue to process level 2 in the lattice structure. The updated lattice structure is shown in Figure 4 after processing the level 1. Because there is no new 1-Web traversal pattern generated and no node deleted, the number of the nodes and the links between the first level and the second level are not changed.

*Algorithm 1. IncWTP (D, min_sup, W, L, InsTID, DelTID, m)*

```
Input: traversal sequence database D, min_sup, web site structure W, lattice structure L,
       insert TID InsTID, delete TID DelTID, maximum level of L m
Output: All Web traversal patterns
k=1;
while(k ≤ m or there are new web traversal patterns generated in level k)
  for each node n in level k
    if(the node n are marked)
      the node n and all the links related to the node can be deleted;
      the nodes in level (k+1) which have links with node n are marked;
    if(node n contains any TID in DelTID)
      delete TIDs contained in DelTID and decrease the support count from n;
  for each inserted user sequence u
    decompose u into several qualified traversal sequences with length k;
    for each decomposed traversal sequence s
      if(s is contained in a node n of the level k)
        add u's TID and increase the support count in the node n;
      else if(all qualified (k-1)-sub-sequences of s are web traversal patterns)
        new node n_s contains s is generated in the level k;
        add u's TID and increase the support count in the node n_s;
  if(the support of a node n_m is less than min_sup)
    all the links between node n_m and the nodes in level (k+1) are deleted;
    the nodes in level (k+1) which have links with node n_m are marked;
    if(the support count of a node n_0 in level k is equal to 0)
      the node n_0 and all the links related to the node can be deleted;
  for each traversal sequence t_s in level k
    if(the support of t_s ≥ min_sup)
      WTP_k = WTP_k ∪ {t_s}; /* WTP_k is the set of all the web traversal patterns */
  NewWTP_k = WTP_k – OriWTP_k; /* OriWTP_k is the set of original web traversal
  patterns and NewWTP_k is the set of new web traversal patterns */
  output all the web traversal patterns in level k;
  CandidateGen (NewWTP_k , OriWTP_k);
  k++;
```

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*Algorithm 2. CandidateGen (NewWTPk , OriWTPk)*

```
for each new web traversal pattern x in NewWTP_k
    for each new web traversal pattern y in NewWTP_k
        if(x and y can be joined)
            generate a n ew c andidate ( k+1)-traversal sequence a nd s tore t he n ew
            candidate in set C;
    for each original web traversal pattern z in OriWTP_k
        if(x and z can be joined)
            generate a n ew c andidate ( k+1)-traversal sequence a nd s tore t he n ew
            candidate in set C;
for each candidate (k+1)-traversal sequence c in C
    count support and record the user sequences' TIDs which contain c from D;
    create a new node n_c which contains c;
    for each node n_s in the kth level which contains a qualified k-sub-sequence of c
create a link between n_s and n_c;
```

According to the deleted sequences' TIDs, the TID 1 and TID 2 are deleted and the support count is decreased in each node of level 2. Then, the inserted user sequence TID 7 is decomposed into length 2 traversal sequences. The TID 7 is added and the support count is increased to each node which contains one of the decomposed 2-traversal sequences. Finally, we can find the traversal sequence <EA> terns out to be 2-Web traversal pattern and the original Web traversal patterns <AD>, <BC>, and <BD> are not Web traversal patterns after updating the database. The five traversal sequences <ABC>, <ABD>, <BCD>, <BCE>, and <DAD> are marked. Figure 5 shows the lattice structure after processing the inserted and deleted traversal sequences in level 2. The sequence with double line is the new Web traversal pattern. After generating the new 2-Web traversal patterns, the two new candidate traversal sequences <CEA> and <EAB> are generated. Similarly, the last level is processed and the lattice structure about level 3 is updated. Figure 6 shows the final result in our example in which the sequences with solid line are the Web traversal patterns.

## Algorithm for Interactive Web Traversal Pattern Mining

In this section, we propose an algorithm, *IntWTP*, for the maintenance of Web traversal patterns when the previous minimum support is changed. For our algorithm *IntWTP*, if the new *min_sup* is larger than the original *min_sup*, then all the traversal sequences whose supports are no less than the new *min_sup* in the lattice structure are Web traversal patterns. If the new *min_sup* is smaller than the original *min_sup*,

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then our algorithm *IntWTP* mines the Web traversal patterns from the first level to the last level in the lattice structure. For each level $k$ ($k \geq 1$), the $k$-Web traversal patterns are generated. There are two main steps in each level $k$: In the first step, the traversal sequences in level $k$ are checked: if the support of a traversal sequence is no less than the new *min_sup*, but less than the original *min_sup*, then the traversal sequence is a new Web traversal pattern. Hence, all the new Web traversal patterns can be generated according to the new *min_sup* and original *min_sup*. In this step, all the $k$-Web traversal patterns including original Web traversal patterns and new Web traversal patterns can be obtained.

In the second step, the candidate ($k$+1)-traversal sequences will be generated. The new Web traversal patterns in level $k$ can be joined by themselves to generate new candidate ($k$+1)-traversal sequences. Besides, the original Web traversal patterns in level $k$ are also joined with the new Web traversal patterns to generate the other new candidate ($k$+1)-traversal sequences. The original $k$-Web traversal patterns need not be joined each other, because they are joined before. After generating the new candidate ($k$+1)-traversal sequences, the database needs to be scanned to obtain the support count and the TID information for each new candidate ($k$+1)-traversal sequence $c$. The new node which contains $c$ is created and inserted into the ($k$+1)th level of the lattice structure. The links between the nodes which contain the qualified length $k$ sub-sequences of $c$ in the $k$th level and the new node which contains $c$ in the $k$th level are created in the lattice structure. If there is no Web traversal pattern generated (including original Web traversal patterns), then the mining process terminates.

Our interactive mining algorithm *IntWTP* is shown in algorithm 3, which is the c++ like algorithm. In algorithm 3, *Ori_min_sup* denotes the original *min_sup* and *New_min_sup* denotes the new *min_sup*. All the Web traversal patterns will be outputted as the results. The following shows an example for our interactive mining algorithm *IntWTP*.

For the previous example (see Table 1 and Figure 3), we first increase the *min_sup* from 50% to 70%. Because the *min_sup* is increased, we just traverse the lattice structure once and output the traversal sequences whose supports are greater than or equal to 70% (i.e., support counts are no less than 4). In this example, <AB>, <C>, and <D> are the Web traversal patterns. If we decrease the *min_sup* from 50% to 40% (i.e., the minimum support count is 2), then new Web traversal patterns may be generated. First of all, we scan the first level (the lowest level) of the lattice structure. Because no new 1-Web traversal patterns are generated, we scan the second level of the lattice structure. In this level, we find that the traversal sequences <AD>, <BC>, <DB> and <DE> tern out to be 2-Web traversal patterns. In Figure 7, the sequences in level 2 with double line are the new Web traversal patterns. After finding the new 2-Web traversal patterns, the new candidate 3-traversal sequences can be generated. In this example, candidate 3-traversal sequences <ADB> and

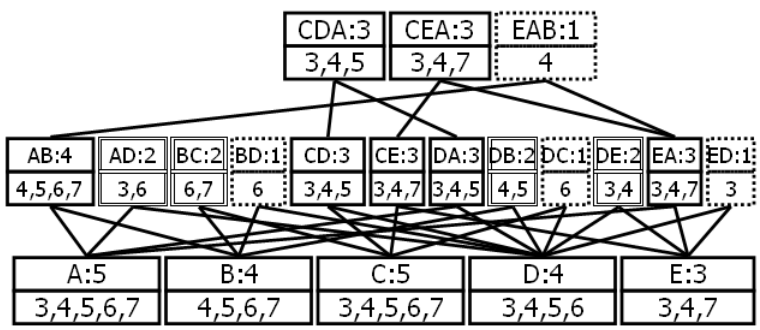
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*Algorithm 3. IntWTP (D, New_min_sup, Ori_min_sup, W, L)*

```
Input: traversal sequence database D, new min_sup New_min_sup, original min_sup
       Ori_min_sup, web site structure W, lattice structure L
Output: All Web traversal patterns
if(New_min_sup < Ori_min_sup)
    k=1; C=ϕ;
    while(there are web traversal patterns in level k)
        find original web traversal patterns OriWTP_k base on Ori_min_sup and new
        web traversal patterns NewWTP_k base on New_min_sup;
        output all the web traversal patterns in level k;
        CandidateGen (NewWTP_k , OriWTP_k)
        k++;
Ori_min_sup = New_min_sup;
```

<ADE> are generated by joining the new 2-Web traversal patterns themselves. The other candidate 3-traversal sequences <ADA>, <BCE>, <DEA>, <ABC>, <CDB>, <CDE> and <DAD> are generated by joining new 2-Web traversal patterns and original 2-Web traversal patterns. In Figure 8, the sequences in level 3 with double line are the new Web traversal patterns. After we finding the new 3-Web traversal patterns, the candidate 4-traversal sequence <CDEA> is generated. Figure 9 shows the final result of the lattice structure when *min_sup* is decreased to 40%. In Figure 9, the sequences with solid line are the Web traversal patterns.

*Figure 7. After processing level 2 of the lattice structure in Figure 3*




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*Figure 8. After processing level 3 of the lattice structure in Figure 7*

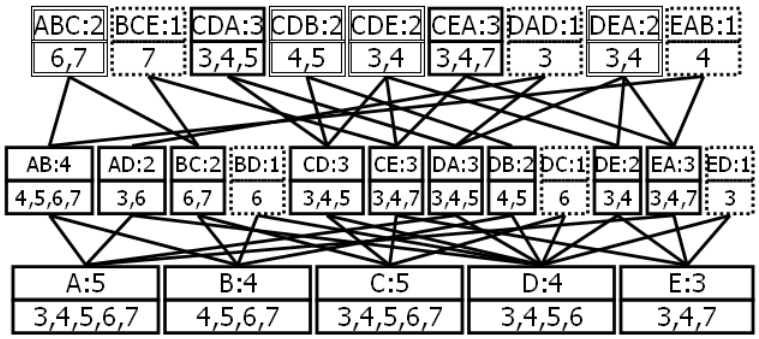
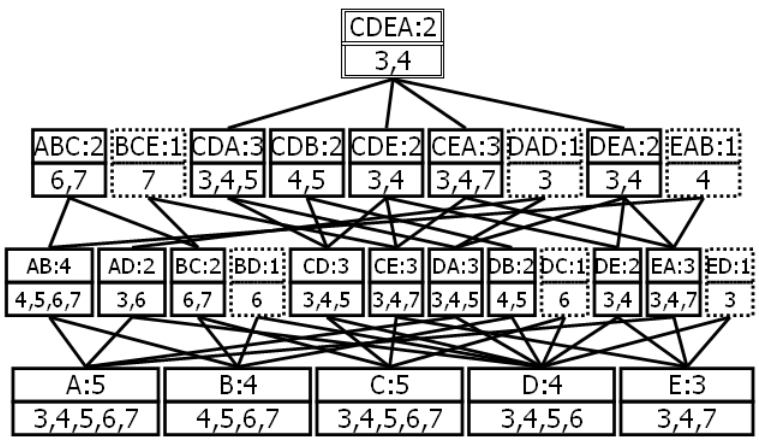



*Figure 9. The final lattice structure*




# Experimental Results

Because there are no incremental and interactive mining algorithms on finding Web traversal patterns currently, we use the algorithm *MFTP* (Yen, 2003), which is also used to find the Web traversal patterns to compare with our algorithm *IncWTP* and *IntWTP*. We implement the algorithm *IncWTP* and *IntWTP* in C language and perform the experiments on a PC with a 1.3GHz Intel Pentium-4 processor, 512 MB RAM, and Windows XP Professional platform.

The procedure of the synthetic dataset generation is shown as follows: First, the Web site structure is generated. The number of Web pages is set to 300 and the average number of out-links for each page is set to 15. According to the Web site

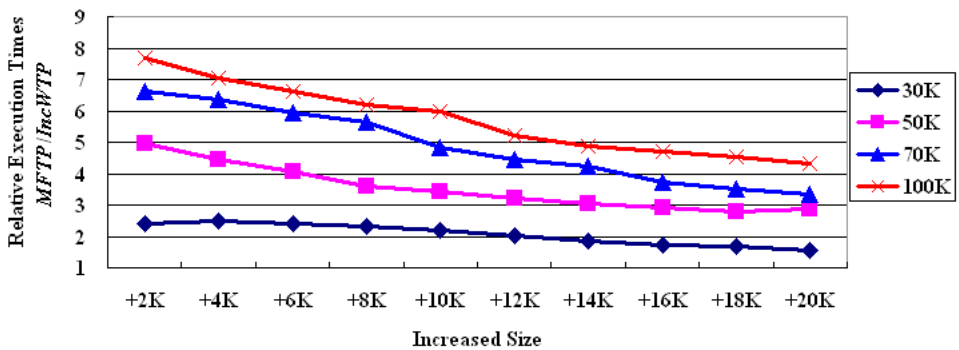
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structure, the potential Web traversal patterns are generated. The average number of Web pages for each potentially Web traversal patterns is set to 6, the total number of potentially Web traversal patterns is set to 2,500 and the maximum size of potentially Web traversal pattern is set to 10. After generating potentially Web traversal patterns, the user sequences are generated by picking the potentially Web traversal patterns from a Poisson distribution and the other pages are picked at random. The average size (the number of pages) per user sequence in database is set to 15 and the maximum size of the user sequences in database is set to 25. We generate four synthetic datasets in which the numbers of user sequences are set to 30K, 50K, 70K and 100K, respectively. In the following, we present the experimental results on the performance of our approaches.

## Performance Evaluation for Incremental Web Traversal Pattern Mining

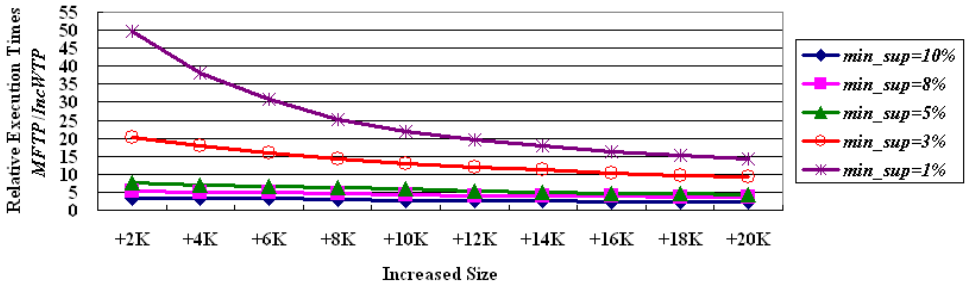
In the experiments, the four original datasets are increased by inserting 2K, 4K, 6K, 8K, 10K, 12K, 14K, 16K, 18K, and 20K user sequences. In the first experiment, the *min_sup* is set to 5%. Figure 10 shows the relative execution times for *MFTP* and *IncWTP* on the four synthetic data sets. In Figure 10, we can see that our algorithm, *IncWTP*, outperforms the *MFTP* algorithm, since our algorithm uses the lattice structure and Web site structure to prune a lot of candidate sequences and keeps the previous mining results such that just inserted user sequences need to be scanned for most of the candidate sequences. The performance gap increases when the size of original database increases. This is because when the size of original database increases, *MFTP* algorithm is worse than *IncWTP* algorithm in terms of the number of candidate traversal sequences and the size of database need to be

*Figure 10. Relative execution times for MFTP and IncWTP (min_sup = 5%)*




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*Figure 11. Relative execution times for MFTP and IncWTP (Dataset = 100K)*

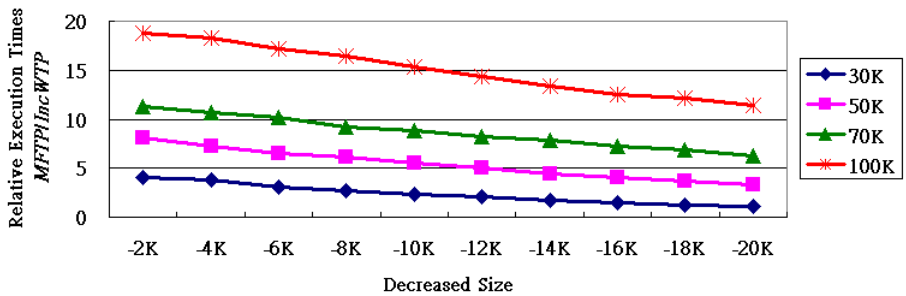



scanned, since *MFTP* algorithm must re-find all the Web traversal patterns from the whole updated database. However, most of the new Web traversal patterns are generated just from the inserted user sequences for *IncWTP* algorithm. Moreover, the less the number of inserted user sequences, the less the generated new candidate sequences for our algorithm. Hence, the performance gap increases as the number of inserted user sequences decreases.

In the second experiment, we use a synthetic data set in which the numbers of user sequences is 100K, and the *min_sup* is set to 10%, 8%, 5%, 3%, and 1%, respectively. Figure 11 shows the relative execution times for *MFTP* and *IncWTP*, in which we can see that our algorithm *IncWTP* outperforms *MFTP* algorithm significantly. The lower the *min_sup*, the more the candidate sequences generated for *MFTP* algorithm. *MFTP* needs to spend a lot of time to count a large number of candidate sequences from the whole updated database. For our algorithm *IncWTP*, just few new candidate sequences are generated for different minimum support. Hence, the performance gap increases as the minimum support threshold decreases.

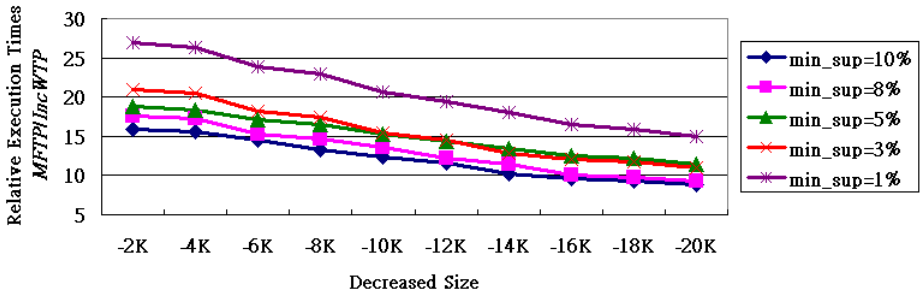
In the third experiment, the *min_sup* is set to 5%. We also use the four synthetic data sets in the first experiment. These original data sets are decreased by deleting

*Figure 12. Relative execution times for MFTP and IncWTP (min_sup = 5%)*




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*Figure 13. Relative execution times for MFTP and IncWTP (Dataset = 100K)*




2K, 4K, 6K, 8K, 10K, 12K, 14K, 16K, 18K, and 20K user sequences. Figure 12 shows the relative execution times for *MFTP* and *IncWTP* on the four synthetic data sets, in which we can see that our algorithm *IncWTP* is also more efficient than *MFTP* algorithm. The more the deleted user sequences, the smaller the size of the updated database. Hence, the performance gap decreases as the number of deleted user sequences increases, since the size of the database needs to be scanned and the number of candidate sequences decrease for *MFTP* algorithm. For our algorithm, there are few or no new candidates generated when the user sequences are deleted from original database, we just need to update the lattice structure for the deleted user sequences when the number of deleted user sequences is small. Hence, *IncWTP* still outperforms *MFTP* algorithm.

In the fourth experiment, we also use the synthetic data set in which the number of user sequences is 100K. The *min_sup* is set to 10%, 8%, 5%, 3%, and 1%, respectively. Figure13 shows the relative execution times for *MFTP* and *IncWTP* on the synthetic data set. In Figure13, we can see that our algorithm *IncWTP* outperforms *MFTP* algorithm significantly. The performance gap increases as the minimum support threshold decreases, since the number of candidate sequences and the whole updated database need to be scanned for the large number of candidate sequences for *MFTP* algorithm. For *IncWTP* algorithm, just the deleted user sequences need to be scanned when the minimum support threshold is large.

## Performance Evaluation for Interactive Web Traversal Pattern Mining

We use a real world user traversing data and generate five synthetic datasets to evaluate the performance of our interactive mining algorithm *IntWTP*. This real database is a networked database. It stores information for renting DVD movies. There are 82 Web pages in the Web site. We collect the user traversing data from 02/18/2001 to 02/24/2001 (7 days), and there are 428,596 log entries in this original database.

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*Figure 14. Execution times on real database*

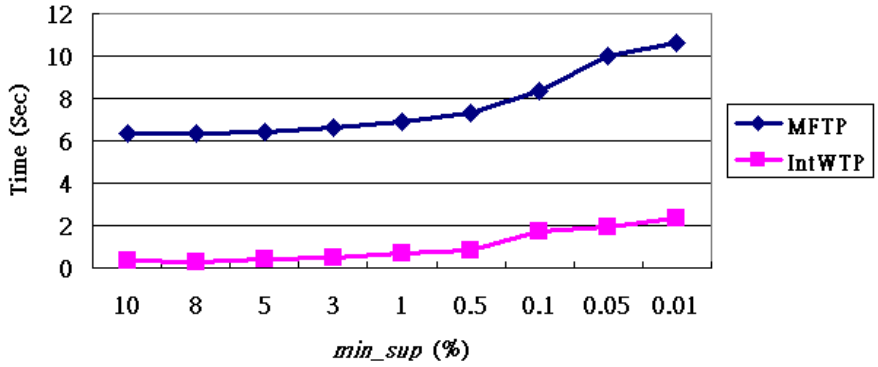
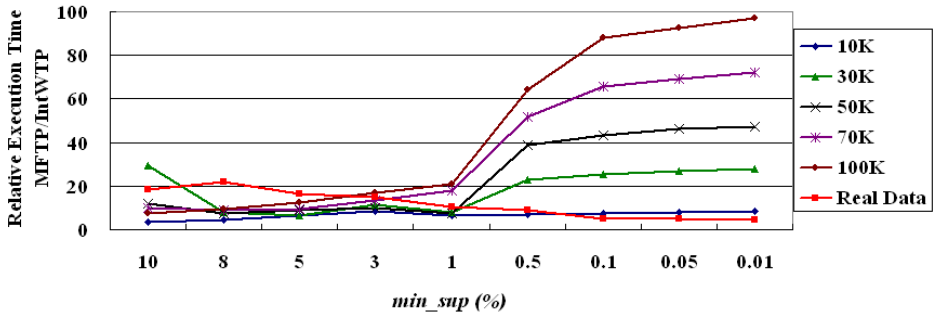



*Figure 15. Relative execution times for MFTP and IntWTP*




Before mining the Web traversal patterns, we need to transform these Web logs into the traversal sequence database. The steps are listed as follows. Because we want to get meaningful user behaviors, the log entries referred to as images are not important. Thus, all log entries with access filename suffix like .JPG, .GIF, .SME, and .CDF are removed. Then, we organize the log entries according to the user's IP address and time limit. After these processes, we can obtain the Web traversal sequence database like Table 1.

According to these steps, we organize the original log entries into 12,157 traversal sequences. The execution times for our interactive mining algorithm *IntWTP* and *MFTP* algorithm are shown in Figure14. In the synthetic datasets, we set the number of Web pages to 300, and generate five datasets with 10K, 30K, 50K, 70K, and 100K user sequences, respectively. The relative execution times for *IntWTP* algorithm and *MFTP* algorithm are shown in Figure15. The initial *min_sup* is set to 20%. Then, we continually decrease the *min_sup* from 10% to 0.01%.

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*Table 3. Relative storage space for lattice size and database size*

| | | Lattice level | | | | | | | | | | SUM (LatticeSize/DBSize) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | LatticeSize/DBSize | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| min_sup (%) | 20 | 0.95 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.95 |
| | 10 | 0.95 | 0.16 | 0.04 | 0.02 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1.17 |
| | 8 | 0.95 | 0.32 | 0.18 | 0.09 | 0.03 | 0.02 | 0.00 | 0.00 | 0.00 | 0.00 | 1.59 |
| | 5 | 0.95 | 0.66 | 0.43 | 0.32 | 0.20 | 0.10 | 0.02 | 0.01 | 0.00 | 0.00 | 2.69 |
| | 3 | 0.95 | 0.84 | 0.61 | 0.48 | 0.30 | 0.17 | 0.08 | 0.02 | 0.00 | 0.00 | 3.45 |
| | 1 | 0.96 | 1.01 | 0.98 | 0.86 | 0.72 | 0.47 | 0.28 | 0.14 | 0.06 | 0.02 | 5.50 |
| | 0.5 | 0.96 | 1.04 | 1.03 | 0.92 | 0.78 | 0.51 | 0.29 | 0.15 | 0.06 | 0.02 | 5.76 |
| | 0.1 | 0.96 | 1.06 | 1.07 | 0.95 | 0.80 | 0.52 | 0.30 | 0.15 | 0.06 | 0.02 | 5.89 |
| | 0.05 | 0.96 | 1.06 | 1.07 | 0.95 | 0.80 | 0.52 | 0.30 | 0.15 | 0.06 | 0.02 | 5.89 |
| | 0.01 | 0.96 | 1.06 | 1.08 | 0.97 | 0.81 | 0.53 | 0.30 | 0.15 | 0.06 | 0.02 | 5.94 |

From Figure 14 and Figure 15, we can see that our algorithm, *IntWTP*, outperforms the *MFTP* algorithm significantly, since our algorithm uses the lattice structure to keep the previous mining results and Web site structure to prune a lot of candidate sequences such that just new generated candidate sequences need to be counted. Besides, the performance gap increases as the minimum support threshold decreases or the database size increases, because when the minimum support decreases or the database size increases, the number of the candidate sequences increases, and the number of database scans also increases, such that the performance is degraded for *MFTP* algorithm. However, for our algorithm *IntWTP*, original Web traversal patterns can be ignored and just few new candidate sequences need to be counted. Hence, the mining time can be reduced dramatically.

Moreover, we also do the experiment on the storage space for lattice structure size and database size. We use the synthetic dataset with 100K user sequences. Table 3 shows the ratio of the space occupied by lattice structure to the space occupied by the database for each level. In Table 3, the size of level 2 and level 3 of the lattice structure are slightly larger than the database size when the minimum support is decreased to 1%. In the other cases, the sizes of the lattice structure are smaller than the database size for each level. Because *IntWTP* algorithm discovers Web traversal patterns level-by-level, it will not cause the memory be broken when we just load one level of lattice structure into memory.

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## Conclusion and Future Work

In this chapter, we propose incremental and interactive data mining algorithms *IncWTP* and *IntWTP* for discovering Web traversal patterns when the user sequences are inserted into and deleted from original database and when the minimum support is changed. In order to avoid re-finding the original Web traversal patterns and re-counting the original candidate sequences, our algorithms use lattice structure to keep the previous mining results such that just new candidate sequences need to be computed. Hence, the Web traversal patterns can be obtained rapidly when the traversal sequence database is updated and users can adjust the minimum support threshold to obtain the interesting Web traversal patterns quickly. Besides, the Web traversal patterns related to certain pages or maximal Web traversal patterns can also be obtained easily by traversing the lattice structure.

However, the Web site structure may be changed. In the future, we shall investigate how to use the lattice structure to maintain the Web traversal patterns when the pages and links in the Web site structure are changed. Besides, the number of Web pages and the user sequences will grow up all the time. The lattice structure may become too large to fit into memory. Hence, we shall also investigate how to reduce the storage space and partition the lattice structure such that all the information can fit into memory for each partition.

## Acknowledgment

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## References

Chen, M. S., Huang, X. M., & Lin, I. Y. (1999). Capturing user access patterns in the Web for data mining. *Proceedings of the IEEE International Conference on Tools with Artificial Intelligence* (pp. 345-348).

Chen, M. S., Park, J. S., & Yu, P. S. (1998). Efficient data mining for path traversal patterns in a Web environment. *IEEE Transaction on Knowledge and Data Engineering, 10*(2), 209-221.

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Chen, S. Y., & Liu, X. (2005). Data mining from 1994 to 2004: An application-orientated review. *International Journal of Business Intelligence and Data Mining, 1*(1), 4-21.

Cheng, H., Yan, X., & Han, J. (2004). IncSpan: Incremental mining of sequential patterns in large database. *Proceedings of 2004 International Conference on Knowledge Discovery and Data Mining* (pp. 527-532).

Cooley, R., Mobasher, B., & Srivastava, J. (1997). Web mining: Information and pattern discovery on the world wide Web. *Proceedings of the IEEE International Conference on Tools with Artificial Intelligence* (pp. 558-567).

EL-Sayed, M., Ruiz, C., & Rundensteiner, E. A. (2004). FS-miner: Efficient and incremental mining of frequent sequence patterns in Web logs. *Proceedings of ACM International Workshop on Web Information and Data Management* (pp. 128-135).

Lee, Y. S., Yen, S. J., Tu, G. H., & Hsieh, M. C. (2004). Mining traveling and purchasing behaviors of customers in electronic commerce environment. *Proceedings of IEEE International Conference on e-Technology, e-Commerce and e-Service* (pp. 227-230).

Lee, Y. S., Yen, S. J., Tu, G. H., & Hsieh, M. C. (2003). Web usage mining: Integrating path traversal patterns and association rules. *Proceedings of International Conference on Informatics, Cybernetics, and Systems* (pp. 1464-1469).

Lin, M. Y., & Lee, S. Y. (2002). Improving the efficiency of interactive sequential pattern mining by incremental pattern discovery. *Proceedings of the Hawaii International Conference on System Sciences* (pp. 68-76).

Ngan, S. C., Lam, T., Wong, R. C. W., & Fu, A. W. C. (2005). Mining n-most interesting itemsets without support threshold by the COFI-tree. *International Journal of Business Intelligence and Data Mining, 1*(1), 88-106.

Parthasarathy, S., Zaki, M. J., Ogihara, M., & Dwarkadas, S. (1999). Incremental and interactive sequence mining. *Proceedings of the 8th International Conference on Information and Knowledge Management* (pp. 251-258).

Pei, J., Han, J., Mortazavi-Asl, B., & Zhu, H. (2000). Mining access patterns efficiently from Web logs. *Proceedings of the Pacific-Asia Conference on Knowledge Discovery and Data Mining* (pp. 396-407).

Pei, J., Han, J., Mortazavi-Asl, B., Pinto, H., Chen, Q., Dayal, U., & Hsu, M. C. (2001). PrefixSpan: Mining sequential patterns efficiently by prefix-projected pattern growth. *Proceeding of International Conference on Data Engineering* (pp. 215-224).

Pei, J., Han, J., Mortazavi-Asl, B., Wang, J., Pinto, H., Chen, Q., Dayal, U., & Hsu, M. C. (2004). Mining sequential patterns by pattern-growth: The prefixspan

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approach. *IEEE Transactions on Knowledge and Data Engineering, 16*(11), 1424-1440.

Sato, K., Ohtaguro, A., Nakashima, M., & Ito, T. (2005). The effect of a Web site directory when employed in browsing the results of a search engine. *International Journal on Web Information System, 1*(1), 43-51.

Srivastava, J., Cooley, R., Deshpande, M., & Tan, P. N. (2000). Web usage mining: discovery and applications of usage patterns from Web data. *SIGKDD Explorations* (pp. 12-23).

Velásquez, J., Ríos, S., Bassi, A., Yasuda, H., & Aoki, T. (2005). Towards the identification of keywords in the Web site text content: A methodological approach. *International Journal on Web Information System, 1*(1), 53-57.

Xiao, Y., Yao, J. F., & Yang, G. (2005). Discovering frequent embedded subtree patterns from large databases of unordered labeled trees. *International Journal of Data Warehousing and Mining, 1*(2), 44-66.

Yen, S. J. (2003). An efficient approach for analyzing user behaviors in a Web-based training environment. *International Journal of Distance Education Technologies, 1*(4), 55-71.

Yen, S. J., & Lee, Y. S. (2006). An incremental data mining algorithm for discovering Web access patterns. *International Journal of Business Intelligence and Data Mining, 1*(3), 288-303.

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*Section II*

# Clustering and Classification

## Chapter V

# Determination of Optimal Clusters Using a Genetic Algorithm

Tushar, Indian Institute of Technology, Kharagpur, India
Shibendu Shekhar Roy, Indian Institute of Technology, Kharagpur, India
Dilip Kumar Pratihar, Indian Institute of Technology, Kharagpur, India

## Abstract

*Clustering is a potential tool of data mining. A clustering method analyzes the pattern of a data set and groups the data into several clusters based on the similarity among themselves. Clusters may be either crisp or fuzzy in nature. The present chapter deals with clustering of some data sets using the fuzzy c-means (FCM) algorithm and the entropy-based fuzzy clustering (EFC) algorithm. In the FCM algorithm, the nature and quality of clusters depend on the pre-defined number of clusters, level of cluster fuzziness, and a threshold value utilized for obtaining the number of outliers (if any). On the other hand, the quality of clusters obtained by the EFC algorithm is dependent on a constant used to establish the relationship between the distance and*

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*similarity of two data points, a threshold value of similarity, and another threshold value used for determining the number of outliers. The clusters should ideally be distinct and at the same time compact in nature. Moreover, the number of outliers should be as minimal as possible. Thus, the previous problem may be posed as an optimization problem, which will be solved using a genetic algorithm (GA). The best set of multi-dimensional clusters will be mapped into 2-D for visualization using a self-organizing map (SOM).*

# Introduction

Clustering is a powerful tool of data mining. Cluster analysis aims to search and analyze the pattern of a data set and group them into several clusters based on their similarity among themselves. It is done in such a way that the data points belonging to a cluster are similar in nature and those belonging to difficult clusters have a high degree of dissimilarity. There exist a number of clustering techniques and those are broadly classified into *hierarchical* and *partitional* methods.

Hierarchical methods iteratively either merge a number of data points into one cluster (called agglomerative method) or distribute the data points into a number of clusters (known as divisive method). An agglomerative method starts with a number of clusters that is equal to the number of data points so that each cluster contains one data point. At each iteration, it merges the two closest clusters into one and ultimately one cluster will be formed consisting of all the data points of them. A divisive method begins with a single cluster containing all the data points. It iteratively divides the data points into more number of clusters and ultimately each cluster will contain only one data point.

The aim of using the partitional methods is to partition a data set into some disjoint subsets of points, such that the points lying in each subset are as similar as possible. Partitional methods of clustering are further sub-divided into hard clustering and fuzzy clustering techniques. In hard clustering, the developed clusters will have their well-defined boundaries. Thus, a particular data point will belong to one and only one cluster. On the other hand, in fuzzy clustering, a particular data point may belong to the different clusters with different membership values. It is obvious that the sum of membership values a data point with various clusters will be equal to 1.0.

This chapter deals with fuzzy clustering. There exist a number of fuzzy clustering algorithms and out of those, the fuzzy c-means (FCM) algorithm (Bezdek, 1981; Dunn, 1974) is the most popular and widely used one due to its simplicity. The performance of the FCM algorithm depends on the number of clusters considered, level of fuzziness and others. However, it has the following disadvantages:

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1. The number of fuzzy clusters is to be pre-defined by the user
2. It may get stuck into the local minima

To overcome the previous drawbacks, attempts have been made to determine optimal clusters using a genetic algorithm (GA) (Goldberg, 1989) along with the FCM algorithm (Hruschkra, Campello, & de Castro, 2004). In the present chapter, an approach has been developed by combining the FCM algorithm with a GA that can automatically determine the number of clusters and at the same time, can decrease the probability of the clustering algorithm for being trapped into the local minima. The working principle of a binary-coded GA has been explained in Appendix A. For a data set, the clusters are said to be optimal, if they are distinct and at the same time, compact in nature, after ensuring the condition that there is no outliers. The distinctness among the developed clusters and compactness among different elements of a cluster are expressed in terms of Euclidean distance. For a set of clusters to be declared as distinct, average Euclidean distance between the cluster centers should be as high as possible. On the other hand, a cluster is said to be compact, if the average Euclidean distance among the elements of that cluster is minimized.

More recently, an entropy-based fuzzy clustering (EFC) algorithm has been proposed by Yao, Dash, Tan, and Liu (2000), where the number of clusters and their qualities are dependent on a number of parameters, such as the constant used to relate similarity between two data points with the Euclidean distance between them, similarity threshold value, threshold value used for declaration of outliers (if any). A set of the previous optimal parameters may be determined by using a GA so that the best set of clusters can be obtained. The present chapter deals also with a GA-based optimization of the above parameters of an entropy-based fuzzy clustering algorithm.

The effectiveness of the proposed technique has been tested on two data sets related to Tungsten Inert Gas (TIG), welding (Ganjigatti, 2006; Juang, Tarng, & Lii, 1998), and abrasive flow machining (AFM) (Jain & Adsul, 2000; Jain & Jain, 2000) processes. A self-organizing map (SOM) (Haykin, 1999; Kohonen, 1995) is used to reduce the dimension of the multi-dimensional data into 2-D for visualization. The working principle of the SOM has been explained in Appendix B. Thus, the best set of clusters can be visualized in 2-D.

The rest of the text is organized as follows: Section 2 explains the clustering algorithms used in the present study. The method of calculating the fitness of a GA solution is discussed in Section 3. The results are stated and explained in Section 4 and conclusions are drawn in Section 5.

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# Clustering Algorithms

The working principle of two clustering techniques, namely fuzzy c-means (FCM) algorithm and entropy-based fuzzy clustering (EFC) algorithm have been explained below, in detail.

## Fuzzy C-Means Algorithm

Fuzzy c-means (FCM) algorithm is one of the most popular fuzzy clustering techniques, in which the data points have their membership values with the cluster centers that will be updated iteratively (Bezdek, 1981; Dunn, 1974). Let us consider *M*-dimensional *N* data points represented by $x_i$ ($i$ = 1, 2, 3, …, *N*), which are to be clustered. The FCM algorithm consists of the following steps:

- **Step 1:** Assume the number of clusters to be made (i.e., *C*, where $2 \le C \le N$).
- **Step 2:** Select an appropriate level of cluster fuzziness f > 1.
- **Step 3:** Initialize the $N \times C$ sized membership matrix [*U*], at random, such that $U_{ij} \in [0,1]$ and $\sum_{j=1}^{C} U_{ij} = 1.0$, for each *i*.

- **Step 4:** Calculate the k[th] dimension of j[th] cluster center $CC_{jk}$ using the following expression:

$$CC_{jk} = \frac{\sum_{i=1}^{N} U_{ij}^{f} x_{ik}}{\sum_{i=1}^{N} U_{ij}^{f}} \quad (1)$$

- **Step 5:** Calculate the Euclidean distance between i[th] data point and j[th] cluster center like the following:

$$D_{ij} = \left\| (x_i - CC_j) \right\| \quad (2)$$

- **Step 6:** Update fuzzy membership matrix [*U*] according to $D_{ij}$. If $D_{ij} > 0$, then:

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$$U_{ij} = \frac{1}{\sum_{c=1}^{C} \left( \frac{D_{ij}}{D_{ic}} \right)^{\frac{2}{f-1}}} \tag{3}$$

If $D_{ij} = 0$, then the data point will coincide with j[th] cluster center $CC_j$ and it will have the full membership value, i.e., $U_{ij} = 1.0$.

- **Step 7:** Repeat Steps 4 through 6, until the changes in [*U*] come out to be less than some pre-specified values.

Using this algorithm, the boundaries of the developed clusters will be fuzzy in nature and there could be some overlapping of two or more clusters.

A parameter γ (in percentage) may be introduced to check the validity of the clusters. If the number of data points contained in a cluster becomes greater than or equal to: $\frac{\gamma N}{100}$, it will be declared as a valid cluster, otherwise the said points will be known as the outliers.

## Entropy-Based Fuzzy Clustering Algorithm

Entropy-based fuzzy clustering (EFC) is an iterative approach, in which the entropy values of the data points are calculated first and then the data point with the minimum entropy value is selected as the cluster center (Yao et al., 2000). Here, the data points are clustered based on the threshold value of similarity. The data points, which are not selected inside any of the clusters, are termed as the outliers. The principle of EFC is explained below.

Let us consider *N* data points in *M*-dimensional [*T*] hyperspace, where each data point $X_i$ ($i$ = 1, 2, 3, …, *N*) is represented by a set of *M* values (i.e., $X_{i1}$, $X_{i2}$, $X_{i3}$, …, $X_{iM}$). Thus, the data set can be represented by an $N \times M$ matrix. A column-wise normalization of the data is done for representing each variable in the range of [0, 1].

The Euclidean distance between any two data points (e.g., *i* and *j*) is determined as follows:

$$D_{ij} = \sqrt{\sum_{k=1}^{M} \left( X_{ik} - X_{jk} \right)^2} \tag{4}$$

Now, similarity between the two points (i.e., *i* and *j*) can be calculated like the following:

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$$S_{ij} = e^{-\alpha D_{ij}}, \tag{5}$$

where $\alpha$ is a numerical constant. Thus, the similarity value between any two points will lie in the range of 0.0 to 1.0. The value of $\alpha$ is determined based on the assumption that the similarity $S_{ij}$ becomes equal to 0.5, when the distance between them (i.e., $D_{ij}$) becomes equal to the mean distance $\overline{D}$, which is represented as follows:

$$\overline{D} = \frac{1}{{}^{N}C_{2}} \sum_{i=1}^{N} \sum_{j=1}^{N, j<i} D_{ij} \tag{6}$$

From equation (5), $\alpha$ can be calculated as follows:

$$\alpha = -\frac{\ln 0.5}{\overline{D}} \tag{7}$$

Now, the entropy of each data point ($E_i$) is calculated with respect to other data points like the following:

$$E_i = -\sum_{j \in X}^{j \neq i} \left(S_{ij} \log_2 S_{ij}\right) + \left(1 - S_{ij}\right) \log_2 \left(1 - S_{ij}\right) \tag{8}$$

The EFC algorithm has been discussed next.

It consists of the following steps:

- **Step 1:** Calculate $E_i$ ($i$ = 1, 2, 3, …, $N$) for each $X_i$ lying in [$T$] hyperspace.
- **Step 2:** Determine minimum $E_i$ and declare $X_{i,Min}$ as the cluster center.
- **Step 3:** Put $X_{i, Min}$ and the data points having similarity with $X_{i,Min}$ greater than $\beta$ (threshold value for similarity) into a cluster and remove them from [$T$].
- **Step 4:** Check whether [$T$] hyperspace is empty. If yes, terminate the program, else go to Step 2.

In this algorithm, $E_i$ is calculated in such a way that a data point that is far away from the rest of the data points, may also be selected as a cluster center. To prevent

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such an odd situation, a threshold value γ (in %) is introduced. If the number of data points present in a cluster becomes greater than or equal to $\frac{\gamma N}{100}$, we declare it as a valid cluster. Otherwise, these data points will be considered as the outliers.

## GA-Fitness Calculation

In this study, a binary-coded GA has been used. Ten bits are assigned to represent each real variable. To improve the performance of EFC algorithm, three real variables, such as α, β, and γ are optimized using the GA, whereas two real variables- f and γ have been varied to optimize the performance of the FCM algorithm. The aim of the present study is to obtain the set of optimal clusters, such that they are compact but at the same time, distinct too. Moreover, the number of outliers should be as minimal as possible. The compactness of a cluster is determined by calculating the average Euclidean distance of the members from its center. The lower the value of average Euclidean distance, the more compact will be the cluster. The distinctness of the developed clusters is decided by calculating the average Euclidean distance of the cluster centers. The clusters are said to be more distinct, if the average Euclidean distance among their centers is more. The above problem has been formulated as a maximization problem, in which the fitness of the GA-string has been expressed as follows:

$$f' = \frac{1}{1+\frac{1}{c}\sum_{k=1}^{c}\left(\frac{1}{n_k}\sum_{i=1}^{n_k}\sqrt{\sum_{j=1}^{M}\left(x_{ij}-v_{kj}\right)^2}\right)} + \frac{1}{{}^{c}C_2}\sum_{i=1}^{c}\sum_{k=1,k\neq i}^{c}\sqrt{\sum_{j=1}^{M}\left(v_{ij}-v_{kj}\right)^2} - O, \tag{9}$$

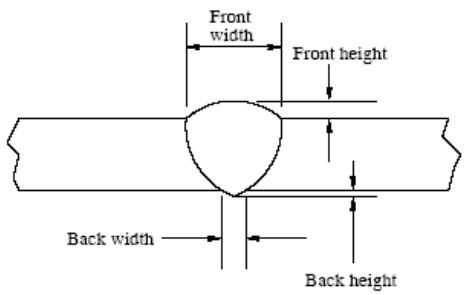
where c represents the number of clusters, $n_k$ indicates the number of points in $k^{th}$ cluster, $M$ denotes the dimension of the data point, $x_{ij}$ represents the $j^{th}$ dimension of $i^{th}$ data point, $v_{kj}$ indicates the $j^{th}$ dimension of $k^{th}$ cluster center and O denotes the number of outliers.

## Results and Discussion

The performances of EFC and FCM clustering techniques have been tested on the data sets related to the above two processes and compared between themselves as follows.

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*Figure 1. A schematic diagram showing the weld bead geometry (Ganjigatti, 2006)*




## Tungsten Inert Gas (TIG) Welding Process

A Bead-on-Plate (BOP) welding was carried out on an aluminum plate using a TIG welding process (Juang et al., 1998). A schematic diagram showing the weld bead geometry (Ganjigatti, 2006)

Figure 1 shows the schematic diagram of weld bead-geometry, which is described using the parameters like front height (FH), front width (FW), back height (BH), and back width (BW), which are dependent on the input process parameters, namely welding speed (A), wire feed rate (B), % cleaning (C), arc gap (D), and welding current (E). Ganjigatti (2006) carried out conventional regression analysis using the data obtained from the literature (Juang et al., 1998) and the following relationships were derived:

$$FH = -17.2504+0.62018A+4.6762B+0.086647C+7.4479D+0.043108E-0.18695AB-0.005792AC-0.22099AD-0.0029123AE+0.0018129BC-1.8396BD+0.019139BE-0.058577CD+0.0017885CE-0.035219DE+0.0014061ABC+0.062296ABD+0.00020568ABE+0.0022313ACD-6.76\times10^{-6}ACE+0.011409ADE+0.0060975BCD-0.0013628BCE-0.0030BDE-0.00027533CDE-0.00042377ABCD+0.0189\times10^{-3}ABCE-0.0459\times10^{-3}ABDE-8.76\times10^{-7}ACDE-0.00037623BCDE-6.87\times10^{-6}ABCDE \quad (10)$$

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$$FW = -329.6758+8.2539A+167.1041B+5.8187C+101.4624D+3.9953E-4.0707AB-0.14141AC-2.5489AD-0.099144AE-2.9150BC-54.1378BD-1.9883BE-1.8510CD-0.068644CE-1.2150DE+0.069989ABC+1.3175ABD+0.048568ABE+0.044112ACD+0.0016977ACE+0.03081ADE+0.93986BCD+0.034547BCE+0.6524BDE+0.022294CDE-0.02226ABCD-0.83924\times10^{-3}ABCE-0.015943ABDE-0.54259\times10^{-3}ACDE-0.011294BCDE+0.27183\times10^{-3}ABCDE \quad (11)$$

$$BH = 20.7999-0.38305A-3.5745B+0.10795C-9.3284D-0.092436E+0.0058295AB-0.0054309AC+0.16652AD+0.00049091AE-0.11145BC+2.2936BD-0.016822BE-0.009152CD-0.0036834CE+0.057568DE+0.0044163ABC-0.023731ABD+0.0014568ABE+0.0015578ACD+0.00012402ACE-0.00055232ADE+0.026228BCD+0.0023696BCE-0.0041BDE+0.00095CDE-0.0013485ABCD-0.0769\times10^{-3}ABCE-0.00031487ABDE-3.93\times10^{-5}ACDE-0.00068428BCDE+0.025\times10^{-3}ABCDE \quad (12)$$

$$BW = -179.4354+4.1209A+104.7708B+4.1113C+52.8753D+2.4368E-2.5474AB-0.094617AC-1.2695AD-0.057292AE-2.2272BC-34.1677BD-1.2973BE-1.3856CD-0.050824CE-0.71979DE+0.052044ABC+0.81877ABD+0.032125ABE+0.031846ACD+0.0012161ACE+0.01818ADE+0.76844BCD+0.026839BCE+0.4353BDE+0.017557CDE-0.017848ABCD-0.64265\times10^{-3}ABCE-0.0.010785ABDE-0.00042119ACDE-0.0093502BCDE+0.22457\times10^{-3}ABCDE \quad (13)$$

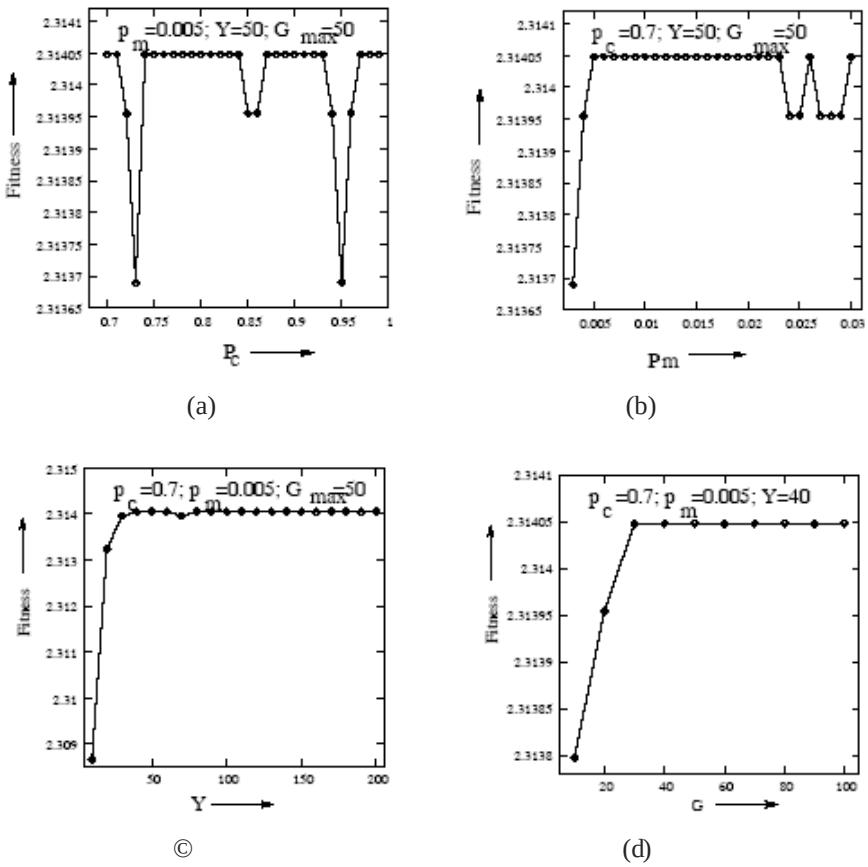
Using the previous equations, one thousand data related to the input-output relationship of the process have been generated by randomly selecting the values of the input variables lying within their respective ranges.

## Entropy-Based Fuzzy Clustering

The performance of the EFC algorithm depends on the parameters like $\alpha$, $\beta$, and $\gamma$. An attempt has been made in the present study to cluster the above data in an optimal sense. As the performance of a GA is dependent on its parameters, namely probability of crossover $p_c$, probability of mutation $p_m$, population size Y and maximum number of generations $G_{max}$, a parametric study is conducted to determine the set of

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*Figure 2. Results of the GA-parametric study (a) fitness vs. pc, (b) fitness vs. pm, (c) fitness vs. population size Y, (d) fitness vs. maximum number of generations Gmax*

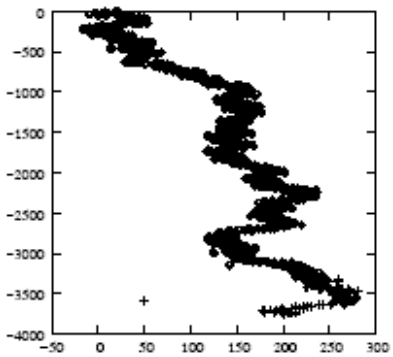



optimal GA-parameters, in which one parameter has been varied at a time, keeping the others unaltered. Figure 2 shows the results of the GA-parametric study. The following GA-parameters are found to yield the best results:

| | |
|---|---|
| Crossover probability | = 0.7 |
| Mutation probability | = 0.005 |
| Population size | = 40 |
| Maximum number of generations | = 30 |

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*Figure 3. Optimal set of clusters on TIG data obtained using EFC algorithm*

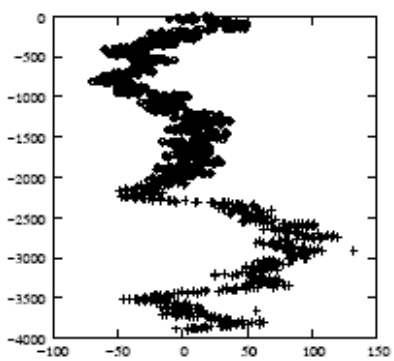



During the optimization, the parameters -$\alpha$, $\beta$, and $\gamma$ are varied in the ranges of (0.6, 2.0), (0.1, 0.99), and (0.02, 0.1), respectively. The optimal values of $\alpha$, $\beta$, and $\gamma$ are found to be equal to 1.356305, 0.261818 and 0.037361, respectively. The set of optimal clusters as previously obtained is shown in Figure 3, after reducing their dimensions to two (using a SOM). The first and second clusters contain 881 and 119 data points, respectively.

## Fuzzy C-Means Clustering

The performance of the FCM algorithm is dependent on the parameters like level of fuzziness f, number of clusters, and threshold value used to check the validity of the clusters $\gamma$. As two clusters are obtained using the EFC algorithm, the number of

*Figure 4. Optimal set of clusters on TIG data obtained using FCM algorithm*




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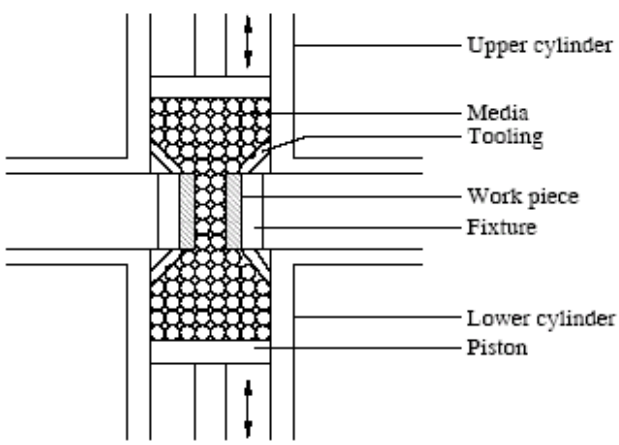
clusters to be made is kept fixed to two only, in the FCM algorithm. The optimal values of f and γ have been determined using a GA. The following GA-parameters are seen to yield the best results:

| | |
|---|---|
| Crossover probability | = 0.7 |
| Mutation probability | = 0.005 |
| Population size | = 50 |
| Maximum number of generations | = 50 |

The parameters f and γ have been varied in the ranges of (1.1, 10.0) and (0.02, 0.1), respectively, during optimization carried out using a GA. The optimal values of f and γ are found to be equal to 1.437243 and 0.040567, respectively. Once the clusters are formed, the higher dimensional data points are mapped into lower dimension using the SOM. The set of optimal clusters obtained using this algorithm is shown in Figure 4. The first cluster contains 504 data points, whereas the second cluster consists of 496 points.

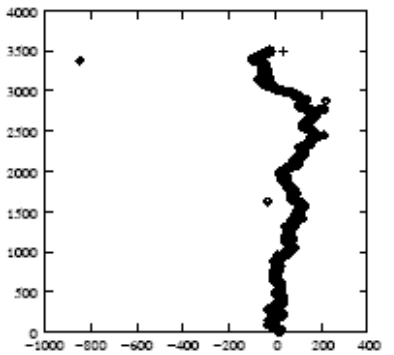
It is to be noted that both the algorithms are able to yield compact as well as distinct clusters. However, the FCM algorithm is able to distribute the data points into two clusters almost equally, whereas the EFC algorithm has put 88.1% of the data points into one cluster and the remaining points into another. As one thousand data points are created at random, it may be realistic to think that they will be distributed almost equally into the clusters and the FCM algorithm is able to hit that target.

*Figure 5. A schematic diagram showing the AFM (Jain et al., 2000)*




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*Figure 6. Optimal set of clusters on AFM data obtained using EFC algorithm*




## Abrasive Flow Machining (AFM) Process

It is a machining process used to polish the surface of a material (Jain et al., 2000). Here, an abrasive-laden pliable semisolid compound is forced to pass to-and-fro across the surface to be machined (refer to Figure 5).

It is to noted that material removal rate (MRR in mg/min) and roughness ($R_a$ in μm) of the machined surface mainly depend on four input parameters, namely media flow speed $v$ (cm/min), percentage concentration of abrasive $c$, abrasive mesh size $d$ (μm) and number of cycles $n$. The following empirical equations for MRR and surface roughness were obtained (Jain et al., 2000):

$$MRR = 5.285\times10^{-7}v^{1.6469}c^{3.0776}d^{-0.9371}n^{-0.1893} \quad (14)$$

$$R_a = 282751.0\times v^{-1.8221}c^{-1.3222}d^{0.1368}n^{-0.2258} \quad (15)$$

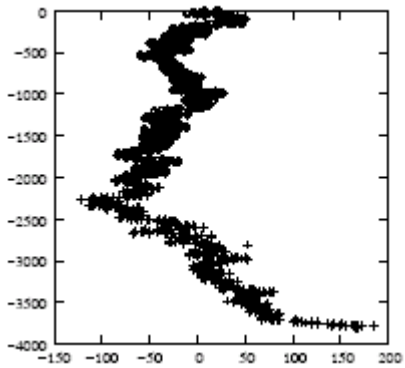
where $40.0 \le v \le 85.0$; $33.0 \le c \le 45.0$; $100.0 \le d \le 240.0$; $20 \le n \le 120$.

## Entropy-Based Fuzzy Clustering

A GA is used to optimize the parameters of the EFC algorithm. A detailed parametric study is carried out and the following GA-parameters are found to give the best results: $p_c = 0.8$, $p_m = 0.12$, $Y = 50$, $G_{max} = 100$. The optimal values of $\alpha$, $\beta$, and $\gamma$ of the algorithm are obtained as 1.709873, 0.10261 and 0.020938, respectively and only two clusters containing 979 and 21 data points are obtained. The higher

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*Figure 7. Optimal set of clusters on AFM data obtained using FCM algorithm*




dimensional clustered data are mapped into 2-D using the SOM. Figure 6 shows the set of optimal clusters obtained using this algorithm.

## Fuzzy C-Means Clustering

The GA-parametric study has yielded the following optimal parameters: $p_c = 0.86$, $p_m = 0.005$ Y = 50, $G_{max} = 100$. As only two clusters are formed using the EFC algorithm, the number of clusters has been set equal to two, in this algorithm. The optimized values of f and $\gamma$ are found to be equal to 1.1 and 0.08303, respectively. The set of optimal clusters determined by this algorithm is shown in Figure 7. The two clusters are found to contain 503 and 497 data points, respectively.

It is interesting to note that the FCM algorithm has yielded the better set of clusters compared to that obtained by the EFC algorithm. It is important to note that a few points are found to deviate considerably with respect to the other points lying in the clusters, as shown in Figures 3, 4, 6, and 7, although both the algorithms are able to yield the outliers-free clustering. It might have happened due to the technique of final mapping (refer to Figure 10) adopted in the self-organizing map.

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*Figure 8. A schematic diagram showing the working cycle of a GA*

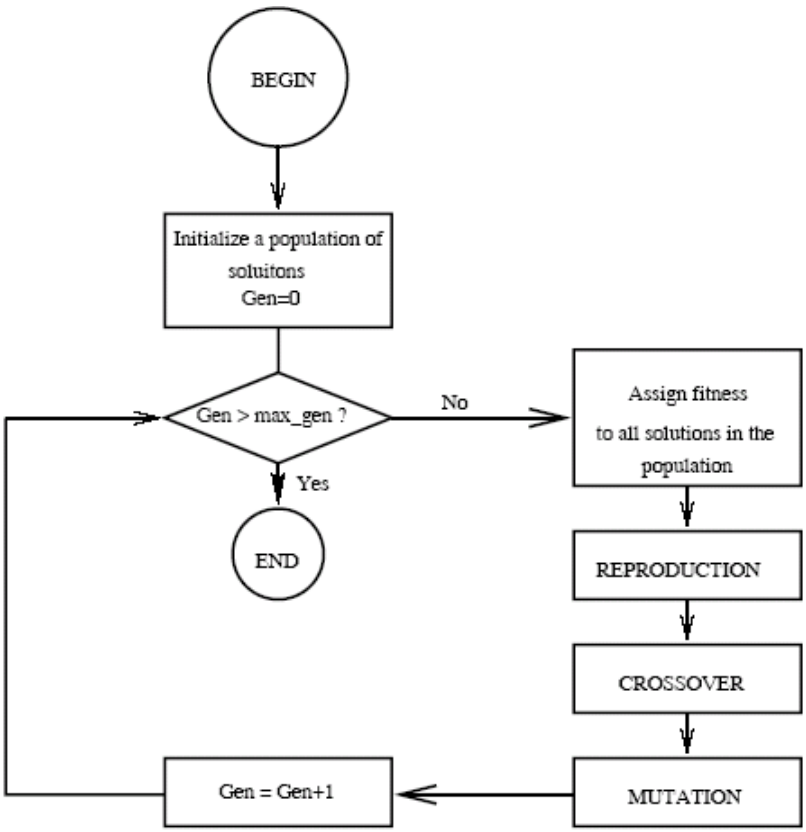
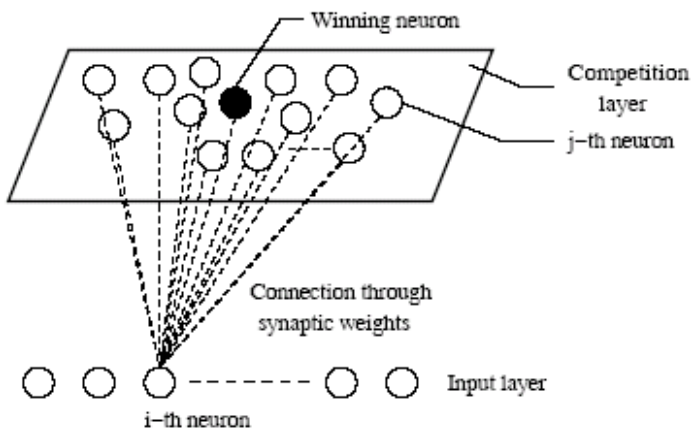
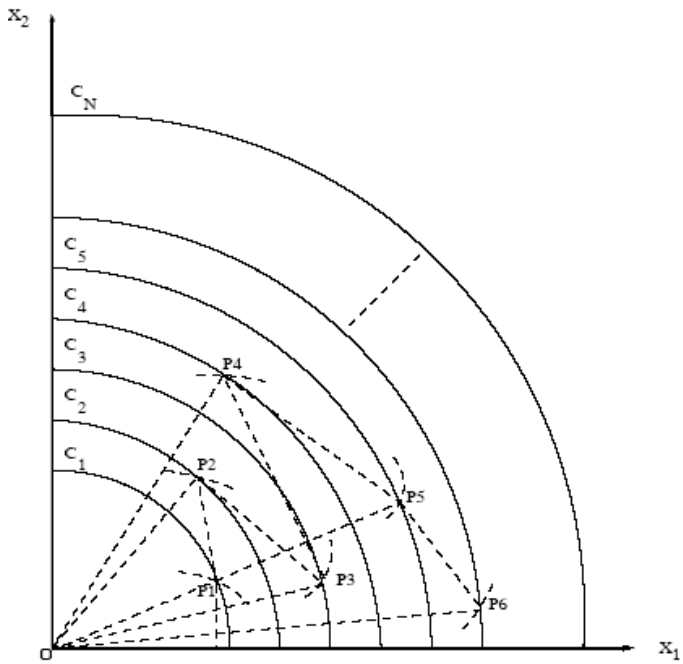



*Figure 9. A schematic diagram of a self-organizing map (Dutta & Pratihar, 2006)*




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*Figure 10. Final mapping of the data in 2-D (Dutta et al., 2006)*




# Conclusion

In the present study, the parameters of two clustering algorithms have been optimized using a GA, so that they can cluster a data set in an optimal sense. The obtained clusters are declared to be the optimal ones, if they are distinct as well as compact in nature. Moreover, the algorithm should be able to cluster as many data points as possible. The performances of two algorithms, namely FCM and EFC have been tested on the data sets related to two physical processes, namely TIG welding and AFM processes. Both the algorithms are able to generate the compact as well as distinct clusters after ensuring zero outliers. However, the FCM algorithm is able to distribute the data points almost equally into two clusters. As the clustering is done on the randomly generated data related to the above physical processes, it may be expected that the data points will be more or less equally distributed into the clusters. This particular fact has been revealed by the FCM algorithm but the EFC algorithm has failed to do so. Thus, the FCM algorithm has performed better than the EFC algorithm. However, their performances may be dependent on the data sets to be clustered.

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# References

Bezdek, J. C. (1981). *Pattern recognition with fuzzy objective function algorithms*. New York: Plenum Press.

Dunn, J. C. (1974). A fuzzy relative of the ISODATA process and its use in detecting compact well-separated clusters. *Journal of Cybernetics*, *3*, 32-57.

Dutta, P., & Pratihar, D. K. (2006). Some studies on mapping methods. *International Journal on Business Intelligence and Data Mining, 1*(3), 347-370.

Ganjigatti, J. P. (2006). *Application of statistical methods and fuzzy logic techniques to predict bead geometry in welding*. PhD thesis, Indian Institute of Technology, Kharagpur, India.

Goldberg, D. E. (1989). *Genetic algorithms in search, optimization, machine learning*. Reading, MA: Addison-Wesley.

Haykin, S. (1999). *Neural networks*. Pearson Education.

Hruschkra, E. R., Campello, R. J. G. B., & de Castro, L. N. (2004). Evolutionary search for optimal fuzzy C-means clustering. *Proceedings of the FUZZ-IEEE Conference* (pp. 685-690). Budapest, Hungary.

Jain, V. K., & Adsul, S. G. (2000). Experimental investigations into abrasive flow machining. *International Journal of machine Tools and Manufacture*, *40*, 1003-1021.

Jain, R.K., & Jain, V.K. (2000). Optimum selection of machining conditions in abrasive flow machining using neural networks. *Journal of Materials Processing Technology*, *108*, 62-67.

Juang, S. C., Tarng, Y. S., & Lii, H. R. (1998). A comparison between the back-propagation and counter-propagation networks in the modeling of the TIG welding process. *Journal of Materials Processing Technology*, *75*, 54-62.

Kohonen, T. (1995). *Self-organizing maps*. Heidelberg: Springer-Verlag.

Yao, J., Dash, M., Tan, S. T., & Liu, H. (2000). Entropy-based fuzzy clustering and fuzzy modeling. *Fuzzy Sets and Systems, 113*, 381-388.

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# Appendix

## Binary-Coded GA

Genetic algorithm (GA) is a population-based search and optimization technique, which works based on the mechanism of natural genetics and Darwin's principle of natural selection (Goldberg, 1989). It was introduced by Prof. John Holland of the University of Michigan, USA in the year 1965. The working principle of a GA is explained with the help of Figure 8 in the form of a flowchart.

There are several versions of the GA such as binary-coded GA, real-coded GA, messy GA and others. In the present study, a binary-coded GA is used. The string of a binary-coded GA will look the following:

1010011110011….1000101110010

The variables (discrete and/or real in nature) of the problem to be optimized are represented using the binary-string. The GA starts with a population of initial solutions, chosen at random. The fitness value (objective function value) of each solution in the population is determined. The population of solutions is then modified using three main operators, namely reproduction, crossover, and mutation, to create a new population of solutions. Reproduction operator selects good strings from the population using fitness information to form a mating pool. There exist a number of reproduction schemes, namely proportionate selection, ranking selection, tournament selection, and others. The strings present in the mating pool form the mating pairs called the parents and each pair consists of two binary strings. In crossover, there is an exchange of properties between two binary strings and two children solutions are produced by one mating pair. There are various types of crossover operators in use, such as single-point crossover, two-point crossover, multi-point crossover, uniform crossover and others. The crossover operator is mainly responsible for the search of new strings. To reduce the chance of destructing the already-found good strings, crossover is usually performed with a high probability value slightly smaller than one. Mutation operator changes 1 to 0 and vice versa with a small probability. Mutation is used for obtaining a local change around the current solution. In brief, reproduction operator selects good strings and crossover operator recombines two good strings to hopefully create better strings. The mutation operator alters a string locally to create a new string. After reproduction, crossover, and mutation are applied to the whole population of solutions, one generation of a GA is completed. The GA is run until the termination criterion is reached. It is important to mention that there could be different termination criteria, namely a maximum number of generations, a desired accuracy in the solution and others.

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## Self-Organizing Map

Self-organizing map (SOM) (also known as Kohonen network) is a dimensionality reduction technique proposed by Kohonen (1995), which is useful for data visualization. It is a special type of neural network, which works based on unsupervised and competitive learning. Figure 9 shows the schematic diagram of a self-organizing map. The SOM can be viewed as a non-linear generalization of principal component analysis, which can preserve the topology in mapping process. There are two layers in SOM, namely input and competition layers. The multivariate data are fed to the input layer, which are to be mapped to a lower dimensional space. The number of neurons in the competition layer is kept equal to that of data points present in the input layer. Each multivariate data present in the input layer is connected to all the neurons in the competition layer through some synaptic weights. The neurons in the competition layer undergo three basic operations, such as *competition*, *cooperation*, and *updating*, in stages. In the competition stage, the neurons in the competition layer compete among themselves to be the winner. In the next stage, a neighborhood surrounding the winning neuron is identified for cooperation among themselves. The winning neuron along with its neighbors is updated in the third stage. The above three stages are explained below in detail.

### *Competition*

Let us consider $N$ points (neurons) in the input layer and each point has m dimensions. A particular input neuron $X_i$ can be represented as follows:

$$X_i = [x_{i1}, x_{i2}, ....., x_{im}]^T \text{ where } i = 1, 2, \ldots, N.$$

Let us also assume that the synaptic weight vector of neuron j lying in the competition layer is indicated like the following:

$$W_j^i = [w_{j1}^i, w_{j2}^i, ..., w_{jm}^i]^T \text{ where } j = 1, 2, \ldots, N.$$

To find the best match for the input data vector $X_i$, the minimum of the Euclidean distances between $X_i$ and $W_j^i$ is determined. Let us represent the neuron lying in the competition layer that has the best match with the input vector $X_i$ by $n$. Thus, the Euclidean distance between n and $X_i$ can be expressed as follows:

$$n(X_i) = \text{Minimum of } \sqrt{(X_i - W_j^i)^2} \text{ where } j = 1,2,....,N .$$

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## *Cooperation*

The winning neuron determines its topological neighborhood of excited neurons for cooperation. They will try to update their synaptic weights through the cooperation. The neighborhood function is assumed to follow a Gaussian distribution as given next:

$$h_{j,n(X_i)}(t) = \exp\left(-\frac{d^2_{j,n(X_i)}}{2\sigma_t^2}\right) \text{ where } t = 0, 1, 2, \ldots$$

where $d_{j,n(Xi)}$ represents the lateral distance between the winning neuron n and the excited neuron j, $\sigma_t$ indicates the value of standard deviation at $t^{th}$ iteration, which is expressed as $\sigma_t = \sigma_0 \exp\left(-\frac{t}{\tau}\right)$, where $\sigma_0$ is the initial value of standard deviation and $\tau$ denotes the maximum number of iterations. In this study, $\sigma_0$ and $\tau$ are assumed to be equal to 1.0 and 100, respectively. It is important to note that the topological neighborhood shrinks with the number of iterations.

## *Updating*

The synaptic weights of the winning neuron and the excited neurons lying in its neighborhood are updated using the rule given below:

$$W_j^i(t+1) = W_j^i(t) + \eta(t) h_{j,n(X_i)}(t)[X_i - W_j^i(t)],$$

where $\eta(t)$ represents the learning rate lying between 0 and 1, which also generally decreases with iteration t. the above procedure is adopted to determine the winning neuron corresponding to each input data.

## *Final Mapping*

The higher dimensional (input) data are to be mapped into the lower dimension (output) after keeping their topological information intact. In higher dimensional space, the Euclidean distances of the winning neurons from the origin are calculated and this information is utilized to draw a number of circular arcs (equal to the number of winning neurons) in 2-D (refer to Figure 10).

In 2-D, the winning neurons are then located such that the distance between any two neurons is kept the same as that in higher dimensional space. Thus, the topological information of the higher dimensional data is kept unaltered in 2-D.

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## Chapter VI

# K-Means Clustering Adopting rbf-Kernel

ABM Shawkat Ali, Central Queensland University, Australia

## Abstract

*Clustering technique in data mining has received a significant amount of attention from the machine learning community in the last few years as one of the fundamental research areas. Among the vast range of clustering algorithms, K-means is one of the most popular clustering algorithm. In this research, we extend the K-means algorithm by adding well known radial basis function (rbf) kernel and find better performance than classical K-means algorithm. It is a critical issue for rbf kernel; how can we select a unique parameter for optimum clustering task. This chapter will provide a statistical-based solution on this issue. The best parameter selection is considered on the basis of prior information of the data by the maximum likelihood (ML) method and nelder-mead (N-M) simplex method. A rule based meta-learning approach is then proposed for automatic rbf kernel parameter selection. We consider 112 supervised data set and measure the statistical data characteristics using basic statistics, central tendency measure, and entropy-based approach. We split these data characteristics using the well-known decision tree approach to generate the rules. Finally, we use the generated rules to select the unique parameter value for rbf kernel and then adopt in K-means algorithm. The experiment has been demonstrated with 112 problems and 10 fold cross validation methods. Finally, the proposed algorithm can solve any clustering task very quickly with optimum performance.*

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# Introduction

Data mining can quite often be defined as a useful hidden knowledge extraction process from a huge database. Basically, two types of techniques, supervised and unsupervised, are using to extract this knowledge. When the problem is not predefined, then the researcher always chooses the unsupervised technique to solve their problems. Now a days, a good number of unsupervised techniques introduced by researchers and are free for use. K-means algorithm is an old unsupervised technique but still it is a popular technique. The job of unsupervised technique is called clustering. In 1967, MacQueen (1967) developed the K-means clustering algorithm for classification and analysis of multivariate observations. Since then, while unsupervised techniques have been studied extensively in the areas of statistics, machine learning, and data mining (Zalane, 2007), the K-means algorithm has been applied to many problem domains, including the area of data mining, and has become one of the most used clustering algorithms. Matteucci (2007) even said that the K-means algorithm is one of the simplest unsupervised learning algorithms that solve the well-known clustering problem. Recently we found after adding the kernel (Vapnik, 1995) components with K-means algorithm, it became a more popular and powerful unsupervised technique (Dhillon, Guan, & Kulis, 2004, 2005; Kulis, Basu, Dhillon, & Mooney, 2005; Zhang & Rudnicky, 2002). In general, kernel function implicitly defines a non-linear transformation that maps the data from their original space to a high dimensional feature space where the data are expected to be more separable. As a result, the kernel methods may achieve better performance by working in the new space (Zhang et al., 2002). Kernel method is comfortable for both linear and non-linear space. Moreover, it can handle any high dimensional data in their transformation space. Three types of classical kernels namely linear, polynomial, and rbf are introduced initially with kernel-based learning algorithms. Among these, rbf kernel is quite popular and many popular kernels for a specific problem are available today (Cheng, Saigo, & Baldiet, 2006; Ou, Chen, Hwang, & Oyang, 2003). The critical issue of rbf kernel is to select unique parameter within a range of values. We proposed a rule-based methodology for rbf kernel parameter selection using statistical data characteristics (Ali & Smith, 2005). In this research, we introduced this method with classical K-means algorithm. First, we do clustering 112 supervised problems by K-means algorithm. The data are considering from two different sources (Blake & Merz, 2002; Lim, 2002). After that, we implemented the rbf kernel with automated parameter selection in K-means algorithm and perform the clustering task with the similar data.

This chapter is organized as follows: In the next section we shall provide the theoretical frameworks regarding K-means clustering, rbf kernel, and it's automated parameter selection with statistical formulation and measures. Then we shall describe the analyses of the experimental results. Finally, we conclude our research toward the end of this chapter.

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# Theoretical Background

K-means learning algorithm constructs the partitions among the data instances into a selected number of clusters under some optimization measures. For instance, we often want to minimize the sum of squares of the Euclidean distance between the instances and the centroids (Zhang & Rudnicky, 2002). K-means algorithms basically frequently change the centroids and set the optimum one. Similarly, there have been different options for distance measurement. The centroids identification and distance measure depends on the quality of the data. In this respect, we introduce rbf kernel and its automated parameter setting for K-means algorithm. In the following section, first we describe the kernel K-means algorithm and then the new addition rbf kernel with automated parameter selection.

## Kernel K-Means Algorithm

Let us consider a data set with $N$ samples $\mathbf{x}_1$, $\mathbf{x}_2$, ..., $\mathbf{x}_N$. K-means algorithm aims to partition the $N$ samples into $K$ clusters, $C_1$, $C_2$, ..., $C_K$, and then returns the centre of each cluster, $\mathbf{m}_1$, $\mathbf{m}_2$, ..., $\mathbf{m}_K$, as the representatives of the data set. Thus, a $N$-point data set is compressed to a $K$-point "code book" (Zhang & Rudnicky, 2002). The batch mode kernel K-Means clustering algorithm using Euclidean distance works as follows:

1. Assign $\delta(\mathbf{x}_i, C_k)(1 \le i \le N, 1 \le k \le K$ with initial value, forming $K$ initial clusters $C_1$, $C_2$, ..., $C_K$.
2. For each cluster $C_K$, compute $\|C_K\|$ and g($C_K$).
3. For each training sample $\mathbf{x}_i$ and cluster $C_k$, compute $f(\mathbf{x}_i, C_k)$. And then assign $\mathbf{x}_i$ to the closest cluster:

$$\delta(\mathbf{x}_i, C_k) = \begin{cases} 1 & f(\mathbf{x}_i, C_k) + g(C_k) < f(\mathbf{x}_i, C_j) + g(C_j) \\ & for \ \ all \ \ j \neq k \\ 0 & otherwise \end{cases} \quad (1)$$

4. Repeat step 2 and 3 until converge.
5. For each cluster $C_k$, select the sample that is closest to the centre as the representative of $C_k$.

$$\mathbf{m}_k = \underset{x_i \ that \ \delta(\mathbf{x}_i, C_k)=1}{\arg\min} D(\varphi(\mathbf{x}_i), \mathbf{z}_k) \quad (2)$$

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The main difference of the kernel K-means clustering and traditional K-means is described in equation (2). The $\phi(\mathbf{x}_i)$ is used to represent the kernel function, which is replaced by rbf kernel in our implementation.

## rbf Kernel and Parameter Selection Estimation

The popular radial basis function (rbf) kernel is such as:

$$K(x_i, x_j) = \exp\left(-\frac{\left\|x_i - x_j\right\|^2}{2h^2}\right) \text{ where the rbf parameter } h > 0 \qquad (3)$$

The most common method to choose the best rbf kernel parameter is manual trial-and-error selection within the range 0.2-1. A graphical comparison among different parameter values and then effect on rbf kernel is explained in Figure 1 for a binary class synthetic problem. The rectangular and the cross signs indicate the two different classes of the problem.

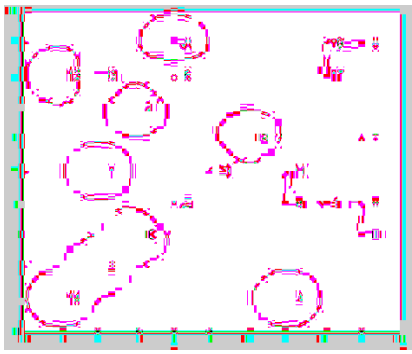
The rbf kernel with width 0.8 and 1 classifies all patterns perfectly with a single optimal hyperplane. But the other parameters for the rbf kernel construct several optimal boundaries to classify the all patterns. It is interesting to observe from Figure 1 how each rbf kernel width generates the optimal hyperplane, and how certain parameter values for the kernel are quite limited in their ability to find the optimal hyperplane for highly non-separable data. It is a time consuming task for large datasets to manually select the parameter value from a certain range. Moreover, there is no guarantee that the best value is in this predefined range.

In an attempt to automate this process, we will examine two different rbf width estimation methods, namely maximum likelihood (ML) (Ross, 2000) and nelder-mead (N-M) simplex method (Lagarias, Reeds, Wright, & Wright, 1998), present comparative performance results, and then attempt to gain insight into which method should be used for certain datasets.

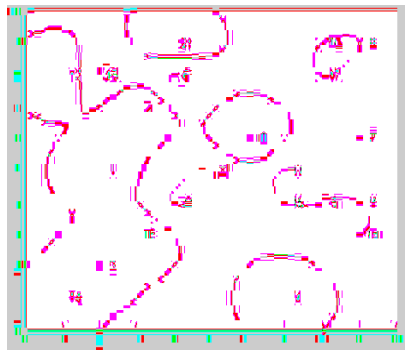
We consider ML and N-M simplex method to estimate best rbf kernel width. The ML method estimates the variance of the data set and then the normalized variance is considered as the width of the rbf kernel. The N-M simplex method searches for the appropriate variance from the transformed data and then selects this value as the best width for rbf kernel. This non-constrained optimization process is a faster method than some other constrained optimization methods. In this following section, we will explain both methods with performance evaluation results. We summarize both of these methods from Ross (2000) and Lagarias, et al. (1998), which are well studied in the statistical community, but have not been applied to rbf kernel parameter estimation before to the best of our knowledge.

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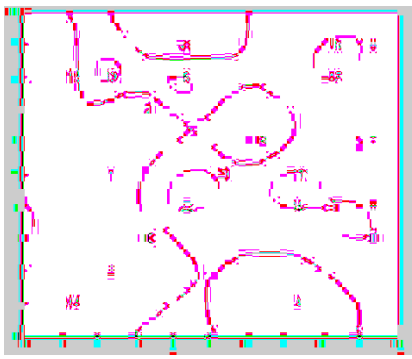
*Figure 1. A pictorial view of the rbf kernel performance on an artificial dataset with different width effect. The cross and rectangular sign indicates the two classes of data. The middle continuous lines (for width 1) of the above graphs represent the optimal hyperplane for classification.*



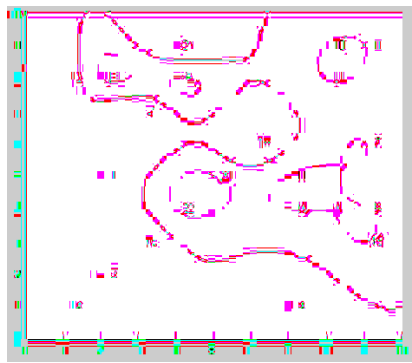
(a) rbf width =0.2: 3 classification errors



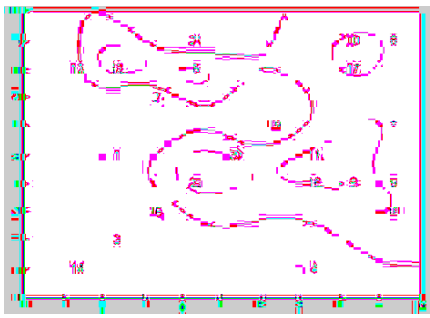
(b) rbf width =0.4: 1 classification errors



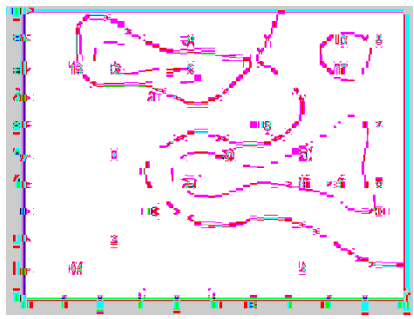
(c) rbf width =0.6: 0 classification errors



(d) rbf width =0.8: 0 classification errors



(e) rbf width =1: 0 classification errors



(f) rbf width =1.2: 0 classification errors

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## Maximum Likelihood Method

Maximum likelihood method has been a very popular parameter estimation method in the statistical community for many years. Let us consider, $X_1$, ..., $X_n$ to be independent, prior normal distribution with unknown mean ($\mu$) and standard deviation ($\sigma$) (Ross, 2000). Now, the density function is as follows:

$$f(x_1,\cdots,x_n \mid \mu,\sigma) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma}} \exp\left[\frac{-(x_i-\mu)^2}{2\sigma^2}\right] = \left(\frac{1}{2\pi}\right)^{n/2} \frac{1}{\sigma^n} \exp\left[\frac{-\sum_{1}^{n}(x_i-\mu)^2}{2\sigma^2}\right] \quad (4)$$

The logarithm of the likelihood density function is given by:

$$\log f(x_1,\cdots,x_n \mid \mu,\sigma) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{\sum_{1}^{n}(x_i-\mu)^2}{2\sigma^2} \quad (5)$$

Now, after differentiating with respect to $\mu$ and $\sigma$, we can write:

$$\frac{d}{d\mu}\log f(x_1,\cdots,x_n \mid \mu,\sigma) = \frac{\sum_{1}^{n}(x_i-\mu)^2}{\sigma^2} \quad (6)$$

$$\frac{d}{d\sigma}\log f(x_1,\cdots,x_n \mid \mu,\sigma) = -\frac{n}{\sigma} + \frac{\sum_{1}^{n}(x_i-\mu)^2}{\sigma^3} \quad (7)$$

By equating these above two equations to zero, we find the maximum likelihood is obtained when the width $\sigma$ of the rbf kernel is:

$$\sigma = \left(\frac{\sum_{n=1}^{n}(x_i-\hat{\mu})^2}{n}\right)^{1/2}, \text{ where } \hat{\mu} = \frac{\sum_{i=1}^{n} x_i}{n}. \quad (8)$$

Now, the calculated $\sigma$ value can be considered as the smoothing parameter *h* of rbf kernel. Since $\sigma$ and rbf kernel parameter *h* are both serving as variance measures, we approximate *h* using equation (8).

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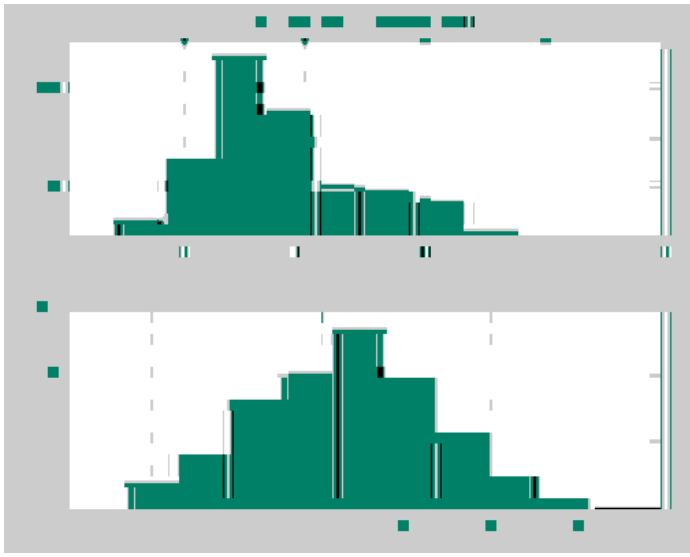
The effectiveness of σ (= 0.1) from maximum likelihood estimation on wdbc dataset is shown in Figure 2, which made the nonnormally distributed wdbc dataset as a normally distributed dataset.

## Nelder-Mead Simplex Method

In our previous study, we observed the rbf kernel performs best if the data is normally distributed (Ali & Smith, 2003). We assume the data distribution is normal if the interquantile range of the data set is close to 1.3. The N-M simplex method is suitable for finding the parameter to transform data into normal distribution. Reshaping the problem, one can find the best smoothing parameter for rbf kernel *h* so that the data is effectively transformed.

The N-M simplex method for unconstrained optimization has been used extensively to solve parameter estimation problems over a few decades. It is still the method of choice in statistics, engineering, and the physical and medical sciences due to its ease of use. This method does not require derivatives and is often claimed to be robust for problems with discontinuous attribute values (Lagarias, et al. 1998). First, we transform the data by following a box-cox transformation (Gentle, 2002) to produce data, which follows a normal distribution more closely than the original data:

*Figure 2. The effect of (σ = 0.1) on wdbc dataset. The suitable σ value makes the non nonnormally distributed wdbc dataset as a normally distributed dataset.*



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$$\tau(X;\lambda)=\begin{cases}(X^{\lambda}-1)/\lambda & \text{if } \lambda \neq 0\\ \log X & \text{if } \lambda = 0\end{cases} \tag{9}$$

The previous transformation can be used only for positive response variables. Box and cox (Gentle, 2002) suggested the transformation for negative elements variable as follows:

$$\tau(X;\lambda)=\begin{cases}\dfrac{(X+\delta)^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0\\ \log(X+\delta) & \text{if } \lambda = 0\end{cases} \tag{10}$$

Now our aim is to find the appropriate value of λ, which can be considered as similar to the width of the rbf kernel since they are both measures of variance. We use N-M simplex method to find out the best value of λ.

Each iteration of the N-M method begins with a simplex, specified by its *n+1* vertices and the associate function values. One or several test points are computed corresponding to their function values, and then the iteration terminates with a new simplex such that the function values at its vertices satisfy some form of descent condition compared with previous simplex.

One iteration of the N-M simplex algorithm consists of the following steps:

1. **Order:** Order and re-label the *n*+1 vertices as $X_1$, ..., $X_{n+1}$ such that $f(X_1) \leq, \cdots, \leq f(X_{n+1})$. Since we want to maximize, we refer to $X_1$ as the *best* vertex or point, to $X_{n+1}$ as the *worst* point, and to $x_n$ as the *next-worst* point. Let $\overline{X}$ refer to the centroid of the *n* best points in the vertex.

2. **Reflect:** Compute the *reflection* point $X_r$, $X_r = \overline{X} + \rho(\overline{X} - X_{n+1})$, where ρ is a parameter.

   Evaluate $f(X_r)$. If $f_1 \leq f_r < f_n$, accept the reflected point $X_r$ and terminate the iteration.

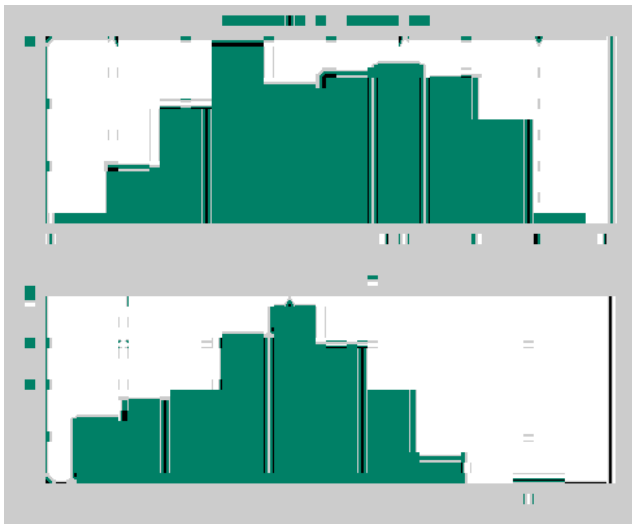
3. **Expand:** If $f_r < f_1$, compute the *expansion* point $x_e$, $X_e = \overline{X} + \chi(X_r - \overline{X})$, where χ is a parameter.

   If $f_e < f_r$ accept $X_e$ and terminate the iteration; otherwise (i.e., if $f_e \geq f_r$) accept $X_r$ and terminate the iteration.

4. **Contract:** If $f_r \geq f_n$, perform a contraction between $\overline{X}$ and the better of $X_{n+1}$ and $X_r$:

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*Figure 3. The effect of Box-Cox transformation on wine dataset (λ = 0.5). The suitable λ value makes the non nonnormally distributed wine dataset as a normally distributed dataset.*



$X_c = \overline{X} - g(\overline{X} - X_{n+1})$, where γ is a parameter.

If $f_c < f_{n+1}$ accept $X_C$ and terminate the iteration.

5. **Shrink simplex:** Evaluate *f* at the *n* new vertices for *i* = 1, ..., *n*.

$V_i = X_1 + \varsigma(X_i - X_1)$, where ς is a parameter.

Now the highest vertex point is considered as the optimum value of λ and is also considered as the smoothing parameter *h* of rbf kernel. For the four coefficients $(\rho, \chi, \eta \text{ and } \varsigma)$, the standard values reported in Lagarias et al. (1998) have been adopted.

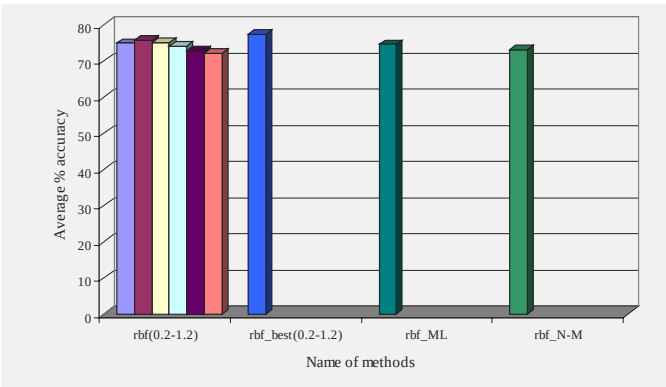
The effectiveness of λ with box-cox transformation on wine dataset is shown in Figure 3. The suitable value of λ is 0.5, which converted the nonnormally distributed wine dataset into a normally distributed dataset.

## rbf Width Estimation Algorithms Performance: Accuracy

The average test set classification performance of rbf kernel with parameter 0.2-1.2, rbf_best (best performance manually selected from width 0.2-1.2), best width approximation by ML, and NM methods is shown in Figure 4.

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*Figure 4. Average test set accuracy for different rbf kernel parameter fitting methods for problems satisfying the rule*




The ML and N-M methods showed close performance with the best rbf accuracy found through exhaustive search of width range 0.2 to 1.2. Both methods showed average higher accuracy than some of individual rbf width performance. For large datasets (more than 1000 samples) rbf_best showed average accuracy 77.52%, ML and N-M methods showed 75.94% and 71.41%. The rbf_best means some data sets are more suitable with rbf kernel than others kernel. The ML showed better performance than N-M method. On the other hand for small datasets (less than 1000 samples) rbf_best showed average accuracy 69.76% and N-M methods showed 67.26% and 64.98%. The ML again showed better performance than N-M method for small dataset. The ML method predicted the best width for rbf kernel for 29.31% of the datasets where rbf kernel is expected to be best. On the other hand N-M method predicted the best width for 31.03% of the datasets. We observed that 24.13% of the datasets have best width outside the range of 0.2 to 1.2. For many of the datasets ML and N-M methods predicted the same rbf width among the 112 problems. The rbf kernel performance with datasets better suited to other (non-rbf) kernels is shown in Figure 5.

## rbf Width Estimation Algorithms Performance: Computational Time

The computational performance to determine the best rbf width using the three methods: rbf_best (exhaustive search of width 0.2-1.2) and estimation by ML and N-M methods, is shown in Table 1.

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*Figure 5. Average test set accuracy for different rbf kernel parameter fitting methods for problems not satisfying the rule*

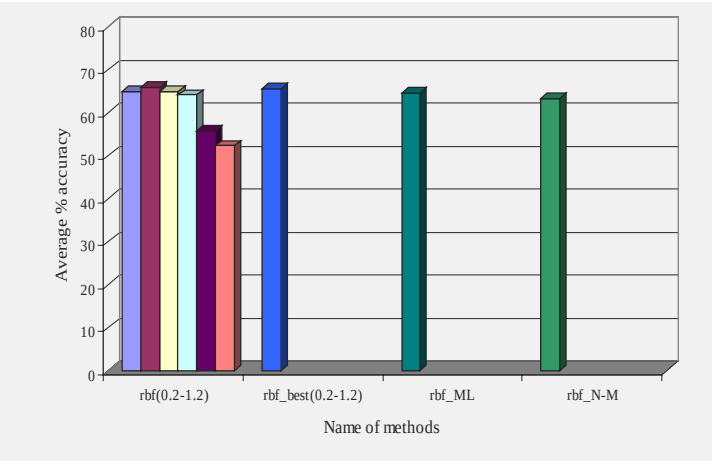



*Table 1. Average computational performance for different rbf kernel width estimation methods*

| Average computational time in Sec. | rbf_best | rbf_ML | rbf_N-M |
|---|---|---|---|
| | 1269.39 | 0.5336 | 5.1350 |

The exhaustive best width search method needed extremely higher computational time than ML and N-M methods. It selects the rbf width one by one from a range 0.2-1.2 to train the SVM rbf model. But both ML and N-M methods estimate the best rbf width for SVM by simply implementing equation (8) and a simple iteration of equation (10) respectively that estimates the likely performance of the SVM model without the need to build such models. Therefore, ML and N-M methods show superior computational performance compared to the exhaustive search method.

## Best rbf Width Methods Significance Test

The t-test results for different rbf kernel width estimation methods are summarized in Table 2. We considered the base kernel as rbf_best. The test input was the percentage of correct classification for all width estimation methods.

The outputs of $H = 0$ in Table 2 indicated we may not reject the null hypothesis that both methods are equally significant. Alternatively, $H = 1$ means we may reject the null hypothesis. The rbf_best with rbf width 1-1.2 showed significant performance difference. The lower values of the significance level suggested rejecting the null

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*Table 2. Results of the t-test of all methods for rbf width selection*

| Algorithms | Hypothesis | Significance | Confidence Interval | |
|---|---|---|---|---|
| | *H* | $\delta$ | *CI* | |
| rbf_best vs rbf_0.2 | 0 | 0.4539 | -7.6381 | 3.4257 |
| rbf_best vs rbf_0.4 | 0 | 0.4368 | -7.8229 | 3.3904 |
| rbf_best vs rbf_0.6 | 0 | 0.2093 | -9.4083 | 2.0726 |
| rbf_best vs rbf_0.8 | 0 | 0.0732 | -11.1674 | 0.5058 |
| rbf_best vs rbf_1 | 1 | 0.0346 | -12.2829 | -0.4664 |
| rbf_best vs rbf_1.2 | 1 | 0.0171 | -13.1628 | -1.3012 |
| rbf_best vs rbf_ML 0 | 0 | 0.4064 | -7.8649 | 3.1961 |
| rbf_best vs rbf_NM 0 | 0 | 0.0746 | -11.2024 | 0.5342 |

hypothesis. The 95% confidence intervals for these methods are highly positively skewed are shown in Table 2. But, the rbf_best with rbf width 0.2-0.8, ML and N-M methods showed no significant performance difference. The higher values of the significance level suggested accepting the null hypothesis. The 95% confidence intervals for these kernels are highly balance skewed as shown in Table 2. The ML and N-M methods give results comparable to exhaustive search, but are much faster to implement.

The average percentage of classification performance and significance testing has shown that classification accuracy depends on particular rbf kernel width selection. We observe from both of these best rbf widths estimation methods the best width could be outside the range 0.2-1.2. Any single method is not always best to estimate the best rbf width for all problems. So, we need a method to provide a priori information about which best width estimation method is suitable for which classification problem with SVM. This will be further pursued in the next section.

# Datasets Characteristics Measurement

Each dataset can be described by a number of simple, statistical, and information theoretical measures (Smith, Woo, Ciesielski, & Ibrahim, 2001, 2002). We average some statistical measures over all the variables and take these as a global measure of the dataset characteristics.

## Simple Measures

Some simple measures for data set characteristics are shown in Table 3. These are the dimensions of each problem, the number of minority and majority samples and the nature of the variables.

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*Table 3. Simple measures for characterization of each dataset*

| Measure | Notation |
|---|---|
| # of attributes | *a* |
| # of samples | *s* |
| % of minority class | *c_min* |
| % of majority class | *c_max* |
| % of binary variables | *b_var* |
| % of discrete variables | *d_var* |
| % of continuous variables | *c_var* |
| % of missing values | *m_per* |

## Statistical Measures

Descriptive statistics can be used to summarize the relevant characteristics of any large dataset. In this section, we present some statistical measurement to summarize a dataset. Let us consider a random variable $X$ as a function from a sample space $\Omega$ with the real numbers $\Re$ where $X : \Omega \to \Re : \omega_n \mapsto \mathbf{x}_n$ with $\omega_n \in \Omega$ and $\mathbf{x}_n$ a realization of the variable $X$.

- **Mean:** The sample mean estimates the population mean, commonly notated as $\overline{X}$ . It is a measure of location in the same variable. It also considers all outliers values during the location measure. It may not appear representative of the central region for skewed datasets. It is especially useful as being representative of the whole sample for identify the characteristics of a variable by a few numbers (Harnett & Horrell, 1998).
- **Median:** The median is the middle point through the ordered variable, so that half of the data points in the variable will lie below the median, and half will be above. It is a good descriptive statistical measure of the location which works well for skewed data or data with outliers (Rasmussen, 1992).
- **Mode:** The mode is used to indicate the point (or points) on the scale of measures where the population is centered. It is the score in the population that occurs most frequently (Kvanli, Guynes, & Pavur, 1996).
- **Mad:** The mad estimates the mean absolute deviation of a dataset (Sachs, 1984).
- **Variance:** The variance is use to characterize the dispersion among the measures in a given population. It calculates the mean of the scores, and then measures the amount that each score deviates from the mean and then squares that devia-

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tion for a given population. Numerically, the variance equals the average of the squared deviations from the mean (Anderson, Sweeney, Williams, Harrison, & Rickard, 1992).

- **Geometric mean (GM):** The geometric mean is the nth root of the product of the sequence. The geometric mean of a sequence $\{X_i\}_{i=1}^{n}$ is:

$$GM = \left[\prod_{i=1}^{n} X_i\right]^{1/n}$$

- **Harmonic mean (HM):** The harmonic mean is the inverse of the mean of the inverse elements of a dataset. The harmonic mean $HM(X_1, ..., X_n)$ of $n$ points $X_i$ is:

$$HM = \frac{n}{\sum_{i=1}^{n} \frac{1}{X_i}}$$

- **Trim mean (TM):** The trim mean measures the arithmetic mean of a sample excluding the specified trim fraction from the same variable. The trim fraction is user dependent parameter. We consider 20 for this parameter value over the experiments. The trimmed mean is a robust estimate of the center location of a sample. For outliers datasets, the trimmed mean is a more appropriate estimation of the center of the dataset.
- **Standard deviation ($\sigma$):** The standard deviation measures the spread of a set of data as a proportion of its mean. The larger the standard deviation indicates the distribution is more widely spread. It is calculated by taking the square root of the variance and is generally symbolized by $\sigma$.

$$\sigma = \left(\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2\right)^{\frac{1}{2}}, \text{ where } \overline{X} = \frac{1}{n}\sum X_i, \ n \text{ is the sequence length.}$$

- **Prctile:** Prctile calculates a value for a variable that is greater than a certain percentage in the same variable. We consider the percentage value is 90 (i.e., the 90$^{th}$ percentile) and average this value over all variables.
- **Interquartile range (IQR):** The IQR is used as a robust measure of scale and measures the distance between the 25th and the 75th percentile (Mandenhall & Sincich, 1995). The hypothesis is, if the variables are approximately normal, then $\frac{IQR}{\sigma} \approx 1.3$, where $\sigma$ is the standard deviation of the population. Another name of 25$^{th}$ percentile is semi interquartile range (SIQR).
- **Max. and Min. Eigenvalue:** Eigenvalues are a special set of scalars associated with a matrix that are sometimes also known as characteristic roots, proper values, or latent roots of the matrix (Marcus & Minc, 1988).

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Let us consider *A* be a linear transformation represented by a matrix **A**. If there is a vector $\mathbf{X} \in \Re^n \neq 0$ such that:

$$\mathbf{AX} = \lambda \mathbf{X}$$

where λ is called the eigenvalues of **A** with corresponding eigenvector **X.**

- **Index of dispersion (*ID*):** The larger value of *ID* indicates the dataset is widely scattered, otherwise it is closely clustered (Craft, 1990).

$$ID = \frac{k\left(N^2 - \sum f^2\right)}{N^2(c-1)}$$

where $N$ = the number of data points, $c$ = the number of categories of the variables and $\sum f^2$ = the sum of the squared frequencies over the categories.

- **Center of gravity (CG):** The *CG* measures the Euclidean norm between minority and majority classes. The minimum value indicates the closeness between groups and the maximum indicates the dispersion between groups.

$$CG = \sqrt{(a_{i,j} - b_{i,j})}$$

where *a* and *b* belongs to the center points of these two groups.

- **Skewness:** Skewness is a descriptive statistical measure about the normality of a dataset. When one tail of the distribution is longer than the other, it indicates the dataset is either highly positive or negatively skewed (Shao, 1999). It can be defined as follows:

$$skewness(\mathbf{x}) = \frac{\frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_k - \bar{\mathbf{x}})^3}{\sigma^3}$$

- **Kurtosis:** Any symmetric distribution could deviate from the normal distribution due to a heavy tail (Rice, 1995). The deviation is measured by the coefficients of kurtosis as follows:

$$kurtosis(\mathbf{x}) = \frac{\frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_k - \bar{\mathbf{x}})^4}{\sigma^4}$$

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- ***Z*-score:** The *Z*-score is sometimes called standardized unit. It indicates how far and in what direction a variable deviates from its distribution's mean, expressed in units of its distribution's standard deviation. The *Z*-scores are especially informative when the distribution to which they refer is normal. If the value of *Z* is positive, it means the variable *X* is above its mean and if Z is negative, then *X* is below its mean (Cryer & Miller, 1994). It can also measure the outliers of a dataset. If the value of a *Z*-score is greater than 3, it indicates that the data distribution has outliers (Tamhane & Dunlop, 2000). It can be defined as follows:

$$Z - score = \frac{X - \overline{X}}{\sigma}$$

- **Correlation coefficient:** The correlation coefficient (*r*) is most widely used in descriptive statistics to summarize a relationship between continuous variables (*X* and *Y*). It is a measure of the degree to which the relationship follows a straight line. Numerically, *r* can assume any value between -1 and +1 depending upon the degree of the relationship. Plus and minus one indicate perfect positive and negative relationships whereas zero indicates that the X and Y values do not co-vary in any linear fashion (Cryer et al., 1994; Tamhane et al., 2000).

$$r = \frac{s_{xy}}{s_x s_y} = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i - \overline{X}}{\sigma_x}\right)\left(\frac{Y_i - \overline{Y}}{\sigma_y}\right)$$

- **Canonical correlation:** Correlation coefficients can be interpreted by the square root of the eigenvalues of a matrix. Because the correlations pertain to the canonical variates, so they are called canonical correlations (Gnanadesikan, 1997).
- **Normal cumulative distribution function:** For each element of *X*, $p_{norm_cdf}$ computes the *cdf* at *X* of the normal distribution with mean $\mu$ and variance $\sigma$ as follows:

$$p_{norm_cdf} = F(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{\frac{-(x-\mu)^2}{2\sigma^2}} dt$$

- **Chi-square cumulative distribution function:** For each element of *X*, $p_{chis_cdf}$ computes the *cdf* at *X* of the chi-square distribution with $\nu$ degrees of freedom as follows:

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$$p_{chis_cdf} = F(x|\nu) = \int_0^x \frac{t^{(\nu-2)/2} e^{-t/2}}{2^{\nu/2}\Gamma(\nu/2)} dt$$

where Γ(.) is the gamma function.

## Distance-Based Measures

The distance-based measures calculate the dissimilarity between samples. We measure the euclidean, city block, and mahalanobis distance between each pair of observations for each data set. Let us consider the data matrix *D*, which is treated as *n* row vectors: $\mathbf{x}_1$, $\mathbf{x}_2$, ..., $\mathbf{x}_n$. The various distances between the vectors $\mathbf{x}_r$ and $\mathbf{x}_s$ are defined as followed (Gentle, 2002; Hair, Anderson, Tatham, & Black, 1998):

- **Euclidean distance:** The euclidean distance is sometimes called the $L_2$ norm. It simply calculates the geometric distance of the samples in the multivariate data space as follows:

$$ed_{rs}^2 = (\mathbf{x}_r - \mathbf{x}_s)(\mathbf{x}_r - \mathbf{x}_s)^T$$

- **Mahalanobis distance:** The mahalanobis distance calculates the geometric distance of the samples like Euclidean but by constructing the covariance matrix rather than diagonal matrix as follows:

$$md_{rs}^2 = (\mathbf{x}_r - \mathbf{x}_s)V^{-1}(\mathbf{x}_r - \mathbf{x}_s)^T$$

  where V is the covariance matrix.

- **City block distance:** The city block distance calculates the sum of the absolute differences between the values of the samples in the multivariate data space as follows:

$$cbd_{rs} = \sum_{j=1}^{n} |\mathbf{x}_{rj} - \mathbf{x}_{sj}|$$

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## Information Theoretical Measures

The quality of the relationships in the data can also be assessed using information theoretical measures. We represent these measures formulation and explanation from Smith et al., 2001, 2002):

- **Mean entropy of variables:** Entropy is a measure of randomness in a variable. The entropy *H(X)* of a discrete random variable *X* is calculated in terms of $q_i$ (the probability that *X* takes on the *i*th value). We average the entropy over all the variables and take this as a global measure of the entropy of the variables:

$$H(X) = -\sum q_i \log q_i$$
$$\overline{H}(X) = p^{-1} \sum H(X_i)$$

- **Entropy of classes:** This is similar to the entropy of variables, except that the randomness in class assignment is measured; where $\pi_i$ is the prior probability of class $A_i$.

$$H(C) = -\sum_i \pi_i \log \pi_i$$

- **Mean mutual entropy of class and variables:** For a measure of common information or entropy shared between the two variables, if $p_{ij}$ denotes the joint probability of observing class $A_i$ and the *j*-th value of variable *X*, if the marginal probability of class $A_i$ is $\pi_i$, and if the marginal probability of variable *X* taking on its *j*-th value of $q_j$, then the mutual information and its mean over all variables is defined as:

$$M(C, X) = \sum_{ij} p_{ij} \log\left(\frac{p_{ij}}{\pi_i q_i}\right)$$
$$\overline{M}(C, X) = p^{-1} \sum_i M(C, X_i)$$

- **Equivalent number of variables (ENV):** This is the ratio between the class entropy and the average mutual information:

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$$ENV = \frac{H(C)}{\overline{M}(C,X)}$$

- **Noise-signal ratio (NSR):** A large NSR implies that a dataset contains much irrelevant information (noise) and could be condensed without affecting the performance of the model.

$$NSR = \frac{\overline{H}(X) - \overline{M}(C,X)}{\overline{M}(C,X)}$$

Finally we use the induction algorithm C5.0 (Windows version See5, http://www.rulequest.com/see5-info.html) to generate the rules to describe rbf kernel parameter is suitable for which type of problem, given the dataset characteristics and the performance of rbf kernel on each dataset. C5.0 has two tuning parameter c and m to generate the best rule. We also examine the rules by 10 fold cross validation (10FCV) performances. Therefore, we finalized the rules for the rbf width with maximum likelihood (ML) method and nelder-mead (N-M) simplex method.

## Rules for rbf_ML Method

The best rules for rbf_ML method are generated with c = 85% and m = 2 as in Table 4.

**Rule # 1.** IF ($md_{rs}$ <= 4.3713 AND $y_{bio_pdf}$ > 2.7036) OR (mean > 49.0052 AND skewness > -1.7911 AND $y_{gama_pdf}$ <= 0.00030981) OR ($p_{chis_cdf}$ <= 30.3236) THEN we should choose rbf_ML method for rbf kernel width estimation.

*Table 4. Confusion matrix based on 10FCV results for the rbf_ML method selection rule*

| Data condition satisfied | | rbf_ML method best | |
|---|---|---|---|
| | | Y | N |
| | Y | 2.7 | 0.2 |
| | N | 0.3 | 2.6 |

*Accuracy = 91.38%*

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*Table 5. Confusion matrix based on 10FCV results for the rbf_N-M method selection rule*

| Data condition satisfied | | rbf_N-M method best | |
|---|---|---|---|
| | | Y | N |
| | Y | 2.9 | 0.2 |
| | N | 0.4 | 2.3 |

*Accuracy = 89.66%*

## Rules for rbf_N-M Method

The best rules for rbf_NM method are generated with c = 80% and m = 2 in Table 5.

**Rule # 2.** IF (skewness > -0.4284 AND $y_{norm_pdf}$ <= 0.23362 AND $y_{ray_pdf}$ <= 2.3497) OR ($y_{bio_pdf}$ > 2.7036) THEN we should choose rbf_NM method for rbf kernel width estimation.

The generated rules for rbf kernel show around 90% accuracy. Individually we observed rbf_ML approximation method showed slightly better performance than rbf_N-M approximation, although rbf_ML was superior for problems not best suited to rbf kernel. These rules might be useful to determine which rbf width approximation method is most appropriate for which problem.

# Experimental Outcome

Finally, we adopted the previous rule with rbf kernel and implemented this idea with classical K-means algorithm. We observed proposed algorithm performed very well in some of cases. We found the classical K-means algorithm failed to cluster the average number of instances are 40%. On the other hand kernel-based K-means algorithm performed 34%. The result is quite acceptable. The details data set description and the individual performance of the classical K-means algorithm and the kernel-based K-means algorithm is summarized in Table 6.

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*Table 6. The data sets description, the number of incorrectly clustered instances by classical K-means algorithm and proposed kernel based K-means algorithm*

| # Data set | Data set name | # Instances | # Attributes | # Class | # Instances Incorrectly clustered by K-means | # Instances Incorrectly clustered by rbf kernel K-means |
|---|---|---|---|---|---|---|
| 1 | abalone | 1253 | 9 | 3 | 521 | 508 |
| 2 | adp | 1351 | 12 | 3 | 659 | 659 |
| 3 | adult+stret | 20 | 5 | 2 | 4 | 4 |
| 4 | adult-stret | 20 | 5 | 2 | 4 | 4 |
| 5 | allbp | 840 | 7 | 3 | 154 | 40 |
| 6 | ann1 | 1131 | 7 | 3 | 439 | 370 |
| 7 | ann2 | 1028 | 7 | 3 | 423 | 358 |
| 8 | aph | 909 | 19 | 2 | 8 | 8 |
| 9 | art | 1051 | 13 | 2 | 57 | 54 |
| 10 | australian | 690 | 15 | 2 | 302 | 314 |
| 11 | balance-sca | 625 | 5 | 3 | 282 | 282 |
| 12 | bcw | 699 | 10 | 2 | 207 | 147 |
| 13 | bcw_noise | 683 | 19 | 2 | 239 | 239 |
| 14 | bld | 345 | 7 | 2 | 155 | 143 |
| 15 | bld_noise | 345 | 16 | 2 | 164 | 149 |
| 16 | bos | 910 | 14 | 3 | 374 | 416 |
| 17 | bos_noise | 506 | 26 | 2 | 238 | 203 |
| 18 | breast-canc | 286 | 7 | 2 | 95 | 115 |
| 19 | breast-canc | 699 | 10 | 2 | 207 | 147 |
| 20 | bupa | 345 | 7 | 2 | 145 | 143 |
| 21 | c | 1500 | 16 | 2 | 710 | 685 |
| 22 | cleveland-heart | 303 | 14 | 5 | 146 | 141 |
| 23 | cmc | 1473 | 10 | 3 | 844 | 844 |
| 24 | crx | 490 | 16 | 2 | 212 | 224 |
| 25 | dar | 1378 | 10 | 5 | 303 | 322 |
| 26 | dhp | 1500 | 8 | 2 | 313 | 314 |
| 27 | DNA-n | 1275 | 61 | 3 | 613 | 613 |
| 28 | dna | 2000 | 61 | 3 | 1162 | 1066 |
| 29 | dna_noise | 2000 | 81 | 3 | 1221 | 1078 |
| 30 | dph | 590 | 11 | 2 | 211 | 213 |
| 31 | echocardiogram | 131 | 8 | 2 | 38 | 39 |
| 32 | flare | 1389 | 11 | 2 | 400 | 400 |
| 33 | german | 1000 | 25 | 2 | 300 | 300 |
| 34 | glass | 214 | 10 | 6 | 148 | 133 |
| 35 | h-d | 303 | 14 | 2 | 72 | 72 |
| 36 | hayes-roth | 132 | 6 | 3 | 60 | 60 |
| 37 | hea | 270 | 14 | 2 | 64 | 64 |
| 38 | hea_noise | 270 | 21 | 2 | 120 | 114 |
| 39 | heart | 270 | 14 | 2 | 64 | 64 |
| 40 | hepatitis | 155 | 20 | 2 | 66 | 70 |
| 41 | horse-23 | 368 | 23 | 2 | 136 | 136 |
| 42 | horse-colic | 368 | 28 | 2 | 244 | 244 |
| 43 | house-votes-84 | 435 | 17 | 2 | 207 | 190 |
| 44 | hyp | 2847 | 16 | 2 | 297 | 133 |
| 45 | hypothyroid | 1265 | 26 | 2 | 154 | 160 |
| 46 | iris | 150 | 5 | 3 | 6 | 6 |
| 47 | khan | 1063 | 6 | 2 | 451 | 422 |
| 48 | kr-vs-kp | 1279 | 37 | 2 | 611 | 611 |
| 49 | labor-neg | 40 | 17 | 2 | 14 | 14 |
| 50 | led-noise | 1047 | 10 | 10 | 918 | 918 |
| 51 | lenses | 24 | 6 | 3 | 19 | 8 |
| 52 | letter-a | 1334 | 17 | 2 | 256 | 161 |

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*Table 6. continued*

| # Data set | Data set name | # Instances | # Attributes | # Class | # Instances Incorrectly clustered by K-means | # Instances Incorrectly clustered by rbf kernel K-means |
|---|---|---|---|---|---|---|
| 53 | lung-cancer | 32 | 57 | 2 | 15 | 15 |
| 54 | lymphography | 148 | 19 | 8 | 91 | 91 |
| 55 | mha | 1269 | 9 | 4 | 441 | 259 |
| 56 | monk1 | 556 | 7 | 2 | 276 | 276 |
| 57 | monk2 | 601 | 7 | 2 | 287 | 287 |
| 58 | monk3 | 554 | 7 | 2 | 266 | 266 |
| 59 | mushroom | 1137 | 12 | 2 | 462 | 323 |
| 60 | nettalk_str | 1141 | 8 | 5 | 627 | 524 |
| 61 | page-blocks | 1149 | 11 | 5 | 478 | 124 |
| 62 | pendigits-8 | 1399 | 17 | 2 | 217 | 168 |
| 63 | pha | 1070 | 10 | 5 | 452 | 380 |
| 64 | phm | 1351 | 12 | 3 | 1023 | 279 |
| 65 | phn | 1500 | 10 | 2 | 402 | 407 |
| 66 | pid | 532 | 8 | 2 | 166 | 170 |
| 67 | Pima | 768 | 9 | 2 | 261 | 266 |
| 68 | poh | 527 | 12 | 2 | 27 | 27 |
| 69 | post-operative | 90 | 9 | 3 | 39 | 35 |
| 70 | primary-tum | 339 | 18 | 2 | 117 | 117 |
| 71 | pro | 1257 | 13 | 2 | 407 | 191 |
| 72 | promoter | 106 | 58 | 2 | 45 | 45 |
| 73 | pvro | 590 | 19 | 2 | 539 | 51 |
| 74 | rph | 1093 | 9 | 2 | 235 | 227 |
| 75 | satimage | 1351 | 11 | 6 | 953 | 989 |
| 76 | shuttle-landing control | 15 | 7 | 2 | 8 | 7 |
| 77 | sick-euthyroid | 1582 | 16 | 2 | 259 | 272 |
| 78 | sma | 409 | 8 | 4 | 274 | 107 |
| 79 | smo | 1855 | 9 | 2 | 697 | 749 |
| 80 | smo_noise | 1855 | 16 | 2 | 910 | 736 |
| 81 | sonar | 208 | 61 | 2 | 97 | 98 |
| 82 | splice | 1589 | 61 | 3 | 860 | 765 |
| 83 | t_series | 62 | 3 | 2 | 20 | 17 |
| 84 | tae | 151 | 6 | 3 | 91 | 91 |
| 85 | tae_noise | 151 | 11 | 2 | 62 | 51 |
| 86 | thy | 1887 | 22 | 3 | 785 | 585 |
| 87 | thynoise | 1132 | 11 | 3 | 740 | 733 |
| 88 | tic-tac-toe | 958 | 10 | 2 | 332 | 332 |
| 89 | titanic | 2201 | 4 | 2 | 493 | 493 |
| 90 | tmris | 100 | 4 | 2 | 3 | 3 |
| 91 | tqr | 1107 | 12 | 2 | 326 | 272 |
| 92 | trains-transformed | 10 | 17 | 2 | 4 | 1 |
| 93 | va-heart | 200 | 9 | 4 | 128 | 130 |
| 94 | veh | 846 | 19 | 4 | 534 | 584 |
| 95 | veh_noise | 761 | 31 | 4 | 561 | 544 |
| 96 | votes_noise | 391 | 31 | 2 | 151 | 151 |
| 97 | waveform | 5000 | 22 | 2 | 2474 | 1696 |
| 98 | waveform_noise | 5000 | 41 | 2 | 2442 | 2044 |
| 99 | wdbc | 569 | 31 | 2 | 172 | 170 |
| 100 | wine | 178 | 14 | 3 | 53 | 103 |
| 101 | wpbc | 199 | 34 | 2 | 47 | 47 |
| 102 | xaa | 94 | 19 | 4 | 67 | 63 |
| 103 | xab | 94 | 19 | 4 | 59 | 61 |
| 104 | xac | 94 | 19 | 4 | 54 | 57 |
| 105 | xad | 94 | 19 | 4 | 54 | 58 |
| 106 | xae | 94 | 19 | 4 | 59 | 62 |

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*Table 6. continued*

| # Data set | Data set name | # Instances | # Attributes | # Class | # Instances Incorrectly clustered by K-means | # Instances Incorrectly clustered by rbf kernel K-means |
|---|---|---|---|---|---|---|
| 107 | xaf | 94 | 19 | 4 | 51 | 57 |
| 108 | xag | 94 | 19 | 4 | 60 | 62 |
| 109 | xah | 94 | 19 | 4 | 59 | 59 |
| 110 | xai | 94 | 19 | 4 | 63 | 58 |
| 111 | yha | 1601 | 10 | 2 | 437 | 435 |
| 112 | zoo | 101 | 17 | 7 | 55 | 55 |

# Conclusion

In this research, we have widely investigate empirically how to fit rbf kernel with automated best parameter for K-means clustering algorithm. We proposed a simple rule for rbf kernel parameter selection based on data set information. This method is faster than the trial-and-error based kernel selection with K-means. We observed the suitable rbf parameter could be out of the range 0.2 and 1.2, which is the commonly tested range on the literature. The estimated higher width increased the kernel performance accuracies for some specific cases. The rbf_ML and rbf_N-M methods are very fast to estimate the best rbf width. We examined the generated rules by 10 fold cross validation evaluation. All generated rules shown higher efficiency rating. The main benefit of our methodology based on meta-learning is that we can achieve higher accuracy for some clustering problems and significant savings in time for at least similar accuracy for all problems. Some one can do further study with other kernels and parameter selection methods with K-means algorithm.

# References

Ali, S., & Smith, K. A. (2005). Kernel width selection for SVM classification-a meta-learning approach. *International Journal of Data Warehousing and Mining*, 78-97.

Ali, S., & Smith, K. A. (2003). Automatic parameter selection for polynomial kernel. *Proceedings of the IEEE International Conference on Information Reuse and Integration* (pp. 243-249). USA.

Anderson, D. R., Sweeney, D. J., Williams, T. A., Harrison, N. J., & Rickard, J. A. (1992). *Statistics for business and economics* (Australian ed.). West Publishing Company, USA.

Blake, C., & Merz, C. J. (2002). *UCI repository of machine learning databases*. Retrieved from http://www.ics.uci.edu /~mlearn/MLRepository.html

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Cheng, J., Saigo, H., & Baldi, P. (2006). Large-scale prediction of disulphide bridges using kernel methods, two-dimensional recursive neural networks, and weighted graph matching. *Proteins: Structure, Function, Bioinformatics*, *62*(3), 617-629. .

Craft, J. L. (1990). *Statistics and data analysis for social workers* (2nd ed.). F. E. Peacock publishers, USA.

Cryer, J. D., & Miller, R. B. (1994). *Statistics for business: Data analysis and modeling* (2nd ed.). Wadsworth, Inc. USA.

Dhillon, I. S., Guan, Y., & Kulis, B. (2004). Kernel k-means, spectral clustering and normalized cuts. *Proceedings of the 10th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining(KDD)*, (pp. 551-556).

Dhillon, I. S., Guan, Y., & Kulis, B. (2005). A fast kernel-based multilevel algorithm for graph clustering. *Proceedings of the 11th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)* (pp. 629-634).

Gentle, J. E. (2002). *Elements of computational statistics.* New York: Springer-Verlag.

Gnanadesikan, R. (1997). *Methods for statistical data analysis of multivariate observations* (2nd ed.). John Wiley & Sons.

Hair, J. F., Anderson, R. E., Tatham, R. L., & Black, W. C. (1998). *Multivariate data analysis* (5th ed). NJ: Prentice-Hall International, Inc.

Harnett, D. L., & Horrell, J. F. (1998). *Data, statistics, and decision models with excel.* John Wiley & Sons, Inc.

Kulis, B., Basu, S., Dhillon, I. S., & Mooney, R. (2005). Semi-supervised graph clustering: A kernel approach. *Proceedings of the 22nd International Conference on Machine Learning (ICML)* (pp. 457-464).

Kvanli, A. H., Guynes, C. S., & Pavur, R. J. (1996). *Introduction to business statistics: A computer integrated, data analysis approach* (4th ed.). West Publishing Company.

Lagarias, J. C., Reeds, J. A., Wright, M. H., & Wright, P. E. (1998). Convergence properties of the nelder-mead simplex method in low dimensions. *SIAM Journal of Optimization, 9*, 112-147.

Lim, T. S. (2002). *Knowledge discovery central, datasets.* Retrieved from http://www.KDCentral.com/

MacQueen, J. B. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1, pp. 281-297). Berkeley, University of California Press.

Mandenhall, W., & Sincich, T. (1995). *Statistics for engineering and the sciences* (4thed.). Prentice Hall.

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Marcus, M., & Minc, H. (1988). *Introduction to linear algebra*. Dover, New York.

Matteucci, M. (2007). *A tutorial on clustering algorithms*. Retrieved from April 27, 2007, from http://www.elet.polimi.it/upload/matteucc/Clustering/tutorial html/

Ou, Y. Y., Chen, C. Y., Hwang, S. C., & Oyang, Y. J. (2003). Expediting model selection for support vector machines based on data reduction. *Proceedings of IEEE International Conference on Systems, Man and Cybernetics* (Vol. 1, pp. 786-791).

Rasmussen, S. (1992). *An introduction to statistics with data analysis*. Brooks/Cole Publishing Company.

Rice, J. A. (1995). *Mathematical statistics and data analysis* (2nd ed.). Duxbury Press.

Ross, S. M. (2000). *Introduction to probability and statistics for engineers and scientists* (2nd ed.). London: Academic Press.

Sachs, L. (1984). *Applied statistics: A handbook of techniques.* Springer-Verlag, (p. 253).

Shao, J. (1999). *Matematical statistics.* New York: Springer-Verlag.

Smith, K. A., Woo, F., Ciesielski, V., & Ibrahim, R. (2001). Modelling the relationship between problem characteristics and data mining algorithm performance using neural networks. In C. Dagli et al. (Eds.), *Smart Engineering system design: Neural networks, fuzzy logic, evolutionary programming, data mining, and complex systems* (Vol. 11, pp. 357-362). ASME Press.

Smith, K. A., Woo, F., Ciesielski, V., & Ibrahim, R. (2002). Matching data mining algorithm suitability to data characteristics using a self-organising map. In A. Abraham and M. Koppen (Eds.), *Hybrid information systems* (pp. 169-180). Heidelberg: Physica-Verlag.

Tamhane, A. C., & Dunlop, D. D. (2000). *Statistics and data analysis.* Prentice Hall.

Vapnik, V. (1995). *The nature of statistical learning theory* (1st ed.). New York: Springer-Verlag.

Zalane, O. R. (2007). Principles of knowledge discovery in databases. Retrieved April 27, 2007, from htttp://www.cs.ualberta.ca/%Ezaiane/courses/cmput690/slides/

Zhang, R., & Rudnicky, A. I. (2002). A large scale clustering scheme for kernel K-Means. *Proceedings of the 16th International Conference on Pattern Recognition* (Vol. 4, pp. 289-292).

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## Chapter VII

# Advances in Classification of Sequence Data

Pradeep Kumar, University of Hyderabad, Gachibowli, India*
P. Radha Krishna, University of Hyderabad, Gachibowli, India
Raju S. Bapi, University of Hyderabad, Gachibowli, India
T. M. Padmaja, University of Hyderabad, Gachibowli, India

## Abstract

*In recent years, advanced information systems have enabled a collection of increasingly large amounts of data that are sequential in nature. To analyze huge amounts of sequential data, the interdisciplinary field of knowledge discovery in databases (KDD) is very useful. The most important step within the process of KDD is data mining, which is concerned with the extraction of the valid patterns. Recent research focus in data mining includes stream data mining, sequence data mining, Web mining, text mining, visual mining, multimedia mining, and multi-relational data mining. Sequence data may be discrete or continuous in nature. Most of the research on discrete sequence data concentrated on the discovery of frequently occurring patterns. However, comparatively less amount of work has been carried out in the area of discrete sequence data classification. In this chapter, data taxonomy is introduced with a review of the state of art for sequence data classification. The usefulness of embedding partial subsequence information extracted using sliding window technique into traditional classifier like kNN has been demonstrated. kNN*

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*has been tested with various vector based distance/similarity metrics. Further, with the use of $S^3M$ similarity metric, the full subsequence information embedded in the data sequences is extracted. The experimental data taken is DARPA'98 IDS benchmark dataset collected from UCIML dataset repository. The chapter closes by pointing out various application areas of sequence data and also the open issues in sequence data classification problem.*

# Introduction

In recent years, the amount of data that is collected by advanced information systems has increased tremendously. The resulting data volume is too large to be examined manually and even the methods for automatic data analysis based on classical statistics and machine learning often face problems when processing large, dynamic data sets consisting of complex data. The interdisciplinary field of knowledge discovery in databases (KDD) helps in analyzing these large volumes of data. KDD employs methods at the cross-post of machine learning, statistics, and database systems. Fayyad, Shapiro, and Smyth (1996) define KDD as follows:

*Knowledge discovery in databases is the non-trivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data.*

According to this definition, data is a set of facts that is somehow accessible in electronic form. The term patterns indicate models and regularities, which can be observed within the data. Patterns have to be valid (i.e., they should be true on new data with some degree of certainty). A novel pattern is not previously known or trivially true. The potential usefulness of patterns refers to the possibility that they lead to an action providing a benefit. A pattern is understandable if it is interpretable by a human user. KDD is a process comprising several steps that are perhaps repeated over several iterations.

The core step of KDD is called *data mining*. Data mining can be defined as an activity that extracts new and nontrivial information contained in large databases. The goal is to discover hidden patterns, unexpected trends or other non-obvious relationships in the data using a combination of techniques such as, machine learning, statistics, and databases.

A huge amount of data is collected every day in the form of sequences. These sequential data are valuable sources of information not only to search for a particular value or event at a specific time, but also to analyze the frequency of certain events or sets of events related by particular temporal/sequential relationship. Examples

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include Web server logs, online transaction logs, alarms/events, and performance measurements generated by distributed computer systems and by telecommunication networks and financial transaction data. In general, sequential data can be either categorical (event streams) or numerical (time series). Both categorical as well as numerical data have been studied exhaustively in data mining. Many aspects of sequential data, like considering their order of occurrence, scalar nature and so on have proven their importance in emerging applications and posed several challenges calling for more research.

We next list some of the frequently used terminology with their definition in context to sequences.

- An **alphabet** generally denoted by $\sum$, is a set of letters, or a set of numbers, etc.
- A **sequence** is a list of symbols (or events) from the alphabet arranged in a linear fashion, such that the order of the members is well defined. For example, sequences $< B,K,M >$ and $< K,M,B >$ defined over set of letters (alphabet) are different as the order of the symbols in both sequences is not the same. Sequences can be definite, as in the above example, or infinite, such as the sequence of all even positive integers $< 2, 4, 6, \ldots >$. The number of symbols (or events) determines the length of the sequence.
- A **subsequence** of a sequence is a new sequence, which is formed from the original sequence by deleting some of the elements without disturbing the relative positions of the remaining elements. For example, suppose that $\sum$ is an alphabet and $S$ is a sequence generated from $\sum$, denoted as $< s_1, s_2, s_3, \ldots, s_k >$ where $k \in N$ denotes the index position of symbols in sequence $S$. A subsequence of $S$ is a sequence formed from it by considering symbols in strictly increasing order in the index set $k$. For example, $< s_1, s_3 >$ and $< s_2, s_3 >$ are subsequences of $S$.

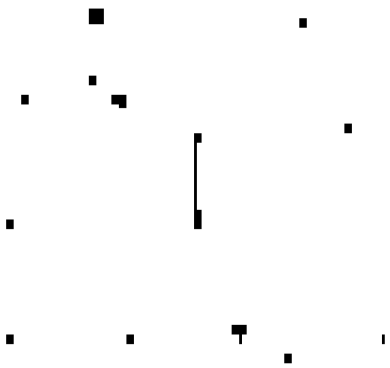
## Data Taxonomy

Data is a representation of facts, concepts, or instructions in a formalized manner suitable for communication, interpretation, or processing by humans or by automated means. Figure 1 shows the classification of data. The bold line in the figure shows the line of study for current work.

Data may be *sequential* or *non sequentia*l in nature. *Non-sequential data* are those data where order of occurrence is not important. Ex: Market Basket data, value on dice in a binomial experiment. *Sequential data* are those data where order of occurrence is important to consider. Sequential data can be ordered with respect to time or some other dimension such as, space. For example, Web logs, protein and

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*Figure 1. Data taxonomy. (The bold line indicates the nature of sequential data considered in this work.)*



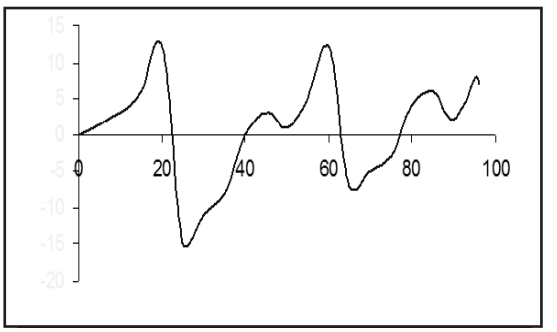
DNA sequences, system calls recorded by an intrusion detection system, speech and so on. Further, sequential data can be classified as *temporal* or *non-temporal*. *Temporal* data are those data, which have time stamp attached to it. Examples of such data are stock market data, meteorological data, and so on. *Non-temporal* data are those which are ordered with respect to some other dimension other than time such as for example, space. Examples of such data are Web logs, sequence of system calls, and so on. Both temporal and non-temporal data can be further classified as *discrete* and *continuous*. *Discrete* data describes observations/elements that have a definite possible domain. Figure 2 shows the representation of discrete data generated from the set of alphabet of letters. Examples of discrete temporal sequential data include time logs of caller in a call center. Examples of discrete non-temporal data include system calls, Web logs. *Continuous* data are observations/elements

*Figure 2. Representation of discrete data*

| Symbols from the set of letter alphabet, Σ | T | T | A | T | E | C | T | E | T | A | P | Y | R | T | T | C | H | G | B | D | E | W | S | L | K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Index position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

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*Figure 3. Representation of continuous data*




that take value from a continuous domain. Generally, continuous data come from measurements. Figure 3 shows the representation of continuous data. Continuous data can accept any value between the defined range where as discrete data accepts defined set from the domain. Examples of continuous temporal data include stock market data and weather data. Examples of continuous non-temporal data include image data and handwriting.

This chapter mainly concentrates on the classification tasks of discrete non-temporal sequential data. Study of continuous temporal sequence data is out of scope of this chapter. However, wherever it is relevant we have given pointers the major references from the world of continuous sequence data.

## Classification

The task of classification is to learn a function that maps data to the available set of classes. A classifier learns from a training set containing a large number of already mapped objects for each class. The training objects are considered to be labelled with the name of the class they belong to. Classification is also called *supervised learning* because it is directed by these labelled objects.

Classification is an extensively researched topic in data mining and machine learning. The main hurdle to leveraging the existing classification methods is that these assume record data with a fixed number of attributes. In general, people dealing with sequence data use to convert the data into non-sequential data and then apply traditional classification algorithm. Commonly used classification algorithms are decision trees, k nearest neighbors, support vector machines, and bayes classifiers.

Here, we briefly outline the basic classification algorithms. Though the methods to achieve classification vary based on the dataset and the model used, all the classifiers

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have something in common. The commonality is that they divide the given object space into disjunctive sections that are mapped to a given class.

## *Decision Trees*

Decision trees are powerful and popular tools for classification and prediction. Decision trees are more popular because in contrast to the neural network based approach they generate rules. These rules can easily be interpreted so that we can understand them. These rules can also be transformed into a database access language like SQL in order to retrieve quickly the records falling into a particular category.

A decision tree is a tree with the following characteristics:

- Each inner node corresponds to one attribute
- Each leaf is associated with one of the classes
- An edge represents a test on the attribute of its parent node

For classification, the attribute values of a new object are tested beginning with the root. At each node, the data object can pass only one of the tests that are associated with the departing edges. The tree is traversed along the path of successful tests until a leaf is reached. Multiple algorithms have been proposed in the literature for constructing decision trees (Breiman, Friedman, Olshen, & Stone, 1984; Gehrke, Ramakrishnan, & Ganti, 1998, Quinlan, 1993). Generally, these algorithms split the training set recursively by selecting an attribute. The best splitting attribute is determined with the help of quality criteria. Examples of such quality criteria are information gain, gini index, Shannon information theory, and statistical significance tests. The advantages of decision trees are that they are very robust against attributes that are not correlated to the classes because those attributes will not be selected for a split. Another more important feature is the induction of rules. Each path from the root to a leaf provides a rule that can be easily interpreted by a human user. Thus, decision trees are often used to explain the characteristics of classes. The drawback of decision trees is that they are usually not capable of considering complex correlations between attributes because they only consider one attribute at a time. Thus, decision trees often model correlated data by complex rules which tend to overfit.

## *k-Nearest Neighbor Classification*

k-Nearest neighbor classifiers are based on the idea that an object should be predicted to belong to the same class as the objects in the training set with the biggest simi-

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larity. To use kNN classification, it is required to have a suitable similarity search system in which the training data is stored. Classification is done by analyzing the results of a kNN query. The simplest way to determine a classification result of a kNN classifier is the majority rule. The objects in the query result are counted for each class and the class having the majority count is predicted to be the class of the test object.

Another method is to consider the distances to the object to weigh the impact of each neighboring object. Thus, a close object contributes more to the decision than an object having a large distance.

kNN classifiers use the class borders of the objects within the training set and thus do not need any training or model building apriori. As a result, kNN classifiers cannot be used to gain explicit class knowledge to analyze the structure of the classes. kNN classification is also known as a lazy learning algorithm. The parameter that determines the neighborhood size, k, is very important to the classification accuracy achieved by a kNN classifier. If k is chosen too small, classification tends to be very sensitive to noise and outliers. On the other hand, a too large value for k might extend the result set of the k nearest neighbor by objects that are too far away to be similar to the classified object.

Since kNN classification works on the training data, the classification time depends on the efficiency of the underlying similarity search system. In the case of large training sets, linear search becomes very inefficient. Suitable index structures can offer a better solution to this problem (Berchtold, Keim, & Kriegel,1996; Ciaccia, Patella, & Zezula, 1997; Lin, Jagadish, & Faloutsos, 1994). Deleting unimportant objects from the training dataset also helps in speeding up the kNN classification algorithm (Brighton & Mellish, 2002). kNN classification algorithms can be sped up by building the centroid for the objects of each class and using only the centroids and nearest neighbor classification (Han & Karypis, 2000). This approach is rather simple and is mostly applied for text data classification.

### *Support Vector Machines*

Support vector machines (SVMs) were first introduced in 1995 for classification of input feature vectors (Cortes & Vapnik, 1995). SVMs distinguish the objects of two classes by linear separation, which is achieved by determining a separating hyperplane in the object space (Burges, 1998, Christianini & Shawe-Taylor, 2000). The idea of SVMs is to find the hyperplane providing the maximum level of generalization and thus avoiding overfit as much as possible. The vectors in the training set that have minimal distance to the maximum margin hyperplane are called support vectors. The location of the maximum margin hyperplane does only depend on these support vectors and thus, the classifier was named support vector machine.

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To determine the exact position of the maximum margin hyperplane and to find out the support vectors, a dual optimization problem is formulated which can be solved by algorithms like sequential minimal optimization (Platt, 1998).

The problem of linear separation is that there is not always a hyperplane that is able to separate all training instances. Therefore, two improvements for SVMs have been introduced that enable SVMs to separate almost any kind of data. The first improvement is the introduction of soft margins. The idea of soft margins is to penalize, but not prohibit classification errors while finding the maximum margin hyperplane. Thus, the maximum margin hyperplane does not necessarily separate all training instances of both classes. If the margin can be significantly increased, the better generalization can outweigh the penalty for a classification error on the training set. The second improvement is the introduction of kernel functions. For many real-world applications, it is not possible to find a hyperplane that separates the objects of two classes with sufficient accuracy. To overcome this problem the feature vectors are mapped into a higher dimensional space by introducing additional features that are constructed out of the original ones. Since this mapping is not linear, hyperplanes in the so-called kernel spaces provide much more complicated separators in the original space. This way the data in the original space is separated in a non linear fashion. An important characteristic of the use of kernel functions is that the calculation of a maximum margin hyperplane in the kernel space is not much more expensive than in the original space. The reason for this effect is that it is not necessary to calculate the feature vectors in the kernel space explicitly. Since the optimization problem calculating the maximum margin hyperplane does only use a scalar product in the feature space, it is enough to replace this scalar product with a so-called kernel function to calculate the maximum margin hyperplane in the kernel space. SVMs have been extended to multi-class problems (Platt, Cristianini, & Shawe-Taylor, 2000).

However, the training of SVMs tends to take large periods of time, especially for multi-class variants calculating many binary separators. The models built by SVMs do not provide any explicit knowledge that might help to understand the nature of the given classes.

### *Bayes Classifier*

Bayesian classifiers are based on the assumption that the objects of a class can be modelled by a statistical process (Mitchell, 1997). Each data object has its origin in the process of a given class with a certain probability and each process of a class generates objects with a certain probability called prior probability. To decide which class is to be predicted for a given object, it is necessary to determine the probability called the posterior probability of the object. It describes the probability that an ob-

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ject has its origin in that particular class. To determine the a posteriori probability, the rule of Bayes is used.

In bayesian, classifier training is very fast. Also the model designed is simple and intuitive. In bayesian, classifier error is minimized subject to the assumptions of independence of attributes and data satisfying distribution model. The major drawback of the bayesian model is that it assumes a normal distribution of patterns. In addition, it can not model disjoint class boundaries automatically.

There are several additional approaches that are used for classification. For example, neural networks are a very powerful direction of machine learning trying to rebuild the learning mechanism that can be found in the human brain. Among several other tasks, neural networks are applicable to classification. There are multiple variants of neural networks. An overview of neural networks can be found in many available publications (Bharat & Henzinger, 1998). Another approach to classification is inductive logical programming (ILP). ILP is often used for data mining in multi-relational data and employs search algorithms that find logical clauses that are valid in a database of facts (Muggleton & Raedt, 1994).

All the techniques previously described used to first convert the sequential data into non-sequential data. Converting a sequential data into non-sequential data leads to the loss of information. This information might be vital for many applications like anomaly detection, text mining and so on. Since all the above described techniques are well established and accepted hence a modification should be done at the data preprocessing or before feeding it into the classifier. For the distance-based classifier like kNN, the distance function should be adopted or devised, which should capture the sequential information embedded in the sequences.

In this chapter, we have demonstrated the application of the kNN classification algorithm for the intrusion detection domain. In this chapter, first we demonstrate the generally adopted technique (i.e., sliding window technique for extraction of subsequences). Results of sliding window experiments adhere that while performing classification of sequence data ignoring order or content information may lead to poor classification accuracy. Hence, we utilized a similarity measure, $S^3M$ for classification of system calls. Better results were achieved with the $S^3M$. $S^3M$ considers both order and content information while computing similarity between two sequences (Kumar, Rao, Krishna, Bapi, & Laha, 2005b).

## State of the Art

Last decade has seen explosive growth in sequential data (Xu & Wunsch, 2005). Sequential data often comprise sequences of variable length and with many other distinct characteristics such as dynamic behaviors, time constraints, etc (Gusfield,

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1997; Sun & Giles, 2001). Sequential data arise from various applications like DNA sequencing in molecular biology and speech processing. Other vital areas of application of sequences include, text mining, medical diagnosis, stock market, customer transactions, Web mining, and robot sensor analysis (Durbin, Eddy, Krogh, & Mitchison, 1999, Sun et al., 2001). In this section, we review the latest literature available in the area of classification of sequence data.

Sequence data classification problem is to train a model using given a set of classes C and a number of example sequences in each class in order to predict the class to which the unlabelled sequence belongs to. The applicability of sequence classification problem is in several real-life applications, for example:

I. Predict the family of protein or DNA sequence from the given set of protein or DNA families
II. Anomaly detection from the set of user sessions
III. Classify the new utterance to the right word using the several utterances of a set of words

Sequence data possess serial correlations and since traditional classification algorithms assume independence of attributes, traditional methods are not suitable for sequence classification tasks. The other important feature of sequence data which makes it important to be considered while capturing is variable length with a special notion of order.

The sequence classification task can be performed by ignoring the order of attributes and by aggregating the elements over the sequence (Chakrabarti, 2002). Over the years, sequence classification applications have seen the use of both pattern based as well as model based methods. In a typical pattern-based method, prototype feature sequences are available for each class. Then the classifier searches over the space of all prototypes, for the one that is closest (or most similar) to the feature sequence of the new pattern. The prototypes and the given features vector sequences are of different lengths. Thus, in order to score each prototype sequence against the given pattern, sequence aligning methods like dynamic time warping are needed. Various pattern based approach for sequence data classification has been designed (Graepel, Herbrich, Sdorra, & Obermayer, 1999; Jacobs, Weinshall, & Gdalyahu, 2000; Jain & Zongker, 1997; Pekalska & Duin, 2001). Literatures available in the field of model-based sequence data classification includes a huge list (Cheng & Klein-Seetharaman, 2005; McCallum & Nigam, 1998; Yan, Dobbs, Honavar, & Dobbs, 2004).

The existing methods of sequence data classification has been categorized into three types namely, generative classifiers, boundary-based classifiers, and, distance/kernel based classifiers (Sarawagi, 2005). Examples of boundary-based classifiers are decision trees, neural network, and linear discriminants like Fisher's. Algorithms of

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this class require the data to have a fixed set of attributes so that each data instance can be treated as a point in a multi-dimensional space. The training process partitions the space into regions for each class. As all the class in this approach as well defined boundaries, hence predicting for the class of an instance is done by seeing under which class boundary the instance falls.

A number of methods have been applied for embedding sequences in a fixed dimensional space in the context of various applications (Sarawagi, 2003, 2005). The simplest of these ignore the order of attributes and aggregate the elements over the sequence. This is mainly applied in text classification where a document that is logically a sequence of words is commonly converted as a vector where each word is a dimension and its coordinate value is the aggregated count or score of the word in the document (Chakrabarti, 2002).

The sliding window technique is the most common technique for sequence classification. In the sliding window technique for a fixed parameter called the window size L, corresponding L-grams dimensions of elements are created. Thus, if the domain size of elements is m, the number of possible dimensions are $mL$. The sliding window approach has been applied to classify sequences of system calls as intrusions or not (Hofmeyr, Forrest, & Somayaji, 1998; Lee & Stolfo, 1998; Warrender, Forrest, & Pearlmutter, 1999). The main shortcoming of the sliding window method is that it creates an extremely sparse space. A technique to address the shortcoming of sliding window technique is proposed by Leslie, Eskin, Weston, and Noble (2004).

Another approach to deal with sequences is to respect the global order in determining a fixed set of properties of the sequence. For categorical elements, an example of such order-sensitive derived features is the number of symbol changes or the average length of segments with the same element. For real-valued elements, examples are properties like fourier coefficients, wavelet coefficients, and autoregressive coefficients. Parameters of the auto regression moving average (ARMA) model can help distinguish between sober and drunk driver (Deng, Moore, & Nechyba, 1997). In their work, they showed that sober drivers have a large coefficient of the second and the third coefficient indicating steadiness. While in the case of, drunk drivers exhibit close to zero values of the second and third coefficients indicating erratic behavior. A wide variety of time series applications in various areas can be found in literatures (Laxman & Sastry, 2005).

The second class of classifiers are generative classifiers. Generative classifiers require a generative model of the data in each class. For each class $i$, during the training phase a generative model $M_i$ is constructed to maximize the likelihood over all training instances in the class $i$. Thus, $M_i$ models the probability $\boldsymbol{Pr}(x|ci)$ of generating any instance $x$ in class $i$. Also, we estimate the prior or background probability of a class $\boldsymbol{Pr}(c_i)$ as the fraction of training instances in class $i$. For predicting the class of an instance $x$, we apply Bayes rule to find the posterior probability $\boldsymbol{Pr}(c_i|x)$ of each class. The class with the higher probability is chosen as the winner.

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Generative classifiers have been extensively applied for classification tasks. They can be extended for sequence data classification provided one can design a distribution that can adequately model the probability of generating a sequence while being trainable with realistic amounts of training data. Various models are available to do so namely, independent model, first order markov model, higher order markov model, variable memory markov model, and hidden markov model (Sarawagi 2005).

The third class of classifiers is the kernel-based classifier**.** Examples of kernel-based classifiers includes support vector machines, radial basis functions, and nearest neighbor classifiers. Kernel-based classifiers are powerful classification techniques. They require a function $K(x_i, x_j)$ that intuitively defines the similarity between two instances and satisfies symmetric and positive definite property (Burges, 1998).

Kernel-based classifiers like SVMs can be utilized for sequence classification, provided we can design appropriate kernel functions that take as input two data points and output a real value that roughly denotes their similarity. For nearest neighbor classifiers it is not necessary for the function to satisfy the above two kernel properties but the basic structure of the similarity functions is often shared. We now discuss examples of similarity/kernel functions proposed for sequence data.

A common approach is to first embed the sequence in a fixed dimensional space using methods discussed for boundary based classifiers and then compute similarity using well-known functions like Euclidean or a dot-product. The mismatch coefficients for categorical data were used with a dot-product to perform protein classification using SVMs (Leslie et al., 2004) .

Another interesting technique is to define a fixed set of dimensions from intermediate computations of a structural generative model. Then, superimpose a suitable distance function on these dimensions. Fisher's kernel is an example of such a kernel, which has been applied to the task of protein family classification (Jaakkola, Diekhans, & Haussler, 1999). A lot of work has been done on building specialized hidden Markov models for capturing the distribution of proteins sequences within a family (Durbin et al., 1999). The Fisher's kernel provides a mechanism of exploiting these models for building kernels to be used in powerful discriminative classifiers like SVMs.

A number of sequence-specific similarity measures have also been proposed. For real-valued elements, these include measures like the dynamic time warping method (Rabiner & Juang, 1993), and for categorical data these includes measures like the edit distance and the more general Levenstein distance (Bilenko, Mooney, Cohen, Ravikumar, & Fienberg, 2003), and sequence alignment distances like BLAST and PSI-BLAST protein data.

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# Classification of System Calls Using Subsequence Information

In this section, we illustrate the methodology of extracting sequential information from sets, thus making it useful for vector-based distance/similarity measures. We considered subsequences of fixed sizes: 1, 2, 3, … . This fixed size subsequence is called a window. This window is slided over the sequence to find unique subsequences of a fixed length over the whole sequence. Frequency count of each subsequence is recorded. Consider a sequence which consists of traces of system calls as shown in Fig 4. For a total length of sequence of 12 and with a sliding window size w of 3, we will have (12 - 3 + 1 =)10 subsequences of size 3. These 10 subsequences of size 3 are as follows

*execve open mmap, open mmap open, mmap open mmap, open mmap mmap, mmap mmap mmap, mmap mmap mmap, mmap mmap mmap, mmap mmap open, mmap open mmap, open mmap exit*

From among these 10 subsequences, unique subsequences and their frequencies (number in the Braces) are as follows:

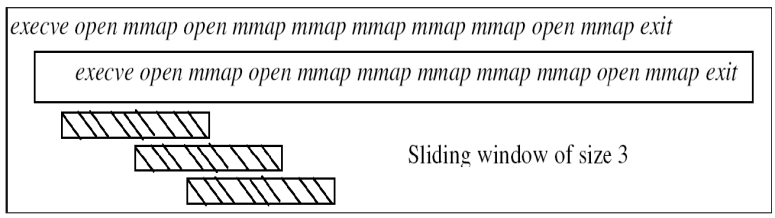
*execve open mmap (1)*
*mmap open mmap (2)*
*open mmap open (1)*
*mmap mmap open (1)*
*open mmap mmap (1)*
*open mmap exit (1)*
*mmap mmap mmap (3)*

With these encoded frequencies for subsequences, we can apply any vector-based distance/similarity measure, thus incorporating partial sequential information with vector space representation. Various distance measures used in this experimentation are euclidian distance (Qian et al., 2004), jaccard similarity (Gludici, 2003), cosine similarity (Qian et al., 2004), and binary weighted cosine similarity measure (BWC) (Rawat, Gulati, Pujari, & Vemuri, 2006).

Detecting and using subsequence pattern information from a sequential dataset plays a significant role in many areas including intrusion detection, monitoring for malicious activities, and molecular biology (Kumar & Spafford, 1994, Pevzner &

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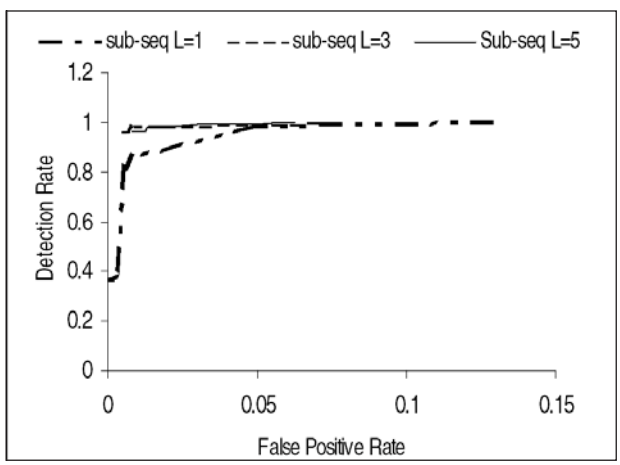
*Figure 4. Sequence of system call trace*




Waterman, 1995). An observed pattern may be significant or not depending on its likelihood of occurrence. In order to detect real intrusion as well as to avoid false alarms simultaneously, a proper choice of threshold and length of subsequence must be done. Keeping threshold value too low may lead for high false alarms and too high threshold values may allow many attacks undetected. A compromise in threshold value may allow an appropriate size for the sliding window. The traditional kNN classification algorithm (Dasarathy, 1991; Han & Kamber, 2001,) with suitable distance/similarity measure can be used to build an efficient classifier. An approach based on kNN classifier is proposed by Liao and Vemuri (2002b). The proposed methodology consists of two phases namely training and testing phase.

Dataset D consists of m sessions. Each session is of variable length. Initially in training phase, all the unique subsequences of size s are extracted from the whole dataset. Let n be the number of unique subsequences of size w, generated from the dataset D. A matrix C of size m × n is constructed where $C_{ij}$ reflects the count of $j^{th}$

*Figure 5. ROC curve for euclidian distance measure using kNN classification for k =5*




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*Figure 6. ROC curve for jaccard similarity measure using kNN classification for k =5*

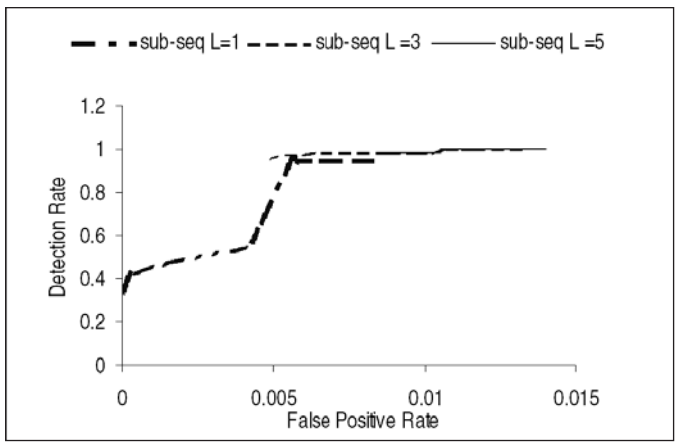
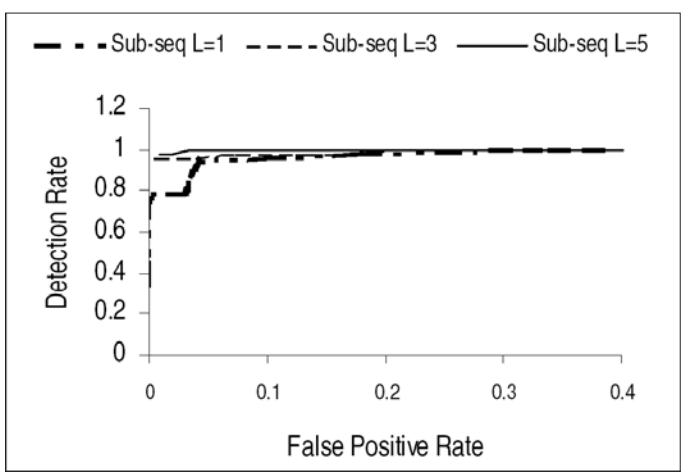



*Figure 7. ROC curve for cosine similarity measure using kNN classification for k =5*

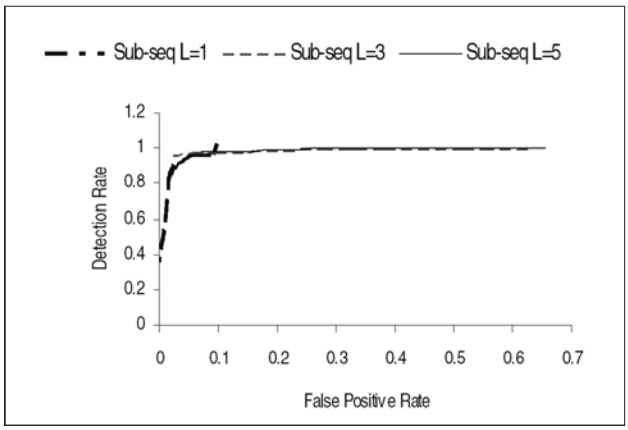



unique subsequence in the $i^{th}$ session. A distance/similarity measure is constructed by applying distance/similarity measure over the C matrix. The model is trained with the dataset consisting of normal sessions.

In testing phase, whenever a new session P comes to the classifier, the algorithm looks first for the presence of any new subsequence of size s. If a new subsequence is found, the new session is marked as abnormal. When there is no new subsequence

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*Figure 8. ROC curve for BWC similarity measure using kNN classification for k =5*




in new session P, the similarity of the new session is computed with respect to all the sessions. If similarity between any session in training set and new session is equal to 1, then it is marked as normal. Otherwise, k sessions that are of similar to the new session P in the training dataset are gathered. From these k maximum values, the average similarity for k-nearest neighbors is calculated. If the average similarity value is greater than user defined threshold value (τ) the new session P is marked as normal, else it is marked as abnormal.

All the experiments quoted in this chapter is implemented in Java 1.4 and were performed on 2.4 GHz, 256 MB RAM, Pentium-IV machine running on Microsoft Windows XP 2002.

## Experimental Results

Experiments were conducted using k-nearest neighbor classifier with euclidean distance, jaccard similarity, cosine similarity, and BWC similarity measures (Kumar et al., 2005a). Each distance/similarity measure was individually experimented with kNN classifier on DARPA'98 IDS dataset (Laboratory). DARPA'98 IDS dataset consists of TCPDUMP and BSM audit data. The network traffic of an Air Force Local Area Network was simulated to collect TCPDUMP and BSM audit data. The audit logs contain seven weeks of training data and two weeks of testing data. There were 38 types of network-based attacks and several realistic intrusion scenarios conducted in the midst of normal background data. For experimental purpose, 605 unique processes

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were used as a training dataset, which were free from any type of attack. Testing was conducted on 5285 normal processes. In order to test the detection capability of the proposed approach, we incorporated 55 intrusive sessions into our test data. For kNN classification experiments, k = 5 was considered. Experiments were carried out with various distance/similarity measures as discussed earlier (euclidean distance measure, jaccard similarity measure, cosine similarity measure and BWC similarity measure) at different subsequence lengths (sliding window size L). Here, L = 1 means that no sequential information is captured whereas, for L > 1 some amount of order information across elements of the data is preserved. To analyze the efficiency of classifier, ROC curve is used. ROC curve depicts the relationship between false positive rate (FPR) and detection rate (DR). It curve gives an idea of the trade off between FPR and DR achieved by a classifier. Detection rate is the ratio of number of intrusive sessions (abnormal) detected correctly to the total number of intrusive sessions. False positive rate is defined as the number of normal sessions detected as abnormal divided by the total number of normal sessions. The threshold values determine how c lose the given session is to the training dataset containing all normal sessions. If the average similarity score obtained for the new session is below the threshold value, it will be classified as abnormal.

ROC curves for euclidean distance, jaccard similarity, cosine similarity, and BWC similarity measures are shown in Figures 5, 6, 7, and 8, respectively. For the experimentation purpose, the value for k in kNN classification algorithm was fixed at 5. It can be observed from Figures 5, 6, 7, and 8 that as the sliding window size increases from L =1 to L = 5, high DR (close to ideal value of 1) is observed with all the distance/similarity measures. This indicates the usefulness of incorporating subsequence information.

Rate of increase in false positive is less for jaccard similarity measure (0.005- 0.015) as compared to the other distance/similarity measures such as cosine (0.1-0.4), euclidian (0.05-0.15), and BWC (0.1-0.7) similarity measures. Thus, we have less false alarms generated in the case of jaccard similarity measure in comparison to other distance/similarity measures.

It was observed that, as per the trend, the FPR is increasing with increasing subsequence lengths for all the four measures. We also observed that as the subsequence length is increased a low false positive value at early threshold values were obtained irrespective of all the four distance/ similarity measure used. We conducted the experiments upto subsequence length of 11 and similar trend was observed. However, for higher subsequence length values ( L > 6) the detection rate to the false positive rate was saturated.

The idea of looking at short sequences of behavior is quite general and might be applicable to several other security problems. For example, people have suggested applying the technique to several other computational systems (Hofmeyr et al., 1998).

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Our approach is similar to other approaches. Teng, Chen, and Lu (1990) considered sequencing information. Their approach differs from our approach that they look at the domain of user behavior, and use a probabilitistic approach for detecting anomalies. Because our results are sufficiently promising, the added complexity of using probabilities seems unnecessary. It is possible that our simple deterministic approach is successful because our data is well structured. If this is the case, it may well be that probabilities are necessary in less structured domains, such as user behavior. Hofmeyr et al. (1998) commented that the window length size of at least 6 is required to detect the intrusion. Later Tan and Maxion (2002) argued and gave explanation on why the length of window size be considered as 6 to detect the anomalous behavior. The experiments in this work adhere to the results of Tan et al. (2002) work for DARPA'98 IDS dataset. In our experiments, it was observed that the window length of 5 is still enough to detect the intrusive sessions.

## $S^3M$- Similarity Metric for Sequential Data

Results of the last section establish that in order to find patterns in sequences, it is necessary to not only look at the items contained in sequences but also the order of their occurrence. In order to account for both kinds of information, a measure called sequence and set similarity measure ($S^3M$), is utilized (Kumar et al., 2005b). $S^3M$ consists of two parts: one that quantifies the composition of the sequence (set similarity) and the other that quantifies the sequential nature (sequence similarity). Sequence similarity is defined as the order of occurrence of item sets within two sequences. Length of longest common subsequence (LLCS) with respect to the length of the longest sequence determines the sequence similarity aspect across two sequences. Set similarity (jaccard similarity measure) is defined as ratio of the number of common item sets and the number of unique item sets in two sequences. Thus, $S^3M$ is constituted as a function of both set similarity and sequence similarity and is defined as below:



where $p+q=1$ and $p, q \geq 0$

Here, p and q determine the relative weights to be given for order of occurrence (sequence similarity) and to content (set similarity), respectively. In practical applications, user could specify these parameters.

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# Classification of Sequences Using S$^3$M

The goal of intrusion detection is to monitor network assets to detect anomalous behavior and misuse. Recently the area of computer and network security has attracted many researchers. The notion of intrusion detection was born in the early 1980s by Anderson (1980). Since then, several pivotal events in intrusion detection system technology have advanced intrusion detection to its current state.

There are two basic approaches for detecting intrusion. The first approach, *anomaly detection*, attempts to define and characterize the correct static form of data and/or acceptable dynamic behavior. In effect, it searches for an anomaly in either stored data or in the system activity. IDS utilizing anomaly detection include Tripwire (Kim & Spafford, 1994), Self-Nonself (Forrest, Hofmeyr, Somayaji, & Longstaff, 1996), NIDES (next generation intrusion detection expert system ) (Anderson, Frivold, Tamaru, & Valdes, 1994), ADAM (audit data analysis and mining) (Liao et al., 2002b), IDDM (intrusion detection using data mining) (Abraham, 2001), and eBayes (Valdes & Skinner, 2000).

The second approach is called *misuse detection*, which involves characterizing known ways to penetrate a system in the form of a pattern. Rules are defined to monitor system activity essentially looking for the pattern. The pattern may be static bit string or may describe a suspect set or sequence of events. The rules may be engineered to recognize an unfolding or partial pattern. Intrusion detection systems utilizing misuse detection include MIDAS (multics intrusion detection and alerting system) (Sebring, Shellhouse, Hanna, & Whitehurst, 1988), STAT (state transition analysis tool for intrusion detection) (Ilgun, Kemmerer, & Porras, 1995), JAM (java agents for meta learning) (Lee et al., 1998, Lee et al., 2001; Liao & Vemuri, 2002a,) MADAM ID (mining audit data for automated models for intrusion detection) (Lee & Xiang, 2001) and automated discovery of concise predictive rules for intrusion detection (Lee, 1999).

In some cases, they are combined in a complementary way in a single intrusion detector. There is a consensus in the community that both approaches continue to have value. Systems taking the combination of both the approaches include NADIR (network anomaly detection and intrusion reporter) (Hochberg et al., 1993), NSTAT (network state transition analysis tool) (Kemmerer, 1998), GrIDS (grid intrusion detection) (Staniford-Chen et al., 1996) and EMERALD (event monitoring enabling responses to anomalous live disturbances ) (Porras & Neumann, 1997).

Formally, the intrusion detection problem on system calls or command sequences can be defined as follows:

Let $P = \{s_1, s_2, s_3, \ldots, s_m\}$ be the set of system calls where m denotes the total number of system calls that occur in sequences. A training set D is defined as containing several sequences of normal behavior. The goal in this case, is to devise a binary classification scheme that maps the incoming sequence to the class of normal session or

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to the class of intrusive session. In general, dealing with sequences is difficult hence each sequence is coded with vector notation. The coding could be binary indicating the presence or absence of a system call in the session or it may be the frequency of each system call in sequences. The other way of vector encoding could be in terms of time stamp associated with the system calls. However, all these techniques does not consider the order information embedded within the session.

A scheme based on the kNN Classifier has been proposed by Liao et al. (2002b), in which each process is treated as a document and each system call as a word in that document. Processes are converted into vectors and cosine similarity measurement is used to calculate similarity among them. A similar approach was followed by Rawat et al. (2006), in which they introduced a new similarity measure, termed binary weighted cosine (BWC) metric. This new similarity measure considers both the number of shared system calls between two processes as well as frequencies of those calls. We extended the work of Liao et al. (2002b) and Rawat et al. (2006) by adopting the same text based classification scheme for system calls with a new similarity measure called, $S^3M$. In this work, we used the classical kNN classification algorithm with $S^3M$. Algorithm 1 presents the algorithm for classification of system calls.

**Algorithm 1: k-Nearest neighbor algorithm for classification of system calls**

**Input:**

$S$ = Set of sessions

$k$ = Number of nearest neighbors

$P$ = New session (Class to be determined)

Sim = $S^3M$ similarity measure

$\tau$ = User specified similarity threshold

Output:

Class label of session $P$

Begin

Training

Construct Training sample $T$ from $S$ consisting of normal sessions.

Classification

**For** session $P$

**do**

Calculate $Sim(P, T^j)$ //$T_j$ is the *jth* session of test set $T$, $1 \leq j \leq |T|$.

Calculate average similarity, AvgSim, for $k$ nearest neighbors

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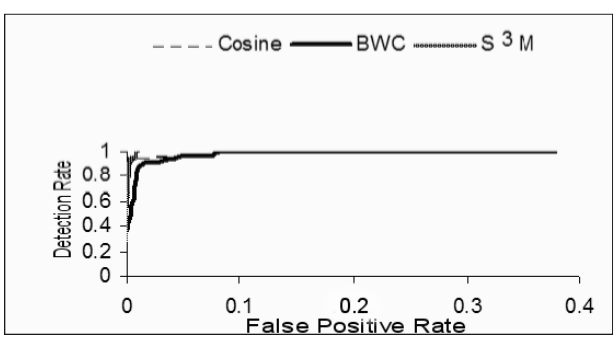
If (AvgSim >τ)
Mark *P* as normal
Else
Mark *P* as abnormal
End For
End

## Experimental Results

Experiments were conducted using k-nearest neighbor classifier with $S^3M$. DARPA'98 IDS dataset was used for the experimental purpose. We performed the comparative analysis of $S^3M$ with cosine and BWC similarity measure.

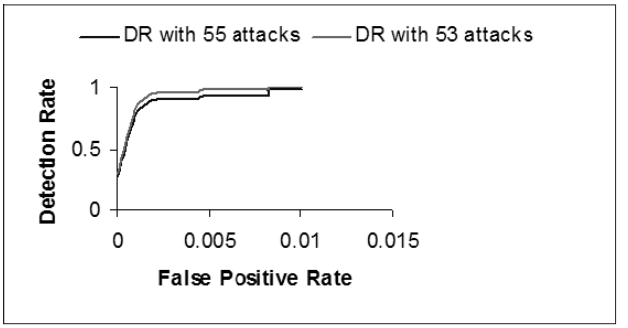
Figure 9 shows the comparative ROC curves for DARPA'98 IDS dataset with the three measures. The experiments were conducted for k = 5. *k* is the nearest neighbor parameter of the *kNN* classification algorithm. The sequence controlling parameter, *p*, for the $S^3M$ measure is taken as 0.5. As can be seen from the figure 9 ROC curve due to $S^3M$ measure has high DR than cosine and BWC similarity measures at low FPR. As can be observed from the figure that the detection rate of 1 was achieved at FPR of 0.01 in the case of $S^3M$ measure where as in the case of *cosine* measure it is achieved at FPR of 0.12. With the *BWC* similarity measure the DR of 1 was achieved at 0.08 FPR value. These values indicate that with $S^3M$, kNN classification algorithm performed well over *cosine* and *BWC* similarity measures. Both the *cosine* as well as *BWC* similarity measures use the vector encoding of the sequences before feding into the *kNN* classifier, thus losing the sequential information embedded in them. Whereas with the $S^3M$ the sequences are not encoded into the vector rather the similarity is computed directly across two sequences keeping both the sequential as well as content information into account.

*Figure 9. Comparative ROC curves for cosine, ureBWC, and S3M measures for k =5*




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*Figure 10. Comparative ROC graph with reduced and without reduced attacks*




Further, as quoted in the chapter (Liao et al., 2002b, Rawat et al., 2006), they achieved the DR =1 after removing two attacks from 55 attacks where as DR of 1 was achieved with $S^3M$ without removing any attacks. There were two attacks included in the testing data namely, 4:5it162228loadmodule and 5:5itfdformatchmod, which could not be detected by Rawat et al. (2006) whereas Liao et al., (2002b) scheme detected them at a lower threshold value. Rawat et al. (2006) quoted that though the attack *4.5it162228loadmodule* was launched, it failed to compromise the system thus making the process to seem normal. In case of the second one, *5.5itfdformatchmod*, the attack was not launched in early stage. Hence, they argued that the attack has not manifested and thus may not be detected by their scheme as they are normal processes. Rawat et al. reported results on experiments after removing these two attacks from the testing data set.

To compare with the Rawat et al. (2006) results we also removed these two attacks from our test set and experiments were conducted. Figure 10 shows the comparative ROC curves with the full and reduced dataset of abnormal attacks. The curve shows that with the reduced abnormal test set $S^3M$ performed more better hence showing its effectiveness. The decrease in the performance on non-reduced dataset is minimal and of course on the reduced dataset the performance is superior. For *DARPA'98 IDS* dataset we found that the order of occurrence of system calls plays a key role in determining the nature of the session.

From the results presented in this section, we can conclude that the $S^3M$ could form an integral part of an intrusion detection system that incorporates sequence classification. Experiments with *DARPA'98 IDS* dataset for k = 10 were also conducted. A similar trend and results were recorded. In this section, we have presented evidence that combining the sequential and content information of system calls results in good discrimination between normal and abnormal sessions.

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# Application Areas of Sequence Data Classification

Data mining finds application in a wide and diverse range of business, scientific, and engineering scenarios. For example, large databases of loan applications are available which record different kinds of demographic and transactional information about the customers. These databases can be mined for interesting and hidden patterns leading to defaulting of loans, which can in turn help in determining whether a new loan application should be accepted or rejected. Data mining can be helpful in the field of medical science by analyzing the medical records of the patients of a region by making early predictions for the potential outbreak of infectious disease or analysis of customer transactions for market research applications, etc. The list of application areas for data mining is large and is bound to grow rapidly in the years to come. A growing body of literature is available discussing the techniques for data mining and its applications (Grossman, Kamath, Kegelmeyer, Kumar, & Namburu, 2001; Han et al., 2001; Ian & Witten, 1999). Supervised learning from sequence data has many possible applications. Some of the applications of sequence learning tasks are as follows:

- **Speech recognition (Gold & Morgan, 2000; Rabiner et al., 1993):** Speech recognition is often defined as the automatic transcription of human utterances as sequence of words. In the training phase, given a large number of utterances with their correct transcriptions, the task is to learn the mapping from the acoustic signal to the word sequence, so as to later be able to recognize new, previously unknown utterances.
- **Automatic (machine) translation (Douglas, Lorna, Siety, Humphreys, & Sadler, 1994):** The task is to learn the mapping from a sequence of words in one language to a sequence of words in another language, which can be viewed as a sequence prediction problem with categorical variables in the input and output space.
- **Gesture prediction (Darrell & Pentland, 1993; Starner, 1995; Yamato, Ohya, & Ishii, 1992):** In gesture (or human body motion) recognition, video sequences containing hand or head gestures are classified according to the actions they represent or the messages they seek to convey. The gestures or body motions may represent, for example, one of a fixed set of messages like waving hello, goodbye, and so on. There could be the different strokes in a tennis video, or in other cases, they could belong to the dictionary of some sign language and so on.
- **Hand-writing recognition (Chen, Kundu, & Zhou, 1994; Nag, Wong, & Fallside, 1986):** Hand-writing recognition can today be done to some extent by many hand-held computers. A large number of words are written by an

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appropriate number of people with different writing styles such as, the movement of the pen is recorded using a digitizer board. Based on the sequential data describing the relationship between pen movement and the corresponding words or letters, models are built that are used to recognize new words.

- **Stock market prediction (Mittermayer, 2004):** The problem of predicting the value of stocks or currencies can be formulated as a sequence prediction problem. Historical data provides the example sequences. For example, the stock price index recorded over time is used to train a model for stock price forecasting.
- **Anomaly detection (Lee & Stolfo, 1998; Somayaji & Forrest, 2000):** The anomaly detection problem has been widely studied in the computer security literature. In the anamoly detection problem, one has to identify the sequence of system calls, which may be of abnormal nature than the available set of normal sessions.
- **Protein or DNA structure prediction (Durbin et al., 1999):** Biologists use the classification technique to find the family of new protein or DNA sequence from the available set of proteins and DNA families. BLAST, PAM, and FASTA are the commonly used algorithms in this domain.

All of the sequence predictions problems discussed are of different nature. Any specific approach to solving them first involves a unification and abstraction of all the specific problems to one scientific core problem, which allows us to use established scientific methods and knowledge from other scientific areas. These areas are as follows:

- **Pattern recognition (Berger, 1985; Bishop, 1995; Duda, Hart, & Stork, 2001):** Pattern recognition is also referred to as learning from examples. Pattern recognition often involves the definition of stochastic models like neural networks or Hidden Markov Models, which are trained on training data and tested on unseen test data. These methods rely on the practical aspects of probability theory and are strongly related to coding and compression.
- **Information theory (Hamming, 1980):** Information theory defines consistent ways of measuring the amount of information in data, called entropy. Many problems of pattern recognition, which are formulated to maximize a probability, can often also be formulated to maximize the amount of information flow through some channel.

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# Summary and Future Directions

The area of KDD deals with analyzing large data collections to extract interesting, potentially useful, hidden, and valid patterns. Data mining is the most important step within the process of KDD. The data being analyzed are either sequential or non-sequential in nature. Generally, sequential data when considered for classification task are converted into frequency vectors thus losing sequential information. Therefore, data mining algorithms should be modified to handle sequential data without losing sequential information embedded in them. This chapter contributes to the development of classification algorithms for sequential data. Initially, a widely adopted technique for sequence data classification was demonstrated. Results adhere that for classification of sequential data, sequence information along with content information is important. $S^3M$ has been utilized, which considers both the order of occurrence as well as the content information while computing similarity between them. Sequence classification task was carried out in intrusion detection system on benchmark dataset DARPA 98 IDS.

Usefulness of sequence as well as the content information for classifying sequential data were demonstrated using sliding window technique. Later, we described a similarity measure that is a linear combination of the length of the longest common subsequence and jaccard similarity measures. LCS calculation is computationally intensive as it uses dynamic programming approach. Hence, a better and efficient combination or an alternative measure capturing sequentiality is to be explored.

Usefulness of $S^3$ M measure for intrusion detection by classifying system calls was demonstrated in the chapter. The results obtained were interesting for mainly two reasons. Firstly, the technique shows the importance of considering sequential information in classifying the intrusion. Secondly, the techniques might be applicable to security problems in other computational settings.

Although the results presented for system call classification are suggestive, much more testing needs to be completed to validate the approach. In particular, extensive testing on a wider variety of programs being subjected to large numbers of different kinds of intrusions is desirable. For each of these programs, ideally results should be in both controlled environment (in which we could run large numbers of intrusions) and in live user environments.

As the DARPA'98 IDS dataset is collected from *solaries* operating system, the testing of the approach should also be done in other operating systems such as Windows NT.

However, there are some problems associated with collecting data in live user environments. Most operating systems do not have robust tracing facilities, hence data may be collected in standardized environments for testing purpose combining with the datasets collected from real time system.

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$S^3M$ measure is for discrete sequential data. In the same line, it needs to be modified and tested for continuous sequential data also.

We showed the effectiveness and efficiency of $S^3M$ in intrusion detection domain however the work can be extended to any other domains were data exhibit sequentiality in nature.

## References

Abraham, T. (2001). *IDDM: Intrusion detection using data mining techniques*. Technical Report DSTO-GD-0286, DSTO Electronics and Surveillance Research Laboratory, Salisbury, South Australia, Australia.

Anderson, D., Frivold, T., Tamaru, A., & Valdes, A. (1994). *Next-generation intrusion detection expert system (NIDES), software user's manual, beta-update release*. Technical Report SRI-CSL-95-07, Computer Science Laboratory, SRI International, Ravenswood Avenue, Menlo Park, CA.

Anderson, J. P. (1980). *Computer security threat monitoring and surveillance*. Technical Report CONTRACT 79F296400, James P. Anderson Company, Fort Washington, PA.

B. Y. M. Cheng, J. G. C., & Klein-Seetharaman, J. (2005). Protein classification based on text document classification techniques. *Proteins*, *58*(1), 955-970.

Berchtold, S., Keim, D. A., & Kriegel, H. P. (1996). The X-tree: An index structure for high-dimensional data. In T. M. Vijayaraman, A. P. Buchmann, C. Mohan, & N. L. Sarda (Eds.), *Proceedings of the 22nd International Conference on Very Large Databases* (pp. 28-39). San Francisco, USA. Morgan Kaufmann Publishers.

Berger, J. O. (1985). *Statistical decision theory and bayesian analysis* (2nd ed.). New York: Springer-Verlag.

Bharat, K., & Henzinger, M. R. (1998). Improved algorithms for topic distillation in a hyperlinked environment. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval* (pp. 104-111), New York, USA. ACM Press.

Bilenko, M., Mooney, R., Cohen, W., Ravikumar, P., & Fienberg, S. (2003). *Adaptive name-matching in information integration*. IEEE Intelligent Systems.

Bishop, C. M. (1995). *Neural networks for pattern recognition*. Oxford University Press.

Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees*. New York: Chapman and Hall.

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Brighton, H., & Mellish, C. (2002). Advances in instance selection for instance-based learning algorithms. *Data Mining and Knowledge Discovery, 6*(2), 153-172.

Burges, C. J. C. (1998). A tutorial on support vector machines for pattern recognition. *Data Mining and Knowledge Discovery, 2*(2), 121-167.

Chakrabarti, S. (2002). *Mining the Web: Discovering knowledge from hypertext data*. San Francisco: Morgan Kaufmann.

Chen, M. Y., Kundu, A., & Zhou, J. (1994). Off-line handwritten word recognition using a hidden markov model type stochastic network. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 16*(5), 481-496.

Christianini, N., & Shawe-Taylor, J. (2000). An introduction to support vector machines and other kernel based learning methods. Cambridge University Press.

Ciaccia, P., Patella, M., & Zezula, P. (1997). M-tree: An efficient access method for similarity search in metric spaces. In *Proceedings of the 23rd International Conference on Very Large Databases* (pp. 426-435). Athens, Greece.

Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning, 20*(3), 273-297.

Darrell, T., & Pentland, A. (1993). Space-time gestures. In *IEEE Conference on Computer Vision and Pattern Recognition* (pp. 335-340). IEEE Computer Society Press.

Deng, K., Moore, A., & Nechyba, M. (1997). Learning to recognize time series: Combining arma models with memory-based learning. *IEEE Int. Symp. on Computational Intelligence in Robotics and Automation* (Vol. 1, pp. 246-250).

Douglas, A., Lorna, B., Siety, M., Humphreys, R. L., & Sadler, L. (1994). Machine translation: An introductory guide. London: Blackwells-NCC.

Duda, R. O., Hart, P. E., & Stork, D. G. (2001). Pattern classification (2nd ed.). New York: John Wiley and Sons.

Durbin, R., Eddy, S., Krogh, A., & Mitchison, G. (1999). *Biological sequence analysis: Probabilistic models of proteins and nucleic acids*. New York: Cambridge University Press.

Fayyad, U. M., Shapiro, G. P., & Smyth, P. (1996). Knowledge discovery and data mining: Towards a unifying framework. In The *2nd International Conference on Knowledge Discovery and Data Mining* (pp. 82-88). Portland, Oregon. AAAI Press.

Forrest, S., Hofmeyr, S. A., Somayaji, A., & Longstaff, T. A. (1996). A sense of self for unix processes. In *Proceedings of the 1996 IEEE Symposium on Security and Privacy* (pp. 120-128). Washington, DC, USA. IEEE Computer Society.

Gehrke, J., Ramakrishnan, R., & Ganti, V. (1998). RainForest—A framework for fast decision tree construction of large datasets. In *Proceedings of the 24th*

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*International Conference on Very Large Databases* (pp. 416-427). San Francisco: Morgan Kaufmann Publishers.

Gold, B., & Morgan, N. (2000). Speech and audio signal processing: Processing and perception of speech and music. New York: John Wiley & Sons.

Graepel, T., Herbrich, R., Sdorra, P. B., & Obermayer, K. (1999). Classification on pairwise proximity data. In *Proceedings of the 1998 Conference on Advances in Neural Information Processing Systems II* (pp. 438-444), Cambridge, MA, USA. MIT Press.

Grossman, R. L., Kamath, C., Kegelmeyer, P., Kumar, V., & Namburu, R. R. (2001). *Data mining for scientific and engineering applications.* Kluwer Academic Press.

Gusfield, D. (1997). *Algorithms on strings, trees, and sequences: Computer science and computational biology.* New York: Cambridge University Press.

Hamming, R. W. (1980). *Coding and information theory*. Englewood Cliffs, NJ: Lawrence Erlbaum Associates, Prentice-Hall.

Han, E. H., & Karypis, G. (2000). Centroid-based document classification: Analysis and experimental results. In *Principles of Data Mining and Knowledge Discovery* (pp. 424-431). Lyon, France: Springer-Verlag.

Han, J., & Kamber, M. (2001). *Data mining: Concepts and techniques*. Morgan Kaufmann.

Hochberg, J., Jackson, K., Stallings, C., McClary, J. F., DuBois, D., & Ford, J. (1993). NADIR: An automated system for detecting network intrusion and misuse. *Computers and Security, 12*(3), 235-248.

Hofmeyr, S. A., Forrest, S., & Somayaji, A. (1998). Intrusion detection using sequences of system calls. *Journal of Computer Security, 6*(3), 151-180.

Ian, E. F., & Witten, H. (1999). Data mining: Practical machine learning tools and techniques with java implementations. Morgan Kaufmann Series in Data Management Systems.

Ilgun, K., Kemmerer, R. A., & Porras, P. A. (1995). State transition analysis: A rule-based intrusion detection approach. *IEEE Transactions on Software Engineering, 21*(3), 181-199.

Jacobs, D. W., Weinshall, D., & Gdalyahu, Y. (2000). Classification with nonmetric distances: Image retrieval and class representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 22*(6), 583-600.

Jain, A. K., & Zongker, D. (1997). Representation and recognition of handwritten digits using deformable templates. *IEEE Transactions on Pattern Analysis and Machine Intelligence, 19*(12), 1386-1391.

Jaakkola, T., Diekhans, M., & Haussler, D. (1999). Using the fisher kernel method to detect remote protein homologies. ISMB, 149-158.

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Kemmerer, R. A. (1998). *NSTAT: A model-based real-time network intrusion detection system*. Technical Report TRCS-97-18, Department of Computer Science, University of California, Santa Barbara.

Kim, G. H., & Spafford, E. H. (1994). The design and implementation of tripwire: A le system integrity checker. In *Proceedings of the 2nd ACM Conference on Computer and Communications Security* (pp. 18-29). New York, USA. ACM Press.

Kumar, P., Rao, M. V., Krishna, P. R., & Bapi, R. S. (2005a). Using sub-sequence information with kNN for classification of sequential data. In *Distributed Computing and Internet Technology* (pp. 536-546). Springer Berlin- Heidelberg. LNCS.

Kumar, P., Rao, M. V., Krishna, P. R., Bapi, R. S., & Laha, A. (2005b). Intrusion detection system using sequence and set preserving metric. In *Intelligence and Security Informatics* (pp. 498-504). Springer Berlin Heidelberg. LNCS.

Kumar, S., & Spafford, E. (1994). *An application of pattern matching in intrusion detection*. Technical Report 94-013, Department of Computer Sciences, Purdue University.

Laboratory, M. L. Intrusion Detection Datasets. http://www.ll.mit.edu/IST/ideval/data/data index.html.

Laxman, S., & Sastry, P. S. (2005). A survey of temporal data mining. SADHANA, *Academy Proceedings in Engineering Sciences*.

Lee, W. (1999). *A data mining framework for constructing features and models for intrusion detection systems*. PhD thesis, Dept of Computer Science, Columbia University.

Lee, W., & Stolfo, S. (1998). Data mining approaches for intrusion detection. In *Proceedings of the 7th USENIX Security Symposium* (pp. 79-93). San Antonio, TX.

Lee, W., & Xiang, D. (2001). Information-theoretic measures for anomaly detection. In *IEEE Symposium on Security and Privacy* (pp. 130-143). Oakland, CA, USA. IEEE Computer Society.

Lee, W., Stolfo, S. J., Chan, P. K., Eskin, E., Fan, W., Miller, M., Hershkop, S., & Zhang, J. (2001). Real time data mining-based intrusion detection. In The *2nd DARPA Information Survivability Conference and Exposition* (pp. 85-100), Anaheim, California.

Leslie, C., Eskin, E., Weston, J., & Noble, W. S. (2004). Mismatch string kernels for discriminative protein classification. *Bioinformatics, 20*(4), 467-476.

Liao, Y., & Vemuri, V. (2002a). Use of k-nearest neighbor classifier for intrusion detection. *Computers and Security, 21*(5), 439-448.

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Liao, Y., & Vemuri, V. R. (2002b). Using text categorization techniques for intrusion detection. In *Proceedings of the 11th USENIX Security Symposium* (pp. 51-59), Berkeley, CA, USA. USENIX Association.

Lin, K. I., Jagadish, H. V., & Faloutsos, C. (1994). The TV-tree: An index structure for high-dimensional data. *Very Large Databases Journal, 3*(4), 517-542.

McCallum, A., & Nigam, K. (1998). A comparison of event models for naive bayes text classification. In *Workshop on Learning for Text Categorization* (pp. 41-48). Madison, WI. AAAI Press.

Mitchell, T. M. (1997). *Machine learning*. Mc Graw Hill.

Mittermayer, M. A. (2004). Forecasting intraday stock price trends with text mining techniques. In *HICSS '04: Proceedings of the Proceedings of the 37th Annual Hawaii International Conference on System Sciences (HICSS'04)--Track 3* (pp. 30064.2/1-30064.2/10). Washington, DC, USA. IEEE Computer Society.

Muggleton, S., & Raedt, L. D. (1994). Inductive logic programming: Theory and methods. *Journal of Logic Programming, 19*(20), 629-679.

Nag, R., Wong, K. H., & Fallside, F. (1986). Script recognition using hidden markov models. In *IEEE International Conference on Acoustics, Speech, Signal Processing* (pp. 2071-2074).

Pekalska, E., & Duin, R. (2001). Automatic pattern recognition by similarity representations. *Electronic Letters, 37*(3), 159-160.

Pevzner, P., & Waterman, M. S. (1995). Multiple filtration and approximate pattern matching. *Algorithmica, 13*(1/2), 135-154.

Platt, J., Cristianini, N., & Shawe-Taylor, J. (2000). Large margin DAGS for multiclass classification. In S. A. Solla, T. K. Leen, & K. R. Mueller (Eds.), *Advances in neural information processing systems* (pp. 547-553). Cambridge: MIT Press.

Platt, J. C. (1998). *Sequential minimal optimization: A fast algorithm for training support vector machines*. Technical Report MSR-TR-98-14, Microsoft Research Technology, Redmond, Washington.

Porras, P. A., & Neumann, P. G. (1997). EMERALD: Event monitoring enabling responses to anomalous live disturbances. In *Proceedings of the 20th NIST-NCSC National Information Systems Security Conference* (pp. 353-365). Baltimore: NIST/NCSC.

Quinlan, J. R. (1993). *C4.5: Programms of machine learning*. San Mateo, CA: Morgan Kaufmann.

Rabiner, L., & Juang, B. H. (1993). *Fundamentals of speech recognition*. Englewood Cliffs, NJ: Prentice Hall.

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Rawat, S., Gulati, V. P., Pujari, A. K., & Vemuri, V. R. (2006). Intrusion detection using processing techniques with a binary-weighted cosine metric. *Journal of Information Assurance and Security, 1*(1), 43-58.

Sarawagi, S. (2005). Sequence data mining. In Advanced Methods for Knowledge Discovery from Complex Data. Springer Verlag.

Sarawagi, S. (2003). Sequence data mining techniques and applications. In *The 19th International Conference on Data Engineering* (ICDE'03).

Sebring, M. M., Shellhouse, E., Hanna, M., & Whitehurst, R. (1988). Expert systems in intrusion detection: A case study. In The *11th National Computer Security Conference* (pp. 74-81), Baltimore, MD.

Staniford-Chen, S., Cheung, S., Crawford, R., Dilger, M., Frank, J., Hoagland, J., Levitt, K., Wee, C., Yip, R., & Zerkle, D. (1996). GrIDS: A graph-based intrusion detection system for large networks. In *Proceedings of the 19th National Information Systems Security Conference* (pp. 361-370). Baltimore, Maryland.

Starner, T. (1995). *Visual recognition of American sign language using hidden markov models*. Master's thesis, Program in Media Arts.

Sun, R., & Giles, C. L. (2001). Sequence learning: Paradigms, algorithms, and applications. *Lecture Notes in Artificial Intelligence*, Springer.

Tan, K. M. C., & Maxion, R. A. (2002). "Why 6?" defining the operational limits of stide, an anomaly-based intrusion detector. In *SP'02: Proceedings of the 2002 IEEE Symposium on Security and Privacy* (pp. 188-201). Washington, DC, USA. IEEE Computer Society.

Teng, H. S., Chen, K., & Lu, S. C. Y. (1990). Security audit trail analysis using inductively generated predictive rules. In *Proceedings of the 6th Conference on Artificial Intelligence Applications* (pp. 24-29). Piscataway, NJ, USA. IEEE Press.

Valdes, A., & Skinner, K. (2000). Adaptive, model-based monitoring for cyber attack detection. In The *3rd International Workshop on Recent Advances in Intrusion Detection* (pp. 80-82), London. Springer-Verlag.

Warrender, C., Forrest, S., & Pearlmutter, B. (1999). Detecting intrusions using system calls: Alternative data models. IEEE Symposium on Security and Privacy.

Yamato, J., Ohya, J., & Ishii, K. (1992). Recognizing human action in timesequential images using a hidden markov model. In *IEEE International Conference on Computer Vision and Pattern Recognition* (pp. 379-385). Champaign, IL.

Yan, C., Dobbs, D., Honavar, V., & Dobbs, D. (2004). A two-stage classifier for identification of protein interface residues. *Bioinfromatics, 20*(S1), i371-i378.

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# Endnote

* The author is currently working with SET Labs, Infosys Technologies Limited, Bangalore, INDIA.

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## Chapter VIII

# Using Cryptography for Privacy-Preserving Data Mining

Justin Zhan, Carnegie Mellon University, USA

## Abstract

*To conduct data mining, we often need to collect data from various parties. Privacy concerns may prevent the parties from directly sharing the data and some types of information about the data. How multiple parties collaboratively conduct data mining without breaching data privacy presents a challenge. The goal of this chapter is to provide solutions for privacy-preserving k-nearest neighbor classification, which is one of the data mining tasks. Our goal is to obtain accurate data mining results without disclosing private data. We propose a formal definition of privacy and show that our solutions preserve data privacy.*

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# Introduction

Recent advances in data collection, data dissemination, and related technologies have inaugurated a new era of research where existing data mining algorithms should be reconsidered from the point of view of privacy preservation. The term privacy is used frequently in ordinary language, yet there is no single definition of this term. The concept of privacy has broad historical roots in sociological and anthropological discussions about how extensively it is valued and preserved in various cultures (Schoeman, 1984). Historical use of the term is not uniform and there remains confusion over the meaning, value, and scope of the concept of privacy. Privacy refers to the right of users to conceal their personal information and have some degree of control over the use of any personal information disclosed to others (Ackerman, Cranor, & Reagle, 1999). Particularly, in this chapter, the privacy preservation means that multiple parties collaboratively get valid data mining results while disclosing no private data to each other or any party who is not involved in the collaborative computations.

The need for privacy is sometimes due to law (e.g., for medical databases) or can be motivated by business interests. However, there are situations where the sharing of data can lead to mutual benefit. Despite the potential gain, this is often not possible due to the confidentiality issues which arise. It is well documented (Epic, 2003) that the unlimited explosion of new information through the Internet and other media has reached a point where threats against the privacy are very common and they deserve serious thinking.

Let us consider an example. There are several hospitals involved into a multi-site medical study. Each hospital has its own data set containing patient records. These hospitals would like to conduct data mining over the data sets from all of hospitals with the goal of more valuable information would be obtained via mining the joint data set. Due to privacy laws, one hospital cannot disclose their patient records to other hospitals.

How can these hospitals achieve their objective? Can privacy and collaborative data mining coexist? In other words, can the collaborative parties somehow conduct data mining computations and obtain the desired results without compromising their data privacy?

We show that privacy and collaborative data mining can be achieved at the same time. The goal of this chapter is to present technologies to solve privacy-preserving collaborative data mining problems over large data sets with reasonable efficiency.

The contributions of this chapter contain the following: (1) a proposed formal definition of privacy for privacy-preserving collaborative data mining, (2) a solution for k-nearest neighbor classification with vertical collaboration, and (3) the efficiency analysis to show the performance scaling up with various factors such

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as the number of parties involved in the computation, the encryption key size, the size of data set, etc.

## Related Work

In this section, we describe the state of the art in privacy-preserving data mining techniques. To protect actual data from being disclosed, one approach is to alter the data in a way that actual individual data values cannot be recovered, while certain computations can still be applied to the data. Due to the fact that the actual data are not provided for the mining, the privacy of data is preserved. This is the core idea of randomization-based techniques. The random perturbation technique is usually realized by adding noise or uncertainty to actual data such that the actual values are prevented from being discovered. Since the data no longer contains the actual values, it cannot be misused to violate individual privacy. Randomization approaches were first proposed by Agrawal and Srikant (2000) to solve the privacy-preserving data mining problem. Specifically, they addressed the following question. Since the primary task in data mining is the development of models about aggregated data, can they develop accurate models without access to precise information in individual data records? The underlying assumption is that a person will be willing to selectively divulge information in exchange of useful information that such a model can provide. Agrawal and Aggarwal (2001) showed that the EM algorithm converges to the maximum likelihood estimate of the original distribution based on the perturbed data. They showed that when a large amount of data is available, the EM algorithm provides robust estimates of the original distribution. Evfimievski, Srikant, Agrawal, and Gehrke (2002) presented a framework for mining association rules from transactions consisting of categorical items where the data has been randomized to preserve privacy of individual transactions. While it is feasible to recover association rules and preserve privacy using a straightforward uniform randomization, the discovered rules can unfortunately be exploited to find privacy breaches. They analyzed the nature of privacy breaches and proposed a class of randomization operators that are much more effective than uniform randomization in limiting the breaches. Du and Zhan (2003) proposed a technique for building decision trees using randomized response techniques, which were developed in the statistics community for the purpose of protecting surveyees' privacy. The randomization-based methods have the benefits of efficiency. However, the drawbacks are that post-randomization data mining results are only an approximation of pre-randomization results. There are some randomization level control parameters. It has been experimentally shown that for certain scenarios under the control of a randomization parameter, the accuracy of the results can achieve a certain level in the case of both decision tree classification (Agrawal et al., 2000) and association rule mining (Evfmievski et. al. 2002; Rizvi

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& Haritsa, 2002). Furthermore, the randomization-based method may be invalid to protect data privacy in certain scenarios. Kargupta, Datta, Wang, and Sivakumar (2003) analyzed the privacy of random perturbation techniques and showed how to attack the privacy by using random matrix-based data filtering techniques. Although Evfmievski et al. (2003) showed how to limit privacy breaches while using randomization for the privacy-preserving data mining, there are still concerns that randomization-based approaches may allow an attacker to reconstruct distributions and also give out much information about the original data values.

Following the idea of secure multiparty computation, people designed privacy-oriented protocols for the problem of the privacy-preserving collaborative data mining. Lindell and Pinkas (2000) first introduced a secure multi-party computation technique for classification using the ID3 algorithm, over horizontally partitioned data. Specifically, they consider a scenario in which two parties owning confidential databases wish to run a data mining algorithm on the union of their databases, without revealing any unnecessary information. Du and Zhan (2002) proposed a protocol for making the ID3 algorithm privacy-preserving over vertically partitioned data. Lin, Clifton, and Zhu (2004) proposed a secure way for clustering using the EM algorithm over horizontally partitioned data. Kantarcioglu and Clifton (2002) described protocols for the privacy-preserving distributed data mining of association rules on horizontally partitioned data. Vaidya and Clifton presented protocols for privacy-preserving association rule mining over vertically partitioned data (Vaidya & Clifton, 2002) and provided a solution for building a decision tree without compromising data privacy (Vaidya & Clifton, 2005).

Encryption is a well-known technique for preserving the confidentiality of sensitive information. Comparing with other techniques described, a strong encryption scheme can be more effective in protecting the data privacy. An encryption system normally requires that the encrypted data should be decrypted before making any operations on it. For example, if the value is hidden by a randomization-based technique, the original value will be disclosed with certain probability. If the value is encrypted using a semantic secure encryption scheme (Paillier, 1999), the encrypted value provide no help for attacker to find the original value. One of the schemes is the homomorphic encryption, which was originally proposed in Rivest, Adleman, and Dertouzos (1978), with the aim of allowing certain computations, performed on encrypted data without preliminary decryption operations. To date, there are many such systems. Homomorphic encryption is a very powerful cryptographic tool and has been applied in several research areas such as electronic voting, online auction, etc. (Wright & Yang 2004) is based on homomorphic encryption where Wright et al. applied homomorphic encryption to the Bayesian networks induction for the case of two parties. Zhan, Matwin, and Chang (2005) proposed a cryptographic approach to tackle collaborative association rule mining among multiple parties. In this chapter, we will apply homomorphic encryption (Paillier, 1999) and digital envelope techniques (Chaum, 1985) to the privacy-preserving data mining and use

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them to design privacy-oriented protocols for privacy-preserving k-nearest neighbor classification problem. The preliminary results have been presented in Zhan and Matwin (2006).

Next, we will provide our building blocks, which contain the notion of security, definition of privacy, homomorphic encryption, digital envelope technique, and attack models that we consider.

# Building Blocks

## Homomorphic Encryption

The concept of homomorphic encryption was originally proposed in Rivest et al. (1978). Since then, many such systems have been proposed. In this chapter, we base our privacy-oriented protocols on Paillier (1999), which is semantically secure. A cryptosystem is homomorphic with respect to some operation $*$ on the message space if there is a corresponding operation $*'$ on the ciphertext space such that $e(m)*'e(m') = e(m * m')$. In this chapter, we utilize the following property of the homomorphic encryption functions: $e(m_1) \times e(m_2) = e(m_1 + m_2)$ where $m_1$ and $m_2$ are the data to be encrypted. Because of the property of associativity, $e(m_1 + m_2 + \cdots + m_n)$ can be computed as $e(m_1) \times e(m_2) \times \cdots \times e(m_n)$ where $e(m_i) \neq 0$. That is

$$e(m_1 + m_2 + \cdots + m_n) = e(m_1) \times e(m_2) \times \cdots \times e(m_n)$$

## Digital Envelope

A digital envelope (Chaum, 1985) is a random number (or a set of random numbers) only known by the owner of private data. To hide the private data in a digital envelope, we conduct a set of mathematical operations between a random number (or a set of random numbers) and the private data. The mathematical operations could be addition, subtraction, multiplication, etc. For example, assume the private data value is v. There is a random number R which is only known by the owner of v. The owner can hide v by adding this random number (e.g., v + *R*).

## Goal-Oriented Attack Model

Secure communication in networks is often goal-oriented. For example, there are two people (e.g., Alice and Bob) who want to talk to each other. Suppose that Alice

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would like to send message m to Bob, the goal of Alice is to let Bob receive message m; the goal of Bob is to receive m from Alice. In the meanwhile, they want to keep the communication secure so that network attackers cannot see m or modify it in a meaningful way. To achieve this goal, they apply encryption to message m.

Privacy-preserving collaborative data mining is also goal-oriented. The fundamental goal of privacy-preserving collaborative data mining in our framework is to obtain the accurate results while keeping the data private. In other words, we need to obtain accurate data mining results without letting each party see other parties' private data. This is an ideal case. In fact, the data mining results may disclose partial information about the private data. Therefore, a practical goal of privacy preservation is to hide private data during the mining stage for a given privacy-preserving data mining problem.

In this chapter, we define our attack model as *goal-oriented attack model*. In this model, all the collaborative parties need to follow their goals. The basic goal of collaborative data mining is to obtain desired data mining results which are shared among the collaborative parties. The attacks can be applied as long as they follow their basic goal and the purpose of the attacks of one party (or a group of parties) is to gain useful information about the data of the other party (or the other group of parties).

## Privacy-Preserving Collaborative Data Mining Problems

Our approaches can be used for the typical privacy-preserving collaborative data mining problems: Multiple parties, each having a private data set which is denoted by $DS_1$, $DS_2$, ... and $DS_n$ respectively, want to collaboratively conduct k-nearest neighbor classification on the concatenation of their data sets. Because they are concerned about their data privacy or due to the legal privacy rules, neither party is willing to disclose its actual data set to others.

In this chapter, we consider the privacy-preserving collaborative k-nearest neighbor classification problem, which is defined as follows:

There are two types of collaborative models. In the vertical collaboration, diverse features of the same set of data are collected by different parties. In the horizontal collaboration,

diverse sets of data, all sharing the same features, are gathered by different parties. The collaborative model that we consider is the *vertical collaboration*.

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# Privacy-Preserving K-Nearest Neighbor Classification

## Background

The k-nearest neighbor classification (Cover & Hart, 1968) is an instance-based learning algorithm that has been shown to be very effective for a variety of problem domains. The objective of k-nearest neighbor classification is to discover k nearest neighbors for a given instance, then assign a class label to the given instance according to the majority class of the k nearest neighbors. The algorithm assumes that all instances correspond to points in the n-dimensional space. The key element of this scheme is the availability of a similarity measure that is capable of identifying neighbors. The nearest neighbors of an instance are defined in terms of a distance function such as the standard Euclidean distance. More precisely, let an arbitrary instance x be described by the feature vector $< a_1(x), a_2(x), \cdots, a_r(x) >$, where $a_i(x)$ denotes the value of the ith attribute of instance x. Then the distance between two instances $x_i$ and $x_j$ is defined as $dist(x_i, x_j)$, where:

$$dist(x_i, x_j) = \sqrt{\sum_{q=1}^{r} (a_q(x_i) - a_q(x_j))^2}$$

In this section, we use the square of the standard Euclidean distance to compare the different distances.

$$dist^2(x_i, x_j) = \sum_{q=1}^{r} (a_q(x_i) - a_q(x_j))^2$$

## K-Nearest Neighbor Classification Procedure

We consider learning discrete-valued target functions of the form $f : R^{\gamma} \rightarrow V$, where V is the finite set $v_1, v_2, \cdots, v_s$. The following is the procedure for building a k-nearest neighbor classifier.

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**Algorithm 1.**

1. Training algorithm

- For each training example (x, f(x)), add the example to the list training-examples.

2. Classification algorithm: Given a query instance $x_q$ to be classified,

- Let $x_1, x_2, \cdots, x_k$ denote the k instances from training-examples that are nearest to $x_q$.

- Return $\hat{f}(x_q) \leftarrow \arg\max_{v \in V} \sum_{i=1}^{k} \delta(v, f(x_i))$, where $\delta(a,b) = 1$ if a=b and $\delta(a,b) = 0$ otherwise.

To build a k-nearest neighbor classifier, the key point is to privately obtain k nearest instances for a given point. Next, we will provide privacy-oriented protocols for the scenarios of vertical collaboration.

## Privacy-Preserving Protocols for Vertical Collaboration

In vertical collaboration, each party holds a subset of attributes for every instance. Given a query instance $x_q$, we want to compute the distance between $x_q$ and each of the N training instances. Since each party holds only a portion (i.e., partial attributes) of a training instance, each party computes their portion of the distance (called the distance portion) according to her attribute set. For example, suppose that each instance has 10 attributes (e.g., $A_1, A_2, \cdots, A_{10}$) and there are 4 parties in the collaboration with $P_1$ holding $\{A_1, A_2, A_3\}$, $P_2$ holding $\{A_4, A_5, A_6\}$, $P_3$ holding $\{A_7, A_8\}$, and $P_4$ holding $\{A_9, A_{10}\}$ respectively. Then $P_1$ computes:

$$dist^2(x_i, x_j) = \sum_{q=1}^{3} (a_q(x_i) - a_q(x_j))^2,$$

$P_2$ computes

$$dist^2(x_i, x_j) = \sum_{q=4}^{6} (a_q(x_i) - a_q(x_j))^2,$$

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$P_3$ computes

$$dist^2(x_i, x_j) = \sum_{q=7}^{8} (a_q(x_i) - a_q(x_j))^2,$$

and $P_4$ computes

$$dist^2(x_i, x_j) = \sum_{q=9}^{10} (a_q(x_i) - a_q(x_j))^2.$$

To decide the k nearest neighbors of $x_q$, all the parties need to sum their distance portions together. For example, assume that the distance portions (the square of the standard Euclidean distance) for the first instance are $s_{11}, s_{12}, \cdots, s_{1n}$; and the distance portions (the square of the standard Euclidean distance) for the second instance are $s_{21}, s_{22}, \cdots, s_{2n}$. To compute whether the distance between the first instance and $x_q$ is larger than the distance between the second instance and $x_q$, we need to compute whether $\sum_{i=1}^{n} s_{1i} \geq \sum_{i=1}^{n} s_{2i}$. How can we obtain this result without compromising data privacy? A naive solution is that those parties disclose their distance portions to each other, and they can then easily decide the k nearest neighbors by comparing the summation of their distance portions. However, the naive solution will lead to private data disclosure. The reasons are as follows:

- **Multi-query problem:** One party can make multiple queries, and if he gets the distance portions from each query, he can then identify the private data. Let us use an example to illustrate this problem. Assume that the query instance contains two non-zero values (e.g., $x_q$ = 1.2, 4.3, 0, ..., 0), and $P_1$ holds the first two attributes. Then, the query requester can learn the private values of $P_1$ with two queries. First, he uses $x_q$ to get a distance value denoted as dist. He uses another $x'_q$ = 5.6, 4.8, 0, ..., 0 to get another distance value {\bf dist'. He can solve the following two equations to get the first and second elements (denoted by $y_1$ and $y_2$) of $x_i$ which are supposed to be private:

$$dist_1^2 = (y_1 - 1.2)^2 + (y_2 - 4.3)^2$$

$$dist_2^2 = (y_1 - 5.6)^2 + (y_2 - 4.8)^2$$

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How do we privately compute the k nearest neighbors for a given instance? Next, we develop a privacy-oriented protocol to tackle this challenge. For illustration purpose, let us first introduce some notations.

**Notation 1.**

- **n:** The total number of parties.
- **N:** The total number of instances that dataset contains.
- **Forward transmission:** The encrypted data are transmitted in the following sequence: $P_1, P_2, \cdots, P_n$, and $P_{n-1}$.
- **Backward transmission:** The encrypted data are transmitted in the following sequence: $P_{n-1}, P_{n-2}, \cdots, P_1$.
- R and R' are random numbers.

- **Highlight of protocol:** Without loss of generality, assume $P_1$ has a private distance portion between the query point and the *ith* training instance, $s_{il}$, and a private distance portion between the query point and the *jth* training instance, $s_{jl}$, for $i, j \in [1, N], i \neq j, l \in [1, n]$. The problem is to decide whether:

$$\sum_{l=1}^{n} s_{il} \leq \sum_{l=1}^{n} s_{jl}$$

for $i, j \in [1, N](i \neq j)$ and select the k smallest values, without disclosing each distance portion. In our protocol, we randomly select a key generator (e.g., $P_n$). The parties first seal their private data in a digital envelope, and apply homomorphic encryptions to their data to compute $e(\sum_{l=1}^{n} s_{il})$ for $i \in [1, N]$ in step I. They then compute $e(\sum_{l=1}^{n} - s_{jl})$ for $j \in [1, N]$ in step II. Finally, they compute the k nearest neighbors in step III.

We present the formal protocol as follows:

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**Protocol**

**Step I: Compute** $e(\sum_{l=1}^{n} s_{il})$ **for** $i \in [1, N]$

1. Key and random number generation

(a) $P_n$ generates a cryptographic key pair(e, d) of a semantically-secure homomorphic encryption scheme and publishes its public key e.

(b) $P_1$ generates N random numbers $R_{il}$, for all $i \in [1, N], l \in [1, n]$.

2. Forward transmission

(a) $P_1$ computes $e(s_{i1} + R_{i1})$, for $i \in [1, N]$, and sends them to $P_2$.

(b) $P_2$ computes $e(s_{i1} + R_{i1}) \times e(s_{i2} + R_{i2}) = e(s_{i1} + s_{i2} + R_{i1} + R_{i2})$ according to Equation 1, where $i \in [1, N]$, and sends them to $P_3$.

(c) Repeat 2a and 2b until $P_{n-1}$ obtains $e(s_{i1} + s_{i2} + \cdots + s_{i(n-1)} + R_{i1} + R_{i2} + \cdots + R_{i(n-1)})$, for all $i \in [1, N]$.

(d) $P_n$ computes $e(s_{in})$, $i \in [1, N]$, and sends them to $P_{n-1}$.

3. Backward transmission

(a) $P_{n-1}$ computes $e(-R_{i(n-1)})$, for $i \in [1, N]$ and sends them to $P_{n-2}$.

(b) $P_{n-2}$ computes $e(-R_{i(n-1)}) \times e(-R_{i(n-2)}) = e(-R_{i(n-1)} - R_{i(n-2)})$ according to Equation 1, $i \in [1, N]$, and sends them to $P_{n-3}$.

(c) Repeat 3a and 3b until $P_1$ obtains $e_{i1} = e(-R_{i1} - R_{i2} - \cdots - R_{i(n-1)})$, for all $i \in [1, N]$.

(d) $P_1$ sends $e_{i1}$, for $i \in [1, N]$, to $P_{n-1}$.

4. Computation of $e(\sum_{l=1}^{n} s_{il})$, for $i \in [1, N]$.

(a) $P_{n-1}$ computes:

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$e_{i(n-1)} = e(s_{i1} + s_{i2} + \cdots + s_{i(n-1)} + R_{i1} + R_{i2} + \cdots + R_{i(n-1)}) \times e(s_{in})$

$= e(s_{i1} + s_{i2} + \cdots + s_{i(n-1)} + s_{in} + R_{i1} + R_{i2} + \cdots + R_{i(n-1)})$ according to Equation 1, $i \in [1, N]$.

(b) $P_{n-1}$ computes $e_{i(n-1)} \times e_{i1} = e(\sum_{l=1}^{n} s_{il})$, for $i \in [1, N]$ and $l \in [1, n]$.

Step II: Compute $e(\sum_{l=1}^{n} - s_{jl})$ for $j \in [1, N]$

1. Random number generation

(a) $P_1$ generates N random numbers $R'_{jl}$, for all $j \in [1, N]$, $l \in [1, n]$.

2. Forward transmission

(a) $P_1$ computes $e(-s_{j1} + R'_{j1})$, for $j \in [1, N]$, and sends them to $P_2$.

(b) $P_2$ computes $e(-s_{j1} + R'_{j1}) \times e(-s_{j2} + R'_{j2}) = e(-s_{j1} - s_{j2} + R'_{j1} + R'_{j2})$ according to Equation 1, where $j \in [1, N]$, and sends them to $P_3$.

(c) Repeat 2a and 2b until $P_{n-1}$ obtains $e(-s_{j1} - s_{j2} - \cdots - s_{j(n-1)} + R'_{j1} + R'_{j2} + \cdots + R'_{j(n-1)})$, for all $j \in [1, N]$.

(d) $P_n$ computes $e(-s_{jn})$, $j \in [1, N]$, and sends them to $P_{n-1}$.

3. Backward transmission

(a) $P_{n-1}$ computes $e(-R'_{j(n-1)})$, for $j \in [1, N]$ and sends them to $P_{n-2}$.

(b) $P_{n-2}$ computes $e(-R'_{j(n-1)}) \times e(-R'_{j(n-2)}) = e(-R'_{j(n-1)} - R'_{j(n-2)})$ according to Equation 1, $j \in [1, N]$, and sends them to $P_{n-3}$.

(c) Repeat 3a and 3b until $P_1$ obtains $e_{j1} = e(-R'_{j1} - R'_{j2} - \cdots - R'_{j(n-1)})$, for all $j \in [1, N]$.

(d) $P_1$ sends $e_{j1}$, for $j \in [1, N]$, to $P_{n-1}$.

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4. Computation of $e(\sum_{l=1}^{n} - s_{jl})$, for $j \in [1, N]$

(a) $P_{n-1}$ computes:

$$e_{j(n-1)} = e(-s_{j1} - s_{j2} - \cdots - s_{j(n-1)} + R'_{j1} + R'_{j2} + \cdots + R'_{j(n-1)}) \times e(-s_{jn})$$

$$= e(-s_{j1} - s_{j2} - \cdots - s_{j(n-1)} - s_{jn} + R'_{j1} + R'_{j2} + \cdots R'_{j(n-1)})$$ according to Equation 1, $j \in [1, N]$.

(b) $P_{n-1}$ computes $e_{j(n-1)} \times e_{j1} = e(\sum_{l=1}^{n} - s_{jl})$ according to Equation 1, for $j \in [1, N]$ and $l \in [1, n]$.

Step III: Compute the k Nearest Neighbors

1. $P_{n-1}$ computes $e(\sum_{l=1}^{n} s_{il}) \times e(\sum_{l=1}^{n} - s_{jl}) = e(s_{il} - s_{jl})$ according to Equation 1, for $i, j \in [1, N]$, and collects the results into a sequence $\mathsf{y}$ which contains N(N-1) elements.
2. $P_{n-1}$ randomly permutes this sequence and obtains the permuted sequence denoted by $\psi_1$, then sends $\psi_1$ to $P_n$.
3. $P_n$ decrypts each element in sequence $\psi_1$. He assigns the element +1 if the result of decryption is not less than 0, and -1, otherwise. Finally, he obtains a +1/-1 sequence denoted by $\psi_2$.
4. $P_n$ sends $\psi_2$ to $P_{n-1}$ who computes k smallest elements as the next section shows. They are the k nearest neighbors for a given query instance $x_q$. He then decides the class label for $x_q$.

*Table 1. An index table of number sorting*

| | $t_1$ | $t_2$ | $t_3$ | …. | $t_n$ |
|---|---|---|---|---|---|
| $t_1$ | +1 + | 1 | -1 … | . | -1 |
| $t_2$ | -1 + | 1 | -1 … | . | -1 |
| $t_3$ | +1 + | 1 | +1 … | . | …. |
| …. … | . | …. … | . | …. … | . |
| $t_n$ | +1 + | 1 | -1 … | . | +1 |

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*Table 2. An example of sorting*

| | $t_1$ | $t_2$ | $t_3$ | $t_4$ | Weight |
|---|---|---|---|---|---|
| $t_1$ | +1 - | 1 | -1 - | 1 | -2 |
| $t_2$ | +1 + | 1 | -1 + | 1 | +2 |
| $t_3$ | +1 + | 1 | +1 + | 1 | +4 |
| $t_4$ | +1 - | 1 | -1 + | 1 | 0 |

**The correctness analysis:** To show that the protocol correctly finds the k-nearest neighbors for a given query instance $x_q$, we analyze it step by step. In step I, $P_{n-1}$ obtains $e(\sum_{l=1}^{n} il)$ for $i \in [1, N]$. In step II, $P_{n-1}$ gets $e(\sum_{l=1}^{n} - s_{jl})$ for $j \in [1, N]$. We will not provide detailed explanations for these two steps since the explanations exactly follow each sub-step. The key issue is that $P_{n-1}$ actually obtains the k-nearest neighbors in step III. This property directly follows the discussion of next section.

## How to Compute k-Nearest Neighbors

$P_{n-1}$ is able to remove permutation effects from $\psi_2$ (the resultant sequence is denoted by $y_3$) since she has the permutation function that she used to permute $\psi$, so that the elements in $\psi$ and $\psi_3$ have the same order. It means that if the qth position in sequence $\psi$ denotes $e(t_i - t_j)$ where $t_i$ denotes $\sum_{l=1}^{n} s_{il}$ and $t_j$ denotes $\sum_{l=1}^{n} s_{jl}$, then the $qth$ position in sequence $\psi_3$ denotes the result of $t_i - t_j$. We encode it as +1 if $t_i \geq t_j$, and as -1 otherwise.

$P_{n-1}$ has two sequences: one is the $\psi$, the sequence of $e(t_i - t_j)$, for $i, j \in [1, n](i > j)$, and the other is $\psi_3$, the sequence of +1/-1. The two sequences have the same number of elements. $P_{n-1}$ knows whether or not $t_i$ is larger than $t_j$ by checking the corresponding value in the $\psi_3$ sequence. For example, if the first element $\psi_3$ is -1, $P_{n-1}$ concludes $t < t_j$. $P_{n-1}$ examines the two sequences and constructs the index table (Table 1) to sort $t_i, i \in [1, n]$.

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In Table 1, +1 in entry ij indicates that the value of the row (e.g., $t_i$ of the $t_j$ row) is not less than the value of a column (e.g., $t_j$ of the *jth* column); -1, otherwise. $P_{n-1}$ sums the index values of each row and uses this number as the weight of that row. She then sorts the sequence according to the weight.

To make it clearer, Let us illustrate it by an example. Assume that: (1) there are 4 elements with $t_1 < t_4 < t_2 < t_3$; (2) the sequence $\psi$ is $[e(t_1 - t_2), e(t_1 - t_3), e(t_1 - t_4), e(t_2 - t_3), e(t_2 - t_4), e(t_3 - t_4)]$. The sequence $\psi_3$ will be (-1, -1, -1, -1, +1, +1). According to $\psi$ and $\psi_3$, $P_{n-1}$ builds the Table 2. From the table, $P_{n-1}$ knows $t_3 > t_2 > t_4 > t_1$ since $t_3$ has the largest weight, $t_2$ has the second largest weight, $t_4$ has the third largest weight, $t_1$ has the smallest weight. Therefore, the 1-nearest neighbor is the first instance (corresponding to $t_1$); the 2-nearest neighbors are the first instance and the fourth instance (corresponding to $t_4$), etc.

Next, we provide the communication cost as well as the computation cost.

- **The communication complexity analysis:** Let us use $\alpha$ to denote the number of bits of each ciphertext and $\beta$ to denote the number of bits of each plaintext. Normally, it is assumed that $\beta < \alpha$. n is the total number of parties and N is the total number of records.

The total communication cost consists of (1) the cost of $2\alpha nN$ from step I; (2) the cost of $2\alpha nN$ from step II; (3) the cost of $aN(N-1) + bN(N-1) + \frac{3}{2}an^2 + a(n-1)$ from step III.

- **The computation complexity analysis:** The following contributes to the computational cost: (1) The generation of one cryptographic key pair, (2) The generation of 2N random numbers, (3) The total number of 4nN encryptions, (4) The total number of $N^2 + 4nN + 3N$ multiplications, (5) The total number of N(N-1) decryptions, (6) 2nN additions, (7) gNlog(N) for sorting N numbers where g is a constant number.

**Theorem 2.** Protocol 1 preserves data privacy.

For the purpose of proof, let us introduce the following notations:

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- We use $ADV_{P_i}$ to denote $P_i$'s advantage to gain access to the private data of any other party via the component protocol.
- We use $ADV_S$ to denote the advantage of one party to gain the other party's private data via the component protocol by knowing the semantic secure encryptions.
- We use $VIEW_{P_i}$ to denote the extra information that $P_i$ obtains via the component protocol.

**Proof 2.** We have to show that

$\Pr(T \mid CP - \Pr(T)) \leq \varepsilon$, for $T = T_{P_i}, i \in [1, n]$, and CP = Protocol 1.

According to the above notations,

$ADV_{P_n} = \Pr(T_{P_i} \mid VIEW_{P_n}, \Pr otocol1) - \Pr(T_{P_i} \mid VIEW_{P_n})$, where $i \neq n$.

$ADV_{P_i} = \Pr(T_{P_j} \mid VIEW_{P_i}, \Pr otocol1) - \Pr(T_{P_j} \mid VIEW_{P_i})$, where $i \neq n, j \neq i$.

The information that $P_i$, where $i \neq n$, obtains from other parties is encrypted by e that is semantic secure. Thus, $ADV_{P_i} = ADV_S$.

In order to show that privacy is preserved according to our definition, we need to know the value of the privacy level $\mathrm{e}$. We set $\mathrm{e} = \max(ADV_{P_n}, ADV_{P_i}) = \max(ADV_{P_n}, ADV_S)$.

Then:

$\Pr(T_{P_i} \mid VIEW_{P_n}, \Pr otocol1) - \Pr(T_{P_i} \mid VIEW_{P_n}) \leq \varepsilon, i \neq n,$

$\Pr(T_{P_j} \mid VIEW_{P_i}, \Pr otocol1) - \Pr(T_{P_j} \mid VIEW_{P_i}) \leq \varepsilon, i \neq n, j \neq i,$ which completes the proof. (Note that all the information that $P_n$ obtains from other parties is

$e_{i(n-1)} \times e_{j(n-1)} = e(\sum_{l=1}^{n} s_{jl} - \sum_{l=1}^{n} s_{jl})$, where $i, j \in [1, N]$ but in a random order.

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# Discussion

Privacy-preserving data mining has generated many research successes. However, we have not yet accepted definitions of privacy and a challenging research question in the area of privacy-preserving data mining is to better understand and define privacy. In this chapter, we propose a formal definition of privacy. The idea is to express the potential disclosure of private data due to collaborative data mining as the advantage that one party gains via collaboration to obtain the private data of other parties. We measure that as the probability difference $\Pr(T \mid PPDMS) - \Pr(T)$ (i.e., the probability that private data T is disclosed with and without privacy-preserving data mining schemes being applied. We use the definition to measure the privacy level for our solution.

We have proposed to use homomorphic encryption and digital envelope techniques to achieve collaborative data mining without sharing the private data among the collaborative parties. Our approach has wide potential impact in many applications. In practice, there are many environments where privacy-preserving collaborative data mining is desirable. For example, several pharmaceutical companies have invested significant amount of money conducting genetic experiments with the goal of discovering

meaningful patterns among genes. To increase the size of the population under study and to reduce the cost, companies decide to collaboratively mine their data without disclosing their actual data because they are only interested in limited collaboration; by disclosing the actual data, a company essentially enables other parties to make discoveries that the company does not want to share with others. In another field, the success of homeland

security aiming to counter terrorism depends on combination of strength across different mission areas, effective international collaboration and information sharing to support a coalition in which different organizations and nations must share some, but not all, information. Information privacy thus becomes extremely important and our technique can be applied.

In particular, we provide a solution for k-nearest neighbor classification with vertical collaboration in this chapter. We provide efficient analysis for our solution. The solution is not only efficient but also provides decent privacy protection under our definition. In the future, we would like to examine other privacy-preserving collaborative data mining tasks and implement a privacy-preserving collaborative data mining system.

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# References

Ackerman, M., Cranor, L., & Reagle, J. (1999). Privacy in ecommerce: Examining user scenarios and privacy preferences. In *Proceedings of the ACM Conference on Electronic Commerce* (pp. 1-8). Denver, Colorado, USA, November.

Agrawal, D., & Aggarwal, C. (2001). On the design and quantification of privacy preserving data mining algorithms. In *Proceedings of the 20th ACM SIGACT-SIGMODSIGART Symposium on Principles of Database Systems* (pp. 247-255). Santa Barbara, CA, May 21-23.

Agrawal, R., & Srikant, R. (2000). Privacy-preserving data mining. In *Proceedings of the ACM SIGMOD Conference on Management of Data* (pp. 439-450). ACM Press, May.

Chaum, D. (1985). Security without identification. *In Communication of the ACM, 28*(10), 1030-1044.

Cover, T., & Hart, P. (1968). Nearest neighbor pattern classification. In *IEEE Transaction of Information Theory*, Vol. 13, pp. 21-27, January, 1968.

Du, W., & Zhan, Z. (2003). Using randomized response techniques for privacy-preserving data mining. In *Proceedings of the 9th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Washington, DC, USA. August 24-27.

Du, W., & Zhan, Z. (2002). Building decision tree classifier on private data. In *IEEE Intentional Workshop on Privacy, Security, and Data Mining*, Maebashi City, Japan, December 9.

Epic (2003). Privacy and human rights an international survey of privacy laws and developments. Retrieved from www.epic.org

Evfimievski, A., Gehrke, J. E., & Srikant, R. (2003). Limiting privacy breaches in privacy preserving data mining. In *Proceedings of the 22nd ACMSIGACT-SIGMOD-SIGART Symposium on Principles of Database Systems (PODS 2003).* San Diego, CA, June.

Evfmievski, A., Srikant, R., Agrawal, R., & Gehrke, J. (2002). Privacy preserving mining of association rules. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 217-228). Edmonton, Alberta, Canada, July 23-262.

Kantarcioglu, M., & Clifton, C. (2002). Privacy-preserving distributed mining of association rules on horizontally partitioned data. In *The ACM SIGMOD Workshop on Research Issues on Data Mining and Knowledge Discovery (DMKD'02)* (pp. 24-31). Madison, WI, June.

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Kantarcioglu, M., & Clifton, C. (2004). Privacy preserving data mining of association rules on horizontally partitioned data. In Transactions on Knowledge and Data Engineering, IEEE Computer Society Press, Los Alamitos, CA, 2004.

Kargupta, H., Datta, S., Wang, Q., & Sivakumar, K. (2003). On the privacy preserving properties of random data perturbation techniques. In *Proceedings of the 3rd IEEE International Conference on Data Mining (ICDM'03)*, Melbourne, FL, November 19-22.

Lin, X., Clifton, C., Zhu, M. (2004). Privacy preserving clustering with distributed em mixture modeling. In Knowledge and Information Systems, 2004.

Lindell, Y., & Pinkas, B. (2000). Privacy preserving data mining. In *Advances in Cryptology - Crypto2000, Lecture Notes in Computer Science* (Vol. 1880).

Paillier, P. (1999). Public-key cryptosystems based on composite degree residuosity classes. In *Advances in Cryptography--EUROCRYPT '99* (pp. 223-238). Prague, Czech Republic.

Rivest, R., Adleman, L., & Dertouzos, M. (1978). On data banks and privacy homomorphisms. In R. A. DeMillo et al. (Eds.), *Foundations of secure computation* (pp. 169-179). Academic Press.

Rizvi, S., & Haritsa, J. (2002). Maintaining data privacy in association rule mining. In *Proceedings of the 28th VLDB Conference*, Hong Kong, China.

Schoeman, F. D. (1984). Philosophical dimensions of privacy. In Cambridge University Press.

Vaidya, J., & Clifton, C. (2005). Privacy-preserving decision trees over vertically partitioned data. In *19th Annual IFIP WG 11.3 Working Conference on Data and Applications Security*. Nathan Hale Inn, University of Connecticut, Storrs, CT, U.S.A., August 7-10.

Vaidya, J., & Clifton, C. (2002). Privacy preserving association rule mining in vertically partitioned data. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Edmonton, Alberta, Canada, July 23-26.

Wright, R., & Yang, Z. (2004). Privacy-preserving Bayesian network structure computation on distributed heterogeneous data. In *Proceedings of the 10th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*.

Zhan, Z., & Matwin, S. (2006). A crypto-approach to privacy-preserving data mining. In *IEEE International Conference on Data Mining workshop on Privacy Aspect of Data Mining*, December 18-22, Hong Kong

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Zhan, J., Matwin, S., & Chang, L. (2005). Privacy-preserving collaborative association rule mining. In The *19th Annual IFIP WG11.3 Working Conference on Data and Applications Security*, Nathan Hale Inn, University of Connecticut, Storrs, CT.

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*Section III*

# Domain Driven and Model Free

## Chapter IX

# Domain Driven Data Mining

Longbing Cao, University of Technology, Sydney, Australia
Chengqi Zhang, University of Technology, Sydney, Australia

## Abstract

*Quantitative intelligence-based traditional data mining is facing grand challenges from real-world enterprise and cross-organization applications. For instance, the usual demonstration of specific algorithms cannot support business users to take actions to their advantage and needs. We think this is due to quantitative intelligence focused data-driven philosophy. It either views data mining as an autonomous data-driven, trial-and-error process, or only analyzes business issues in an isolated, case-by-case manner. Based on experience and lessons learned from real-world data mining and complex systems, this article proposes a practical data mining methodology referred to as domain-driven data mining. On top of quantitative intelligence and hidden knowledge in data, domain-driven data mining aims to meta-synthesize*

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*quantitative intelligence and qualitative intelligence in mining complex applications in which human is in the loop. It targets actionable knowledge discovery in constrained environment for satisfying user preference. Domain-driven methodology consists of key components including understanding constrained environment, business-technical questionnaire, representing and involving domain knowledge, human-mining cooperation and interaction, constructing next-generation mining infrastructure, in-depth pattern mining and postprocessing, business interestingness and actionability enhancement, and loop-closed human-cooperated iterative refinement. Domain-driven data mining complements the data-driven methodology, the metasynthesis of qualitative intelligence and quantitative intelligence has potential to discover knowledge from complex systems, and enhance knowledge actionability for practical use by industry and business.*

# Introduction

Traditionally data mining is presumed as an automated process. It produces automatic algorithms and tools with limited or no human involvement. As a result, they lack the capability of adapting to external environment change. Many patterns mined but few are workable in real business. On the other hand, real-world data mining must adapt to dynamic situations in the business world. It also expects actionable discovered knowledge that can afford important grounds to business decision makers for performing appropriate actions.

Unfortunately, mining actionable knowledge is not a trivial task. As pointed out by the panel discussions of SIGKDD 2002 and 2003 (Ankerst, 2002, Fayyad & Shapiro, 2003), it was highlighted as one of the grand challenges for the extant and future data mining. The weakness of existing data mining partly results from the data-driven trial-and-error methodology (Ankerst, 2002), which depreciates the roles of domain resources such as domain knowledge and humans. For instance, data mining in the real world such as crime pattern mining (Bagui, 2006) is highly constraint based (Boulicaut & Jeudy, 2005; Fayyad et al., 2003). Constraints involve technical, economical, and social aspects in the process of developing and deploying actionable knowledge. For actionable knowledge discovery from data embedded with the previous constraints, it is essential to slough off the superficial and captures the essential information from data mining.

Many data mining researchers have realized the significant roles of some domain-related aspects, for instance, domain knowledge and constraints, in data mining. They further develop specific corresponding data mining areas such as constraint-based data mining to solve issues in traditional data mining. As a result, data mining is progressing toward a more flexible, specific, and practical manner with increasing capabilities of tackling real-world emerging complexities. In particular, data min-

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*Table 1. Data mining development*

| Dimension | Key research progress |
|---|---|
| Data mined | • Relational, data warehouse, transactional, object-relational, active, spatial, time-series, heterogeneous, legacy, WWW<br>• Stream, spatiotemporal, multi-media, ontology, event, activity, links, graph, text, etc. |
| Knowledge discovered | • Characters, associations, classes, clusters, discrimination, trend, deviation, outliers, etc.<br>• Multiple and integrated functions, mining at multiple levels, exceptions, etc. |
| Techniques developed | • Database-oriented, association and frequent pattern analysis, multidimensional and OLAP analysis methods, classification, cluster analysis, outlier detection, machine learning, statistics, visualization, etc.<br>• Scalable data mining, stream data mining, spatiotemporal data and multimedia data mining, biological data mining, text and Web mining, privacy-preserving data mining, event mining, link mining, ontology mining, etc. |
| Application involved | • Engineering, retail market, telecommunication, banking, fraud detection, intrusion detection, stock market, etc.;<br>• Specific task-oriented mining<br>• Biological, social network analysis, intelligence and security, etc.<br>• Enterprise data mining, cross-organization mining |

ing is gaining rapid development in comprehensive aspects such as data mined, knowledge discovered, techniques developed, and applications involved. Table 1 illustrates such key research and development progress in KDD.

Our experience (Cao & Dai, 2003a, 2003b) and lessons learned in real world data mining such as in capital markets (Cao, Luo, & Zhang, 2006; Lin & Cao 2006) show that the involvement of domain knowledge and humans, the consideration of constraints, and the development of in-depth patterns are very helpful for filtering subtle concerns while capturing incisive issues. Combining these and other aspects together, a sleek data mining methodology can be developed to find the distilled core of a problem. It can advise the process of real-world data analysis and preparation, the selection of features, the design and fine-tuning of algorithms, and the evaluation and refinement of mining results in a manner more effective to business. These are our motivations to develop a practical data mining methodology, referred to as *domain-driven data mining*.

*Domain-driven data mining* complements data-driven data mining through specifying and incorporating *domain intelligence* into data mining process. It targets the discovery of actionable knowledge that can support business decision-making. Here *domain intelligence* refers to all necessary parts in the problem domain surrounding the data mining system. It consists of domain knowledge, humans, constraints, organization factors, business process, and so on.

*Domain-driven data mining* consists of *a domain-driven in-depth pattern discovery* (DDID-PD) framework. The DDID-PD takes *I³D* (namely interactive, in-depth, iterative, and domain-specific) as real-world KDD bases. I³D means that the discovery of actionable knowledge is an iteratively interactive in-depth pattern discovery process

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in a domain-specific context. I$^3$D is further embodied through (i) mining constraint-based context, (ii) incorporating domain knowledge through human-machine-cooperation, (iii) mining in-depth patterns, (iv) enhancing knowledge actionability, and (v) supporting loop-closed iterative refinement to enhance knowledge actionability. Mining constraint-based context requests to effectively extract and transform domain-specific datasets with advice from domain experts and their knowledge.

In the DDID-PD framework, data mining and domain experts complement each other in regard to in-depth granularity through interactive interfaces. The involvement of domain experts and their knowledge can assist in developing highly effective domain-specific data mining techniques and reduce the complexity of the knowledge producing process in the real world. In-depth pattern mining discovers more interesting and actionable patterns from a domain-specific perspective. A system following the DDID-PD framework can embed effective supports for domain knowledge and experts' feedback, and refines the lifecycle of data mining in an iterative manner.

The remainder of this chapter is organized as follows. Section 2 discusses KDD challenges. Section 3 addresses knowledge actionability. Section 4 introduces domain intelligence. A domain driven data mining framework is presented in section 5. In section 6, key components in domain-driven data mining are stated. Section 7 summarizes some applications domain driven actionable knowledge discovery. We conclude this chapter and present future work in section 8.

## KDD Challenges

Several KDD-related mainstreams have organized forums discussing the actualities and future of data mining and KDD, for instance, the panel discussions in SIGKDD 2002 and 2003. In retrospect of the progress and prospect of existing and future data mining, many great challenges, for instance, link analysis, multi-data sources, and complex data structure, have been identified for the future effort on knowledge discovery. In this chapter, we highlight two of them, mining actionable knowledge and involving domain intelligence. This is based on the consideration that these two are more general significant issues of existing and future KDD. Not only do they hinder the shift from data mining to knowledge discovery, they also block the shift from hidden pattern mining to actionable knowledge discovery. It further restrains the wide acceptance and deployment of data mining in solving complex enterprise applications. Moreover, they are closely related and to some extent create a cause-effect relation that is involving domain intelligence for actionable knowledge discovery.

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## KDD Challenge: Mining Actionable Knowledge

Discovering actionable knowledge has been viewed as the essence of KDD. However, even up to now, it is still one of the great challenges to existing and future KDD as pointed out by the panel of SIGKDD 2002 and 2003 (Ankerst 2002; Cao & Zhang 2006a, 2006b) and retrospective literature. This situation partly results from the limitation of traditional data mining methodologies, which view KDD as a data-driven trial-and-error process targeting automated hidden knowledge discovery (Ankerst 2002; Cao & Zhang 2006a, 2006b). The methodologies do not take into much consideration of the constrained and dynamic environment of KDD, which naturally excludes humans and problem domain in the loop. As a result, very often data mining research mainly aims to develop, demonstrate, and push the use of specific algorithms while it runs off the rails in producing actionable knowledge of main interest to specific user needs.

To revert to the original objectives of KDD, the following three key points have recently been highlighted: comprehensive constraints around the problem (Boulicaut et al., 2005), domain knowledge (Yoon, Henschen, Park, & Makki, 1999), and human role (Ankerst, 2002; Cao & Dai, 2003a; Han, 1999,) in the process and environment of real-world KDD. A proper consideration of these aspects in the KDD process has been reported to make KDD promising to dig out actionable knowledge satisfying real life dynamics and requests even though this is very difficult issue. This pushes us to think of *what knowledge actionablility is* and *how to support actionable knowledge discovery*.

We further study a practical methodology called *domain-driven data mining* for *actionable knowledge discovery* (Cao & Zhang, 2006a, 2006b). On top of the data-driven framework, domain-driven data mining aims to develop proper methodologies and techniques for integrating domain knowledge, human role and interaction, as well as actionability measures into the KDD process to discover actionable knowledge in the constrained environment. This research is very important for developing the next-generation data mining methodology and infrastructure (Ankerst, 2002). It can assist in a paradigm shift from "*data-driven hidden pattern mining*" to "*domain-driven actionable knowledge discovery*," and provides supports for KDD to be translated to the real business situations as widely expected.

## KDD Challenge: Involving Domain Intelligence

To handle the previous challenge of mining actionable knowledge, the development and involvement of domain intelligence into data mining is presumed as an effective means. Since real-world data mining is a complicated process that encloses mix-data, mix-factors, mix-constraints, and mix-intelligence in a domain-specific

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organization, the problem of involving domain intelligence in knowledge discovery is another grand challenge of data mining.

Currently, there is quite a lot of specific research on domain intelligence related issues, for instance, modeling domain knowledge (Anand, Bell, & Hughes, 1995; Yoon et al., 1999), defining subjective interestingness (Liu, Hsu, Chen, & Ma, 2000), and dealing with constraints (Boulicaut et al., 2005). However, there is no consolidation state-of-the-art work on both domain intelligence and the involvement of domain intelligence into data mining. There are a few issues that are essentially necessary to be considered and studied. First, we need to create a clear and operable definition and representation of what domain intelligence is or is substantially significant. Second, a flexible, dynamic, and interactive framework is necessary to integrate the basic components of domain intelligence. Third, complexities come from the quantification of many semi-structure and ill-structure data, as well as the role and cooperation of humans in data mining.

In theory, we actually need to develop appropriate methodologies to support the involvement of domain intelligence into KDD. The methodology of *metasynthesis from qualitative to quantitative* (Dai, Wang, & Tian, 1995; Qian, Yu, & Dai, 1991) for dealing with open complex intelligent systems (Qian et al., 1991) is suitable for this research because it has addressed the roles and involvement of humans especially domain experts, as well as the use and balance of qualitative intelligence and quantitative intelligence in human-machine-cooperated environment (Cao et al., 2003a, 2003b).

# Knowledge Actionablility

## Measuring Knowledge Actionability

In order to handle the challenge of mining actionable knowledge, it is essential to define what is knowledge actionability. Often mined patterns are non-actionable to real needs due to the interestingness gaps between academia and business (Gur & Wallace, 1997). Measuring *actionability* of knowledge is to recognize statistically interesting patterns permitting users to react to them to better service business objectives. The measurement of knowledge actionability should be from perspectives of both objective and subjective.

Let $I = \{i_1, i_2, \ldots, i_m\}$ be a set of items, *DB* be a database that consists of a set of transactions, $x$ is an itemset in *DB*. Let $P$ be an interesting pattern discovered in *DB* through utilizing a model $M$. The following concepts are developed for domain driven data mining.

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- **Definition 1. Technical Interestingness:** The technical interestingness *tech_int*() of a rule or a pattern is highly dependent on certain technical measures of interest specified for a data mining method. *Technical interestingness* is further measured in terms of $\forall x \in I, \exists P : act(P) = f(tech_obj(P) \wedge tech_subj(P) \wedge biz_obj(P) \wedge biz_subj(P))$. *Technical objective measures tech_obj*() and *technical subjective measures tech_sub*().
- **Definition 2. Technical Objective Interestingness:** Technical *objective* measures *tech_obj*() capture the complexities of a link pattern and its statistical significance. It could be a set of criteria. For instance, the following logic formula indicates that an association rule P is technically interesting if it satisfies *min_support* and *min_confidence*. $\forall x \in I, \exists P : x.min_support(P) \wedge x.min_confidence(P) \rightarrow x.tech_obj(P)$
- **Definition 3. Technical Subjective Interestingness:** On the other hand, technical *subjective* measures *tech_subj*(), also focusing and based on technical means, recognize to what extent a pattern is of interest to a particular user needs. For instance, *probability-based belief* (Padmanabhan et al 1998) is developed for measuring the expectedness of a link pattern.
- **Definition 4. Business Interestingness:** The business interestingness *biz_int*() of an itemset or a pattern is determined from domain-oriented social, economic, user preference and/or psychoanalytic aspects. Similar to technical interestingness, business interestingness is also represented by a collection of criteria from both objective *biz_obj*() and subjective *biz_subj*() perspectives.
- **Definition 5. Business Objective Interestingness:** The business objective interestingness *biz_obj*() measures to what extent that the findings satisfy the concerns from business needs and user preference based on objective criteria. For instance, in stock trading pattern mining, *profit* and *roi* (*return on investment*) are often used for judging the business potential of a trading pattern objectively. If *profit* and *roi* (*return on investment*) of a stock price predictor P are satisfied, then P is interesting to trading. $\forall x \in I, \exists P : x.profit(P) \wedge x.roi(P) \rightarrow x.biz_obj(P)$
- **Definition 6. Business Subjective Interestingness:** *Biz_subj*() measures business and user concerns from subjective perspectives such as psychoanalytic factors. For instance, in stock trading pattern mining, a kind of *psycho-index 90%* may be used to indicate that a trader thinks it as very promising for real trading.

A successful discovery of an actionable knowledge is a collaborative work between miners and users, which satisfies both academia-oriented technical interestingness measures *tech_obj*() and *tech_subj*() and domain-specific business interestingness *biz_obj*() and *biz_subj*().

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- **Definition 7. Actionability of a pattern:** Given a pattern *P*, its actionable capability *act*() is described as to what degree it can satisfy both technical interestingness and business one.

If a pattern is automatically discovered by a data mining model while it only satisfies technical interestingness request, it is usually called an (technically) *interesting* pattern. It is presented as $\forall x \in I, \exists P : x.tech_int(P) \rightarrow x.act(P)$

In a special case, if both technical and business interestingness, or a hybrid interestingness measure integrating both aspects, are satisfied, it is called an *actionable* pattern. It is not only interesting to data miners, but generally interesting to decision-makers.

$\forall x \in I, \exists P : x.tech_int(P) \wedge x.biz_int(P) \rightarrow x.act(P)$

Therefore, the work of actionable knowledge discovery must focus on knowledge findings, which can not only satisfy technical interestingness but also business measures. Table 2 summarizes the interestingness measurement of data-driven vs. domain-driven data mining.

## Narrowing Down Interest Gap

To some extent, due to the selection criteria difference, the interest gap between academia and business is inherent. Table 3 presents a view of such interest gap. We classify data mining projects into (1) discovery research projects, which recognize the importance of fundamental innovative research, (2) linkage research projects, which support research and development to acquire knowledge for innovation as well as economic and social benefits, and (3) commercial projects, which develop knowledge that solves business problems. Interest gap is understood from input and output perspective, respectively. *Input* refers to the problem under studied, while *output* mainly refers to algorithms and revenue on problem-solving. Both input and output are measured in terms of academia and business aspects. For instance,

*Table 2. Interestingness measurement of data-driven vs. domain-driven data mining*

| Interestingness | | Traditional data-driven | Domain-driven |
|---|---|---|---|
| Technical | Objective | Technical objective *tech_obj*() | Technical objective *tech_obj*() |
| | Subjective | Technical subjective *tech_subj*() | Technical subjective *tech_subj*() |
| Business | Objective | - | Business objective *biz_obj*() |
| | Subjective | - | Business subjective *biz_subj*() |
| Integrative | | - | Actionability *act*() |

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*Table 3. Interest gap between academia and business*

| | Input | | Output | |
|---|---|---|---|---|
| | Research issues | Business problems | Algorithms | Revenue |
| Discovery research | [+++++] | [] | [+++++] | [] |
| Linkage research | [++] | [+++] | [+++] | [++] |
| Commercial project | [] | [+++++] | [] | [+++++] |

academic output of a project is mainly measured by algorithms, while business output is evaluated according to revenue (namely dollars).

We evaluate the input and output focuses of each type of projects in terms of a five-scale system, where the presence of a + indicates certain extent of focus for the projects. For instance, [+++++] indicates that the relevant projects fully concentrate on this aspect, while less number of + means less focus on the target. The marking in the table shows the gap or even conflict of problem definition and corresponding expected outcomes between business and academia for three types of projects.

Furthermore, the interest gap is embodied in terms of the interestingness satisfaction of a pattern. In the real-world mining, business interestingness *biz_int*() of a pattern may differ or conflict from technical significance *tech_int*() that guides the selection of a pattern. This situation happens when a pattern is originally or mainly discovered in terms of technical significance. In varying real-world cases, the relationship between technical and business interestingness of a pattern *P* may present as one of four scenarios as listed in Table 4. Hidden reasons for the conflict

*Table 4. Relationship between technical significance and business expectation*

| **Relationship Type** | **Explanation** |
|---|---|
| *tech_int*() $\ll$ *biz_int*() | The pattern *P* does not satisfy technical significance but satisfies business expectation |
| *tech_int*() $\gg$ *biz_int*() | The pattern *P* does not satisfy business expectation but satisfies technical significance |
| *tech_int*() $\cong$ *biz_int*() | The pattern *P* satisfies business expectation as well as technical significance |
| *tech_int*() $\hat{=}$ *biz_int*() | The pattern *P* satisfies neither business expectation nor technical significance |

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of business and academic interests may come from the neglectful business interest checking in developing models.

Clearly, the goal of actionable knowledge mining is to model and discover patterns considering both technical and business concerns, and confirming the relationship type *tech_int*() ≅ *biz_int*(). However, it is sometime very challenging to generate decent patterns of bilateral interest. For instance, quite often a pattern with high *tech_int*() creates bad *biz_int*(). Contrarily, it is not a rare case that a pattern with low *tech_int*() generates good *biz_int*(). Rather, patterns are extracted first by checking technical interestingness. Further, they are checked in terms of business satisfaction. Often it is a kind of artwork to tune thresholds of each side and balance the difference between *tech_int*() and *biz_int*(). If the difference is too big to be narrowed down, it is domain users who can better tune the thresholds and difference. Besides the above-discussed work on developing useful technical and business interestingness measures, there are some other things to do to reach and enhance knowledge actionability such as efforts on designing and actionability measures by integrating business considerations, testing actionability, enhancing actionability and assessing actionability in domain-driven data mining process.

## Specifying Business Interestingness

There is only limited research on business interestingness development in traditional data mining. Business interestingness cares about business concerns and evaluation criteria. They are usually measured in terms of specific problem domains by developing corresponding business measures.

Recently, some research is emerging in developing more general business interestingness models. For instance, Kleinberg et al. presented a framework of the microeconomic view of data mining. Profit mining (Wang et al., 2002) defined a set of past transactions and pre-selected target items, a model is further built for recommending target items and promotion strategies to new customers, with the goal of maximizing the net profit. Cost-sensitive learning () is another interesting area on modeling the error metrics of modeling and minimizing validation error.

In our work on capital market mining (Cao, Luo, & Zhang, 2006), we re-define financial measures such as *profit*, *return on investiment* and *sharpe ratio* to measure the business performance of a mined trading pattern in the market. In mining debt-related activity patterns in social security activity transactions, we specify business interestingness in terms of benefit and risk metrics, for instance benefit metrics such as pattern's *debt recovery rate* and *debt recovery amount* are developed to justify the prevention benefit of an activity pattern, while debt risk such as *debt duration risk* and *debt amount risk* measure the impact of a debt-related activity sequence on a debt (Centrelink summary report).

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## Domain Intelligence

Traditionally, data mining only pays attention to and relies on data intelligence to tell a story wrapping a problem. Driven by this strategic idea, data mining focuses on developing methodologies and methods in terms of data-centered aspects, particularly the following issues:

- Data type such as numeric, categorical, XML, multimedia, composite
- Data timing such as temporal, time-series and sequential
- Data space such as spatial and temporal-spatial
- Data speed such as data stream
- Data frequency such as high frequency data
- Data dimension such as multi-dimensional data
- Data relation such as multi-relational, link

On the other hand, domain intelligence consists of qualitative intelligence and quantitative intelligence. Both qualitative and quantitative intelligence is instantiated in terms of domain knowledge, constraints, actors/domain experts and environment. Further, they are instantiated into specific bodies. For instance, constraints may include domain constraints, data constraints, interestingness constraints, deployment constraints and deliverable constraints. To deal with constraints, various strategies and methods may be taken, for instance, interestingness constraints are modeled in terms of interestingness measures and factors, say objective interestingness and subjective interestingness.

In a summary, we categorize domain intelligence in terms of the following major aspects.

**Domain knowledge**

- Including domain knowledge, background and prior information

**Human intelligence**

- Referring to direct or indirect involvement of humans, imaginary thinking, brainstorm
- Empirical knowledge
- Belief, request, expectation

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**Constraint intelligence**

- Including constraints from system, business process, data, knowledge, deployment, etc
- Privacy
- Security

**Organizational intelligence**

- Organizational factors
- Business process, workflow, project management and delivery
- Business rules, law, trust

**Environment intelligence**

- Relevant business process, workflow
- Linkage systems

**Deliverable intelligence**

- Profit, benefit
- Cost
- Delivery manner
- Business expectation and interestingness
- Embedding into business system and process

Correspondingly, a series of major work needs to be studied in order to involve domain intelligence into knowledge discovery, and complement data-driven data mining towards domain driven actionable knowledge discovery. For instance, the following lists some of such tasks:

- Definition of domain intelligence
- Representation of domain knowledge
- Ontological and semantic representation of domain intelligence
- Domain intelligence transformation between business and data mining
- Human role, modeling and interaction
- Theoretical problems in involving domain intelligence into KDD
- Metasynthesis of domain intelligence in knowledge discovery
- Human-cooperated data mining
- Constraint-based data mining
- Privacy, security in data mining

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- Open environment in data mining
- In-depth data mining
- Knowledge actionability
- Objective and subjective interestingness
- Gap resolution between statistical significance and business expectation
- Domain-oriented knowledge discovery process model
- Profit mining, benefit/cost mining
- Surveys of specific areas

## Domain Driven Data Mining Framework

The existing data mining methodology, for instance CRISP-DM, generally supports autonomous pattern discovery from data. On the other hand, the idea of domain driven knowledge discovery is to involve domain intelligence into data mining. The DDID-PD highlights a process that discovers in-depth patterns from constraint-based context with the involvement of domain experts/knowledge. Its objective is to maximally accommodate both naive users as well as experienced analysts, and satisfy business goals. The patterns discovered are expected to be actionable to solve domain-specific problems, and can be taken as grounds for performing effective actions. To make domain-driven data mining effective, user guides and intelligent human-machine interaction interfaces are essential through incorporating both human qualitative intelligence and machine quantitative intelligence. In addition, appropriate mechanisms are required for dealing with multiform constraints and

*Table 5. Data-driven vs. domain-driven data mining*

| Aspects | Traditional data-driven | Domain-driven |
|---|---|---|
| Object mined | Data tells the story | Data and domain (business rules, factors etc.) tell the story |
| Aim | Developing innovative approaches | Generating business impacts |
| Objective | Algorithms are the focus | Systems are the target |
| Dataset | Mining abstract and refined data set | Mining constrained real life data |
| Extendibility | Predefined models and methods | Ad-hoc and personalized model customization |
| Process | Data mining is an automated process | Human is in the circle of data mining process |
| Evaluation | Evaluation based on technical metrics | Business say yes or no |
| Accuracy | Accurate and solid theoretical computation | Data mining is a kind of artwork |
| Goal | Let data to create/verify research innovation; Demonstrate and push the use of novel algorithms discovering knowledge of interest to research | Let data and domain knowledge to tell hidden story in business; discovering actionable knowledge to satisfy real user needs |

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domain knowledge. Table 5 compares major aspects under research of traditional data-driven and domain-driven data mining.

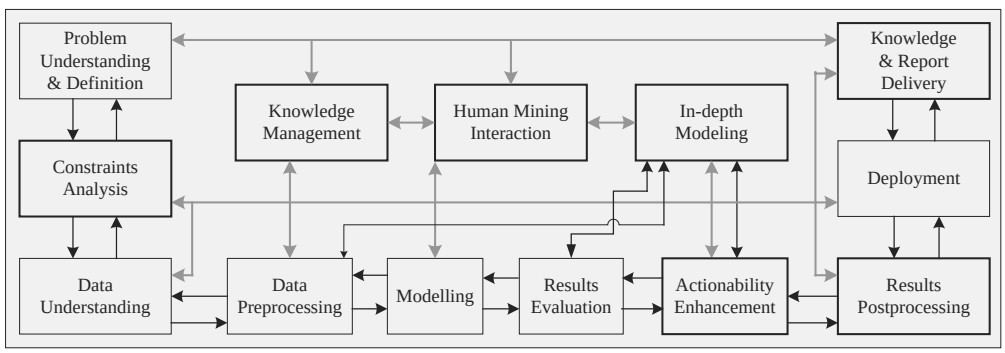
## DDID-PD Process Model

The main functional components of the DDID-PD are shown in Figure 1, where we highlight those processes specific to DDID-PD in thicken boxes. The lifecycle of DDID-PD is as follows, but be aware that the sequence is not rigid, some phases may be bypassed or moved back and forth in a real problem. Every step of the DDID-PD process may involve domain knowledge and interaction with real users or domain experts.

The lifecycle of DDID-PD is as follows, but be aware that the sequence is not rigid, some phases may be bypassed or moved back and forth in a real problem. Every step of the DDID-PD process may involve domain knowledge and the assistance of domain experts.

- P1. Problem understanding
- P2. Constraints analysis
- P3. Analytical objective definition, feature construction
- P4. Data preprocessing
- P5. Method selection and modeling
- P5'. In-depth modeling
- P6. Initial generic results analysis and evaluation

*Figure 1. DDID-PD process model*




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- P7. It is quite possible that each phase from P1 may be iteratively reviewed through analyzing constraints and interaction with domain experts in a back-and-forth manner
- P7': In-depth mining on the initial generic results where applicable
- P8. Actionability measurement and enhancement
- P9. Back and forth between P7 and P8
- P10. Results post-processing
- P11. Reviewing phases from P1 may be required
- P12. Deployment
- P13. Knowledge delivery and report synthesis for smart decision making

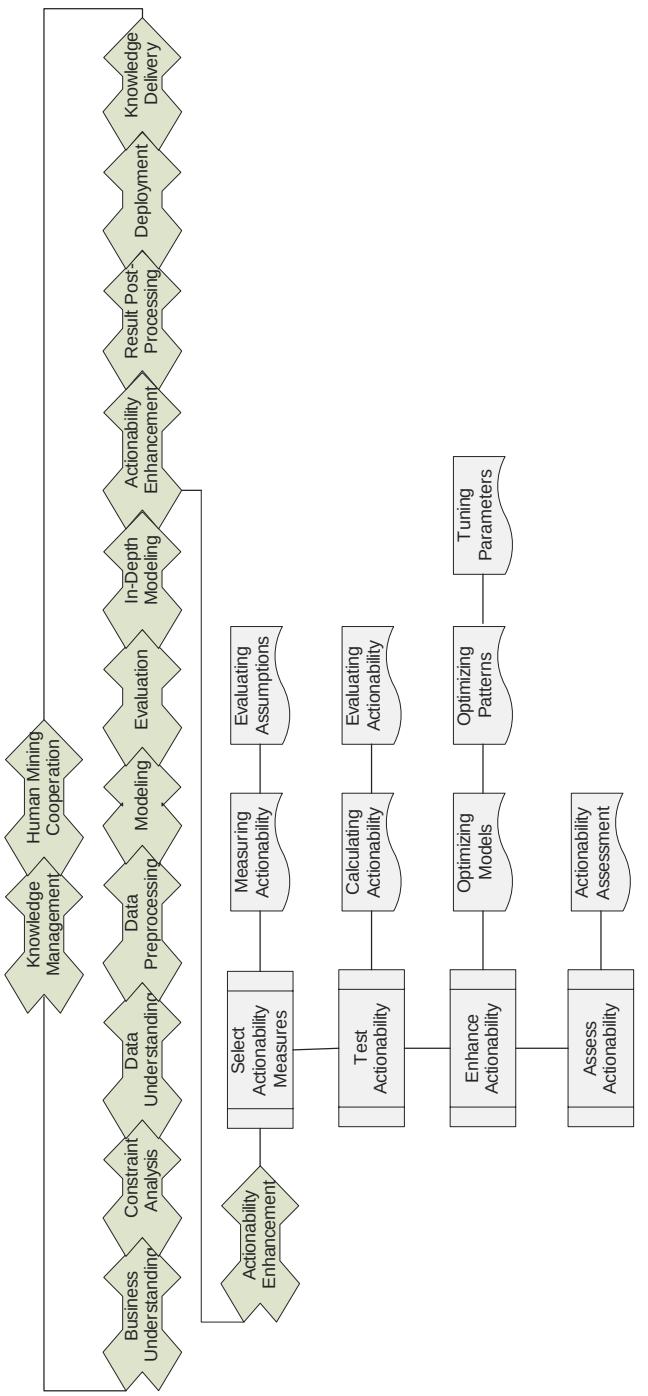
The DDID-PD process highlights the following highly correlated ideas that are critical for the success of a data mining process in the real world. They are:

i. *Constraint-based context*, actionable pattern discovery are based on deep understanding of the constrained environment surrounding the domain problem, data and its analysis objectives.
ii. *Integrating domain knowledge*, real-world data applications inevitably involve domain and background knowledge which is very significant for actionable knowledge discovery.
iii. *Cooperation between human and data mining system*, the integration of human role, and the interaction and cooperation between domain experts and mining system in the whole process are important for effective mining execution.
iv. *In-depth mining*, another round of mining on the first-round results may be necessary for searching patterns really interesting to business.
v. *Enhancing knowledge actionability*, based on the knowledge actionability measures, further enhance the actionable capability of findings from modeling and evaluation perspectives.
vi. *Loop-closed iterative refinement*, patterns actionable for smart business decision-making would in most case be discovered through loop-closed iterative refinement.
vii. *Interactive and parallel mining supports*, developing business-friendly system supports for human-mining interaction and parallel mining for complex data mining applications.

The following section outlines each of them respectively.

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*Figure 2. Actionability enhancement*




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## Qualitative Research

In the field of developing real-world data mining applications, qualitative research is very helpful for capturing business requirements, constraints, requests from organization and management, risk and contingency plans, expected representation of the deliverables, etc. For instance, questionnaire can assist with collecting human concerns and business specific requests. The feedback from business people can improve the understanding of business, data and business people.

In addition, reference models are very helpful for guiding and managing the knowledge discovery process. It is recommended that those reference models in CRISP-DM be respected in domain-oriented real-world data mining. However, actions and entities for domain-driven data mining, such as considering constraints, integrating domain knowledge, should be paid special attention into the corresponding models and procedures. On the other hand, new reference models are essential for supporting components such as in-depth modeling and actionablility enhancement. For instance, the following Figure 2 illustrates the reference model for actionability enhancement.

# Key Components Supporting Domain-Driven Data Mining

In domain-driven data mining, the following seven key components are recommended. They have potential for making KDD different from the existing data-driven data mining if they are appropriately considered and supported from technical, procedural and business perspectives.

## Constraint-Based Context

In human society, everyone is constrained by either social regulations or personal situations. Similarly, actionable knowledge can only be discovered in a constraint-based context such as environmental reality, expectations and constraints in the mining process. Specifically, in Cao and Zhang (2006b), we list several types of constraints which play significant roles in a process effectively discovering knowledge actionable to business. In practice, many other aspects such as data stream and the scalability and efficiency of algorithms may be enumerated. They consist of domain-specific, functional, nonfunctional and environmental constraints. These ubiquitous constraints form a constraint-based context for actionable knowledge discovery. All the above constraints must, to varying degrees, be considered in relevant phases of

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real-world data mining. In this case, it is even called *constraint-based data mining* (Boulicaut et al., 2005; Han 1999).

Some major aspects of domain constraints include the domain and characteristics of a problem, domain terminology, specific business process, policies and regulations, particular user profiling and favorite deliverables. Potential matters to satisfy or react on domain constraints could consist of building domain model, domain metadata, semantics and ontologies (Cao et al., 2006), supporting human involvement, human-machine interaction, qualitative and quantitative hypotheses and conditions, merging with business processes and enterprise information infrastructure, fitting regulatory measures, conducting user profile analysis and modeling, etc. Relevant hot research areas include interactive mining, guided mining, and knowledge and human involvement etc.

Constraints on particular data may be embodied in terms of aspects such as very large volume, ill-structure, multimedia, diversity, high dimensions, high frequency and density, distribution, and privacy, etc. Data constraints seriously affect the development of and performance requirements on mining algorithms and systems, and constitute some grand challenges to data mining. As a result, some popular researches on data constraints-oriented issues are emerging such as stream data mining, link mining, multi-relational mining, structure-based mining, privacy mining, multimedia mining and temporal mining.

What makes this rule, pattern and finding more interesting than the other? In the real world, simply emphasizing technical interestingness such as objective statistical measures of validity and surprise is not adequate. Social and economic interestingness (we refer to business interestingness) such as user preferences and domain knowledge should be considered in assessing whether a pattern is actionable or not. Business interestingness would be instantiated into specific social and economic measures in terms of the problem domain. For instance, *profit*, *return*, and *roi* are usually used by traders to judge whether a trading rule is interesting enough or not.

Furthermore, the delivery of an interesting pattern must be integrated with the domain environment such as business rules, process, information flow, presentation, etc. In addition, many other realistic issues must be considered. For instance, a software infrastructure may be established to support the full lifecycle of data mining; the infrastructure needs to integrate with the existing enterprise information systems and workflow; parallel KDD may be involved with parallel supports on multiple sources, parallel I/O, parallel algorithms, memory storage; visualization, privacy, and security should receive much-deserved attention; false alarming should be minimized.

In summary, actionable knowledge discovery won't be a trivial task. It should be put into a constraint-based context. On the other hand, tricks may not only include how to find a right pattern with a right algorithm in a right manner, they also involve a suitable process-centric support with a suitable deliverable to business.

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## Integrating Domain Knowledge

It is gradually accepted that domain knowledge can play significant roles in real-world data mining. For instance, in cross-market mining, traders often take "*beating market*" as a personal preference to judge an identified rule's actionability. In this case, a stock mining system needs to embed the formulas calculating *market return* and *rule return*, and set an interface for traders to specify a favorite threshold and comparison relationship between the two returns in the evaluation process. Therefore, the key is to take advantage of domain knowledge in the KDD process.

The integration of domain knowledge subjects to how it can be represented and filled into the knowledge discovery process. Ontology-based domain knowledge representation, transformation and mapping between business and data mining system is one of the proper approaches (Cao et al., 2006) to model domain knowledge. Further work is to develop agent-based cooperation mechanisms (Cao et al., 2004; Zhang et al., 2005) to support ontology-represented domain knowledge in the process.

Domain knowledge in business field often takes forms of precise knowledge, concepts, beliefs, relations, or vague preference and bias. Ontology-based specifications build a *business ontological domain* to represent domain knowledge in terms of ontological items and semantic relationships. For instance, in the above example, return-related items include *return*, *market return*, *rule return*, etc. There is *class_of* relationship between *return* and *market return*, while *market return* is *associated* with *rule return* in some form of user specified logic connectors, say beating market if *rule return* is *larger* (>) than *market return* by a threshold $\phi$. We can develop ontological representations to manage the above items and relationships.

Further, business ontological items are mapped to data mining system's internal ontologies. So we build a *mining ontological domain* for KDD system collecting standard domain-specific ontologies and discovered knowledge. To match items and relationships between two domains and reduce and aggregate synonymous concepts and relationships in each domain, ontological rules, logical connectors and cardinality constraints will be studied to support the ontological transformation from one domain to another, and the semantic aggregations of semantic relationships and ontological items intra or inter domains. For instance, the following rule transforms ontological items from business domain to mining domain. Given input item *A* from users, if it is associated with *B* by *is_a* relationship, then the output is *B* from the mining domain: $\forall$ (*A* AND *B*), $\exists$ *B* ::= *is_a*(*A*, *B*) $\Rightarrow$ *B*, the resulting output is *B*. For rough and vague knowledge, we can fuzzify and map them to precise terms and relationships. For the aggregation of fuzzy ontologies, fuzzy aggregation and defuzzification mechanisms (Cao, Luo & Zhang 2006) will be developed to sort out proper output ontologies.

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## Cooperation between Human and Mining System

The real requirements for discovering actionable knowledge in constraint-based context determine that real data mining is more likely to be human involved rather than automated. Human involvement is embodied through the cooperation between humans (including users and business analysts, mainly domain experts) and data mining system. This is achieved through the complementation between human qualitative intelligence such as domain knowledge and field supervision, and mining quantitative intelligence like computational capability. Therefore, real-world data mining likely presents as a human-machine-cooperated interactive knowledge discovery process.

The role of human can be embodied in the full period of data mining from business and data understanding, problem definition, data integration and sampling, feature selection, hypothesis proposal, business modeling and learning to the evaluation, refinement and interpretation of algorithms and resulting outcomes. For instance, experience, metaknowledge and imaginary thinking of domain experts can guide or assist with the selection of features and models, adding business factors into the modeling, creating high quality hypotheses, designing interestingness measures by injecting business concerns, and quickly evaluate mining results. This assistance can largely improve the effectiveness and efficiency of mining actionable knowledge.

Humans often serve on the feature selection and result evaluation. Humans may play roles in a specific stage or during the full stages of data mining. Humans can be an essential constituent of or the centre of data mining system. The complexity of discovering actionable knowledge in constraint-based context determines to what extent human must be involved. As a result, the human-mining cooperation could be, to varying degrees, human-centred or guided mining (Ankerst, 2002; Fayyad, 2003), or human-supported or assisted mining, etc.

To support human involvement, human mining interaction, or in a sense presented as interactive mining (Aggarwal, 2002; Ankerst, 2002), is absolutely necessary. Interaction often takes explicit forms, for instance, setting up direct interaction interfaces to fine tune parameters. Interaction interfaces may take various forms as well, such as visual interfaces, virtual reality technique, multi-modal, mobile agents, etc. On the other hand, it could also go through implicit mechanisms, for example accessing a knowledge base or communicating with a user assistant agent. Interaction communication may be message-based, model-based, or event-based. Interaction quality relies on performance such as user-friendliness, flexibility, runtime capability, representability, and even understandability.

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## Mining In-Depth Patterns

The situation that many mined patterns are interesting to data miners than to business persons has hindered the deployment and adoption of data mining in real applications. Therefore it is essential to evaluate the actionability of a pattern and further discover actionable patterns, namely $\forall P$: $x.tech_int(P) \wedge x.biz_int(P) \rightarrow x.act(P)$, to support smarter and more effective decision-making. This leads to *in-depth pattern mining*.

Mining in-depth patterns should consider how to improve both technical (*tech_int()*) and business interestingness (*biz_int()*) in the above constraint-based context. Technically, it could be through enhancing or generating more effective interestingness measures (Omiecinski, 2003), for instance, a series of research have been done on designing right interestingness measures for association rule mining (Tan et al., 2002). It could also be through developing alternative models for discovering deeper patterns. Some other solutions include further mining actionable patterns on the discovered pattern set. Additionally, techniques can be developed to deeply understand, analyze, select and refine the target data set in order to find in-depth patterns.

More attention should be paid to business requirements, objectives, domain knowledge and qualitative intelligence of domain experts for their impact on mining deep patterns. This can be through selecting and adding business features, involving domain knowledge into modeling, supporting interaction with users, tuning parameters and data set by domain experts, optimizing models and parameters, adding factors into technical interestingness measures or building business measures, improving result evaluation mechanism through embedding domain knowledge and human involvement, etc.

## Enhancing Knowledge Actionability

Patterns which are interesting to data miners may not necessary lead to business benefits if deployed. For instance, a large number of association rules are often found, while most of them might not be workable in business. These rules are *generic patterns* or technically interesting rules. Further actionability enhancement is necessary for generating actionable patterns of use to business.

The measurement of actionable patterns is to follow the actionablilty framework of a pattern discussed in Section 3.1. Both technical and business interestingness measures must be satisfied from both objective and subjective perspectives. For those generic patterns identified based on technical measures, business interestingness needs to be checked and emphasized so that the business requirements and user preference can be put into proper consideration.

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*Actionable* patterns in most cases can be created through rule reduction, model refinement or parameter tuning by optimizing *generic* patterns. In this case, actionable patterns are a revised optimal version of generic patterns, which capture deeper characteristics and understanding of the business, are also called *in-depth* or *optimized* patterns. Of course, such patterns can also be directly discovered from data set with sufficient consideration of business constraints.

## Loop-Closed Iterative Refinement

Actionable knowledge discovery in a constraint-based context is likely to be a closed rather than open process. It encloses iterative feedback to varying stages such as sampling, hypothesis, feature selection, modeling, evaluation and interpretation in a human-involved manner. On the other hand, real-world mining process is highly iterative because the evaluation and refinement of features, models and outcomes cannot be completed once, rather is based on iterative feedback and interaction before reaching the final stage of knowledge and decision-support report delivery.

The above key points indicate that real-world data mining cannot be dealt just with an algorithm, rather it is really necessary to build a proper data mining infrastructure to discover actionable knowledge from constraint-based scenarios in a loop-closed iterative manner. To this end, agent-based data mining infrastructure (Klusch et al., 2003; Zhang et al., 2005) presents good facilities since it provides good supports for both autonomous problem-solving and user modeling and user agent interaction.

## Interactive and Parallel Mining Supports

To support domain-driven data mining, it is significant to develop interactive mining supports for human-mining interaction and evaluate the findings. On the other hand, parallel mining supports are often necessary and can greatly upgrade the real-world data mining performance.

For interactive mining supports, intelligent agents and service-oriented computing are some good technologies. They can support flexible, business-friendly and user-oriented human-mining interaction through building facilities for user modeling, user knowledge acquisition, domain knowledge modeling, personalized user services and recommendation, run-time supports, and mediation and management of user roles, interaction, security and cooperation.

Based on our experience in building agent service-based stock trading and mining system F-Trade (Cao et al., 2004, F-TRADE), an agent service-based actionable discovery system can be built for domain-driven data mining. User agent, knowledge management agent, ontology services (Cao et al., 2006) and run-time interfaces can

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be built to support interaction with users, take users' requests and manage information from users in terms of ontologies. Ontology-represented domain knowledge and user preferences are then mapped to mining domain for mining purposes. Domain experts can help train, supervise and evaluate the outcomes.

Parallel KDD (Domingos, 2003; Taniar et al., 2002) supports involve parallel computing and management supports to deal with multiple sources, parallel I/O, parallel algorithms and memory storage. For instance, to tackle cross-organization transactions, we can design efficient parallel KDD computing and system supports to wrap the data mining algorithms. This can be through developing parallel genetic algorithms and proper processor-cache memory techniques. Multiple master-client process-based genetic algorithms and caching techniques can be tested on different CPU and memory configurations to find good parallel computing strategies.

The facilities for interactive and parallel mining supports can largely improve the performance of real-world data mining in aspects such as human-mining interaction and cooperation, user modeling, domain knowledge capturing, reducing computation complexity, etc. They are some essential parts of next-generation KDD infrastructure.

## Domain-Driven Mining Applications

There are a couple of applications that utilize and strengthen the domain driven data mining research, for instance, domain driven actionable trading pattern mining in capital markets, and involving domain intelligence in discovering actionable activity patterns in social security. In the following, we briefly introduce them, respectively.

The work of actionable trading pattern mining in capital markets consist of the following actions (Cao et al., 2006; Lin et al., 2006). (1) Discovering in-depth trading patterns from generic trading strategy set: There exist many generalized trading rules in financial literature (Sullivan et al., 1999; Tsay, 2005) and trading houses. In a specific market trading, there are huge quantities of variations and modifications of a particular rule by parameterization, for instance, a moving average based trading strategy could be instantiated into MA(2, 50) or MA(10, 50). However, it is not clear to a trader which specific rule is more actionable for his or her particular investment situation. To solve this problem, we use data mining to discover in-depth rules from generic rule set by inputting market microstructure and organizational factors, and adding and checking business performance of a trading rule in terms of business metrics such as return and beating market return. Some other work is on (2) discovering unexpected evidence from stock correlation analysis, (3) mining actionable trading rules and stocks correlation.

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In social security area, some behavior of customers for instance changing address may indicate reasons associating with follow-up debt. We have several projects (Cao & Zhang, 2006) on mining actionable activity patterns from customer contact transactions by involving domain intelligence. For instance, to analyze debtor demographic patterns, domain experts are invited to provide business stories of debt scenarios. Further we do scenario analysis to construct hypothesis and relevant possible attributes. After risk factors and risk groups are mined, business analytic experts again are invited to go through all factors and groups. Some factors and groups with high statistical significance may be pruned further since they are commonsense to business people. We also develop *pattern-debt amount risk* ratio and *pattern-debt duration risk* ratio to measure the impact of a factor or a group on how much debt they may cause or how long the debt may exist. In some cases, the gap and difference between patterns we initially find and those picked up by business analytics are quite big. We then redesign attributes and models to reflect business concerns and feedbacks, for instance, adding customer demographics indicating earnings information in analyzing debt patterns. Domain involvement also benefits deliverable preparation. We generate a full technical report including all findings of interest to us. Our business experts further extract and re-present the findings in a business summary report using their language and fitting into the business process.

## Conclusion and Future Work

Real-world data mining applications have proposed urgent requests for discovering actionable knowledge of main interest to real user and business needs. Actionable knowledge discovery is significant and also very challenging. It is nominated as one of great challenges of KDD in the next 10 years. The research on this issue has potential to change the existing situation where a great number of rules are mined while few of them are interesting to business, and promote the widely deployment of data mining into business.

This paper has developed a new data mining methodology, referred to as *domain-driven data mining*. It provides a systematic overview of the issues in discovering actionable knowledge, and advocates the methodology of mining actionable knowledge in constraint-based context through human-mining cooperation in a loop-closed iterative refinement manner. It is useful for promoting the paradigm shift from *data-driven hidden pattern mining* to *domain-driven actionable knowledge discovery*. Further, progress in studying domain-driven data mining methodologies and applications can help the deployment shift from standard or artificial data set-based testing to real data and business environment based backtesting and development.

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On top of data-driven data mining, domain-driven data mining include almost all phases of the well-known industrial data mining methodology CRISP-DM. However, it has also enclosed some big difference from data-driven methodologies such as CRISP-DM. For instance:

- Some new essential components, such as in-depth modeling, the involvement of domain experts and knowledge, and knowledge actionability measurement and enhancement are included into the lifecycle of KDD,
- In the domain-driven methodology, the phases of CRISP-DM highlighted by thick boxes in Figure 1 are enhanced by dynamic cooperation with domain experts and the consideration of constraints and domain knowledge, and
- Knowledge actionability is highlighted in the discovery process. Both technical and business interestingness must be concerned to satisfy both needs and especially business requests.

These differences actually play essential roles in improving the existing knowledge discovery in a more effective way. Our on-going work is on developing proper mechanisms for representing, transforming and integrating domain intelligence into data mining, and providing mining process specifications and interfaces for easily deploying domain-driven data mining methodology into real-world mining.

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## References

Aggarwal, C. (2002). Towards effective and interpretable data mining by visual interaction. *ACM SIGKDD Explorations Newsletter, 3*(2), 11-22.

Anand, S., Bell, D., & Hughes, J. (1995). The role of domain knowledge in data mining. *CIKM1995* (pp. 37-43).

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Ankerst, M. (2002). Report on the SIGKDD-2002 panel the perfect data mining tool: Interactive or automated? *ACM SIGKDD Explorations Newsletter, 4*(2), 110-111.

Bagui, S. (2006). An approach to mining crime patterns. *International Journal of Data Warehousing and Mining, 2*(1), 50-80.

Boulicaut, J. F., & Jeudy, B. (2005). Constraint-based data mining. In O. Maimon & L. Rokach (Eds.), *The data mining and knowledge discovery handbook* (pp. 399-416). Springer.

Cao, L. (2006). *Domain-driven actionable trading evidence discovery through fuzzy genetic algorithms*. Technical report, Faculty of Information Technology, University of Technology Sydney.

Cao, L., & Dai., R. (2003a). Human-computer cooperated intelligent information system based on multi-agents. *ACTA AUTOMATICA SINICA, 29*(1), 86-94.

Cao, L., & Dai., R. (2003b). Agent-oriented metasynthetic engineering for decision making. *International Journal of Information Technology and Decision Making, 2*(2), 197-215.

Cao, L., et al. (2006). Ontology-based integration of business intelligence. *International Journal on Web Intelligence and Agent Systems, 4*(4).

Cao, L., Luo, & Zhang et. al. (2004). Agent services-based infrastructure for online assessment of trading strategies. *Proceedings of the 2004 IEEE/WIC/ACM International Conference on Intelligent Agent Technology* (pp. 345-349). IEEE Press.

Cao, L., & Zhang, C. (2006a). Domain-driven actionable knowledge discovery in the real world. *PAKDD2006* (pp. 821-830). LNAI 3918.

Cao, L., & Zhang, C. (2006c). Improving centrelink income reporting project. *Centrelink contract research project*.

Chen, S. Y., & Liu, X. (2005). Data mining from 1994 to 2004: An application-orientated review. *International Journal of Business Intelligence and Data Mining, 1*(1), 4-21.

DMP. *Data mining program*. Retrieved from http://www.cmcrc.com/ rd/data_mining/index.html

Domingos, P. (2003). Prospects and challenges for multi-relational data mining. *SIGKDD Explorations, 5*(1), 80-83.

Fayyad, U., & Shapiro, G. (2003). Summary from the KDD-03 panel—Data mining: The next 10 years. *ACM SIGKDD Explorations Newsletter, 5*(2), 191-196.

Gur Ali, O. F., & Wallace, W. A. (1997). Bridging the gap between business objectives and parameters of data mining algorithms. *Decision Support Systems, 21*, 3-15.

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Han, J. (1999). Towards human-centered, constraint-based, multi-dimensional data mining. *An invited talk at Univ. Minnesota*, Minneapolis, Minnesota.

Hu, X., Song, I. Y., Han, H., Yoo, I., Prestrud, A. A., Brennan, M. F., & Brooks, A. D. (2005). Temporal rule induction for clinical outcome analysis. *International Journal of Business Intelligence and Data Mining, 1*(1), 122-136.

Klusch, M., et al. (2003). The role of agents in distributed data mining: Issues and benefits. *Proceeding of IAT03* (pp. 211-217).

Kovalerchuk, B., & Vityaev, E. (2000). *Data mining in finance: Advances in relational and hybrid methods*. Kluwer Acad. Publ.

Lin, L., & Cao, L. (2006). Mining in-depth patterns in stock market. *International Journal of Intelligent System Technologies and Applications* (Forthcoming).

Liu, B., W. Hsu, S. Chen, & Ma, Y. (2000). Analyzing subjective interestingness of association rules. *IEEE Intelligent Systems, 15*(5), 47-55.

Longbing, C., & Chengqi, Z. (2006b). Domain-driven data mining, a practical methodology. *International Journal of Data Warehousing and Mining, 2*(4), 49-65.

Longbing, C., Dan, L., & Chengqi, Z. (2006). Fuzzy genetic algorithms for pairs mining. *PRICAI2006, LNAI4099* (pp. 711-720).

Luo, D., Liu, W., Luo, C., Cao, L., & Dai, R. (2005). Hybrid analyses and system architecture for telecom frauds. *Journal of Computer Science, 32*(5), 17-22.

Manlatty, M., & Zaki, M. (2000). Systems support for scalable data mining. *SIGKDD Explorations, 2*(2), 56-65.

Omiecinski, E. (2003). Alternative interest measures for mining associations. *IEEE Transactions on Knowledge and Data Engineering, 15*, 57-69.

Padmanabhan, B., & Tuzhilin, A. (1998). A belief-driven method for discovering unexpected patterns. *KDD-98* (pp. 94-100).

Pohle, C. *Integrating and updating domain knowledge with data mining.* Retrieved from citeseer.ist.psu.edu/668556.html

Ryan, S., Allan, T., & Halbert, W. (1999). Data-snooping, technical trading rule performance, and the bootstrap. *The Journal of Financial, 54*(5), 1647-1692.

Sullivan, R., Timmermann, A., & White, H. (1999). Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance, 54*, 1647-1691.

Tan, P., Kumar, V., & Srivastava, J. (2002). Selecting the right interestingness measure for association patterns. *SIGKDD'02* (pp. 32-41).

Taniar, D., & Rahayu, J. W. (2002). Chapter 13: Parallel data mining. In H. A. Abbass, R. Sarker, & C. Newton (Eds.), *Data mining: A heuristic approach* (pp. 261-289). Hershey, PA: Idea Group Publishing.

Tsay, R. (2005). *Analysis of financial time series.* Wiley.

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Wang, K., Zhou, S., & Han, J. Profit mining: From patterns to actions. *EBDT 2002*.

Yoon, S., Henschen, L., Park, E., & Makki, S. (1999). Using domain knowledge in knowledge discovery. *Proceedings of the 8th International Conference on Information and Knowledge Management*. ACM Press.

Zhang, C., Zhang, Z., & Cao, L. (2005). Agents and data mining: Mutual enhancement by integration. *LNCS 3505* (pp. 50-61).

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## Chapter X

# Model Free Data Mining[1]

Can Yang, Zhejiang University, Hangzhou, P. R. China
Jun Meng, Zhejiang University, Hangzhou, P. R. China
Shanan Zhu, Zhejiang University, Hangzhou, P. R. China
Mingwei Dai, Xi'an Jiao Tong University, Xi'an, P. R. China

## Abstract

*Input selection is a crucial step for nonlinear regression modeling problem, which contributes to build an interpretable model with less computation. Most of the available methods are model-based, and few of them are model-free. Model-based methods often make use of prediction error or sensitivity analysis for input selection and model-free methods exploit consistency. In this chapter, we show the underlying relationship between sensitivity analysis and consistency analysis for input selection, and then derive an efficient model-free method from our common sense, and then formulate this common sense by fuzzy logic, thus it can be called fuzzy consistency analysis (FCA). In contrast to available methods, FCA has the following desirable properties: (1) it is a model-free method so that it will not be biased on a specific*

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*model, exploiting "what the data say" rather than "what the model say," which is the essential point of data mining—input selection should not be biased on a specific model, (2) it is implemented as efficiently as classical model-free methods, but more flexible than them, and (3) it can be directly applied to a data set with mix continuous and discrete inputs without doing rotation. Four benchmark problems study indicates that the proposed method works effectively for nonlinear problems. With the input selection procedure, the underlying reasons, which affect the prediction are work out, which helps to gain an insight into a specific problem and servers the purpose of data mining very well.*

## Introduction

For real-world problems such as data mining or system identification, it is quite common to have tens of potential inputs to the model under construction. The excessive inputs not only increase the complexity of the computation necessary for building the model (even degrade the performance of the model, which is the curse of dimensionality) (Bellman, 1961; Hastie, Tibshirani, & Friedman, 2001), but also impair the transparency of the underlying model. Therefore, a natural choice of the solution is the number of inputs actually used for modeling should be reduced to the necessary minimum, especially when the model is nonlinear and contains many parameters. Input selection is thus a crucial step for the purposes of (1) removing noises or irrelevant inputs that do not have any contribution to the output; (2) removing inputs that depend on other inputs; and (3) making the underlying model more concise and transparent. However, figuring out which ones to keep and which ones to drop is a daunting task. Large arrays of feature selection methods, like the principal component analysis (PCA), have been introduced in linear regression problems. However, they usually fail to discover the significant inputs in real-world applications, which often involve nonlinear modeling problems.

Input selection thus has drawn great attention in recent years, and some methods have been presented which can be categorized into two groups:

1. Model-based methods, which use a particular model in order to find the significant inputs. In general, model-based methods do input selection mainly by (a) trial and error or (b) sensitivity analysis. They need to try different combinations of inputs to find a good subset for our model to use.
    a. **Trial and error:** This kind of method usually builds up a specified model first with training data, and then checks its prediction accuracy by checking data. Wang (2003) proposed his method for variable importance ranking based on mathematic analysis of approximation accuracy. A relatively

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simple and fast method was proposed in Jang (1996) by using ANFIS. Other methods have been developed for various model structure such as fuzzy inference systems (FIS) (Chiu, 1996), neural networks (NN) (Fernandez & Hernandez, 1999), decision tree (Breiman, 1984), and so on. Some of the model-based methods even employ genetic search, which is aiming at optimizing model structure to produce a concise model structure, but this task is somewhat time consuming (Carlos, 2004).

b. **Sensitivity analysis:** The main idea is that if the output is more sensitive to the variable $x_i$ than $x_j$, we should put as more important variable as $x_i$ and the less important variable as $x_j$. Friedman and Popescu (2005) prefer this natural way to rank input variable importance. The method proposed in Gaweda, Zurada, and Setiono (2002) analyzes each individual input with respect to the output by making use of the structure of Takagi-Sugeno FIS. The sensitivity analysis is also taken into consideration when building a hierarchical system (Wang, 1998).

2. Model-free methods, which do not need to develop models to find the relevant input. Relative few model-free methods are available yet. The philosophy of model-free methods is that input selection will not be biased on different models. For instance, the so-called "Lipschitz coefficients" method (He & Asada, 1993) is computed in order to find the optimal order of an input-output dynamic model. False nearest neighbors (FNN) was originally developed in Kennel, Brown, and Abarbanel (1992) and extended to determine the model structure in Rhodes and Morari (1998), and the method of "Lipschitz coefficients" and FNN were evaluated in Bomberger and Seborg (1998). As a result, these available model-free methods are mainly from the view of consistency analysis.

From a philosophical view, it is generally agreed that variable importance will not vary when choosing different model structures. It is a model-free method that tries to exploit "what really the data say" without the presence of a specified model structure. Thus a model-free method draws more essential characteristics of the data set for input selection, which can be expected not to be biased on a specified model structure and less time-consuming.

## Relationship between Sensitivity Analysis (SA) and Consistency Analysis (CS) for Input Selection

From the brief description in the introduction, except the way of "trial and error," there are two main ways for input selection: SA and CA. What is the relationship

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between them? Are they equivalent in essence? Thus it is necessary for us to investigate the relationship between them before the further discussion of input selection. In the following part of this section, we will show the equivalence of SA and CA in terms of input selection.

Let's recall the definition of sensitivity and consistency first.

- **Sensitivity:** If the output $y$ is more sensitive to the variable $x_i$ than $x_j$ at one point $\boldsymbol{x}^p$ then

$$\left|\frac{\partial y}{\partial x_i}\right|_{x=x^p} > \left|\frac{\partial y}{\partial x_j}\right|_{x=x^p} \tag{1}$$

- **Consistency:** If f is a function such that:

$$y^p = f(\boldsymbol{x}^p) \text{ and } y^q = f(\boldsymbol{x}^q) \text{ then } \boldsymbol{x}^p = \boldsymbol{x}^q \rightarrow y^p = y^q. \tag{2}$$

- **Inconsistency:** If there exist points yp = f(xp) and yq = f(xq) such that yp ≠ yq even xp = xq, then the function f(·) is inconsistency at the point xp.

If the function $f(\cdot)$ is inconsistency at some points, then $f(\cdot)$ is *ill-defined*, otherwise $f(\cdot)$ is *well-defined*.

Now consider a continuous and differentiable function:

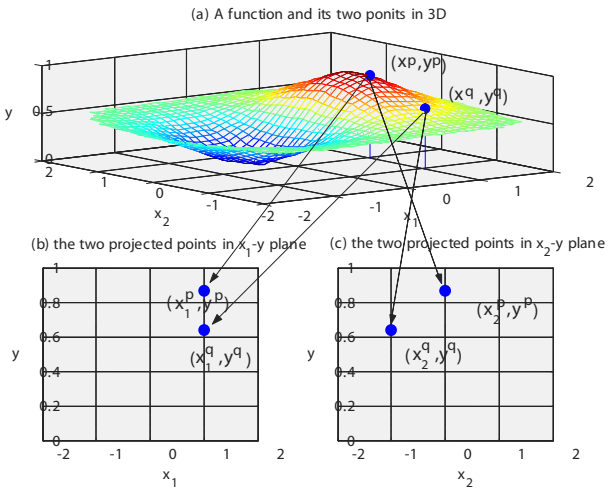
$$y = f(\boldsymbol{x}) = f(x_1, x_2, \ldots, x_n) \tag{3}$$

its definition domain is $[\alpha_1, \beta_1]\times[\alpha_2, \beta_2]\times\ldots\times[\alpha_n, \beta_n]$, and the infinite norm of derivative $\left\|\frac{\partial f}{\partial x_i}\right\|_\infty$ is bounded for any $i = 1, \ldots, n$. Let $\boldsymbol{N}(\boldsymbol{x}^p; r)$ denote the neighbor domain of the point $\boldsymbol{x}^p = [x_1^p, x_2^p, \ldots, x_n^p] \in \mathbf{R}^n$. Then the function $f(\cdot)$ can be expanded within $\boldsymbol{N}(\boldsymbol{x}^p; r)$ in the form of:

$$\Delta y^{\boldsymbol{x}^p + \boldsymbol{x} \in N(\boldsymbol{x}^p; r)} \approx \Delta y_1 + \cdots + \Delta y_n = \frac{\partial f}{\partial x_1}\Delta x_1 + \cdots + \frac{\partial f}{\partial x_n}\Delta x_n \tag{4}$$

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*Figure 1. Demo of inconsistency arising in the process of dimension reduction*




where $\Delta y_i = \dfrac{\partial f}{\partial x_i}\Delta x_i$ . (5)

Now consider the situation when a variable $x_i$ is removed, which means that the $n$ dimension input space of $f(\cdot)$ is reduced to $n$-1 dimension. More clearly, when calculating the function $y = f(\boldsymbol{x}^p) = f(x_1^p, x_2^p, \ldots, x_n^p)$, each input $x_l$ ($l$ = 1, …, $n$ and $l \neq i$) takes the value of $x_l^p$ while $x_i$ is varying freely in the range of $[\alpha_i, \beta_i]$. This is because the variation of $x_i$ cannot be observed from reduced input space with $n$-1 dimension. This process of dimension reduction is shown in Figure 1. Figure 1(a) shows a nonlinear function and its two points $(\boldsymbol{x}^p, y^p)$ and $(\boldsymbol{x}^q, y^q)$ in 3D of $x_1$-$x_2$-$y$ space. Assume $x_2$ is removed such that the corresponding two points of $(\boldsymbol{x}^p, y^p)$ and $(\boldsymbol{x}^q, y^q)$ in $x_1$-$y$ plane are $(x_1^p, y^p)$ and $(x_1^q, y^q)$. In the reduced space, $x_1$-$y$ plane, we can see that $x_1^p = x_1^q$, but $y^p \neq y^q$ (see Figure 1 (b)), while the difference between $x_2^p$ and $x_2^q$ (see Figure 1(c)) cannot be observed in $x_1$-$y$ plane. This is so-called "inconsistency."

To measure the "inconsistency," it is necessary to define an inconsistency degree, which follows the definition of "inconsistency" naturally:

*Inconsistency Degree*: if there exist points $y^p = f(\boldsymbol{x}^p)$ and $y^q = f(\boldsymbol{x}^q)$ such that $y^p \neq y^q$ even $\boldsymbol{x}^p = \boldsymbol{x}^q$, then the inconsistency degree of $f(\cdot)$ at the point $x^p$ is:

$$\Delta y = y^p - y^q. \quad (7)$$

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This definition facilitates the comparison of inconsistency, for example, we can say that $f(\cdot)$ at the point $\boldsymbol{x}^p$ is more inconsistent if $\Delta y$ is larger.

Now consider the input space with $x_i$ is removed so that any points $\boldsymbol{x}^p = [x_1^p, x_2^p, \ldots, x_n^p] \in \mathbf{R}^n$ becames $\boldsymbol{x}^{p,n-1} = [x_1^p, \ldots, x_{i-1}^p, x_{i+1}^p, \ldots, x_n^p] \in \mathbf{R}^{n-1}$. There must be an infinite sequence of points $x_{(k)}^{q,n-1}$, $k = 1, \ldots, \infty$, which tend to the point $\boldsymbol{x}^{p,n-1}$ in the reduced input space, that is, $\left\| x_{(k)}^{q,n-1} - x^{q,n-1} \right\| \to 0$ as $k \to \infty$. But actually in the original input space $\|\boldsymbol{x}^q - \boldsymbol{x}^p\| \to \Delta x_i$ because $\Delta x_i$ can not be observed in the reduced input space as we previously discussed. Thus the difference between the corresponding outputs will not tend to zero, where the inconsistency arises.

According to Eq.(4), yields:

$$\|f(\boldsymbol{x}^q) - f(\boldsymbol{x}^p)\| = \Delta y \overset{\boldsymbol{x}^q = \boldsymbol{x}^p + \ \boldsymbol{x} \in N(\boldsymbol{x}^p; r)}{\approx} \Delta y_1 + \cdots + \Delta y_n = \frac{\partial f}{\partial x_1}\Delta x_1 + \cdots + \frac{\partial f}{\partial x_n}\Delta x_n$$

$$\because \|\boldsymbol{x}^{q,n-1} - \boldsymbol{x}^{p,n-1}\| \to 0, \ \therefore \Delta x_l = 0, \ (l = 1, \ldots, n \text{ and } l \neq i).$$

$$\therefore \|f(\boldsymbol{x}^q) - f(\boldsymbol{x}^p)\| \approx \frac{\partial f}{\partial x_i}\Delta x_i = \Delta y_i \tag{8}$$

According to Eq.(7), Eq.(8) can be considered as the inconsistency degree of $f(\cdot)$ at point $x^p$. Without loss of generality, $\Delta x_i$ can be any given value $\varepsilon$ such that $\boldsymbol{x}^q = \boldsymbol{x}^p + \Delta \boldsymbol{x} \in \boldsymbol{N}(\boldsymbol{x}^p; r)$, where $\Delta \boldsymbol{x} = [0, \ldots, \varepsilon, \ldots, 0]$, then Eq.(8) yields:

$$\frac{\partial f}{\partial x_i}\varepsilon = \Delta y_i(\varepsilon) \tag{9}$$

So with the same procedure, if $x_j$ is removed, the inconsistency degree of $f(\cdot)$ at point $x^p$ is Eq.(10):

$$\|f(\boldsymbol{x}^q) - f(\boldsymbol{x}^p)\| \approx \frac{\partial f}{\partial x_j}\Delta x_j = \Delta y_j \tag{10}$$

$\Delta x_j$ also can be the value $\varepsilon$ because the define domain is continuous, then Eq.(10) yields:

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$$\frac{\partial f}{\partial x_j}\varepsilon = \Delta y_j(\varepsilon) \tag{11}$$

Compare Eq.(9) with Eq.(11) yielding:

$$\left|\frac{\Delta y_i(\varepsilon)}{\Delta y_j(\varepsilon)}\right| = \frac{\left|\frac{\partial f}{\partial x_i}\varepsilon\right|}{\left|\frac{\partial f}{\partial x_j}\varepsilon\right|} = \frac{\left|\frac{\partial f}{\partial x_i}\right|}{\left|\frac{\partial f}{\partial x_j}\right|} \tag{12}$$

From Eq.(12), it can be seen that the inconsistency degree of the function $f(\cdot)$ at point $\boldsymbol{x}^p$ is proportional to the sensitivity of the removed variable. Then we come to the conclusion: *If the variable $x_i$ is more sensitive than the variable $x_j$, then the function $f(\cdot)$ will be more inconsistent when $x_i$ is removed, and vice versa*. Thus it is reasonable for input selection from the perspective of either SA or CA.

# Input Selection Based on CACA

Before proposing our method, we give a brief comment on the two classical model-free methods—"Lipschitz coefficients" method and FNN method, so that it is natural to see the superiority of the proposed method.

## The Two Classical Model-Free Methods: "Lipschitz Coefficients" and FNN

1. "Lipschitz coefficients" works efficiently but its performance is poor if the data set is quite noisy.

   The reason for its poor performance is analyzed next:

   The Lipschitz quotient is defined by

$$q(i, j) = \frac{\| y(i) - y(j) \|}{\| \boldsymbol{x}(i) - \boldsymbol{x}(j) \|},\ i \neq j\ (i = 1,\ldots N, j = 1, \ldots, N) \tag{13}$$

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where $y(l)$ and $x(l)$ ($l = 1,\ldots N$) are respectively output variable and the regressor.

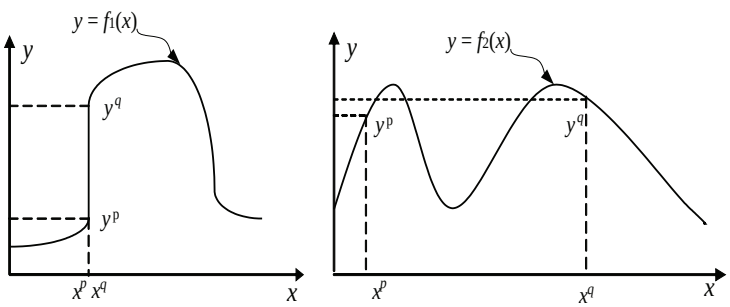
When $\|\boldsymbol{x}(i)\text{-}\boldsymbol{x}(j)\|$ is small, $q(i, j)$ will change greatly if $\|\boldsymbol{x}(i)\text{-}\boldsymbol{x}(j)\|$ changes a little. Thus "Lipschitz coefficients" works badly when the data set is noisy, and usually numerical problem arises when the denominator $\|\boldsymbol{x}(i)\text{-}\boldsymbol{x}(j)\|$ tends to zero. As a result, it can not be applied directly to many applications, which can be seen in the section 4.

2. The problem of FNN is that the threshold constant $R$ needs to be determined before performing FNN. Some methods for determining $R$ are suggested, for example, in Bomberger et al. (1998) the threshold value $R$ should be selected by trial and error method based on empirical rules of thumb, $10 \leq R \leq 50$; the threshold value $R$ is estimated by fuzzy clustering in Feil, Abonyi, and Szeifert (2002), whereas these methods make the original FNN more complicated or time consuming.

## A New Model-Free Method for Input Selection Based on CA

Although SA and CA are equivalent for input selection, methods based on SA usually require building up a model first and then calculating each input's sensitivity based on the model so that they are often regarded as model-based methods. Comparing with the methods based on SA, methods based on CA are model-free so that they are not biased on models. In this section, a new method from the perspective of CA is proposed. Then the basic principle derived from common sense is formulized through fuzzy logic and fuzzy inference, so that the proposed method can be called fuzzy consistency analysis (FCA). FCA could be used directly on the dataset to measure

*Figure 2. (a) An ill-defined function y = f1(x); (b) A well-defined function y = f2(x)*




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the inconsistency of its underlying function. Then the data set, which generates the least inconsistency degree, should be chosen for modeling.

Recall the main idea of the FNN algorithm: *if there is enough information in the regression space to predict the future output, close regressors will lead to close outputs in some sense*. More concisely, this idea can be summarized as a rule:

**Rule 1:** *if* $\mathbf{x}^p$ *and* $\mathbf{x}^q$ *are close to each other, then* $y^p$ *and* $y^q$ *should be close too.*

FCA will be different from FNN algorithm from here. Notice that the word "close" is a fuzzy word, and nobody knows what it exactly means. However, if it is defined by a suitable membership function, then the meaning of this rule can be formulized by fuzzy logic. Let $d_i(p,q)$ denote the distance between $x_i^p$ and $x_i^q$, so that a suitable membership function to depict the word "close" can be defined as:

$$\mu_i(d_i(p,q)) = \exp(-\frac{d_i^2(p,q)}{\sigma_i^2}) \tag{14}$$

where $\sigma_i$ is the standard deviation of $i$-th variable $x_i$. The reason for introducing $\sigma_i$ is the distance of each variable should be normalized so that we can determine the true degree of "closeness." From Eq.(14), if $d_i \to 0$, then $\mu_i \to 1$, thus we can say $x_i^p$ and $x_i^q$ are *very close*; if $d_i \to \infty$, then $\mu_i \to 0$, thus we can say $x_i^p$ and $x_i^q$ are *not close at all*; otherwise we can say $x_i^p$ and $x_i^q$ are *more or less close*. As a result, membership function (14) serves a friendly interface between mathematical calculation and natural language.

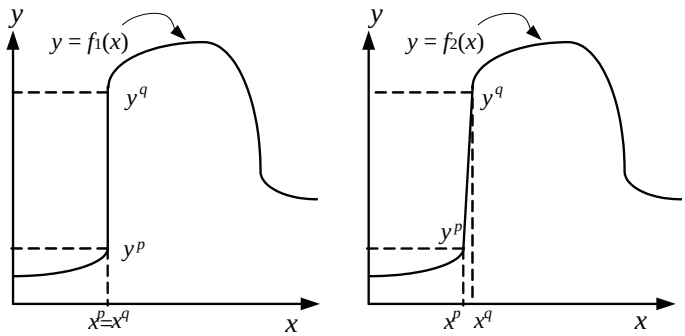
How to interpret this fuzzy rule—**Rule 1**? Theoretically, all the implications, such as Dienes-Rescher implication, Godel implication, Lukasiewicz implication, Zadeh implication, and Mamdani implication etc, can be used for the interpretation of a fuzzy rule, and different implications will generate different results (Wang, 1997). Which implication should be chosen for **Rule 1**? In fact, Rule 1 is not a global rule because **Rule 1'** can not be deduced from **Rule 1**:

*Table 1. Properties of w1(p, q) and w2(p, q)*

| Case | $\mathbf{x}^p$, $\mathbf{x}^q$ | $y^p$, $y^q$ | $w_1(p, q)$ | $w_2(p, q)$ | $f(\cdot)$ |
|---|---|---|---|---|---|
| Case 1 | Close | Close | Close to 1 | Close to 0 | well-defined |
| Case 2 | Close | Far | Close to 0 | Close to 1 | ill-defined |
| Case 3 | Far | Close | Close to 0 | Close to 0 | well-defined |
| Case 4 | Far | Far | Close to 0 | Close to 0 | well-defined |

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*Figure 3. (a) The function y = f1(x) with inconsistency defined by Eq.(3); (b) The function y = f2(x) with the situation described by Rule 2*




**Rule 1'**: *if* $\boldsymbol{x}^p$ *and* $\boldsymbol{x}^q$ *are far away, then* $y^p$ *and* $y^q$ *should be far away too.*

Figure 2 shows the difference between **Rule 1** and **Rule 1'**. Figure 2 (a) shows the function $f_1(\cdot)$ is ill-defined with violation of **Rule 1** and Figure 2 (b) shows the function $f_2(\cdot)$ is well-defined with violation of **Rule 1'**.

So Rule 1 is a local rule and should be interpret as: *if* $\boldsymbol{x}^p$ *and* $\boldsymbol{x}^q$ *are close to each other, then* $y^p$ *and* $y^q$ *should be close too; else NOTHING*.

Then interpretation of **Rule 1** prefers local implication to global implications, so that Mamdani implication should be take into consideration rather than global implications, such as Dienes-Rescher implication, Godel implication, Lukasiewicz implication, Zadeh implication, and etc. Thus the firing strength $w_1(p, q)$ of Rule 1 for a pair of data $(\boldsymbol{x}^p, y^p)$ and $(\boldsymbol{x}^q, y^q)$ is:

$$w_1(p,q) = \prod_{i=1}^{n} \exp\left(-\frac{d_i^2(p,q)}{s_i^2}\right) \exp\left(-\frac{d_y^2(p,q)}{s_y^2}\right) \tag{15}$$

Unfortunately, $w_1(p, q)$ is not a suitable criterion for us to determining whether the function $f(\cdot)$ is well-defined or not(see Table 1). $w_1(p, q)$ can not distinguish Case 2, where $f(\cdot)$ is ill-defined, from Case 3 and Case 4. We modify **Rule 1** to **Rule 2**, which can be considered as an extended definition of inconsistency (see equation (3)):

**Rule 2:** *if* $\boldsymbol{x}^p$ *and* $\boldsymbol{x}^q$ *are close to each other, then* $y^p$ *and* $y^q$ *are not close.*

Figure 3 (a) shows the function $y = f_1(x)$ with inconsistency defined by equation (3) and Figure 3 (b) shows the function $y = f_2(x)$ with the situation described by **Rule 2**, which are quite similar. The firing strength $w_2(p, q)$ of **Rule 2** is given by equation (16), which can be considered as an extended version of inconsistency degree defined by equation (3).

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$$w_2(p,q)=\prod_{i=1}^{n}\exp\left(-\frac{d_i^2(p,q)}{s_i^2}\right)\left(1-\exp\left(-\frac{d_y^2(p,q)}{s_y^2}\right)\right) \tag{16}$$

All the data pairs should be treated equally without discrimination, the sum of $w_2(p, q)$ can be viewed as a total inconsistency degree:

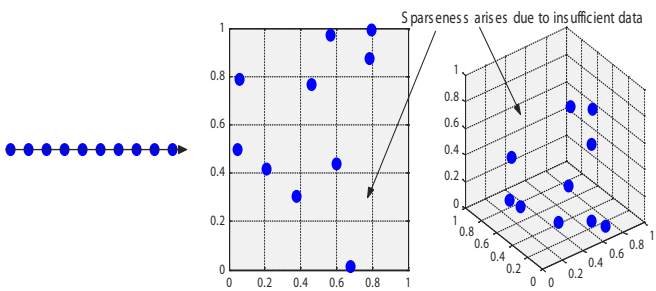
$$w_2=\sum_{p,q}w_2(p,q)=\sum_{p,q}\left(\prod_{i=1}^{n}\exp(-\frac{d_i^2(p,q)}{s_i^2})\left(1-\exp(-\frac{d_y^2(p,q)}{s_y^2})\right)\right) \tag{17}$$

The less inconsistency is, the less $w_2$ will be, thus $w_2$ is an indicator for input selection. For example, if $w_2^{z_1}$ of the data set $\boldsymbol{z}_1=[x_1, x_2, y]$ is less than $w_2^{z_2}$ of the data set $\boldsymbol{z}_2=[x_1, x_3, y]$, then the data set $\boldsymbol{z}_1$ is more suitable for modeling, that is, $[x_1, x_2]$ should be chosen to be input variables rather than $[x_1, x_3]$. Intuitionally, $w_2$ can be interpreted as smoothness degree of a function: the less $w_2$ indicates the smoother function surface, which can be seen in section 4.1. The core code of equation (17) for a data set is provided in the Appendix.

## Searching Strategies

In general, we need to calculate $w_2$ for different combinations of inputs to find a good subset for modeling. To assist in this searching process, several automated strategies have been involved, including exhaustive search, forward selection search, and backward elimination search (see Figure 4).

*Figure 4. Most subspaces of input space will be empty because of insufficient data in high dimension*




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Exhaustive search is the only method guaranteed to find the optimal subset for an arbitrarily complex problem. For most practical situations, however, this method is too slow. Still, if the number of inputs is reasonably small or there is enough time to wait, this may be a viable option.

A forward search starts by trying all possible models that use a single input. The best single input is retained and a search begins on the remaining candidate inputs to become the second input. The input that most improves the model is kept and so on.

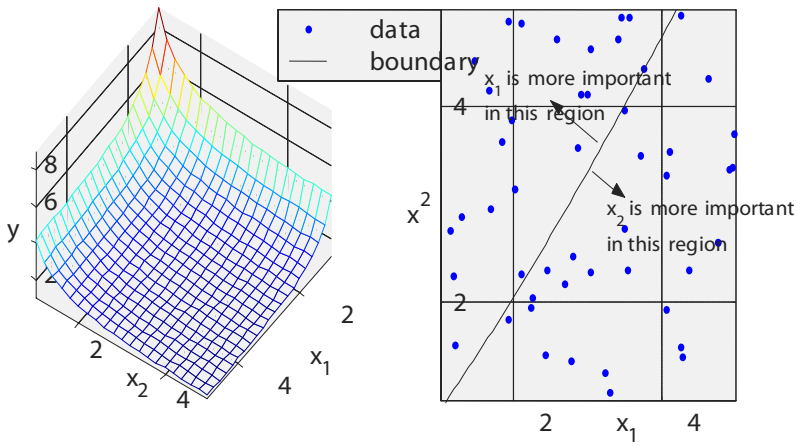
A backward search works exactly like a forward search, except that it moves in the other direction. Backward searching begins with a model that uses all the inputs and then removes input variables one at a time.

## Ending Condition

Curse of dimensionality (Friedman, 1994) should be taken into consideration for ending condition. Generally speaking, for a medium complex problem, we need 10 examples for one input, but 100 examples, for just two inputs. With three inputs, the required number of samples jumps to 1000. Before long, this number grows quite large. If not so, the input space will be so sparse that most subspace of input space will be empty (see Figure 4).

According to this discussion, the ending condition may depend on specific problems such as the complexity of the problem, the available data size $N$ for modeling, and so on. Generally speaking, if $N < 100$, we often pick out one or two inputs for

*Figure 5. (a) The function surface; (b) data and boundary*




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modeling; if $100 < N < 1000$, we often pick out two or three inputs for modeling, and so on. We will show this in the following section.

# Simulation and Application

In this part, four benchmark problems are used to illustrate the efficiency of FCA. A simulated example is firstly considered to illustrate our basic idea, including the procedure of input selection and relationship between SA and CA. Secondly, an automobile MPG (miles per gallon) prediction is used as case study, which is a nonlinear regression problem. Thirdly, a more challenging problem, Boston Housing, is carefully analyzed by FCA. The advantages of FCA are illustrated by the comparison between FCA and the method in Jang (1996) in these two examples. Finally, the well-known Box and Jenkins gas furnace process problem indicates that FCA is also superior to the classical model-based methods.

## Input Selection for a Nonlinear Function

We first consider the example taken from Sugeno and Kang (1988), in which $w_2$ (equation (17)) is illustrated for input selection, and the equivalence between SA and CA analyzed in Section 2 is verified. Consider the nonlinear function:

$$y = (x_1^{-2} + x_2^{-1.5} + 1)^2, \; x_1, x_2 \in [1\ 5] \times [1\ 5] \tag{18}$$

*Table 2(a). Exhaustive search for a single input: The most important input*

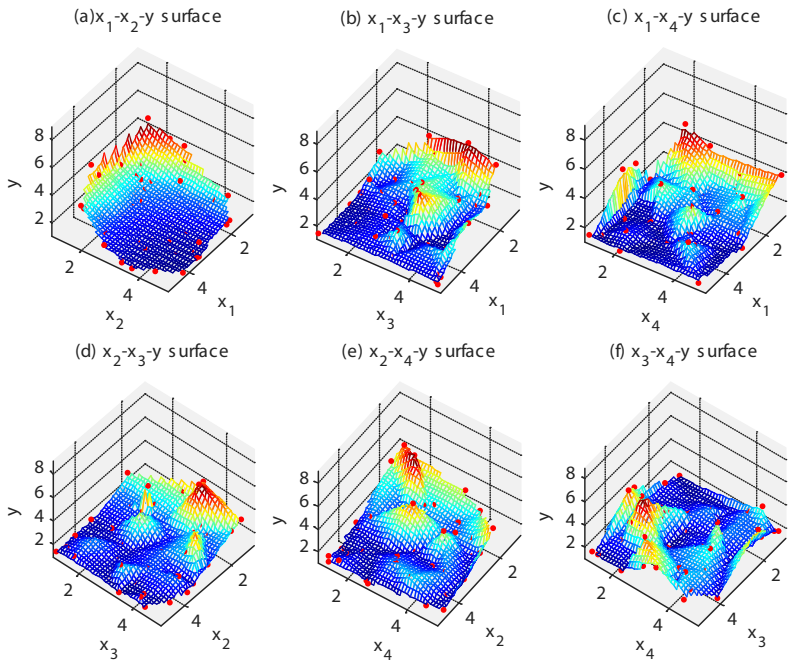
| 1 input | $x_1$ | $x_2$ | $x_3$ | $x_4$ |
|---|---|---|---|---|
| $w_2$ | 209.4779 | ***195.5277*** | 254.4146 | 240.7981 |

*Table 2(b). Exhaustive search for two inputs: The most important two inputs*

| 2 inputs | $x_1, x_2$ | $x_1, x_3$ | $x_1, x_4$ | $x_2, x_3$ | $x_2, x_4$ | $x_3, x_4$ |
|---|---|---|---|---|---|---|
| $w_2$ | ***30.3920*** | 59.8856 | 53.3158 | 58.8568 | 56.9741 | 68.2948 |

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*Figure 6. Function surfaces of different combinations of inputs*




A set of 50 samples are generated for $x_1$ and $x_2$ according to uniform distribution $U[1, 5]$ and the corresponding output $y$ is computed. In addition, two dummy inputs $x_3$ to $x_4$ are added to the set of candidate inputs (these data are also generated randomly according to $U[1, 5]$). The function surface is shown in Figure 5 (a) and the data set of $x_1$ and $x_2$ are shown in Figure 5(b).

Now calculate the derivative for sensitivity analysis:

$$\frac{\partial y}{\partial x_1} = 2(x_1^{-2} + x_2^{-1.5} + 1)(-2)x_1^{-3} \quad \frac{\partial y}{\partial x_2} = 2(x_1^{-2} + x_2^{-1.5} + 1)(-1.5)x_2^{-2.5} \tag{19}$$

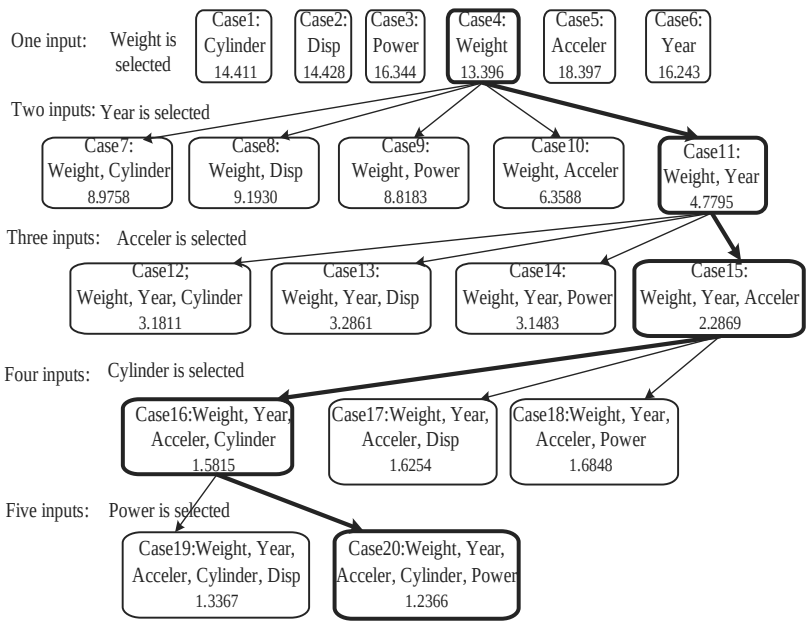
and then yields

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*Table 3. Attributes of MPG prediction and their data type*

| Attributes | Shortened form of attributes | Data type |
|---|---|---|
| No. of cylinders | Cylinder | multi-valued discrete |
| Displacement | Disp | continuous |
| Horsepower | Power | continuous |
| Weight | Weight | continuous |
| Acceleration | Acceler | continuous |
| Model year | Year | Multi-valued discrete |

*Figure 7. FCA for MPG input selection by forward search*




$$\frac{|\frac{\partial y}{\partial x_1}|}{|\frac{\partial y}{\partial x_2}|} = \frac{2x_2^{2.5}}{1.5x_1^3} \tag{20}$$

According to Eq.(1), if $|\frac{\partial y}{\partial x_1}| > |\frac{\partial y}{\partial x_2}|$, then $x_1$ is more sensitive (important) than $x_2$, and vice versa. Let $R_{x_i}$ denote the region in which $x_i$ is more important. The size of

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$R_{x_2}$ is larger than that of $R_{x_1}$ as shown in Figure 5(b), so it is reasonable to rank $x_2$ as the most important input and $x_1$ as the second important input.

Now consider using $w_2$ for input selection with exhaustive search and the results are shown in Table 2.

*Remark*: the value $w_2$ may be different in simulation because the data set is generated randomly. However, the value $w_2$ of $x_2$ is less than that of $x_1$, which happens with the probability that is proportional to the area radio of $R_{x_2}$ to $R_{x_1}$.

From Table 2, it can be seen that the most important input for the function (18) is $x_2$ and the second import input is $x_1$. Searching process stops because other values $w_2$ (e.g., 59.8856) is much larger than that of the inputs [$x_1$, $x_2$] (e.g., 30.3920). Generally speaking, the smooth data set is preferable to non-smooth data set for modeling, which is also the idea behind Regularization Theory (Girosi, 1995). The values $w_2$ in Table 2(b) are the measurements of smoothness degree of the surfaces shown in Figure 6. The surface $x_1$-$x_2$-$y$ is much smoother than other surfaces, thus the inputs [$x_1$, $x_2$] are considered as the most appropriate combination.

## Automobile MPG Prediction

The automobile MPG (mile per gallon) prediction problem is a typical nonlinear regression problem where six attributes (input variables) are used to predict another continuous attribute (output variable). In this case, the six input attributes include profile information about the automobiles (see Table 3).

The data set is available from the UCI repository of machine learning databases and domain theories (ftp://ics.uci.edu/pub/macine-learning-databases/auto-mpg). After removing instances with missing values, the data set was reduced to 392 entries. Our object is then to utilize the data set and FCA to clear out the way that MPG is predicted, which is the purpose of data mining.

*Remark*:

1. The values in Care $j$ ($j$ = 1, …, 36) are training RMSE and test RMSE, respectively.

$$RMSE = \sqrt{1/N \sum_{k=1}^{N} (y_k - \hat{y}_k)}$$

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*Figure 8. FCA for MPG input selection by backward search*

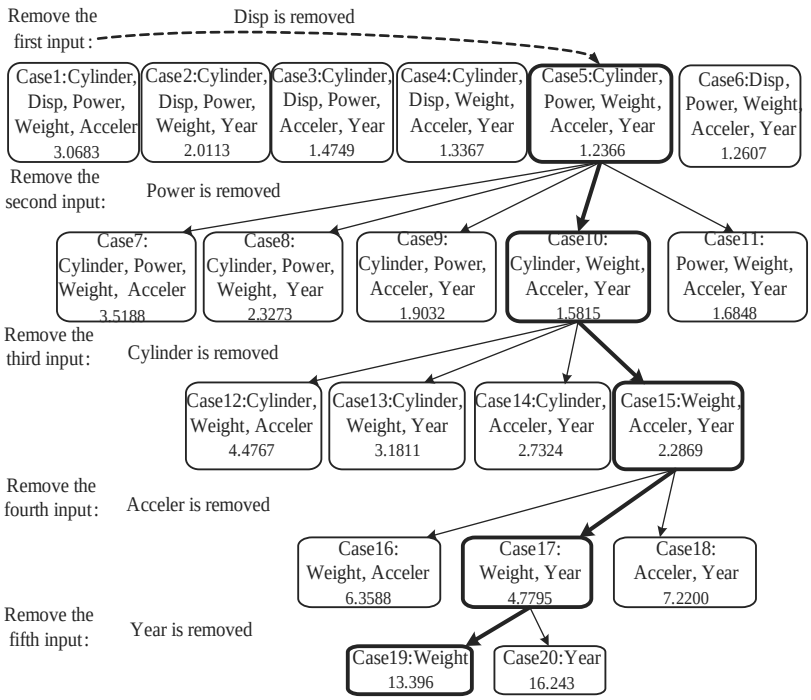
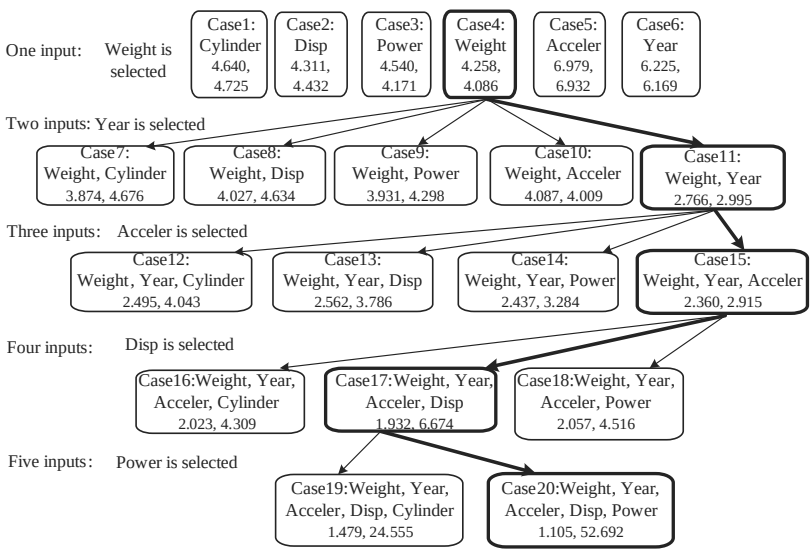



*Figure 9. Jang's method for MPG input selection by ANFIS*




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2. The first 196 are used as training data, and the remaining 196 are used as test data.
3. Only two membership functions are defined for each input in ANFIS.

Input selection serves our goal in two aspects: (1) in prediction study, the relative importance or relevance of the respective input variables to the output variable, which is often of great interest, can be worked out by input selection and (2) it is quite common to suffer from the curse of dimensionality when modeling, where input selection contributes an accurate and concise model.

The results of FCA for input selection are listed in Figure 8 for forward search and Figure 9 for backward search. "Weight," "year," "acceler," "cylinder," and "power" are selected in sequence by the forward search, and they are eliminated in the inverse sequence by the backward search. Thus for MPG input selection, both

*Figure 10. MPG surface by (Wang, 2003)*

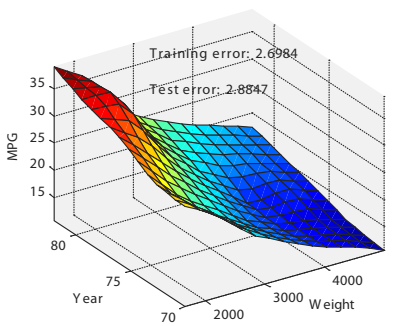
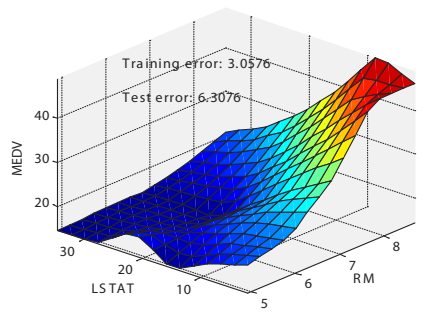



*Figure 11. Boston Housing surface by (Wang, 2003)*




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forward search and backward search come to the same result, in which "weight" is ranked as the first important variable, "year" is the second important variable and so on. Interestingly, there is an extra finding that "weight" turns to be more important than "year" after removing "cylinder" in backward search, which indicates the correlation between "weight" and "year."

For another well-known model-based method proposed by Jang (1996), ANFIS is built to find out a good subset for modeling by the forward search. The choice for the top three important variables is the same: "weight," "year," and "acceler." However, the fourth variable is "disp" that is different from our result "cylinder." In fact, taking "cylinder" as the fourth input for modeling is better than "disp." Although the training error of the input group (weight year acceler disp) is a little smaller than that of the input group (weight year acceler cylinder) (1.932<2.023), its checking error is much larger (6.674>>4.309), which indicates the model have worse generalization capability. The potential problems of this model-based method can observed here: (1) Jang's method only keep eyes on training errors which depend on training data sets, but there is no theory to select data for training until now. (2) The parameters of ANFIS (e.g. the number of membership functions for each variable) should be determined in advance. Those are the reasons for the model-based method leading biased result, but FCA does not have these problems. Following the result by FCA, a better model can be obtained. Please compare the test errors of Case 16 with Case 17 in Figure 9.

Now it is time for us to have a discussion on the meaning of input selection:

1. As a comparison, we first look at the result of linear regression, where the model is expressed as: MPG = $a_0$ + $a_1$ *cylinder +$a_2$ *disp +$a_3$ *power +$a_4$ *weight +$a_5$ *acceler +$a_6$ *year (22) with $\boldsymbol{A}$ = [$a_0$, $a_1$,…, $a_6$] being seven modifiable linear parameters. The optimum values of these linear parameters are obtained directly by least squares method $\boldsymbol{A}$ = [-21.164, -0.363, 0.009, 0.038, -0.008, 0.325, 0.795], and the training and test RMSE are 3.45 and 3.44 which is much worse than ANFIS, whose training and test RMSE are 2.61 and

*Table 4. Attributes of Boston housing data*

| Attributes | Shortened form of attributes | Data type |
|---|---|---|
| No. of cylinders | Cylinder | multi-valued discrete |
| Displacement | Disp | continuous |
| Horsepower | Power | continuous |
| Weight | Weight | continuous |
| Acceleration | Acceler | continuous |
| Model year | Year | Multi-valued discrete |

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2.99, respectively. It can see that irrelative input might impair the modeling accuracy.

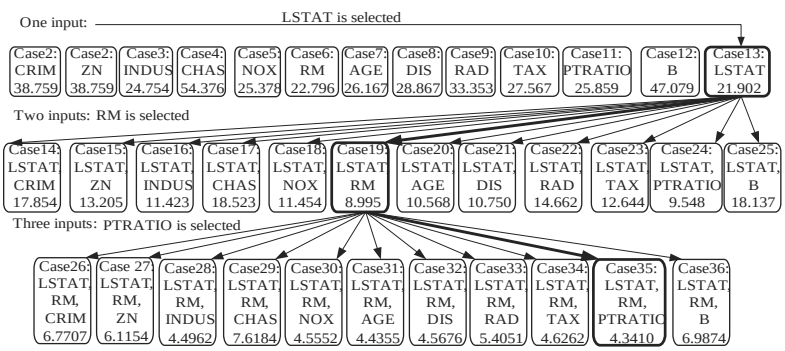
2. Equation (22) only serves the purpose of MPG prediction but tells nothing about which variables are more important than others. Input selection helps us to clearly understand the way that MPG is predicted and extract some knowledge for MPG prediction, which serves the purpose of data mining well. The result of proposed method tells that fuel consumption (MPG) is strongly related to the attribute "weight" not other factors (e.g., vehicle aging ("disp")). Also the attribute "year" tells us the technology development is anther critical reason for fuel saving, the vehicle manufactured by advanced technology is more energy-saving. According to the ending condition in section 3.4, the fuzzy model is obtained by Wang (2003) and its surface is shown in Figure 10. A clearer conclusion comes out: (a) if the vehicle is light and made in later year by advanced technology, its MPG is high and (b) if the vehicle is heavy and made in earlier year, its MPG is low.

## Boston Housing Data

This is a well-known public data set often used to compare the performance of prediction methods. It was first studied by Harrison and Rubinfeld (1978), and later it was used in other publication as well (Quinlan, 1993). It consists of $N$ = 506 neighborhoods in the Boston metropolitan area. For each neighborhood, 14 summary statistics were collected. The goal is to predict the median house value in the respective neighborhoods as function of the 13 input attributes. These variables and their meanings are given in Table 4.

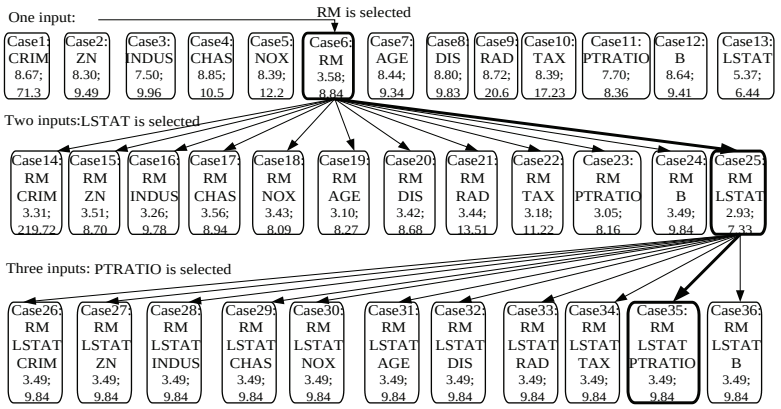
It is a more challenge data mining task, as there are 13 input candidates but only 506 instances are available. A feasible way for Boston Housing data mining is to

*Figure 12. FCA for input selection of Boston Housing problem by forward search*




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*Figure 13. Jang's method for Boston Housing input selection by ANFIS*




rank the input variable importance and then make clear that "MEDV" is strongly influenced by which factors. According to the ending condition in section 3.4, we will pick out three most important variables by FCA.

*Remark*: $w_2(j) = v(j)\times 10^3$, where $v(j)$ is the value in Case $j$ ($j$ = 1, …, 36).

**Remark:**

1. The values in Care $j$ ($j$ = 1, …, 36) are training error and test error, respectively.
2. The first 300 are used as training data, and the remaining 206 are used as test data.
3. Only two membership functions are defined for each input in ANFIS.

The selection process of FCA is shown in Figure 12. "LSTAT," "RM," and "PTRATIO" are picked out in sequence as the three most important inputs. But by the method (Jang, 1996), "RM" is ranked as the first important variable, and "LSTAT" as the second important variable (Figure 13). Here we introduce the result of Friedman et al. (2005) to check which one is more appropriate. From Figure 14, it can seen that "LSTAT" is more important than "RM," which can also be observed from the test errors of Case 6 and Case 13 in Figure 13 (8.84>6.44). The reason for biased result of Jang (1996) has been explained in the last section.

As we can see, the "MEDV" was greatly affected by lower-status people percentage rather than some other factors such as the "ZN" etc. Then we can easily obtain an intuitive conclusion: the higher percentage of lower-status people in an area, the lower the "MEDV" was there. The "MEDV" was also strongly related to the "RM."

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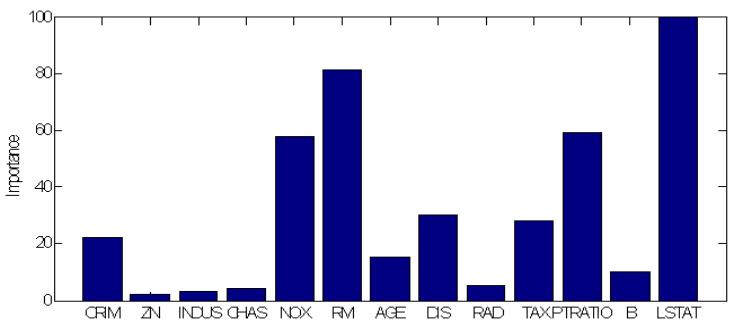
It shows that Americans care about the room number of the dwelling, which somehow reflects the American culture: people eager for private spaces. Interestingly, the "PTRATIO" tell us that the American would like to live where their children go to school conveniently. A model with inputs "LSTAT" and "RM" is obtained by Wang (2003) as shown in Figure 11, a more precise data mining result comes out: (1) if the "LSTAT" is smaller, the "MEDV" changes a little even when the "RM" changes greatly or (2) if the "LSTAT" is larger, the "MEDV" will change a lot due to the change of the "RM." As we can see, the "LSTAT" play fundamental role in predicting the "MEDV," and RM plays more important role when the "LSTAT" is larger.

## Dynamic System Identification

The two model-free methods, "Lipschitz coefficients" and FNN, fail for input selection of MPG and Boston Housing, as they were originally developed for determining the order of NARX models. The denominator $\|\boldsymbol{x}(i) - \boldsymbol{x}(j)\|$ of equation (13) will tend to zero when they are applied for the above two problems. Except that FCA can be more widely used than the two classical model-free methods, it works better than them in dynamic system identification.

In this section, FCA is applied to nonlinear system identification by means of the well-known Box and Jenkins gas furnace data (Box & Jenkins, 1970) as the modeling data set, which is a frequently used benchmark problem (Jang, 1996; Sugeno, 1993). This is a time-series data set for a gas furnace process with gas flow rate $u(t)$ as the furnace input and $CO_2$ concentration $y(t)$ as the furnace output. For modeling purpose, the original data set containing 296 $[u(t)\ y(t)]$ data pairs is reorganized as $[y(t\text{-}1), \ldots, y(t\text{-}4), u(t\text{-}1), \ldots, u(t\text{-}6); y(t)]$. This reduces the number of the instances

*Figure 14. Variable importance ranking by RuleFit reproducing from (Friedman, 2005)*




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*Table 5. FCA for input selection of Box-Jenkins gas furnace*

| $v(i, j)$ | $u(t\text{-}1)$ | $u(t\text{-}2)$ | $u(t\text{-}3)$ | $u(t\text{-}4)$ | $u(t\text{-}5)$ | $u(t\text{-}6)$ |
|---|---|---|---|---|---|---|
| $y(t\text{-}1)$ | 2.2299 | 2.1501 | ***2.0877*** | 2.1706 | 2.5699 | 3.1094 |
| $y(t\text{-}2)$ | 2.5948 | 2.3763 | 2.1514 | 2.1449 | 2.6529 | 3.6185 |
| $y(t\text{-}3)$ | 3.0369 | 2.7369 | 2.3778 | 2.2087 | 2.5874 | 3.5897 |
| $y(t\text{-}4)$ | 3.4022 | 3.0650 | 2.6247 | 2.3129 | 2.5076 | 3.3284 |

*Remark: 1) w2(i, j) = v(i, j)×103, where v(i, j) is the value in row i, column j, e.g. v(1,3) = 2.0877. 2) the corresponding input variables of v(i, j) are in row i, column j, e.g. the corresponding input variables of v(1,3) are [y(t-1), u(t-3)].*

*Table 6. "Lipschitz coefficients" method for Box-Jenkins gas furnace*

| Lipchitz coefficients | | Input delay | | | | | |
|---|---|---|---|---|---|---|---|
| | | 0 | 1 | 2 | 3 | 4 | 5 |
| Output delay | 0 | 57.5802 | 8.0216 | ***5.6209*** | 5.6003 | 5.3211 | 5.6166 |
| | 1 | 15.4634 | 5.8974 | 4.9722 | 4.7678 | 4.8045 | 4.9856 |
| | 2 | 9.1955 | 5.5151 | 4.6846 | 4.4274 | 4.4654 | 4.5843 |
| | 3 | 9.5205 | 5.4620 | 4.8715 | 4.4800 | 4.4723 | 4.5825 |

to 290, out of which the first 145 are used as training data, and the remaining 145 are used as test data. From the reorganized data set, one can see there are ten candidate input variables for modeling. It is reasonable and necessary to select input first to reduce the input dimension. For modeling dynamic process, the inputs selected must contain elements forming both the set of historical furnace outputs {$y(t\text{-}1)$, …, $y(t\text{-}4)$} and the set of historical furnace inputs {$u(t\text{-}1)$, …, $u(t\text{-}6)$}.

*Remark*:

1. $w_2(i, j) = v(i, j)\times 10^3$, where $v(i, j)$ is the value in row $i$, column $j$ (e.g., $v(1,3) = 2.0877$).
2. The corresponding input variables of $v(i, j)$ are in row $i$, column $j$ (e.g., the corresponding input variables of $v(1,3)$ are [$y(t\text{-}1)$, $u(t\text{-}3)$]).

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*Table 7. Comparison between FCA and "Lipschitz coefficients" method*

| Methods | inputs | No. of Rules | No. of parameters <1> | Model errors <2> | Time cost <3> |
|---|---|---|---|---|---|
| FCA | $y(t\text{-}1)$ $u(t\text{-}3)$ | 4 | 12; 12 | 0.1349; 0.5299 | 0.1720 sec |
| "Lipschitz coefficients" method | $y(t\text{-}1)$ $u(t\text{-}1)$ $u(t\text{-}2)$ $u(t\text{-}3)$ | 16 | 24; 80 | 0.0811; 0.5862 | 2.4530 sec |

*Remark: <1> they are No. of nonlinear and linear parameters, respectively. <2> they are training error and test error, respectively. <3> this is tested in MATLAB on the personal computer with Pentium 2.4(CPU), 256MB (memory), Window XP (operating system).*

FCA is applied for selecting two inputs by exhaustive search (see Table 5). [$y(t\text{-}1)$ $u(t\text{-}3)$] can be considered as the most appropriate input subset for modeling. Now let us see the performance of "Lipschitz coefficients" method, which results in a dynamic system structure with zero output delay and two input delays (Table 6). That is, variables [$y(t\text{-}1)$ $u(t\text{-}1)$ $u(t\text{-}2)$ $u(t\text{-}3)$] are all involved for modeling. The limitation of "Lipschitz coefficients" method is clear: the method can possibly select the order of the system (the largest lag for the inputs and outputs), but not an arbitrary subset of regressors in which some lags are missing (such as $u(t\text{-}1)$ $u(t\text{-}2)$ in this example). FNN also inherits this limitation.

*Remark:*

1. They are No. of nonlinear and linear parameters, respectively.
2. They are training error and test error, respectively.
3. This is tested in MATLAB on the personal computer with Pentium 2.4(CPU), 256MB (memory), Window XP (operating system).

Compared to the two classical model-free methods, this example shows some advantages of FCA: (1) The limitation of the classical model-free methods will lead to more complex model for dynamic system (e.g., only two inputs are selected by FCA but "Lipschitz coefficients" method results in four inputs). (2) More detail information is available through FCA (e.g., the result of FCA indicates that the gas furnace process is a first order plus three sampling intervals time-delayed process, but we can not find this in the result of "Lipschitz coefficients" method).

The two selection results can be checked by ANFIS modeling as listed in Table 7. Let ANFIS_1 and ANFIS_2 denote the models based on the selection results of FCA and "Lipschitz coefficients" method, respectively. ANFIS_2 is much complex than

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ANFIS_1 (e.g., NO. of rules and parameters) so that is modeling process is time-consuming. Unfortunately, the more complex model ANFIS_2 does not improve the performance of ANFIS_1. The test error of ANFIS_2 is much larger than that of ANFIS_1, which indicates ANFIS_2 has worse generalization capability than ANFIS_1. Here, we see it again that including more inputs does not mean more accurate model can be obtained. Thus we come to the conclusion: $u(t\text{-}1)$ and $u(t\text{-}2)$ can be removed from the input set and FCA works better.

# Conclusion

Input selection is a feasible solution for curse of dimensionality (Bellman, 1961), thus it has drawn great attention in recent years. Most of the available methods are model-based, and few of them are model-free. Model-based methods often make use of prediction error or sensitivity analysis for input selection and Model-free methods exploit consistency. In this chapter, we showed the underlying relationship between sensitivity analysis (SA) and consistency analysis (CA) for input selection: they are equivalent, and then derive an efficient model-free method from CA. The philosophy and justification of this method is the common sense: similar inputs have similar outputs. Fuzzy logic is employed to formulate the vague expression of the common sense, and then a concise mathematic expression named "$w_2$" is derived for input selection. The "$w_2$" can be considered as an extended version of inconsistency degree theoretically or smoothness degree intuitionally for easier interpretation, so that the "$w_2$" can be applied for input selection directly to evaluate different combinations of inputs, and then the inputs which make the mapping between the input space and output space smoothest are considered as the most appropriate ones. Four examples indicate that FCA has the following merits: (1) it is a model-free method so that it will not be biased on a specific model, (2) it works as efficient as the two classical model-free method, but more flexible than them (e.g., Box-Jenkins gas furnace process), and (3) it can be directly applied to a data set with mix continuous and discrete inputs (e.g., MPG and Boston Housing) without doing rotation.

# References

Bellman, R. E. (1961). *Adaptive control processes.* Princeton University Press.

Breiman, L., & Ihaka, R. (1984). *Nonlinear discriminant analysis via scaling and ACE*. Technical report, University. of California, Berkeley.

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Bomberger, J. D., & Seborg, D. E. (1998). Determination of model order for NARX models directly from input-output data. *Journal of process Control, 8*, 459-468, Oct.-Dec., 1998.

Carlos, A. P. R. (2004). *Coevolutionary fuzzy modeling.* Springer.

Chiu, S. (1996). Selecting input variables for fuzzy models. *Journal of Intelligent and Fuzzy Systems, 4*, 243-256.

Feil, B., Abonyi, J., & Szeifert, F. (2002). Determining the model order of nonlinear input-output systems by fuzzy clustering. In J. M. Benitez, O. Cordon, F. Hoffmann, R. Roy (Eds.), *Advances in soft computing, engineering design, and manufacturing* (pp. 89-98). Springer Engineering Series.

Fernandez R. M., & Hernandez E. C. (1999). Input selection by multilayer feedforward trained networks. *Proceedings of the International Joint Conference on Neural Networks, 3*, 1834-1839.

Friedman, J. H. (1994). An overview of computational learning and function approximation. In Cherkassy, Friedman, & Wechsler (Eds.), *From statistics to neural networks. theory and pattern recognition applications*. Springer-Verlag 1.

Friedman, J. H., & Popescu, B. E. (2005). *Predictive learning via rule ensembles*. Working paper, Stanford University, Otc.

Gaweda, A. E., Zurada, J. M., & Setiono, R. (2002). Input selection in data-driven fuzzy modeling. *IEEE International Conference on Fuzzy Systems, 3*, 1251-1254.

Girosi, F., & Poggio, T. (1995). Regularization theory and neural networks architectures. *Neural Computation, 7*, 219-269.

Hastie, T., Tibshirani, R., & Friedman, J. H. (2001). *The elements of statistical learning*. Springer.

Harrison, D., & Rubinfeld, D. L. (1978). Hedonic prices and the demand for clean air. *Journal of Environ. Economics & Management, 5*, 81-102.

He, X., & Asada, H. (1993). A new method for identifying orders of input-output models for nonlinear dynamic system. *Proceedings of the American Control Conference* (pp. 2520-2523). San Francisco, California USA.

Jang, J. R. (1993). ANFIS: Adaptive-network-based fuzzy inference system. *IEEE Trans. on Systems, Man, and Cybern., 23*(3), 665-685.

Jang, J. R. (1996). Input selection for ANFIS learning. *Proceeding of the IEEE International Conference on Fuzzy System*, New Orleans.

Kennel, M. B., Brown, R., & Abarbanel, H. D. I. (1992). Determining embedding dimension for phase-space reconstruction using a geometrical construction. *Physical Review, A,* 3003-3009.

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Rhodes C., & Morari, M. (1998). Determining the model order of nonlinear input/output systems. *AIChE Journal*, *44*, 151-163.

Quinlan, R. (1993). Combining instance-based and model-based learning. *Proceedings on the 10th International Conference of Machine Learning* (pp. 236-243), University of Massachusetts, Amherst. Morgan Kaufmann.

Sugeno, M., & Kang, G. T. (1988). Structure identification of fuzzy model. *Fuzzy Sets and Systems*, *28*(1), 15-33.

Wang, L. X. (2003). The WM method completed: A flexible fuzzy system approach to data mining. *IEEE Trans. on Fuzzy Systems, 11*(6), December.

Wang, L. X. (1997). *A course in fuzzy systems and control.* NJ: Prentice-Hall.

## Endnote

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# Appendix: The Core Code of FCA

The inputs of the program are a data set and its standard deviation, the output is $w_2$ in Eq.(17). The program can be call in MATLAB in the form of:

w2 = Cal_w2(***data***, ***std***)

where "***data***" is a $N\times(n+1)$ matrix, ***std*** is a $1\times(n+1)$ vector ($N$: Number of data; $n$: input dimension of ***data***).

The program of Mexfunction in MATLAB is given as follows.

```
#include <math.h>
#include "mex.h"
/* Input Arguments */
#define DATA        prhs[0]
#define STD         prhs[1]
/* Output Arguments */
#define OUT         plhs[0]
void mexFunction( int nlhs, mxArray *plhs[], int nrhs, const mxArray *prhs[] )
{
   double *out, *data, *std, *tmp, index_sumx, index_sumy, W2;
   int m, p, n, p2, i, j, k, index;
   // check whether input arguments are 2 and output argument is 1
   // nrhs = no. of right-hand-side arguments
   // nlhs = no. of left hand side arguments
            if (nrhs!=2)
                     mexErrMsgTxt("Cal_w2 requires two input arguments.");

// check dimension
   m = mxGetM(DATA);
   p = mxGetN(DATA);
   n = mxGetM(STD);
   p2= mxGetN(STD);
   if(n!=1)
            mexErrMsgTxt("Wrong input of Std!");
   if(p!=p2)
            mexErrMsgTxt("Matrix sizes mismatch!");
```

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```
// assign memory
   OUT = mxCreateDoubleMatrix(1,1, mxREAL); // a column vector

// obtain pointer of input and output
   out = mxGetPr(OUT);
   data = mxGetPr(DATA);
   std  = mxGetPr(STD);

// computing part
// DATA(i,j) ----data[(j-1)*m+(i-1)]
// STD(1,j)  ----std[j-1]
// OUT(1,1) ---- out[0]

        W2 = 0.0;
        for (i=1; i<=m; i++)
                for (j=i+1; j<=m; j++)
                {
                        index_sumx = 0.0;
                        index_sumy = 0.0;
                        for (k=1; k<=p; k++)
                        {
                                if (k<p)
                                  index_sumx  +=  pow(data[(k-1)*m  +(i-1)]-data[(k-1)*m
+(j-1)],2.0)/(std[k-1]*std[k-1]);
                                else
                                  index_sumy = pow(data[(k-1)*m +(i-1)]-data[(k-1)*m +(j-
1)],2.0)/(std[k-1]*std[k-1]);
                        }
                        W2 += exp(-index_sumx)*(1-exp(-index_sumy));
                }
        out[0] = W2;
}
```

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*Section IV*

# Issues and Applications

## Chapter XI

# Minimizing the Minus Sides of Mining Data

John Wang, Montclair State University, USA
Xiaohua Hu, Drexel University, USA
Dan Zhu, Iowa State University, USA

## Abstract

*This research explores the effectiveness of data mining in a commercial perspective. Statistical issues are specified first. Data accuracy and standardization follow. Diverse problems related to the information used for conducting a data mining research are identified. Also, the technical challenges and potential roadblocks in an organization itself are described. Certainly, minimizing the possible negative side of data mining, hopefully, without interfering its tremendous potentials, is a challenging task we are facing.*

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# Introduction

Data mining (DM) or knowledge discovery in databases (KDD) is an experimental, exploratory, and iterative process that consists of a number of stages. "Knowledge discovery in databases is a new, multidisciplinary field that focuses on the overall process of information discovery in large volumes of warehoused data" (Abramowicz & Zurada, 2001; Pyle, 2003). DM involves searching through databases (DBs) for correlations and/or other non-random patterns. DM has been used by statisticians, data analysts, and the management information systems community, while KDD has been mostly used by artificial intelligence and machine learning researchers. Chen and Liu (2005) demonstrated that the practice of DM is becoming more common in many industries especially in the light of recent trends toward globalization. This is particularly the case for major corporations who are realizing the importance of DM and how it can provide help with the rapid growth and change they are experiencing. Despite the large amount of data already in existence, much information has not been compiled and analyzed. With DM, existing data can be sorted and information utilized for maximum potential. Over the last 40 years, the tools and techniques to process structured information have continued to evolve from databases to data warehouses to data mining (DM). Although DM is still in its infancy, it is now being used in a wide range of industries and for a range of tasks in a variety of contexts (Marshall, McDonald, Chen, & Chung, 2004). There are several sectors that are more interested in DM: banking, medicine, insurance, retailing, and government (Firestone, 2005). The applications of DM are everywhere: from biomedical data (Hu & Xu, 2005) to mobile user data (Goh & Taniar, 2005), from data warehousing (Tjioe & Taniar, 2005) to intelligent Web personalization (Zhou, Cheung, & Fong, 2005), from analyzing clinical outcome (Hu et al., 2005) to mining crime patterns (Bagui, 2006), from mining geographical data (Savary, Gardarin, & Zeitouni, 2006) to an application on XML documents (Messaoud, Boussaid, & Rabaséda, 2006), from tapping the power of text mining (Fan, Wallace, Rich, & Zhang, 2006) to decision trees for mining data streams (Gama, Fernandes, & Rocha, 2006).

Although we fully recognize the importance of DM, another side of the same coin deserves our attention. There is a dark side of DM that many of us fail to recognize and without recognition of the pitfalls of DM, the data miner is proving to fall deep into traps. Coy (1997) noted four pitfalls in DM. The first pitfall is if DM is if performed incorrectly, it can produce "bogus correlations" and generate expensive misinterpretations. The second pitfall is allowing the computer to work long enough until it finds "evidence to support any preconception." The third pitfall is called "story-telling" and says, "a finding makes more sense if there's a plausible theory for it. But a beguiling story can disguise weaknesses in the data." The fourth pitfall that Coy cautions is "using too many variables."

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Other scholars have mentioned three disadvantages of mining a DB: the high knowledge requirement of the user, the choice of the DB, and the usage of too many variables during the process (Chen & Sakaguchi, 2000; Chung, 1999). "The more factors the computer considers, the more likely the program will find relationships, valid or not" (Sethi, 2001, p. 69).

Our research indicated that the pitfalls of DM might be categorized into several groups. This article will first explore the theoretical issues in contrast with statistical inference. Following that, we consider the data related issues, which are the most serious concern. Here we find different problems related to the information used for conducting a DM research. Then, we present the technical challenges regarding pitfalls of DM. Finally, we describe the potential roadblocks in an organization itself. Certainly, there are some social, ethical, and legal issues related to the use of DM, the most important among which is the privacy issue.

# Statistical Issues

## Systematic Patterns vs. Sample-Specific

There are two main goals of time series analysis: (a) identifying the nature of the phenomenon represented by the sequence of observations, and (b) forecasting (predicting future values of the time series variable). Both of these goals require that the pattern of observed time series data is identified and more or less formally described. Once the pattern is established, we can interpret and integrate it with other data. Regardless of the depth of our understanding and the validity of our interpretation of the phenomenon, we can extrapolate (ceteris paribus) the identified pattern to predict future events. Pyle (1999) argued that series data had many of the problems non-series data had. Series data also had a number of own special problems.

Temporal DM pertains to trying to forecast events that may occur in the future. Trends usually are regarded with the incline or decline of something. There could also be seasonal patterns that could be tracked over the years that will allow future outcomes to be predicted. There is also an irregular pattern that has erratic outcomes and follows no true pattern. The shortcoming of temporal DM is when something follows the irregular pattern. For the most part events that follow a pattern continue to follow that course but occasionally an outcome will occur that will be unexpected. These outcomes could be a result of something that the decision maker could not foresee happening in the near future, and may make the outcome of any analysis worthless. Temporal DM would not be able to foresee this happening because its user would not be able to predict this occurrence.

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As DM is used more extensively, caution must be exercised while searching through DBs for correlations and patterns. The practice of using DM is not inherently bad, but the context in how it is used must be observed with a keen and practiced eye. Possibly the most notorious group of data miners are stock market researchers who seek to predict future stock price movement. Basically, past performance is used to predict future results. There are a number of potential problems in making the leap from a back tested strategy to successfully investing in future real world conditions. The first problem is determining the probability that the relationships occurred at random or whether the anomaly may be unique to the specific sample that was tested. Statisticians are fond of pointing out that if you torture the data long enough, it will confess to anything (McQueen & Thorley, 1999).

In describing the pitfalls of DM, Leinweber "sifted through a United Nations CD-ROM and discovered that historically, the single best predictor of the Standard & Poor's 500-stock index was butter production in Bangladesh." The lesson to learn here is that a "formula that happens to fit the data of the past won't necessarily have any predictive value" (Investor Home, 1999). The random walk theory of stock prices suggests that securities prices cannot be forecasted. Successful investment strategies, even those that have been successful for many years, may turn out to be fool's gold rather than a golden chalice.

Given a finite amount of historical data and an infinite number of complex models, uninformed investors may be lured into "overfitting" the data. Patterns that are assumed to be systematic may actually be sample-specific and therefore no value (Montana, 2001). When people search through enough data, the data can be tailored to back any theory. The vast majority of the things that people discover by taking standard mathematical tools and sifting through a vast amount of data are statistical artifacts.

Chase Manhattan keeps DM from going astray by making sure its business units have a big role in the process. Teams of DB marketers and business analysts pose specific questions in order to unearth precious nuggets of information, but not statistically invalid dirt (Stedman, 1997).

## Explanatory Factors vs. Random Variables

The variables used in DM need to be more than variables; they need to be explanatory factors. If the factors are fundamentally sound, there is a greater chance the DM will prove to be more fruitful. We might review the relationship in several, distinct time periods. A common mistake made by those inexperienced in DM is to do "data dredging," that is, simply attempting to find associations, patterns, and trends in the data by using various DM tools without any prior analysis and preparation of the data. Using *multiple comparison procedures* indicates the data can be twisted to indicate a trend if the user feels so inclined (Jensen, 1999). Those who do "data

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dredging" are likely to find patterns that are common in the data, but less likely to find patterns that are rare events, such as fraud.

An old saw, which may fit in the analysis of DM, is "garbage in, garbage out." One of the quirks of statistical analysis is that one may be able to find a factor that seems very highly correlated with the dependent variable during a specific time period, but such a relationship turns out to be spurious when tested in other time periods. Such spurious correlations produce the iron pyrite ("fool's gold") of DM. However, even when adjustments are made for excessive collinearity by removing less explanatory, co-linear variables from the model, many such models have trouble withstanding the test of time (Dietterich, 1999). Just as gold is difficult to detect in the ground because it is a rare and precious metal, so too are low-incidence occurrences such as fraud. The person mining has to know where to search and what signs to look for to discover fraudulent practice, which is where data analysis comes in.

Statistical inference has a great deal to offer in evaluating hypotheses in the search, in evaluating the results of the search and in applying the results. Any correlations found through statistical inference might be considered completely random and therefore not be meaningful. Even worse variables not included in a dataset may obscure relationships enough to make the effect of a variable appear the opposite from the truth (Numerical Machine Learning, 1997). Furthermore, other research indicates that on large datasets with multiple variables, results using statistics can become overwhelming and therefore be the cause of difficulty in interpreting results.

Hypothesis testing is a respectable and sometimes valuable tool to assess the results of experiments. However, it too has difficulties. For example, there may be a problem with the asymmetry in the treatment of the null and alternative hypotheses, which will control the probability of Type I errors, but probability of Type II errors may have to be largely guessed (Berger & Berry, 1988). In practice, the approach is not followed literally: common sense prevails. Rather than setting an $\alpha$ in advance and then acting accordingly, most researchers tend to treat the *p*-value obtained for their data as a kind of standardized descriptive statistic. They report these *p*-values, and then let others draw their own conclusions; such conclusions will often be that further experiments are needed. The problem then is that there is no standard approach to arriving at a final conclusion. Perhaps this is how it should be; but this means that statistical tests are used as a component in a slightly ill-defined mechanism for accumulating evidence, rather than in the tidy cut-and-dried way that their inventors were trying to establish. The rejection/acceptance paradigm also leads to the problem of biased reporting. Usually, positive results are much more exciting than negative ones, and so it is tempting to use low *p*-values as a criterion for publications of results. Despite these difficulties, those who seek rigorous analysis of experimental results will often want to see *p*-values, and provided its limitations are borne in mind, the hypothesis testing methodology can be applied in useful and effective ways.

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As Salzberg (1997) emphasized, an important component of many data mining projects is finding a good classification algorithm, a process that requires very careful thought about experimental design. If not done very carefully, comparative studies of classification and other types of algorithms can easily result in statistically invalid conclusions. Salzberg provided some suggestions about how to avoid the pitfalls suffered by many experimental studies.

## Segmentation vs. Sampling

Segmentation is an inherently different task from sampling. As a segment, we deliberately focus into a subset of data, sharpening the focus of the analysis. But when we sample data, we lose information, because we throw away data not knowing what we keep and what we ignore. Sampling will almost always result in a loss of information, in particular with respect to data fields with a large number of non-numeric values.

It does not make sense to analyze all of variables from a large dataset most of the time because patterns are lost through dilution. To find useful patterns in a large data warehouse, we usually have to select a segment (and not a sample) of data that fits a business objective, prepare it for analysis and then perform DM. Looking at all of the data at once often hides the patterns, because the factors that apply to distinct business objectives often dilute each other.

While sampling may seem to offer a short cut to faster data analysis, the end results are often less than desirable. Sampling was used within statistics because it was so difficult to have access to an entire population. Sampling methods were developed to allow for some rough calculations about some of the characteristics of the population without access to the entire population. This contradicts having a large DB. We build DBs of a huge customer's behavior exactly in order to have access to the entire population. Sampling a large warehouse for analysis almost defeats the purpose of having all the data in the first place (DataMines for Data Warehouses, 2001).

In discussing some pitfalls of DM, the issue of how to avoid them deserves mention. There are unique statistical challenges produced by searching a space of models and evaluating model quality based on a single data sample. Work in statistics on *specification searches* and *multiple comparisons* has long explored the implications of DM, and statisticians have also developed several adjustments to account for the effects of search. Work in machine learning and knowledge discovery related to *overfilling* and *oversearching* has also explored similar themes, and researchers in these fields have also developed techniques such as *pruning* and *minimum description length encoding* to adjust for the effects of the search (Jensen, 1999). However, this "dark side" of DM is still largely unknown to some practitioners, and problems such as *overfilling* and *overestimation* of accuracy still arise in knowledge discovery applications with surprising regularity. In addition, the statistical effects

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of search can be quite subtle, and they can trip up even experienced researchers and practitioners.

A very common approach is to obtain new data or to divide an existing sample into two or more subsamples, using one subsample to select a small number of models and other subsamples to obtain unbiased scores. *Cross-validation* is a related approach that can be used when the process for identifying a "best" model is algorithmic. Sometimes, *incremental induction* is efficient (Hand, Mannila, & Smyth, 2001). A model is developed on a small data sample and, while suggestive of an interesting relationship, it does not exceed a prespecified critical value. Another small sample of data becomes available later, but it is also too small to confer statistical significance to the model. However, the relationship would be significant if considered in the context of both data samples together. This indicates the importance of maintaining both tentative models and links to the original data (Jensen, 2001).

Several relatively simple mathematical adjustments can be made to statistical significance tests to correct for the effects of multiple comparisons. These have been explored in detail within the statistical literature on experimental design. Unfortunately, the assumptions of these adjustments are often restrictive. Many of the most successful approaches are based on computationally intensive techniques such as *randomization* and *resampling* (Saarenvirta, 1999). *Randomization* tests have been employed in several knowledge discovery algorithms. Serious statistical problems are introduced by searching large model spaces, and that unwary analysts and researchers can still fall prey to these pitfalls.

## Data Accuracy and Standardization

Another important aspect of DM is the accuracy of the data itself. It follows that poor data are a leading contributor to the failure of DM. This factor is a major business challenge. The emergence of electronic data processing and collection methods has led some to call recent times as the "information age." However, it may be more accurately termed as "analysis paralysis." Most businesses either posses a large DB or have access to one. These DBs contain so much data that it may be quite difficult to understand what that data are telling us. Goff (1998) introduced an interesting case showing Hewlett-Packard Co. learned early that data warehouses cannot live by operational data alone. Just about all transactions in the market will generate a computer record somewhere. All those data have meaning with respect to making better business decisions or understanding customer needs and preferences. But discovering those needs and preferences in a DB that contains terabits of seemingly incomprehensible numbers and facts is a big challenge (Abbot, 2001).

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## Accuracy and Cohesiveness

A model is only as good as the variables and data used to create it. Many dimensions of this issue apply to DM, the first being the quality and the sources of the data. We repeatedly can find that data accuracy is imperative to very crucial functions. Administrative data are not without problems, however. Of primary concern is that, unlike a purposeful data collection effort, the coding of data is often not carefully quality controlled. Likewise, data objects may not necessarily be defined commonly across DBs nor in the way a researcher would want. One of the most serious concerns is matching records across different DBs in order to build a more detailed individual record. In addition, administrative records may not accurately represent the population of interest, leading to issues similar to sampling and nonresponse bias. Transaction records and other administrative data are volatile, that is rapidly changing over time, so that a snapshot of information taken at one time may not indicate the same relationships that an equivalent snapshot taken at a later time (or using a different set of tables) would reveal. Finally, programs and regulations themselves may introduce bias into administrative records, thus making them unreliable over time (Judson, 2000).

No matter how huge a company's datasets may be, freshness and accuracy of the data are imperative for successful DM. Many companies have stores of outdated and duplicate data, as anyone who has ever received multiple copies of the same catalog can attest. Before a company can even consider what DM can do for business, the data must be clean and fresh, up-to-date, and error and duplication free. Unfortunately this is easier said than done. Computer systems at many large companies have grown rather complicated over the years by encompassing a wide variety of incompatible DBs and systems as each division or department bought and installed what they thought was best for their needs without any thought of the overall enterprise. This may have been further complicated through mergers and acquisitions that may have brought together even more varied systems. "Bad data, duplicate data, and inaccurate and incomplete data are a stumbling block for many companies" (E-commag, 2001).

The first step to achieve proper DM results is to start with the correct raw data. For companies to mine their customer transaction data (which sometimes has additional demographic information) they can figure out an 'ad infinitum' value of revenue for each customer. It is clear here that it is very important for companies to get proper data accuracy for forecasting functions. Poor data accuracy can lead to the risk of poor pro forma financial statements.

With accurate data, one should be able to achieve a single customer view. This will eliminate multiple counts of the same individual or household within and across an enterprise or within a marketing target. Additionally this will normalize business information and data. With an information quality solution, one will be able

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to build and analyze relationships, manage a universal source of information and make more informed business decisions. By implementing an information quality solution across an organization one can maximize the ROI from CRM, business intelligence, and enterprise applications.

If dirty, incomplete, and poorly structured data are employed in the mining, it will make the task of finding significant patterns in the data much harder. The elimination of errors, removal of redundancies and filling of gaps in data, although tedious and time-consuming tasks, are integral to successful DM (Business Lounge, 2001).

## Standardization and Verification

Data in a DB or data store is typically inconsistent and lacking conformity. In some cases, there are probably small variations in the way that even the subscriber's name or address appear in a DB. This will lead to the organization allocating resources based on inaccurate information. This can be a very costly problem for any organization that routinely mails against a large customer DB. An added difficulty of getting these correct data sources for data accuracy is the large amount of the sources themselves. The number of enterprise data sources is growing rapidly, with new types of sources emerging every year. The newest source is, of course, enterprise e-business operations. Enterprises want to integrate clickstream data from their Web sites with other internal data in order to get a complete picture of their customers and integrate internal processes. Other sources for valuable data include ERP programs, operational data stores, packaged and homegrown analytic applications, and existing data marts. The process of integrating these sources into one datasets can be complicated and is made even more difficult when an enterprise merges with or acquires another enterprise.

Enterprises also look to a growing number of external sources to supplement their internal data. These might include prospect lists, demographic and psychographic data, and business profiles purchased from third-party providers (Hoss, 2001). Enterprises might also want to use an external provider for help with address verification, where internal company sources are compared with a master list to ensure data accuracy. Additionally, some industries have their own specific sources of external data. For example, the retail industry uses data from store scanners, and the pharmaceutical industry uses prescription data that are totaled by an outsourced company.

Although data quality issues vary greatly from organization to organization, we can discuss these issues more effectively by referring to four basic types of data quality categories that affect IT professionals and business analysts on a daily basis. These categories--standardization, matching, verification, and enhancement--make up the general data quality landscape (Moss, 2000).

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*Verification* is the process of verifying any other type of data against a known correct source. For example, if a company decided to import some data from an outside vendor, the U.S. Postal Service DB might be used to make sure that the ZIP codes match the addresses and that the addresses were deliverable. If not, the organization could potentially wind up with a great deal of undeliverable mail. The same principle would apply to verifying any other type of data. Using verification techniques, a company ensures that data are correct based on an internal or external data source that has been verified and validated as correct.

*Enhancement* of data involves the addition of data to an existing data set or actually changing the data in some way to make it more useful in the long run. Enhancement technology allows a company to get more from its data by enriching the data with additional information.

## Data Sampling and Variable Selection

Data accuracy can also fall victim to the potential of corruption by means of data sampling. This is not to say that the information is necessarily wrong per se, but in fact the sample of information that you have taken may not be wholly representative of what you are mining for.

Apart from sampling, *summarization* may be used to reduce data sizes (Parsaye & Chignell, 1993). But summarization can cause problems too. In fact, the summarization of the same dataset with two sampling or summarization methods may yield the same result, and the summarization of the same dataset with two methods may produce two different results.

## Outliers and Noise

Outliers are variables that have a value that is far away from the rest of the values for that variable. One needs to investigate if there is a mistake due to some external or internal factor, such that a failing sensor or a human error. If it can be established that the outlier is due to a mistake, Pyle (1999) suggests that it should be corrected by a *filtering* process or by treating it like a missing value.

Noise is simply a distortion to the signal and is something integral to the nature of the world, not the result of a bad recording of values. The problem with noise is that it does not follow any pattern that is easily detected. As a consequence of this there are no easy ways to eliminate noise but there are ways to minimize the impact of noise.

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## Missing Values or Null Values

Companies rely on data for a variety of different reasons--from identifying opportunities with customers to ensuring a smooth manufacturing process--and it is impossible to make effective business decisions if the data are not good. In order to be able to fully assess these solutions, it is helpful to have a thorough understanding of the basic types of data quality problems. As companies look to address the issue of data quality, there are five basic requirements to consider when looking for a total data quality management solution: ease of use, flexibility, performance, platform independence, and affordability (Atkins, 2000).

Because DBs often are constructed for other tasks than DM, attributes important for a DM task could be missing. Data that could have facilitated the DM process might not even be stored in the DB. Missing values can cause big problems in series data and series modeling techniques. There are many different methods for "repairing" series data with missing values such that multiple regression and autocorrelation.

Therefore, data quality control is a serious problem. Data inconsistency and data redundancy might mire any data mining effort. Data quality is a company-wide problem that requires company-wide support. A total data quality solution must be able to address the needs of every individual within an organization from the business analysts to the IT department. To meet the needs of these diverse groups, a solution must integrate with current IT tools and still be easy to use. For companies to truly thrive and achieve successful business intelligence, data quality must be an issue that is elevated to top priority.

# Technical Issues

There are many technical challenges regarding DM. These technical issues can cover a whole spectrum of possibilities. However, in this case, technical issues will be limited to the pitfalls regarding certain DM methods and other general technical issues affecting a standard IT department. The selected technical methods reviewed for pitfalls that are used by DM covered are neural networks, decision trees, genetic algorithms, fuzzy logic, and data visualization. General technical issues covered are with regards to requirements of disaster planning.

When reviewing DM for technical issues, an obvious point that might be initially missed by new users is that DM does not automatically happen by just loading software onto a computer. Many issues must be painstakingly thought out prior to moving forward with DM. There is no single superior DM method that exists among those more than 25 different types of methods being currently available. If there were one specific method that was best and without any disadvantages then

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there would be no need for all of these different types of methods. One company even noted, with regards to DM, that there "is no predictive method that is best for all applications" (International Knowledge Discovery Institute, 1999). Therefore, when this company begins dealings with prospective clients, after having already gained a thorough understanding of the client's problem they will use 8 to 10 different DM methods in an effort to find the best application for the client. Listed next are a few examples of some DM methods generally used.

## The Problems with Neural Networks (NNs)

Neural networks (or neural nets) are computer functions that are programmed to imitate the human brain when it comes to making decision. NNs are one of the oldest and most frequently used techniques in DM. This program is able to choose the best solution for a problem that contains many different outcomes. Even though the knowledge of how human memory works is not known, the way people learn through constant repetition is not a mystery. One of the most important abilities that humans possess though is the capability to infer knowledge. NNs are very useful not because the computer is coming up with a solution that a human would not come up with, but rather that they are able to arrive at that solution much faster. "In terms of application, NNs are found in systems performing image and signal processing, pattern recognition, robotics, automatic navigation, prediction and forecasting, and a variety of simulations" (Jones, 2001).

There are several possible pitfalls regarding NNs. One issue is with regards to learning. A drawback of this process is that learning or training can take a large amount of time and resources to complete. Since results or the data being mined are time critical, this can pose a large problem for the end user because it is so time intensive. Therefore, most articles point out that NNs better suited to learning on small- to medium-sized datasets as it becomes to time inefficient on large sized datasets (Hand, Mannila, & Smyth, 2001).

Also, NNs lack explicitness. As several researchers have noted out, the process it goes through is considered by most to be hidden and therefore left unexplained. One article summarized, "it is almost as if the pattern discovery process is handled within a black box procedure" (Electronic Textbook Statsoft, 2001). This lack of explicitness may lead to lesser confidence in the results and a lack of willingness to apply those results from DM, since there is no understanding of how the results came about. It becomes obvious as the datasets variables increase in size, it will become more difficult to understand how it came to its conclusions.

Another possible weakness is that the program's capability to solve problems could never exceed that of the user's. This means that the computer will never able to produce a solution that a human could not produce if given enough time. As a result, the user has to program problems and solutions into the computer so that

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it can decide what the best solutions are. If the user has no *answer(s)*, the chances are, neither would the computer.

A final drawback of NNs noted in one paper is that "there is a very limited tradition of experience on which to draw when choosing between the nets that are on offer" (Australian National University, 2001). This lack of knowledge or general review on which type of NNs is best, might result in the purchase of application that will not make the best predictions or in general just work poorly.

## The Problems with Decision Trees (DTs)

The DT is a method that uses hierarchy of simple if-then statements in classifying data. This tends to be one of the most inexpensive methods to use. However, it is a faster to use, easier to generate understandable rules, and is simpler to explain since any decision that is made can be understood by viewing the path of decisions. DTs provide an effective structure in which alternative decisions and the implications of taking those decisions can be laid down and evaluated. They also help to form an accurate, balanced picture of the risks and rewards that can result from a particular choice. However, for all the advantages there are some disadvantages or possible pitfalls that users should be aware of.

The problems with DTs can be divided into two categories: algorithmic problems that complicate the algorithm's goal of finding a small tree and inherent problems with the representation. The conventional DT is useful for small problems but quickly becomes cumbersome and hard to read for intermediate-sized problems. In addition, special software is required to draw the tree. Representing all the information on the tree requires writing probability numbers and payoff numbers under the branches or at the ends of the nodes of the tree (KDNuggets, 2001).

In DTs learning, the goal is to arrive at a classification while minimizing the depth of the final tree. Choosing attributes that divide the positive examples from the negative examples does this. Thus, if there is noise in the training set DTs will fail to find a valid tree.

A big headache is that the data used must be interval or categorical. Therefore, any data not received in this format will have to be recoded to this format in order to be used. This recoding could possibly hide relationships that other DM methods would find. Also, *overfitting* can occur whenever there is a large set of possible hypothesis. A technique to deal with this problem is to use DTs *pruning*. Pruning prevents recursive splitting if the relevance of the attribute is low.

Another possible drawback is that the DTs generally represent a finite number of classes or possibilities. It becomes difficult for decision makers to quantify a finite amount of variables. Due to this limitation, the accuracy of the output will be limited to the number of classes selected that may result in a misleading answer. Even

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worse, if the user does try covering an acceptable number of variables as the list of variables increase, the if-then statements created can become more complex.

Finally, there are two other separate drawbacks due to variable systematic risk. First, DTs are not appropriate for estimation. This is based on the fact that complicated, but understandable, rules must be adhered too. Therefore, estimation would not be appropriate. Second, this method is not useful for all types of DM. It is "hard to use DTs for problems involving time series data unless a lot of effort is put into presenting the data in such a way that trends are made visible" (Knowledge Acquisition Discovery Management, 1997).

The newly modified DT will alleviate the current calculation problem. The modified DTs can decrease calculations needed by the traditional DT method by more than 75% in some examples. The modified DTs can be used in the same situations as traditional DTs. It can be used for single level and multi level trees and symmetrical and non-symmetrical decision-making problems. Even with fewer calculations, the modified DT will still maintain the advantages of the traditional trees and make them more efficient than the current state of traditional DTs. The modified DT helps fix the problem with excessive calculations and keeps the advantages of the traditional DT. It becomes more efficient, requires less work, and counters the claim of some researches problems with traditional DTs (Zebda, 2006).

## The Problems with Genetic Algorithms (GAs)

GAs relate to evolutionary computing that solves problems through application of natural selection and evolution. Evolutionary computing is a component of machine learning, "a sub-discipline of artificial intelligence" (Sethi, 2001, p. 33). It is a "general-purpose search algorithm" (Chen, 2001, p. 257) that utilizes rules mimicking a population's natural genetic evolution to solving problems. Fundamentally, they are free of derivatives (or outgrowths) and best utilized in optimization problems with allegories founded in evolutionary operations such as selection, crossover, and mutation.

A chromosome in computer language is a string of binary bits in which the possible solution is encoded. The term "fitness function" is designated to mean the quality of the solution. Beginning with a random accumulation of potential solutions, the search creates what is termed the "gene pool or population" (Sethi, 2001, p. 37). Upon each completion of a search, a new pool is constructed by the GA and the pool has evolved with improved "fitness values." The most desirable solution is chosen from resulting generations of GAs.

Despite these advantages, these algorithms have not yet been applied to very large-scale problems. One possible reason is that GAs require a significant computational

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effort with respect to other methods, when parallel processing is not employed (Sethi, 2001, p. 39).

## The Problems with Fuzzy Logic (FL)

Fuzzy logic, first developed by Zadeh, uses fuzzy set theory. FL, a multi-valued (as opposed to binary) logic developed to deal with imprecise or vague data. Although there are some limitations to using FL, the main reason that it is not used very widely is that many people are misinformed about it. The difficulties of using FL are different than those of using conventional logic. In addition to technical difficulties, lack of knowledge about this rather new field also causes many problems.

How to define a membership functions is a common problem. The rules will be based on trial and error, rather than the designer's intuition and knowledge. Critics argue that this process, in contrast to the rigid mathematical design process of conventional devices, could lead to undesired outcomes under abnormal circumstances (Pagallo & Haussler, 1990).

Lack of knowledge or misinformation about FL are the main reasons it is still a relatively small field. Many critics argue that FL is a *fad*. They feel it will not be useful since there is no known way to guarantee that it will work under all circumstances. Also, many argue that conventional methods work just or nearly as well, so it is not worthwhile to develop new methods (Chen, 2001).

Another concern seems to be the language barrier. Japanese engineers have pursued FL even though the University of California was its birthplace. It seems as if many American researchers have shunned it. Although technical terms often have a neutral connotation, the Japanese term for FL clearly has a positive tone ("*clever*"). Unlike the trendy Japanese term, Americans assign a negative connotation to *fuzzy*. Despite Japanese and worldwide successes, this technology is still somewhat underdeveloped in the U.S.

## The Problems with Data Visualization

This is a method that allows the user to gain a more intuitive understanding of the data. Generally, graphics tools used to better illustrate relationships among data. Data visualization further allows the user to more easily focus and see the patterns and trends amongst the data. However, the major pitfall in this case, is as the volume of data increases it can become increasingly difficult to discern accurate patterns from the datasets. This problem has been resolved in many instances by using 3-dimensional visual representations of the data instead of two-dimensional visual representations. Spinning data to gain a 3-dimensional understanding of pointclouds,

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or the use of projection pursuit are just two examples of visualization technologies that emerged from statistics (Benjamini & Leshno, 2005). And one successful application of this is 3D charting in the stock market.

Viktor and Paquet (2005) showed how virtual reality and collaborative virtual environments may be used to obtain an immersive perspective of the data and the DM process. The human visual perception system can detect patterns within the data that are unknown to a DM tool. Visualizing a model also allows users to explore, argue, and explain the logic behind the model.

In line with Maimon and Rokach (2005), decomposition methods can be used to increase comprehensibility. Instead of getting a single and complicated model, decomposition methods create several sub-models. Smaller models are more appropriate for user-driven DM that is based on visualization techniques. Moreover, if the decomposition structure is induced by automatic means, it can provide new insights about the explored domain.

## Disaster Planning

Finally, disaster recovery plans must be analyzed. This may result in the need to redesign data backup management techniques to accommodate business growth. Once a coherent plan to accommodate the current and future business needs of the corporation has been developed, it must be implemented quickly. Administrators must know that the backup systems that are in place will work each time and that the data are properly protected. As Gardner (2006) claims, "Failing to Plan = Planning to Fail." Data center preparedness, contingency plans, and hot backup locations for data storage and operations are expensive to activate and to maintain. These systems must be reviewed once or twice per year to ensure that it is delivering the protection that the business needs in the most cost-effective manner.

In a nutshell, there are high administrative and maintenance costs to the firm for DM in terms of both time and money. Failure to recognize the investment necessary to maintain the system can lead to data corruption and system failure.

# Organizational Issues

DM in an organization has both benefits and drawbacks. Naturally, the manner in which we interpret data will determine its ultimate benefit. Gathering data is generally not the issue here; there is much data already stored in data warehouses. We need to remember that DM, when misinterpreted, may lead to costly errors. There are a number of organizational factors and issues that also may be drawbacks and

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limit DM's implementation and effectiveness (Wang, 2003). These factors will be discussed in this section.

A recent survey of retail IT indicated that of the companies using DM, 53% attribute no direct benefit to their bottom line from DM. About 20% of respondents indicated that DM has contributed very little, while only 8.4% of the respondents indicated that DM has contributed substantially to profitability. Additionally, 64% of all companies responding indicated that they do not plan to use dedicated technology to manage their customer relationships (Anonymous, 2001).

The lack of bottom line results can be partially attributed to a number of organizational factors. First, start up costs for implementing DM projects are very high, and can create barriers that many corporations choose not to overcome. There are also short-term and long-term administrative costs associated with DM. The information and output provided by DM systems must be delivered to end users within the organization in a meaningful form, otherwise the information may not be useable, or worse, may be used inappropriately. This can actually lead to a reduction in the return on investment (ROI).

Finally, organizations implementing DM systems must view the project as an enterprise-wide endeavor, rather than a department endeavor. Failure to address these problems will limit a firm's ability to successfully implement DM initiatives and recognize the benefits to their profitability.

## Qualifications of Information Technology (IT) Staff

The move by corporations to DM has added to the administrative burdens of, already, overworked IT staffs. Often, DM requires that data be accessible 24 hours a day, 7 days a week, and that it is adequately protected at all times. This is a very difficult task given the shortage of personnel available for around the clock management. Before the project even begins, it is vital that IT staffs qualify the current network hardware and software, particularly in terms of how much data will be held on the servers.

Additionally, they must qualify the amount of data that will be handled on a daily basis, the devices that are already in place to perform data backup and recovery, and the software running those devices. Careful analysis must be given to the current network configuration, the number of servers, their location, where those servers are administered, and where and how the data are backed up. Administrators must also analyze future needs for the corporation, such as employees, functions and business processes.

Successful DM projects require two components: appropriate technical expertise and an appropriate technical infrastructure for accessing the Web. Many problems can be traced back to these two issues. Other pitfalls include inadequate technical

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expertise, inadequate planning, inadequate integration, and inadequate security (Griffin, 2000).

## Requirements of Information Technology Infrastructure (ITIS)

In general, for most corporations that are involved in the process of DM, ITIS is not a major issue because these corporations will already have an ITIS and sufficient hardware/software in place to handle DM algorithms. Please note that the IT structure meant here is that the structure is not limited to hardware and software but also personnel. Another simple point to keep in mind here is that the more data being mined the more powerful ITIS required.

An ITIS does become an issue for companies that do not have one in place. Establishing on will usually involve rather large capital expenditures and that this cost would not be limited to a one-time event. Instead, as software and hardware modifications are made it will be necessary for capital expenditures to continue to be made for maintenance and future upgrades of the existing ITIS. Total cost of ownership (TCO) will be very high. This might become even more of an issue for smaller to medium sized firms that might not have sufficient resources for such measures.

## Accessibility and Usability

Many organizations have experienced serious problems implementing a standard DM project. Most problems do not lie with the technology, but rather with the people using it. To have a successful impact on an organization, a DM system must be speedy, accessible, and user friendly. In today's highly technological society, many of these systems employ very sophisticated tools running against data stored on high-end systems, but it is important that management can interpret the end results. The information desired must be accessible quickly through a PC or Web browser. If the end users have difficulty using a system, or cannot deliver needed information in a short time and be able to resolve their needs, they may abandon the application altogether, and hence the benefits will not be realized.

It is noted that "this historically has been the main reason for the failure of sales force automation systems over the years. Sales representatives, usually too focused on the business at hand, refused to find the time to learn to use what they perceived as a difficult or cumbersome system." Communication of the analysis to decision-makers and customers must formatted to be "decision ready" (Berry & Linoff, 2000).

At the other end of this spectrum, data that is not "decision ready" can be misinterpreted by the end user. If in turn the information is used incorrectly, the effects

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can actually be detrimental to the corporation and actually lead to a decrease in the return on investment (ROI). Examples would include sales people focusing on the wrong customer types, or marketing incorrect products to customers. Both of these effects will often have a negative impact on sales and profitability.

## Affordability and Efficiency

A simple pitfall that must be viewed by potential DM users is with its cost. These can range from a just a few thousand to millions of dollars for both hardware and software. Implementing an effective DM system can be a very complicated and expensive endeavor for a firm. META Group estimates that within the next 2 years, "the top 2000 global firms will spend $250 million each on CRM and DM solutions." On the average, data warehouse users spent an average of $6 million on their data warehouse in 1998. This cost has been rising at an annual rate of 35% since 1996 (Groth, 2001).

Given these costs, corporate executives who may be seeking to cut expenses and contain losses may not want to invest the resources needed to develop such a system. Add to this that this development may take a few years to implement and may not have an immediate, observable impact on the firm's profitability, and corporate reluctance is even clearer. Furthermore, because even with all of the money being spent on DM projects, many companies have not been able to calculate the resulting ROI for these systems. As a result, all too often corporate executives do not view DM projects in the appropriate manner; as long-term strategic investments.

It becomes apparent that expenditures towards maintaining DM through faster hardware and more powerful DM applications will become a constant cash outflow for many companies. As the DB increases in size over time, more capital will be needed towards hardware and software upgrades. While this might not represent a large commitment for companies of substantial size, small to medium sized firms may not be able to afford an ITIS or a DM application and the support it needs over the long term.

Efficiency may be defined as a measure of ability to do a task within certain time and space constraints and should include the bottlenecks that might arise in a system when doing certain tasks (Wasserman, 1999). As a dataset becomes larger, most IT managers will want to reduce bottlenecks in the network to increase speed or supplement processing power through investment in new hardware.

The costs of implementing DM systems that are especially burdensome to small firms may include the costs of intellectual property, licensing, hardware, and software because these represent a much larger percentage of small firm's revenues. This may place a greater burden on the small firm's ability to identify and realize measurable results quickly. Failure to do so may even jeopardize the company's

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very survival. This requires extreme patience, vision, and long-range planning to build a successful system.

Also, to make products more appealing to specific industries, vendors are rolling out versions aimed at vertical markets. Other vendors are integrating data mining with DB products. It is vital that the tool is fitted into a business process that generates clean data (Nadile, 1997). Vendors of desktop query and online analytical processing (OLAP) tools are adding low-end DM products that provide limit capabilities.

## Scalability and Adaptability

Scalability refers to how well a computer system's hardware or software can adapt to increased demands. Since DM tends to work with large amounts of data, scalability of computer system often becomes a major issue. The network and computers on the network must be scalable or large enough to handle increased data flows, otherwise this may bring a network or individual computers on the network to a grinding halt. Also, system solutions need to be able to grow along with evolving user needs in such a way as to not lock the organization into a particular vendor's infrastructure as technology changes.

This is highlighted as an important issue as the volume of data has increased in recent years at a significant rate. One paper further points out, "that some companies already have data warehouses in the terabyte range (e.g., FedEx, UPS, Wal-Mart). Similarly, scientific data is reaching gigantic proportions (e.g., NASA space missions, Human Genome Project)" (Two Crows Corporation, 2001).

Most recent research notes scalability as a possible pitfall of DM. Even in the cases of very simple forms of data analysis, speed and memory become issues. Since hard drive access speed or network speed is not as fast as resident memory, many older DM applications prefer to be loaded into memory. As the datasets become larger or more variables are added, it follows that the amount of memory needed increases. Without this hardware, virtual memory may have to be used. In general terms, virtual memory is a process of using space on the hard drive to serve as actual memory. The problem with virtual memory is that it is slower, which in turn makes DM slower.

## An Enterprise View

Another organizational factor that can be a problem to DM is not viewing the project as an enterprise-wide endeavor. In many failed DM projects, companies viewed DM as an IT project that was relegated to a specific department, rather than an enterprise-wide initiative. Most DM initiatives have been fragmented, implemented

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within departments without a cross--organizational perspective. Given the cost and scope of DM projects, it is essential that all facets of the business have representation in the design of the system. Feelders, Daniels, and Holsheimer (2000) pointed out that successful data mining projects require the involvement of expertise in data mining, company data, and the subject area concerned. The commitment of a subject area expert, data mining expert as well as a data expert to the project is critical for its success.

Customer information is vital to the overall success of sales, marketing, and customer service and this information must be shared across the organization. If one or more departments are excluded from this design process, there will be an inclination to not accept the finished product, or departments will lack the required knowledge to operate them successfully.

Another problem is a gap between what's expected and what's delivered. Draaisma (1998) believes the largest pitfall with DM is not invasion of privacy, a concern he says is overstated, but managing expectations. Our expectations should be reasonable.

## Final Thoughts on Organizational Implications

There are many organizational factors that can serve as drawbacks to successful utilization of DM. Initially, organizations may balk at the high price tag that is associated with DM, and choose to not invest in items such as computer hardware, software, etc. The decision to do so may only come with plausible estimations of ROI, something, as was said above, that is often very difficult to measure. The output provided by DM must be usable information that can be quickly acted upon; otherwise end users will not rely on or effectively use the information.

Finally, DM must be viewed as a company-wide initiative. This way all employees will feel that they have a stake in the systems outcome. Failure to take this route may result in failure of the DM project. Remember that failure rates are often as high as 70%. Companies embarking on such projects must take these issues into account when making decisions.

# Conclusion

DM helps deliver tremendous insights for businesses into the problems they face and aids in identifying new opportunities. It further helps businesses to solve more complex problems and make smarter decisions. DM is a potentially powerful tool for companies, however, more research is needed to measure the benefits of DM.

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If managers are better able to quantify the benefits of DM they will be in a better position to justify its relatively high costs. In concert with better justification and acceptance will come the treatment of DM as a more legitimate and serious tool in organizational development.

With DM, the same process should be applied as with any other organization development tool. Once a DM process has been implemented a company should take the steps needed to monitor the process and obtain feedback from all elements of the process. Monitoring and feedback will give a company the information necessary to determine if it is encountering any of the potential pitfalls and develop corrective strategies to take through the implementation →monitoring →feedback loop. For example, if a company sees lowered costs as a result of the use of information obtained through DM but loses revenues as a result of a loss of customers having concerns privacy issues, then no real grain is made. Through the process of monitoring and feedback firms can better maximize opportunities to increase revenues and lower costs while minimizing the risks posed by any of the downsides of DM.

Despite the attractive proposition of half-automatic and/or fully data analysis, thorough knowledge of the processes behind the data remains vital in avoiding the many pitfalls of DM. DM is still a rapidly changing field. Certainly, minimizing the **possible negative side of data mining**, hopefully without interfering its tremendous potentials, is a challenging task we are facing.

## References

Abbot, D. (2001, November). *Data mining accuracy*. Retrieved from

Abramowicz, W., & Zurada, J. (2001). Knowledge discovery for business information systems. Boston: Kluwer Academic Publishers.

Anonymous. (2001, October). Data mining/CRM: Searching for an ROI, *Chain Store Age, Retail IT 2001*, 24-26.

Atkins, M. E. (2000, August 23). *A jump start in winning: Understanding enterprise data quality management*. Retrieved from http://www.the_modeling agency.com

Australian National University. (2001). *Data mining from a statistical perspective*. Retrieved from http://wwwmaths.anu.edu.au/~johnm/dm/dmpaper.html

Bagui, S. (2006). An approach to mining crime patterns. *International Journal of Data Warehousing and Mining*, *2*(1), 50-80.

Benjamini, Y., & Leshno, M. (2005). Statistical methods for data mining. In O. Maimon, & L. Rokach (Eds.), *The data mining and knowledge discovery handbook* (pp. 565-587). New York: Springer.

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Berger, J. O., & Berry, D. A. (1988). The relevance of stopping rules in statistical inference. In S. Gupta, & J. O. Berger (Eds.), *Statistical decision theory and related topics* (4th ed., Vol. 1, pp. 29-47). NY: Springer-Verlag.

Berry, M., & Linoff, G. (2000). Mastering data mining: The art and science of customer relationship management. New York: Wiley Computer Publishing.

Business Lounge. (2001). *Data mining: A new branch of knowledge discovery*. Retrieved from http://www.bl.com.au/pages/articles/data mining.htm

Chen, L., & Sakaguchi, T. (2000). Disadvantages of data mining. *Information Systems Management, 17*(1), 65.

Chen, S. Y., & Liu, X. (2005). Data mining from 1994 to 2004: An application-orientated review. *International Journal of Business Intelligence and Data Mining, 1*(1), 4-21.

Chen, Z. (2001). *Data mining and uncertain reasoning: An integrated approach*. New York: John Wiley & Sons, Inc.

Chung, H. M. (1999). Data mining. *Journal of Management Information Systems, 16*(1), 11-17.

Cline, K. (1997, January/February). Data mining's promise & pitfalls. *Banking Strategies, 73*(1), 40-44.

Coy, P. (1997, June 16). He who mines data may strike fool's gold. *Business Week,* (3531), 40.

DataMines for Data Warehouses. (2001). Online: http://www.ite.poly.edu/mg/mg693/e0025984/projects/WrightBro/DforD.htm

Dietterich, T. G. (1999). Machine learning. In R. Wilson & F. Keil (Eds.), *The MIT encyclopedia of the cognitive sciences* (pp. 497-498). Cambridge, MA: MIT Press.

Draaisma, M. (1998). Mining for data gold: There are gems of information to be dug up, but perils too. *Marketing Magazine, 103*(37), 18.

E-commag. (2001, December). *Solutions to data inaccuracies*. Retrieved from

Electronic Textbook Statsoft. (2001). *Data mining techniques*. Retrieved from http://www.statsoft.com/textbook/stdatmin.html

Fan, W., Wallace, L., Rich, S., & Zhang, Z. (2006). Tapping the power of text mining. *Communications of the ACM, 49*(9), 77-82.

Feelders, A., Daniels, H., & Holsheimer, M. (2000). Methodological and practical aspects of data mining. *Information & Management, 37*(5), 271-281.

Firestone, J. M. (2005). Mining for information gold. *Information Management Journal, 39*(5), 47-51.

Gama, J., Fernandes, R., & Rocha, R. (2006). Decision trees for mining data streams. *Intelligent Data Analysis, 10*(1), 23-45.

Copyright © 2008, IGI Global. Copying or distributing in print or electronic forms without written permission of IGI Global is prohibited.

Gardner, T. (2006, March 3). Disaster planning and recovery: Details for personal and professional success, DM Direct Newsletter.

Goff, L. (1998, Aug. 24). Surviving the data minefield. *Computerworld, 32*(34), 49-50.

Goh, J., & Taniar, D. (2005). Mining parallel patterns from mobile users. *International Journal of Business Data Communications and Networking, 1*(1), 50-76.

Griffin, J. (2001, March 23). *Four big pitfalls to avoid when Web-enabling your data warehouse and how to avoid them*. Retrieved from

Griffin, J. (2000, July 18). *Fundamental pitfalls to avoid in your data warehouse*. Retrieved from http://www.datawarehouse.com /iknowledge/articles/article.cfm?contentID=1214

Groth, R. (2001). *Data mining: Building competitive advantage.* New York: Prentice Hall.

Gunter, B. (1996). Data mining: Mother lode or fool's gold. *Quality Progress, 29*(4), 113-118.

Hand, D. J., Mannila, H., & Smyth, P. (2001). *Principles of data mining*. Cambridge, MA: MIT Press.

Hoss, D. (2001). *Data mining: Pitfalls*. Retrieved from http://www.datawarehouse.com

Hu, X., & Xu, X (2005). Mining novel connections from online biomedical text databases using semantic query expansion and semantic-relationship pruning. *International Journal of Web and Grid Services, 1*(2), 222-239.

Hu, X., Song, I. Y., Han, H., Yoo, I., Prestrud, A. A., Brennan, M. F., & Brooks, A. D. (2005). Temporal rule induction for clinical outcome, *1*(1), 122-136.

International Knowledge Discovery Institute. (1999, August 21). *What is data mining?* Retrieved from http://www.ikdi.net/data mining.html

Investor Home. (1999, July I). *Data mining*. Retrieved from http://www.investorhome.com/ mining.htm

Jensen, H. (2001). *Using data mining with statistics*. Retrieved from http://www.isi.edu

Jensen, D. (1999). Data snooping, dredging, and fishing: The dark side of data mining. *A SIGKDD99 Panel Report, 1*(2), 52.

Jones, L. (2001). *Neural networks*. Retrieved from http://ei.cs.vt.edu/~NEURALNET/html

Judson, H. (2000, October). *Make data mining work for you*. Retrieved from http://wwwksi.cpsc.ucalgary.ca

KDNuggets. (2001). *Data mining frequently asked questions*. Retrieved from http://www.kdnuggcts.com/publications/index.html#Definitions

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Knowledge Acquisition Discovery Management. (1997). *Introduction to data mining and knowledge discovery*. Retrieved from

Knowledge Discovery & Data Mining. (2001). *The 7th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco.

Laudon, K. C., & Laudon, J. P. (2001). *Essentials of management information systems: Organization & technology in the networked enterprise* (4th ed.). Upper Saddle River, NJ: Prentice Hall.

Maimon, O., & Rokach, L. (2005). Decomposition methodology for knowledge discovery and data mining. In O. Maimon, & L. Rokach (Eds.), *The Data mining and knowledge discovery handbook* (pp. 981-1003). New York: Springer.

Marshall, B., McDonald, D., Chen, H. C., & Chung, W. Y. (2004). EBizPort: Collecting and analyzing business intelligence information. *Journal of the American Society for Information Science and Technology (JASIST), 55*(10), 873-891.

McQueen, G., & Thorley, S. (1999, March/April). Mining fool's gold. *Financial Analysts Journal*, *55*(2), 61-72.

Messaoud, R. B, Boussaid, O., & Rabaséda, S. L. (2006). A data mining-based OLAP aggregation of complex data: Application on XML documents. *International Journal of Data Warehousing and Mining*, *2*(4), 1-26.

Montana, J. C. (2001, October). Data mining: A slippery slope. *Information Management Journal*, *35*, 50-54.

Moss, L. (2000, August 14). *Data warehouse goals and objectives: Part 3: Long-term objectives*. Retrieved from http://www.datawarehouse.com/iknowledge/articles/topic_sub.cfm

Nadile, L. (1997, May 24). Data mining you can afford. *InformationWeek*, (623), 88-92.

Numerical Machine Learning. (1997). *What is data mining?* Retrieved from http://www.cc.gatech.edu/~kingd/datamine/datamine.html

Pagallo, G., & Haussler, D. (1990, March). Boolean feature discovery in empirical learning. *Machine Learning, 5*(1), 71-99. The Netherlands: Springer.

Parsaye, K., & Chignell, M. (1993). Data quality control with SMART databases. *AI Expert, 8*(5), 23-27.

Pyle, D. (2003). *Business modeling and data mining*. San Francisco: Elsevier Science.

Pyle, D. (1999). *Data preparation for data mining.* Morgan Kaufmann.

Saarenvirta, G. (1999, February). Slice and dice. *CMA Management, 73*(1), 26-30.

Salzberg, S. L. (1997). On comparing classifiers: Pitfalls to avoid and a recommended approach. *Data Mining and Knowledge Discovery, 1*(3), 317-328.

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Sauter, V. (1999, June). Intuitive decision-making. *Communications of the ACM*, *42*(6), 109-115.

Savary, L., Gardarin, G., & Zeitouni, K. (2006). GeoCache: A cache for GML geographical data. *International Journal of Data Warehousing & Mining*, *3*(1), 67-88.

Sethi, I. K. (2001). *Data mining: An introduction.* Retrieved from http://www.cs.ubc.ca/nest/temporal.html

Stedman, C. (1997, Dec. 1). Data mining for fool's gold. *Computerworld*, *31*(48), 0-2.

Teo, T. S. H., & Ang, J. S. K. (2000). Building a data warehouse at the housing and development board. *Database for Advances in Information Systems*, *31*, 35-45.

Tjioe, H. C., & Taniar, D. (2005). Mining association rules in data warehouses. *International Journal of Data Warehousing and Mining*, *1*(3), 28-62.

Two Crows Corporation. (2001). *Scalable data mining*. Two Crows White Paper. Retrieved from http://www.twocrows.com/whilep.htm

Viktor, H., & Paquet, E. (2005). Visualization techniques for data mining. In J. Wang (Ed.), *Encyclopedia of data warehousing and mining* (2 Volumes, 1st ed., pp. 1190-1195), Hershey, PA: Idea Group Reference.

Wang, J. (2003, January). *Data mining: Opportunities and challenges*. Hershey, PA: Idea Group Publishing.

Wasserman, M. (1999, January). *Efficiency as an issue in cognitive architectures, trust your data?* Retrieved from http://www.bos.frb.org

Wu, J. (2000). A framework for information quality in a data warehouse. *NCR: Data Warehousing Report*, *3*(4).

Zebda, A. (2006). The use of Bayesian decision analysis and strategy matrix for quality control decision. *The Business Review, Cambridge*, *5*(1), 287-293.

Zhou, B., Cheung, S., & Fong, A. C. M. (2005). A Web usage lattice based mining approach for intelligent Web personalization. *International Journal of Web Information Systems*, *1*(3), 1-9.

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## Chapter XII

# Study of Protein-Protein Interactions from Multiple Data Sources

Tu Bao Ho
Japan Advanced Institute of Science and Technology, Japan

Thanh Phuong Nguyen
Japan Advanced Institute of Science and Technology, Japan

Tuan Nam Tran
Japan Advanced Institute of Science and Technology, Japan

## Abstract

*The objective of this chapter is twofold. First is to provide a survey of computational methods for protein-protein interaction (PPI) study. Second is to introduce our work and results in using inductive logic programming to learn prediction rules for PPI and DDI (domain-domain interactions) from multiple data sources. We show advantages of exploiting various types of data in these important problems of bioinformatics.*

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# Introduction

Proteins are macromolecules made of 20 amino acids arranged in a linear chain, which participate in every process within cells. Many proteins play a key role in biochemical reactions, and structural or mechanical functions, and thus understanding functions of proteins is a main task in molecular biology. Early work has focused on finding protein functions via prediction of protein structures (Bock & Gough, 2001; Marcotte et al., 1999). Recently, detecting protein functions via protein-protein interactions (PPI) has emerged as a new trend in computational biology (Baudot et al., 2006; Chen & Liu, 2005; Chen & Yuan, 2006). Protein-protein interaction study is not only crucial in finding protein functions, but also is a significant task, as protein interactions are one of the most important regulatory mechanisms in cells, and most of the cellular processes are coordinated by specific protein interactions. For example, from the physical association between a novel protein and a well-characterized protein, we can infer the functions of the former.

Discovering protein-protein interactions has been a key problem in molecular biology and bioinformatics. Some good surveys about protein-protein interaction research have been available (Ng & Tan, 2004; Uetz & Vollert, 2006). Generally, there are experimental and computational methods for prediction of protein interactions. The experimental methods are divided into two groups, the traditional and the high-throughput ones. Traditional experimental methods typically include co-immunoprecipitation and synthetic lethal screening. Although the high-throughput experimental detection methods for PPI (typically, yeast two-hybrid, phage display, affinity purification and mass spectrometry, and protein micro-arrays) present many advantages over traditional experimental methods, but they are still tedious, labor-intensive, and usually have high false positive and high false negative rates.

Computational methods for detecting protein interactions, recently developed with various machine learning techniques and various types of available biological data, allow a chance to study more widely and deeply about protein-protein interactions.

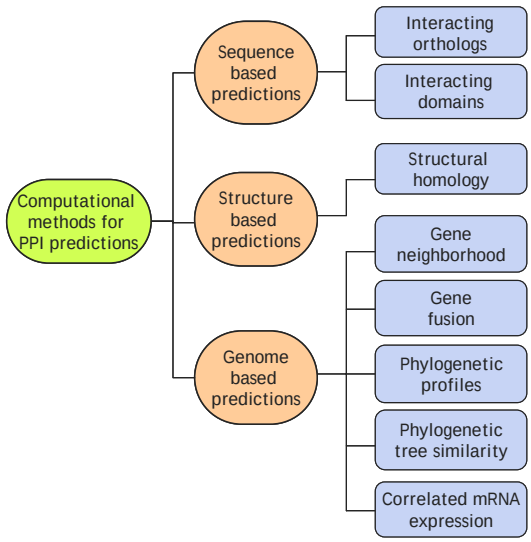
The objective of this chapter is twofold. First is to briefly review the key ideas, advantages, and limitations of computational methods for PPI prediction, some new trends, and potential usage of PPI prediction results. Second is to present our work on PPI and DDI prediction using inductive logic programming on multiple sources of genomic and proteomic data.

## An Overview of Computational Methods for PPI Prediction

The main target of computational methods for PPI prediction is to learn patterns or models from available genomic/proteomic data, which can be used to predict the interaction between given proteins.

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*Figure 1. An overview of computational methods for PPI prediction*




Computational methods for PPI prediction can be viewed in three groups, based on information of protein sequences, structures, or genomes as shown in Figure 1, synthesized from an early tutorial of Ng et al. (2004). Next is a short description of the methods.

The first group consists of methods that exploit information in amino acid sequences of proteins. As the information of amino acid sequences of most proteins is available, these sequence-based prediction methods currently are widely applicable. There are mainly two approaches of sequence-based methods: one is based on interactions of orthologs (i.e., sequence homology across various species (interacting orthologs)) and the other is based on interactions of protein domains (interacting domains).

The key idea of the interacting orthologs approach is that if two proteins, A and B, of a species interact then their similar proteins, A' and B' (orthologs), respectively, in another species are likely to interact. This idea is based on a widely accepted fact that a protein A in a species having a similar sequence of amino acids to a sequence of an annotated protein A' in another species will have very similar properties or functions as A'. A typical work in this approach is to use protein-protein interactions of *S. cerevisiea* to predict the interactions between proteins of *C. elegans* (Matthews et al., 2001).

The key idea of the interacting domain approach is that if two proteins, A and B, contain interacting domains—fundamental structural units of the protein chain—then A and B will interact. This idea is based on the fact that proteins do not interact as a whole but rather only small parts of their sequences, especially the domains, interact. The work in this approach has shown that it is better to use protein domains to

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predict protein interactions, rather than to use the whole protein sequences (Deng, Mehta, Sun, & Chen, 2002; Wojcik & Schachter, 2001).

The second group consists of methods that exploit information on protein structures (Bock et al., 2001; Cockell, Oliva, & Jackson, 2007; Jones & Thornton, 1996). This is based on the fact that the three-dimensional shapes of proteins play a major role in their interactions. The key idea is to exploit structure homology (i.e., if two protein A and B interact, and other two proteins A' and B' have similar structures to A and B, respectively, then A' and B' are likely to interact). Though being a powerful approach to protein interaction prediction, the structure-based methods have to deal with two problems. One is the lack of docking algorithms for predicting large protein molecules, and the other is the difficulty of protein structure determination, especially the tertiary structures (Ng et al., 2004). These difficulties currently limit the usage of this approach, but also encourage further research to solve them.

The third group consists of methods that exploit genomic data, especially gene locality context (gene neighborhood or fusion), phylogenetic context (profiles or tree similarity) and gene expression, to study protein interactions. As various genomes have already been sequenced, the genome-based methods are widely applicable.

The key idea of the gene neighborhood method for predicting protein-protein interactions is that if several genes were conserved as neighbors across genomes, then related proteins have the chance to interact. This idea is based on the fact that a pair of genes has functional linkage, and thus functionally-related proteins, if their orthologs are physically close (gene proximity) in many genomes. Some studies have shown that the gene neighborhood method is quite powerful for bacteria, but may not be directly applicable to other higher eukaryotic species.

The key idea of the gene fusion method for predicting protein-protein interactions is that proteins related to gene fusion are likely to interact. The idea is based on the fact that in evolution, many genes become fused, and thus produced interactions or associated functions of proteins (Marcotte et al., 1999). Similarly, to predict domain-domain interaction for PPI in sequence-based methods, the prediction of domain fusion alternatively offers another way to predict protein interactions. As the gene fusion method does not require genes to be close along the chromosomes, it can be applicable in eukaryotic genomes.

The key idea of phylogenetic profile method for predicting protein-protein interactions is that if two proteins A and B are jointly present or absent in different organisms, then they have the chance to interact or to be functionally linked (Pellegrini et al., 1999; Sun et al., 2005). This idea is based on the fact that during evolution, functionally-linked proteins tended to be preserved or eliminated in a new species. While unable to detect linkages between proteins, this method however can be widely applicable, as various genome sequences have been completed.

The key idea of the phylogenetic tree similarity method for predicting protein-protein interactions is that if two proteins A and B interact, they would have similar series

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of residue changes in the course of evolution, whereas such residue changes do not happen in non-interacting proteins. This idea is based on the fact that a residue change occurring in one protein will imply some compensatory residue change in its interacting partners in order to sustain the interaction, or they will be eliminated. The method searches across various genomes for each pairs of proteins to find their orthologs, then constructs the distance matrix for each protein by comparing their orthologs. If the correlation coefficient between two matrices is high then the two proteins are likely to interact (Goh et al., 2000; Sato, Yamanishi, Horimoto, Kanehisa, & Toh, 2006). This method does not require full-sequenced genomes, but requires a large number of species to search the orthologs of proteins that can be found quickly.

The key idea of correlated mRNA expression method for predicting protein-protein interactions is that if two proteins A and B are encoded by genes having similar expression profiles, then they are likely to interact. Many works have shown the strong relation between protein interaction and gene expression, in particular RNA expression, in different genomes (Grigoriev, 2001; Jansen et al., 2002). This method is widely applicable due to the progress of high-throughput technologies.

The previously mentioned PPI prediction methods are essentially divided into three approaches by the types of information sources (sequences, structures, and genomes). Additionally, several works recently exploit textual information such as gene ontology or PubMed abstracts in MEDLINE to predict protein-protein interactions (Deng, Tu, Sun, & Chen, 2004; Jang et al., 2006).

Typical databases of protein-protein interactions include BIND (Bader et al., 2001), DIP (Salwinski et al., 2004), those obtained by high throughput methods (Ito et al., 2001; Uetz et al., 2000), or domain-domain interactions database Interdom (Ng, Zhang, Tan, & Lin, 2003).

Besides the view of PPI prediction methods by information sources, it is worth noting that various machine learning and data mining methods have been applied in PPI prediction. Some of these methods are popular ones such as cluster analysis (Arnau, Mars, & Marín, 2005), association rules (Oyama, Kitano, Satou, & Ito, 2002), hidden Markov models (Friedrich, Pils, Dandekar, Schultz, & Muller, 2006), and graphical and probabilistic methods such as Bayesian networks (Han, Kim, Seo, & Jang, 2003; Jansen et al., 2003). Also, some recently developed methods have been employed such as kernel methods and support vector machines (Ben-Hur & Noble, 2005; Schölkopf, Tsuda, & Vert, 2004), and conditional random fields (Li et al., 2007).

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# DDI and PPI Prediction from Multiple Data Sources Using Inductive Logic Programming

As commented in the previous section, each approach to PPI prediction has its own advantages and limitations, and each work in the literature usually exploits only one dataset. Probably, those works were limited, as each of the available biological databases only partial contains protein information. In this section, we present our approach that aims to exploit multiple information sources, covering sequence, structure, and genomic and textual data. The computational method we employed is inductive logic programming, which has been recognized as appropriate for exploiting multiple biological data types (Page & Craven, 2003). Our work is performed to predict protein-protein interactions (Tran, Satou, & Ho, 2005) and domain-domain interactions, in short DDI (Nguyen & Ho, 2006a). Because the framework is the same for both PPI and DDI predictions, we will present only the DDI prediction work and will alternatively add PPI prediction results if necessary. In our proposed framework, both PPI and DDI prediction have two main tasks. The first one is extracting data from multiple data sources to construct biologically significant background knowledge (domain knowledge) related to PPI or DDI. The second one is applying ILP method to predict PPI or DDI. The key point is how to construct comprehensively and appropriately, the background knowledge from numerous biological data so that ILP method can infer efficiently PPI or DDI. For PPI prediction, we first constructed background knowledge from multiple genomic data source, whereas the constructed background knowledge of DDI prediction later contained proteomic, genomic and especially protein domain data sources. Our experimental results of DDI prediction (and PPI prediction) demonstrated the better performance of our proposed framework compared with other computational methods.

## Related Work on Prediction of Domain-Domain Interaction

Within a protein, a domain is a fundamental structural unit that is self-stabilizing and often folds independently of the rest of the protein chain. Domains often are named and singled out because they figure prominently in the biological function of the protein they belong to. The domains form the structural or functional units of proteins that partake in intermolecular interactions. Therefore, domain-domain interaction (DDI) problem has biological significance in understanding in depth protein-protein interactions and cellular processes and pathways as well.

Because of the biological roles of protein domains in PPI, a number of domain-based approaches to predict PPI have recently been proposed. Some work has been association methods like (Kim, Park, & Suh, 2002; Sprinzak & Margalit, 2001),

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graph-based method (Wojcik et al., 2001) or a domain combination-based method (Han et al., 2003). The Han et al.'s method uses a combination of sequence similarity search and clustering based on interaction patterns. The purpose of the previous works was to predict and/or to validate protein interactions by studying domain-domain interactions. In their work, the biological roles of DDI in PPI are confirmed; however, they did not take domain-domain interactions into account sufficiently.

Recently, there are several works that not only use protein domains to predict protein interactions, but also attempt to discover DDIs. An integrative approach has been proposed to infer putative domain-domain interactions from three data sources, including experimentally derived protein interactions, protein complexes and Rosetta stone sequences (Ng et al., 2003). The interaction scores for domain pairs in these data sources were obtained with a calculation scheme similar to the association method, by considering frequency of each domain among the interacting protein pairs. The maximum likelihood estimation (MLE) is applied to infer domain interactions by maximizing the likelihood of the observed protein interaction data (Deng, Mehta, Sun, & Chen, 2002). The probabilities of interaction between two domains (only single-domain pairs are considered) are optimized using the expectation maximization (EM) algorithm. Chen et al. (2005) used domain-based random forest framework to predict PPIs. In fact, they used the PPI data from DIP and a random forest of decision trees to classify protein pairs into sets of interacting and non-interacting pairs. Following the branches of trees, they found a number of DDIs. Riley, Lee, Sabatti, and Eisenberg (2005) proposed a domain pair exclusion analysis (DPEA) for inferring DDIs from databases of protein interactions. DPEA features a log odds score, $E_{ij}$, reflecting confidence that domains *i* and *j* interact.

The previously mentioned works mostly use protein interaction data to infer DDI, and all of them have two limitations. First, they used only the protein information (particularly protein-protein interaction data) or the co-occurrence of domains in proteins, and ignored other domain-domain interaction information between the protein pairs. However, DDI also depend on other features of proteins and domains as well—not only protein interactions (Moon, Bhak, Lee, & Lee, 2005; Wilson & Walker, 2005). Second, each of them usually exploited only a single protein database, and none of the single protein databases can provide all the information needed for better DDI prediction.

In this chapter, we present an approach using ILP and multiple genome databases to predict domain-domain interactions. The key idea of our computational method of DDI prediction is to exploit, as much as possible, background knowledge from various databases of proteins and domains for inferring DDI. To this end, we first examined seven most informative genome databases, and extracted more than a hundred thousand possible and necessary ground facts on protein domains. We then employed inductive logic programming (ILP) to infer DDI efficiently. We carried out a comparative evaluation of methods of finding DDI in terms of sensitivity and specificity. By analyzing various produced rules, we found many interesting rela-

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tions between DDI and protein functions, biological pathways, conserved motifs and pattern sites.

Our framework for predicting PPI or DDI from multiple data sources by inductive logic programming consists of two phases:

1. Generating background knowledge as ground facts from multiple genomic and protein domain databases;
2. Learning PPI or DDI predictive rules by ILP from generated ground facts and related protein data.

## Prediction of Domain-Domain Interaction from ILP from Multiple Genomic and Proteomic Databases

### *Generating Background Knowledge from Multiple Genome Databases*

We chose and extracted data from seven genomic and protein domain databases to generate background knowledge[1] with an abundant number of ground facts, and used these ground facts to predict DDI. The following are the seven databases:

1. **Pfam:** Pfam contains a large collection of multiple sequence alignments and profile hidden Markov models (HMM) covering the majority of protein domains.
2. **PRINTS:** A compendium of protein fingerprints database. Its diagnostic power is refined by iterative scanning of a SWISS-PROT/TrEMBL composite.
3. **PROSITE:** Database of protein families and domains. It consists of biologically significant sites, patterns, and profiles.
4. **InterPro:** InterPro is a database of protein families, domains and functional sites in which identifiable features found in known proteins can be applied to unknown protein sequences.
5. **Uniprot:** UniProt (universal protein resource) is the world's most comprehensive catalog of information on proteins, which consists of protein sequence and function data created by combining the information in Swiss-Prot, TrEMBL, and PIR.
6. **MIPS:** The MIPS Mammalian protein-protein interaction database is a collection of manually obtained high-quality PPI data collected from the scientific literature by expert curators.

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7. **Gene ontology (GO):** The three organizing principles of GO are molecular function, biological process, and cellular component.

We integrated domain data and protein data from these seven genome databases: our domain databases (Pfam database, PROSITE database, PRINTS database, and InterPro database) and three protein databases (UniProt database, MIPS database, and gene ontology).

(a) *Extract domain and protein data from multiple genome databases*

The first issue faced is, what kinds of genome databases are suitable for DDI prediction. When choosing data, we are concerned with two points. First is the biological role of that data in domain-domain interaction, and second is the availability of that data.

Denote by $D$ the set of all considered protein domains, $d_i$ a domain in $D$, $p_k$ a protein that consists of some domains $d_i$s , and $P$ the set of such proteins. A domain pair $(d_i,d_j)$ that interacts with each other is denoted by $d_{ij}$, otherwise by $\neg d_{ij}$ . In fact, whether two domains $d_i$ and $d_j$ interact depends on: (i) the domain features of $d_i$ and $d_j$ and, (ii) the protein features of some proteins $p_k$s consisting of $d_i$ and $d_j$ (Wilson et al., 2005). Denote by $df_t^m$ a domain feature t$^{th}$ extracted from the domain database $M$. With different domains, one feature $df_t^m$ may have different values. For example, the domain site and pattern feature extracted from PROSITE database have some values like *Casein kinase II phosphorylation site* or *Anaphylatoxin domain signature*. Denote by $pf_r^l$ a protein feature $r^{th}$ extracted from protein database $L$. Also in different domains, one protein feature $pf_r^l$ may have different values. For example, GO term feature extracted from GO database has some values like *go0006470* or *go0006412*. The extracted domain/protein features are mentioned as biologically significant factors in domain-domain interactions (Moon et al., 2005; Reichmann et al., 2005; Wilson et al., 2005), etc. The combination of both domain features and protein features constructed the considerable background knowledge associated with DDI.

Algorithm 1 shows how to extract data (values) of domain features $df^m$s and protein features $pf^l$s for all domains $d_i$s$\in$D from multiple data sources. Pfam domain accessions are domain identifiers and ORF (open reading frame) names are protein identifiers. We know that one protein can have many domains and one domain can belong to many proteins. Then, each protein identifier is mapped with the identifiers of its own domains. As the result, protein feature values are assigned to domains.

This chapter concentrates on predicting DDIs for *Saccharomyces cerevisiae*—a budding yeast, as the *Saccharomyces cerevisiae* database is available. To map proteins and their own domains, the interacting proteins in DIP database (http://dip.doe-mbi.ucla.edu/), a well-known yeast PPI database, are selected. If one protein has no

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domain, features of that protein are not predictive for domain-domain interactions. If one domain does not belong to any interacting proteins in DIP database, it seems not to have any chance to interact with others. Thus, we excluded all proteins and domains which did not have matching partners (Step 5, Step 6). Having extracted interacting proteins from DIP database, mapping data are more reliable and meaningful. After mapping proteins and their domains, the values of all domain/protein features are extracted (from Step 8 to Step 12).

*Algorithm 1. Extracting protein and domain data from multiple sources*

INPUT:

Set of domains $D \in \{d_i\}$.

Multiple genomic data used for extracting background knowledge ($S^{Pfam}$, $S^{InterPro}$, $S^{PROSITE}$, $S^{PRINTS}$, $S^{UniProt}$, $S^{MIPS}$, $S^{GO}$).

OUTPUT:

Set of domain feature values $Feature^{domain}$.

Set of protein features values $Feature^{protein}$.

1: $Feature^{domain} := \varnothing$, $Feature^{protein} := \varnothing$, $P := \varnothing$.

2: Extract all interacting proteins $p_k$s from DIP database, $P := P \cap \{p_k\}$.

3: for all proteins $p_k$s $\in P$ and domains $d_i$s $\in D$.

4: Map proteins $p_k$s with their own domains $d_i$s by the protein identifiers and the domain identifiers.

5: if a domain $d_i$ does not belong to any protein $p_k$ then $D := D \backslash \{d_i\}$.

6: if a protein $p_k$ does not contain any domain $d_i$ then $P := P \backslash \{p_k\}$.

7: for each $d_i \in D$

8: Extract all values $df_t^m.values$ for domain features $df_t^m$s from the domain database $M$ ($\forall M \in (S^{Pfam}, S^{InterPro}, S^{PROSITE}, S^{PRINTS})$).

9: if $df_t^m.values \notin Feature^{domain}$ then

$Feature^{domain} := Feature^{domain} \cap \{ df_t^m.values \}$.

10: Extract all values $pf_r^l.value$s of protein feature $pf_r^l$s from the protein database $L$

($\forall L \in S^{UniProt}, S^{MIPS}, S^{GO}$)).

11: if $pf_r^l.value\ Feature^{protein}$ then

$Feature^{protein} := Feature^{protein} \cap \{ pf_r^l.value \}$.

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*Table 1. Predicates used as background knowledge in various genomic data*

| Genomic data | Background knowledge predicates | #Ground fact |
|---|---|---|
| **Pfam** | prosite(+Domain, -PROSITE Domain)<br>A domain has a PROSITE annotation number | 1804 |
| | interpro(+Domain, -InterPro Domain)<br>A domain has an InterPro annotation number | 2804 |
| | prints(+Domain, -PRINTS Domain)<br>A domain has a PRINTS annotation number | 1698 |
| | go(+Domain, -GO Term)<br>A domain has a GO term | 2540 |
| **InterPro** | interpro2go(+InterPro Domain, -GO Term)<br>Mapping of InterPro entries to GO | 2378 |
| **PROSITE** | prosite site(+Domain, #prosite site)<br>A domain contains PROSITE significant sites or motifs | 2804 |
| **PRINTS** | motif compound(+Domain, #motif compound)<br>A domain is compounded by number of conserved motifs | 3080 |
| **Uniprot** | haskw(+Domain, #Keyword)<br>A domain has proteins keywords | 13164 |
| | hasft(+Domain, #Feature)<br>A domain has protein features | 8271 |
| | ec(+Domain, #EC)<br>A domain has coded enzyme of its protein | 2759 |
| | pir(+Domain, -PIR Domain)<br>A domain has a Pir annotation number | 3699 |
| **MIPS** | subcellular location(+Domain, #Subcellular Structure)<br>A domain has subcellular structures in which its protein is found. | 10638 |
| | function category(+Domain, #Function Category)<br>A domain belongs to proteins categorized to a certain function category | 11975 |
| | domain category(+Domain, #Protein Category)<br>A domain has proteins categorized to a certain protein category | 5323 |
| | phenotype category(+Domain, #Phenotype Category)<br>A domain belongs to proteins categorized to a certain phenotype category | 8066 |
| | complex category(+Domain, #Complex Category)<br>A domain belongs to proteins categorized to a certain complex category | 7432 |
| **GO** | is a(+GO Term, -GO Term)<br>is a relation between two GO terms | 1009 |
| | part of(+GO Term, -GO Term)<br>part of relation between two GO terms | 1207 |

*continued on following page*

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*Table 1. continued*

| | | |
|---|---|---|
| **Others** | num int(+Domain, #num int)<br>A domain has a number of domain-domain interactions | 804 |
| | ig(+Domain, + Domain, #ig)<br>Interaction generality is the number of domains that interact with just two considered domains | 8246 |
| **Totals** | | **100,421** |

12: return $Feature^{domain}$, $Feature^{protein}$.

(b) *Generating background knowledge*

The data which we extracted from seven databases have different structures: numerical data (for example, the number of motifs), text data (for example, protein function category), mixture of numerical, and text data (for example, protein keywords, domain sites). The extracted data (the values of all domain/protein features) are represented in form of predicates.

Aleph system is applied to induce rules. Note that Aleph uses mode declarations to build the bottom clauses, and a simple mode type is one of: (1) *the input variable (+),* (2) *the output variable (−),* or (3) *the constant term (#)*. In this chapter, target predicate is `interact_domain (+domain, +domain)`. The instance of this relation represents the interaction between two domains. For background knowledge, all domain/protein data are briefly denoted in the form of different predicates. Table 1 shows the list of predicates used as background knowledge for each genomic data. With the twenty background predicates, we obtained a total of 100,421 ground facts associated with DDI prediction.

Extracted domain features (from four databases: Pfam, InterPro, PROSITE, and PRINTS) are represented in the form of predicates. These predicates describe domain structures, domain characteristics, domain functions and protein domain relations. Among them, there are some predicates which are the relations between accession numbers of two databases, for example, *prints (pf00393, pr00076).*

Data from different databases are bound by these predicates. In PRINTS database, motif compound information gives the number of conserved motifs found in proteins and domains. The number of motifs is important in understanding the conservation of protein/domain structures in the evolutionary process (PRINT database). We generated predicate `motif_compound (+Domain,#motif_compound)`. This predicate is predictive for DDI prediction and gives information about the stability of DDI

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(example rules are shown and analyzed in Section 3.3.1). For example: *motif_compound (pr00517, compound(8))*, where *pr00517* is the accession number in PRINTS database and *compound(8)* is the number of motifs.

Protein domains are the basic elements of proteins. Some protein features have a significant effect on domain-domain interactions. These protein features (extracted from three databases Uniprot, MIPS, and GO) are shown in the form of predicates. These predicates describe function categories, subcellular locations, GO terms, etc. For example, most interacting proteins are in the same complexes (Ng and Tan, 2003). The domains of these interacting proteins can interact with each other. As a result, if some domains belong to some proteins categorized in the same complex, they can be predicted to have some domain-domain interactions. The predicate complex `category (+Domain,#Complex Category)` means that a domain belongs to proteins categorized to a certain complex category. For example, for complex *category (pf00400, transcription complexes)*, *pf00400* is the Pfam accession number and *transcription complexes* is the complex category name.

## *Learning DDI Predictive Rules by ILP from Generated Domain and Protein Data*

There have been many ILP systems that were successfully applied to various problems in bioinformatics, such as protein secondary structure prediction (Muggleton, King, & Sternberg, 1992), protein fold recognition (Turcotte, Muggleton, & Sternberg, 1998), and protein-protein interaction prediction (Tran et al., 2005). The proposed ILP framework for predicting DDI from multiple genome databases is described in Algorithm 2.

In the framework, the common procedure of ILP method is presented. Step 2 and Step 3 are for generating positive and negative examples (see Section 3). In Steps 4 to 7, we extract background knowledge including both domain features and protein features (see Section 3.2.1). Aleph system (http://web.comlab.ox.ac.uk/oucl/research/areas/ machlearn/Aleph/) is applied to induce rules in Step 8. Aleph is an ILP system that uses a top-down ILP covering algorithm, taking as input background information in the forms of predicates, a list of modes declaring how these predicates can be chained together, and a designation of one predicate as the head predicate to be learned.

*Algorithm 2. Discovering rules for domain-domain interactions*

INPUT:

The domain-domain interaction database InterDom

Number of negative examples ($\neg d_{ij}$) $N$

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Multiple genomic data used for extracting background knowledge ($S^{Pfam}$, $S^{InterPro}$, $S^{PROSITE}$, $S^{PRINTS}$, $S^{UniProt}$, $S^{MIPS}$, $S^{GO}$).

OUTPUT:

Set of rules $R$ for domain-domain interaction prediction.

1: R := $\varnothing$.
2: Extract positive examples set $S_{interact}$ from InterDom database.
3: Generate negative examples $\neg d_{ij}$s by selecting randomly $N$ domain pairs from $D$ where $\neg d_{ij}$s $\notin S_{interact}$.
4: **for each** domain $d_i \in$ D
5: **call** Algorithm 1 to generate values for features $df_t^m$s from the domain database $M$ ($\forall M \in (S^{Pfam}, S^{InterPro}, S^{PROSITE}, S^{PRINTS})$) and protein features $pf_r^l$s from the protein database $L$ ($\forall L \in S^{UniProt}, S^{MIPS}, S^{GO}$)).
6: Integrate all domain features $df_t^m$s and protein features $pf_r^l$s for generating background knowledge
7: Run Aleph to induce rules $r$.
8: $R := R \cap \{r\}$.
9: **return** $R$.

Aleph is able to use a variety of search methods to find good clauses such as the standard methods of breadth-first search, depth-first search, iterative beam search, as well as heuristic methods requiring an evaluation function. We use the default evaluation function coverage (the number of positive and negative examples covered by the clause) in our work.

## Evaluation of DDI Prediction

### *Experiment Design*

In this chapter, we used 3000 positive examples from InterDom database. Inter-Dom database consists of DDI of multiple organisms (http://interdom.lit.org.sg/). Positive examples are domain-domain interactions in InterDom database, which have score threshold over 100 and no false positives. The set of interacting pairs $S_{interact}$ in Algorithm 2 consists of these domain-domain interactions. Because there is currently no efficient computational and experimental method to detect non-interacting

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domain pairs, the negative examples $\neg d_{ij}$s are randomly generated. A domain pair $(d_i, d_j) \in D$ is considered to be a negative example, if the pair does not exist in the interaction set. In this chapter, we chose different numbers of negatives (500, 1000, 2000, 3000 negative examples). To validate our proposed method, we conducted a 10-fold cross-validation test, comparing cross-validated sensitivity and specificity with results obtained by using AM (Sprinzak et al., 2001) and SVMs methods.

We implemented the AM and SVMs methods in order to compare them with our proposed method. The probability threshold was set to 0.05 for simplicity of comparison. For SVM method, we used SVM$^{light}$ (SVM). The linear kernel with default values of the parameters was used. For Aleph, we selected minpos = 3 and noise = 0 (i.e., the lower bound on the number of positive examples to be covered by an acceptable clause is 3, and no negative examples are allowed to be covered by an acceptable clause). These parameters are the smallest that allow us to induce rules with biological meaning. We also used the default evaluation function coverage which is defined as P − N, where P, and N are the number of positive and negative examples covered by the clause.

## *Analysis of Experimental Results*

Table 2 shows the performance of Aleph compared with AM and SVM methods. Most of our experimental results had higher sensitivity and specificity than AM and SVMs. The sensitivity of a test is described as the proportion of true positives it detects of all the positives, measuring how accurately it identifies positives. On the other hand, the specificity of a test is the proportion of true negatives it detects of all the negatives, and thus is a measure of how accurately it identifies negatives. It can be seen from Table 2 that the proposed method showed high sensitivity and

*Table 2. Performance of Aleph compared with AM and SVMs methods. The sensitivity and specificity are obtained for each randomly chosen set of negative examples. The last column demonstrates the number of rules obtained using our proposed method, with the minimum positive cover set to 3.*

| # Neg | Sensitivity | | | Specificity | | | # Rules |
|---|---|---|---|---|---|---|---|
| | AM | SVM | Aleph | AM | SVM | Aleph | |
| 500 | 0.49±.027 | **0.86**±.010 | 0.83±.016 | 0.54±.074 | 0.24±.004 | **0.61**±.075 | 127 |
| 1000 | 0.57±.018 | 0.63±.074 | **0.78**±.042 | 0.44±.033 | 0.49±.009 | **0.68**±.042 | 173 |
| 2000 | 0.50±.015 | 0.32±.014 | **0.69**±.027 | 0.50±.021 | 0.73±.015 | **0.80**±.018 | 196 |
| 3000 | 0.49±.021 | 0.22±.017 | **0.62**±.027 | 0.53±.022 | 0.81±.013 | **0.84**±.010 | 235 |
| Avg. | 0.51±.020 | 0.51±.029 | **0.73**±.028 | 0.50±.038 | 0.57±.010 | **0.73**±.036 | |

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*Figure 2. Some induced rules obtained with minpos = 3*

```
Rule 1 [Pos cover = 15 Neg cover = 0]
    interact_domain(A, B) :   ig(A, B, C ), C = 5, f unction_category(B, transcription),
    protein_category(A, transcription f actors).
Rule 2 [Pos cover = 20 Neg cover = 0]
    interact_domain(A, B) :   num int(B, C ), gteq(C, 20),
    complex_category(A, scf comlexes).
Rule 3 [Pos cover = 51 Neg cover = 0]
    interact_domain(A, B) :   interpro(B, C ), interpro(A, C ), interpro2go(C, D).
Rule 4 [Pos cover = 23 Neg cover = 0]
    interact_domain(A, B) :   prints(B, C ), motif_compound(C, compound(8)),
    function_category(A, protein synthesis).
Rule 5 [Pos cover = 31 Neg cover = 0]
    interact_domain(A, B) :   prints(B, C ),
    motif_compound(C, compound(13)), haskw(A, cell cycle).
Rule 6 [Pos cover = 29 Neg cover = 0]
    interact_domain(A, B) :-  num int(A, C ), C = 7,
    function_category(B, metabolism), haskw(B, thread structure).
Rule 7 [Pos cover = 32 Neg cover = 0]
    interact_domain(A, B) :   ig(A, A, C ), C = 3,
    function_category(B, cell type differentiation),
    phenotype_category(A, nucleic acid metabolism defects).
Rule 8 [Pos cover = 15 Neg cover = 0]
    interact_domain(A, B) :   phenotype_category(B, conditional phenotypes)
    hasf t(A, domain rna binding rrm).
Rule 9 [Pos cover = 16 Neg cover = 0]
    interact_domain(A, B) :   prosite(B, C ),
    prosite_site(C, tubulin subunits alpha beta and gamma signature).
Rule 10 [Pos cover = 37 Neg cover = 0]
    interact_domain(A, B) :   go(B, C ), is a(C, D), hasf t(A, chain bud site
    selection protein bud5).
```

specificity given a certain number of negative examples. The number of negative examples should be chosen neither too large nor too small to avoid an imbalanced learning problem.

The performance of the method in terms of specificity and sensitivity are also statistically tested in terms of confidence intervals. Confidence intervals give us an estimate of the amount of error involved in our data. To estimate 95% confidence interval for each calculated specificity and sensitivity, we used *t* distribution. The 95% confidence intervals are shown in Table 2.

Besides comparing cross-validated sensitivity and specificity, cross-validated accuracy and precision are considered. And all of our experiment results had higher accuracy and precision compared than AM and SVMs. The average accuracy (0.76) and precision (0.82) of Aleph are higher than both AM method (0.50 and 0.61 respectively) and SVM method (0.62 and 0.70 respectively).

The experimental results have shown that ILP approach predicted DDI with high sensitivity and specificity. Further more, the inductive rules of ILP encouraged us to

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discover many comprehensive relations between DDI and domain/protein features. Analysing our results in comparison with information in biological literature and books, we found that ILP induced rules could be applied to further related studies in biology. The simplest rule covering many examples of positives is the self-interact rule. Many domains tend to interact with themselves (86 domain-domain interactions among positive examples). This phenomenon is reasonable because indeed lots of proteins interact with themselves, and they are included in many of the same domains. Figure 2 shows some other induced rules.

In the set of induced rules, there are (1) rules of only domain features (i.e., Rule 9), rules of only protein features (i.e., Rule 8), and especially rules of a mixture of both domain features and protein features (i.e., Rule 4, Rule 5). In rules, the coverage values presented are the average predictive coverage on the 10 folds.

Related to motif compound feature in domain, we found that the more motifs a domain has, the more interactions the domain has with other domains. This means that domains which have many conserved motifs tend to interact with others. And the interactions of these domains play an important role in forming stable domain-domain interactions in particular, and protein-protein interactions in general (Moon et al., 2005). Rule 4 shows that if we have two domains, one of them with eight motifs, and the other one belonging to proteins categorized in protein synthesis function category, then the two domains interact.

Discovering the rules related to domain sites and domain signatures with predicate `prosite_site(domain,#prosite site)`, we found some significant sites in domain joining in the domain-domain interactions. Rule 9 shows the relation between the accession numbers in Pfam database and PROSITE database, and then the signature information of domain in the PROSITE database. This rule means that if one domain belongs to both Pfam database and PROSITE database and has *tubulin subunits alpha beta and gamma signature*, then it can interact with others. The rules like Rule 9 can be applied to understand protein-protein interaction interfaces and protein structures (Reichmann et al., 2005).

Rule 6 is an example, which infers the relation between DDI and biological pathways. From this rule, if we have an interacting domain pair, one of domains has seven domain-domain interactions, and the other domain belongs to one protein, which has keyword thread structure, we can say that protein functions in a certain metabolic pathway.

Thanks to the inductive rules of ILP, we found many relations between DDI and different domain and protein features. We expect that the combination of these rules will be very useful for understanding DDI in particular and protein structures, protein functions and protein-protein interactions in general.

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# Some Experimental Results of PPI Prediction

Applying the same framework using ILP, we predicted PPI with multiple genomic data sources. With more than 200,000 ground facts extracted from five databases (SWISS-PROT, MIPS, InterPro, GO, Gene Expression), we input these ground facts into the ILP framework to effectively predict PPI.

We used the core data of the yeast interacting proteins database provided by Ito et al.'s as positive examples. Ito et al. (2001) conducted comprehensive analysis using their system to examine two-hybrid interactions in all possible combinations between the 6000 proteins of the budding yeast *Saccharomyces cerevisiae*. Among 4,549 interactions detected using yeast-hybrid analysis, the "core" data consist of 841 interactions with more than two IST hits, accounting for 18.6% of the data. Note that the core data used is a subset of protein-protein interactions of MIPS (http://mips.gsf.de/) database, which is considered as the gold-standard for positive examples in Jansen et al. (2003). A negatives gold-standard is defined similarly in Jansen et al. (2003), in which negative examples are synthesized from lists of proteins in separate subcellular compartments.

We employed our approach to predict protein-protein interactions. We used the core data of Ito et al.'s data set mentioned above as positive examples, selecting at random various numbers of protein pairs whose elements are in separate subcellular compartments as negative examples. Each interaction in the interaction data originally shows a pair of bait and prey ORF (open reading frame) some of which are not found in SWISS-PROT database. After removing all interactions in which either bait ORF or prey ORF is not found in SWISS-PROT, we obtained 602 interacting pairs from the original 841 pairs.

To validate our proposed method, we conducted a 10-fold cross-validation test, comparing cross-validated sensitivity and specificity with those obtained by using AM and SVMs methods. Table 3 shows that our method achieved higher sensitivity and specificity than those of AM and SVMs methods in most cases.

*Table 3. Performance of Aleph with the minimum positive cover set to 2, compared with AM and SVMs methods*

| #Neg | Sensitivity | | | Specificity | | | #Rule |
|---|---|---|---|---|---|---|---|
| | AM | SVMs | Aleph | AM | SVMs | Aleph | |
| 100 | 0.70 | **0.99** | 0.90 | **0.46** | 0.01 | 0.44 | 27 |
| 500 | 0.68 | 0.54 | **0.79** | 0.42 | 0.61 | **0.84** | 63 |
| 1000 | 0.71 | 0.32 | **0.73** | 0.39 | 0.88 | **0.93** | 62 |
| 2000 | **0.69** | 0.26 | **0.69** | 0.38 | 0.95 | **0.96** | 58 |
| 4000 | **0.69** | 0.15 | 0.68 | 0.39 | 0.98 | **0.99** | 68 |

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*Table 4. Evaluation of the proposed method using EPR index. The number of interactions after preprocessing means the number of interactions obtained after removing all inter-actions in which either bait ORF or prey ORF it not found in SWISS-PROT.*

| Data | Number of interactions | | | EPR index | |
|---|---|---|---|---|---|
| | Original | After pre-processing | Proposed | Original | Proposed |
| Ito | 4549 | 3174 | 1925 | 0.1910± 0.0306 | 0.2900 ± 0.0481 |
| Uetz | 1474 | 1109 | 738 | 0.4450± 0.0588 | 0.5290 ± 0.0860 |
| Ito + Uetz | 5827 | 4126 | 2567 | 0.2380± 0.0287 | 0.3170 ± 0.0431 |
| MIPS | 14146 | 10894 | 7980 | 0.5950± 0.0337 | 0.6870 ± 0.0420 |
| DIP | 15409 | 12152 | 8674 | 0.4180± 0.0260 | 0.5830 ± 0.0374 |

Since high-throughput experimental methods may produce false positives, it is essential to assess the reliability of protein-protein interaction data obtained. Deane, Salwinski, Xenarios, and Eisenberg (2002) proposed the expression profile reliability (EPR) index to assess the reliability of measurement of protein interaction. The EPR index estimates the biologically relevant fraction of protein interactions detected in a high throughput screen. For each given data, we retrieved all protein pairs that classified as positive. Table 4 shows the EPR index calculated using the original and our proposed method for a number of well-known protein-protein interaction data. It can be seen that the EPR index of our method is higher than the original one, demonstrating the validity of the proposed method.

# Discussion of PPI Related Problems

In this section, we discuss some challenging problems related to PPI, which recently have been subject interest of many ongoing researches. We concentrated on two kinds of problems related to PPI, the first one is detecting biological features of PPI, and the second one is discovering effects of PPI on biological processes.

## Detecting Biological Features of PPI

With the recent blooming of public proteomic and genomic databases and of course PPI databases, many computational methods have been proposed to give insights into the features of protein-protein interactions. Besides of features of interacting partners, PPI have dynamic and complex features of their own that differentiate one from another, and these features later determine the functionality of PPI in cells. These features consist of biological ones, physical ones, and chemical features,

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etc., and also the combinations like physiological or chemico-physical ones, etc. Among numerous of PPI features, in this chapter, we focus on discussing some very interesting features of PPI and related work. These features are stable and transient features of PPI, sites of PPI, and interfaces of PPI.

Because the number of experimentally determined structures for protein-protein and protein ligand complexes is still quite small, methods for computational prediction of protein-protein interaction sites are becoming increasingly important. Li et al. (2006) developed conditional random fields (CRFs) methods for identification of residues participating in protein-protein interactions. Some other related work predicts protein interaction sites in terms of binding hot-spots in protein-protein and protein-ligand interfaces (Burgoyne & Jackson, 2006), and discovers interaction sites identified from sequence (Ofran & Rost, 2007), etc. Knowing the sites of PPI allows the modification of the activity of such proteins or changing their specificity. In addition, small molecules may be designed that block interactions, thereby blocking infection. Proteins and their interactions are therefore potential drug targets.

In huge and complex networks of protein-protein interactions, some of them are transient, which are temporal and specific to a certain condition or a subset of cellular states, while others are stable, which are maintained throughout most cellular conditions (Chen et al., 2003). To detect transient and stable PPI, Mintseris and Weng (2003) took a multifaceted approach by looking at the structure, function, and sequence evolution of interacting proteins of different types. Nooren and Thornton (2003) studied experimentally validated, weak transient homodimers, as well as functionally validated transient heterodimers, such as intracellular signalling complexes that may be weakly or strongly transient. Study of the stability and transience of PPI often requires much biological knowledge and specific data sources. Although data supporting the discovery of stable and transience PPI has not been widely available yet, computational approaches attempted to solve it, and achieved some considerable results, whereas experimental methods are laborious and expensive.

Associated with PPI sites, PPI interfaces have important implications for problems ranging from rational drug design to analysis of metabolic and signal transduction networks. Many of the residues on PPI interfaces that are critical for binding are likely to be evolutionarily conserved. Therefore, there are many studies that try to discover the characteristics of such interfaces. Several approaches for predicting the interface residues from amino acid sequence, protein structure or both have been explored with varying degrees of success. Some recent work on PPI interfaces are identifying of protein-protein interface residues (Yan, Dobbs, & Honavar, 2004), combining structure and sequence conservation (Ayatuna, Aytuna, Gursoy, & Keskin, 2005), discovering modular architecture of protein-protein binding interfaces (Reichmann, 2005), etc. Understanding characteristics and architectures of PPI interfaces is promising for the design of new PPI, so that target proteins in drug design could be feasibly created.

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## Discovering Effects of PPI on Biological Processes

The interaction between proteins is fundamental to a broad spectrum of biological processes including metabolic pathways, signal transduction pathways, regulatory pathways, DNA replication, and protein synthesis and disease pathogenic mechanisms. In this subsection, we discuss two biologically significant effects of PPI on signal transduction pathways and diseases.

Signal transduction networks are crucial for inter- and intra-cellular signaling. Signals are often transmitted via covalent modification of protein structure. Signal transduction networks are complex pathways that allow the cell to receive, transmit, and act upon molecular signals. These networks are important to the correct functioning of the cell and can produce appropriate outcomes, such as cell division, apoptosis, or differentiation in response to a variety of biological signals (Allen, Fetrow, Daniel, Thomas, & John, 2006). It is believed that PPI are the key elements to cascade STN, so many PPI-based approaches are proposed to discover STN. A statistical model, based on representing proteins as collections of domains or model motifs, which predicts unknown molecular interactions within these biological networks, was first proposed by Gomez, Lo, and Rzhetsky (2001). Steffen, Petti, Aach, D'haeseleer, and Church (2002) developed a computational approach for generating static models of signal transduction networks which utilizes protein-interaction maps generated from large-scale two-hybrid screens and expression profiles from DNA micro-arrays (Steffen et al., 2002). Liu et al. applied a score function that integrated protein-protein interaction data and micro-array gene expression data to predict the order of signaling pathway components (Liu et al., 2004). Concerning on protein modification time-course data, Allen et al. (2006), applied a method of computational algebra to the modeling of signaling networks.

One of the ultimate goals of biological sciences, and certainly one with a high impact on society, is to improve our understanding of the processes and events related to diseases. Molecular biologists traditionally studied the structure and function of individual proteins and genes. However, biological processes are not realized by a single molecule, but rather by the complex interaction of proteins. As a result, the study of diseases and PPI soon became a major area of biomedical research. Researches on PPI and disease are rapidly increasing in recent years. Disease-genes were discovered by topological features in human protein–protein interaction networks (Xu & Li, 2006). Some others were mining disease related proteins from OMIM and protein interaction data for Alzheimer's disease (Chen et al., 2006), exploring the relationship between a disease and protein interactions using structural analysis of the interacting proteins (Singh, Xu, & Berger, 2006; Ye, Li, & Godzik, 2006), and using hybrid Petri nets to model the regulation of human hematopoieses (Troncale et al., 2006 ). The increasing number of computational studies of protein networks, biological pathways, protein-protein, protein-metabolite, and protein-DNA/RNA interactions indicates that it is now feasible to address the connections between

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protein interactions and diseases. Furthermore, the results of such work contribute greatly to pharmaceutical industry and clinical practice.

## Conclusion

In conclusion, this chapter presents one of the most challenging problems in bioinformatics, protein-protein interactions. In addition to experimental methods, computer science came to the scene bringing about another approach, the computational one, to the standing PPI issues. The ILP framework shown here is our effort to solve the problem of DDI and PPI prediction and the experimental results are considerable. With the key roles of PPI in cells, it is promising to use machine learning and data mining techniques to discover useful knowledge from protein-protein interaction data produced by high-throughput biological techniques for further studies in life science.

## References

Albrecht, M., Huthmacher, C., Tosatto, S. C. E., & Lengauer, T. (2005). Decomposing protein networks into domain-domain interactions. *Bioinformatics, 21*(suppl.2), ii220-221.

Allen, E., Fetrow, J., Daniel, L. W., Thomas, S., & John, D. (2006). Algebraic dependency models of protein signal transduction networks from time-series data. *Journal of Theoretical Biology, 238*, 317-330.

Archakov, A. I., Govorun, V. M., Dubanov, A. V., Ivanov, Y. D., Veselovsky, A. V., Lewi, P., & Janssen, P. (2003). Protein-protein interactions as a target for drugs in proteomics. *Proteomics, 3*(4), 380-391.

Arnau, V., Mars, S., & Marín, I. (2005). Iterative cluster analysis of protein interaction data. *Bioinformatics, 21*(3), 364-378.

Aytuna, A. S., Gursoy, A., & Keskin, O. (2005). Prediction of protein-protein interactions by combining structure and sequence conservation in protein interfaces. *Bioinformatics, 21*, 2850-2855.

Bader, G. D., Donaldson, I., Wolting, C., Ouellette, B. F. F., Pawson, T., & Hogue, C. W. V. (2001). BIND—The biomolecular interaction network database. *Nucleic Acids Research, 29*, 242-245.

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Bader, J. S., Chaudhuri, A., Rothberg, J. M., & Chant, J. (2004). Gaining confidence in high-throughput protein interaction networks. *Nature Biotechnology, 22*, 78-85.

Baudot, A., Martin, D., Mouren, P., Chevenet, F., Guenoche, A., Jacq, B., & Brun, C. (2006). "PRODISTIN Web site: A tool for the functional classification of proteins from interaction networks. *Bioinformatics, 22*(2), 248-250.

Bauer, A., & Kuster, B. (2003). Affinity purification-mass spectrometry: Powerful tools for the characterization of protein complexes. *European. Journal of Biochem. 270*(4), 570-578.

Ben-Hur, A., & Noble, W. S. (2005). Kernel methods for predicting protein-protein interactions. *Bioinformatics, 21*(suppl.1), i38-46.

Bock, J. R., & Gough, D. A. (2001). Predicting protein-protein interactions from primary structure. *Bioinformatics, 17*(5), 455-460.

Brown, K. R., & Jurisica, I. (2005). Online predicted human interaction database. *Bioinformatics, 21*(9), 2076-2082.

Burgoyne, N. J., & Jackson, R. M. (2006). Predicting protein interaction sites: binding hot-spots in protein-protein and protein-ligand interfaces. *Bioinformatics, 22*, 1335-1342.

Chen, J. , & Yuan, B. (2006). Detecting functional modules in the yeast protein-protein interaction network. *Bioinformatics, 22*(18), 2283-2290.

Chen, X., & Liu, M. (2006). Domain-based predictive models for protein-protein interaction prediction. *EURASIP Journal on Applied Signal Processing, 2006*, 1-8.

Chen, X., & Liu, M. (2005). Prediction of protein-protein interactions using random decision forest framework. *Bioinformatics, 21*(24), 4394-4400.

Chen, Y., & Xu, D. (2003). Computational analyses of high-throughput protein-protein interaction data. *Current Protein and Peptide Science, 4*(3),159-180.

Chen, J. Y., Shen, C., & Sivachenko, A. Y. (2006). Mining alzheimer disease relevant proteins from integrated protein interactome data. *Pacific Symposium on Biocomputing, 11*, 367-378.

Cockell, S. J., Oliva, B., & Jackson, R. M. (2007). Structure-based evaluation of in silico predictions of protein-protein interactions using comparative docking. *Bioinformatics*, btl661.

Deane, C. M., Salwinski, L., Xenarios, I., & Eisenberg, D. (2002). Protein interactions: Two methods for assessment of the reliability of high-throughput observations. *Mol Cell Proteomics*, M100037-MCP200.

Deng, M., Mehta, S., Sun, F., & Chen, T. (2002). Inferring domain-domain interactions from protein-protein interactions. *Genome. Res. 12*(10), 1540-1548.

Copyright © 2008, IGI Global. Copying or distributing in print or electronic forms without written permission of IGI Global is prohibited.

Deng, M., Tu, Z., Sun, F., & Chen, T. (2004). Mapping gene ontology to proteins based on protein-protein interaction data. *Bioinformatics, 20*(6), 895-902.

Friedrich, T., Pils, B., Dandekar, T., Schultz, J., & Muller, T. (2006). Modelling interaction sites in protein domains with interaction profile hidden Markov models. *Bioinformatics, 22*(23), 2851-2857.

Goh, C. S., Bogan, A. A. and et al., M. J. (2000). Co-evolution of proteins with their interaction partners. *J. Mol. Biol. 299*(2), 283-293.

Gomez, S. M., Lo, S., & Rzhetsky, A. (2001). Probabilistic prediction of unknown metabolic and signal-transduction networks. *Genetics, 159*(3), 1291-1298.

Grigoriev, A. (2001). A relationship between gene expression and protein interactions on the proteome scale: Analysis of the bacteriophage T7 and the yeast Saccharomyces cerevisiae. *Nucleic Acids Res. 29*(17), 3513-3519.

Han, D., Kim, H. S., Seo, J., & Jang, W. (2003). A domain combination based probabilistic framework for protein-protein interaction prediction. *Genome Inform. Ser. Workshop Genome Inform*, 250-259.

Ito, T., Chiba, T., Ozawa, R., Yoshida, M., Hattori, M., & Sakaki, Y. (2001). A comprehensive two-hybrid analysis to explore the yeast protein interactome. *Proc. Natl. Acad. Sci.* USA 98, 4569-4574.

Jang, H., Lim, J., Lim, J., Park, S., Lee, K., & Park, S. (2006). Finding the evidence for protein-protein interactions from PubMed abstracts. *Bioinformatics, 22*(14), 220-226.

Jansen, R., Yu, H., Greenbaum, D., Kluger, Y., Krogan, N. J., Chung, S., Emili, A., Snyder, M., Greenblatt, J. F., & Gerstein, M. (2003). A bayesian networks approach for predicting protein-protein interactions from genomic data. *Science, 302*(5644), 449-453.

Jennifer, A., Dahesh, S., Haynes, J., Andrews, B., & Davidson, A. (2004). Protein-protein interaction affinity plays a crucial role in controlling the Sho1p-mediated signal transduction pathway in Yeast. *Molecular Cell, 14*, 813-823.

Jones, S., & Thornton, J. (1996). Principles of protein-protein interactions derived from structural studies. *Proceedings of the National Academy of Science* (U.S.A) (Vol. 93, pp. 13-20).

Kim, R., Park, J., & Suh, J. (2002). Large scale statistical prediction of protein-protein interaction by potentially interacting domain (PID) pair. *Genome Inform. Ser. Workshop Genome Inform*, 48-50.

Li, H., Li, J., & Wong, L. (2006). Discovering motif pairs at interaction sites from protein sequences on a proteome-wide scale. *Bioinformatics, 22*(8), 989-996.

Li, M., Wang, X., Lin, L., & Liu, T. (2007). Protein-protein interaction site prediction based on conditional random fields, *Bioinformatics*, btl660.

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Liang, Z., Xu, M., Teng, M., & Niu, L. (2006). NetAlign: A Web-based tool for comparison of protein interaction networks. *Bioinformatics, 22*(17), 2175-2177.

Liu, Y., & Zhao, H. (2005). A computational approach for ordering signal transduction pathway components from genomics and proteomics data. *BMC Bioinformatics 5*.

Ma, X., Lee, H., Wang, L., & Sun, F. (2007). CGI: A new approach for prioritizing genes by combining gene expression and protein-protein interaction data. *Bioinformatics, 23*(2), 215-221.

Marcotte, E. M., Pellegrini, M. and et al., H. L. N. (1999). Detecting protein function and protein-protein interactions from genome sequences. *Science, 285*(5428), 751-753.

Marcotte, E. M., Xenarios, I., & Eisenberg, D. (2001). Mining literature for protein-protein interactions. *Bioinformatics, 17*(4), 359-363.

Matthews, L. R., Vaglio, P. and et al., J. R. (2001). Identification of potential interaction networks using sequence-based searches for conserved protein-protein interactions or "interologs." *Genome Res. 11*(12), 2120-2126.

Mintseris, J., & Weng, Z. (2005). Structure, function, and evolution of transient and obligate protein-protein interactions. *PNAS, 102*(31), 10930-10935.

Moon, H. S., Bhak, J., Lee, K., & Lee, D. (2005). Architecture of basic building blocks in protein and domain structural interaction networks. *Bioinformatics, 21*(8), 1479-1486.

Morrison, J. L., Breitling, R., Higham, D. J., & Gilbert, D. R. (2006). A lock-and-key model for protein-protein interactions. *Bioinformatics, 22*(16), 2012-2019.

Muggleton, S., King, R. D., & Sternberg, M. J. E. (1992). Protein secondary structure prediction using logic-based machine learning. *Protein Engeneering, 6*(5), 647-657.

Murakami, Y., & Jones, S. (2006). SHARP2: Protein-protein interaction predictions using patch analysis. *Bioinformatics, 22*(14), 1794-1795.

Ng, S., Zhang, Z., Tan, S., & Lin, K. (2003). InterDom: A database of putative interacting protein domains for validating predicted protein interactions and complexes. *Nucleic Acids Res, 31*(1), 251-254.

Ng, S. K., & Tan, S. H. (2004). Discovering protein-protein interactions. *Journal of Bioinformatics and Computational Biology, 1*(4), 711-741.

Nguyen, T. P., & Ho, T. B. (2006a). Prediction of domain-domain interactions using inductive logic programming from multiple genome databases. *Discovery Science DS'06*, October 7-10, Barcelona, Springer LNAI 4256 (pp. 185-196).

Nguyen, T. P., & Ho, T. B. (2006b). Discovering signal transduction networks using signaling domain-domain interactions. *Genome Informatics 2006*, Yokohama, December 18-20, Universal Academic Press (pp. 35-45).

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Nooren, I. M. A., & Thornton, J.M. (2003). Structural characterisation and functional significance of transient protein-protein interactions. *J. Mol. Biol. 325*, 991-1018.

Ofran, Y., & Rost, B. (2007). ISIS: Interaction sites identified from sequence. *Bioinformatics, 23*, e13-16.

Oyama, T., Kitano, K., Satou, K., & Ito, T. (2002). Extracting of knowledge on protein-protein interaction by association rule discovery. *Bioinformatics, 18*(5), 705-714.

Page, D., & Craven, M. (2003). Biological applications of multi-relational data mining. *SIGKDD Explorations, 5*(1), 69-79.

Pagel, P., Kovac, S., Oesterheld, M., Brauner, B., Dunger-Kaltenbach, I., Frishman, G., Montrone, C., Mark, P., Stumpflen, V., Mewes, H., Ruepp, A., & Frishman, D. (2005). The MIPS mammalian protein-protein interaction database. *Bioinformatics, 21*(6), 832-834.

Pellegrini, M., Marcotte, E. M. and et al., M. J. T. (1999). Assigning protein functions by comparative genome analysis: Protein phylogenetic profiles. *Proc. Natl. Acad. Sci. USA* 96(8), pp. 4285-4288.

Reichmann, D., Rahat, O., Albeck, S., Meged, R., Dym, O., & Schreiber, G. (2005). From the cover: The modular architecture of protein-protein binding interfaces. *PNAS, 102*(1), 57-62.

Riley, R., Lee, C., Sabatti, C., & Eisenberg, D. (2005). Inferring protein domain interactions from databases of interacting proteins. *Genome Biology, 6*(10), R89.

Robert, R., Christopher, L., Chiara, S., & Eisenberg, D. (2005). Inferring protein domain interactions from databases of interacting proteins. *Genome Biology, 6*(R89), 895-902.

Salwinski, L., Miller, C. S., Smith, A. J., Pettit, F. K., Bowie, J. U., & Eisenberg, D. (2004). DIP: The database of interacting proteins: 2004 update. *Nucleic Acids Research, 32*, 449-451.

Sato, T., Yamanishi, Y., Horimoto, K., Kanehisa, M., & Toh, H. (2006). Partial correlation coefficient between distance matrices as a new indicator of protein-protein interactions. *Bioinformatics, 22*(20), 2488-2492.

Schölkopf, B., Tsuda, K., & Vert, J. P. (2004). *Kernel methods in computational biology.* The MIT Press.

Singh, R., Xu, J., & Berger, B. (2006). Struct2net: Integrating structure into protein-protein interaction prediction. *Pacific Symposium on Biocomputing, 11*, 403-414.

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Spellman, P. T., Sherlock, G. and et al., M. Q. Z. (1998). Comprehensive identification of cell cycle-regulated genes of the yeast Saccharomyces cerevisiae by microarray hybridization. *Molecular Biology of the Cell, 9*(12), 3273-3297.

Sprinzak, E., & Margalit, H. (2001). Correlated sequence-signatures as markers of protein-protein interaction. *Journal of Molecular Biology, 311*, 681-692.

Steffen, M., Petti, A., Aach, J., D'haeseleer, P., & Church, G. (2002). Automated modelling of signal transduction networks. *BMC Bioinformatics, 3*(34).

Sun, J., Xu, J., Liu, Z., Liu, Q., Zhao, A., Shi, T., & Li, Y. (2005). Refined phylogenetic profiles method for predicting protein-protein interactions. *Bioinformatics, 21*(16), 3409-3415.

Tran, T., Satou, K., & Ho, T. B. (2005). Using inductive logic programming for predicting protein-protein interactions from multiple genomic data. in *PKDD*, pp. 321-330.

Troncale, S., Tahi, F., Campard, D., Vannier, J., & Guespin, J. (2006). Modeling and simulation with hybrid functional Petri nets of the role of interleukin-6 in human early haematopoiesis. *Pacific Symposium on Biocomputing, 11*, 427-438.

Turcotte, M.., Muggleton, S. H., & Sternberg, M. J. E. (1998). Protein fold recognition. *Proceedings of the 8th International Workshop on Inductive Logic Programming (ILP-98)* (pp. 53-64).

Uetz, P., Giot, L., Cagney, G., Mansfield, T. A., Judson, R. S., Knight, J. R., Lockshon, D., Narayan, V., Srinivasan, M., Pochart, P., Qureshi-Emili, A., Li, Y., Godwin, B., Conover, D., Kalbfleisch, T., Vijayadamodar, G., Yang, M., Johnston, M., Fields, S., & Rothberg, J. M. (2000). A comprehensive analysis of protein-protein interactions in Saccharomyces cerevisiae. *Nature, 403*(6770), 623-627.

Uetz, P., & Vollert, C. S. (2006). Protein-protein interactions. *Encyclopedic Reference of Genomics and Proteomics in Molecular Medicine (ERGPMM)*, Springer Verlag, Part 16: 1548-1552.

Wilson, K., & Walker, J. (2005). *Principle and techniques of biochemistry and molecular biology. Cambridge University Press.*

Wojcik, J., & Schachter, V. (2001). Protein-protein interaction map inference using interacting domain profile pairs. *Bioinformatics, 17*(suppl-1), S296-305.

Xenarios, I., & Eisenberg, D. (2001). Protein interaction databases. *Current Opinion in Biotechnology, 12*, 334-339.

Xu, J., & Li, Y. (2006). Discovering disease-genes by topological features in human protein-protein interaction network. *Bioinformatics, 22*(22), 2800-2805.

Yan, C., Dobbs, D., & Honavar, V. (2004). A two-stage classifier for identification of protein-protein interface residues. *Bioinformatics, 20*, i371-378.

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Ye, Y., Li, Z., & Godzik, A. (2006). Modeling and analyzing three-dimensional structures of human disease proteins. *Pacific Symposium on Biocomputing, 11*, 439-450.

Yu, H., Paccanaro, A., Trifonov, V., & Gerstein, M. (2006). Predicting interactions in protein networks by completing defective cliques. *Bioinformatics, 22*(7), 823-829.

## Endnote

[1] The term "background knowledge" is used here in terms of language of inductive logic programming.

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## Chapter XIII

# Data Mining in the Social Sciences and Iterative Attribute Elimination

Anthony Scime, SUNY Brockport, USA
Gregg R. Murray, SUNY Brockport, USA
Wan Huang, SUNY Brockport, USA
Carol Brownstein-Evans, SUNY Brockport, USA

## Abstract

*Immense public resources are expended to collect large stores of social data, but often these data are under-examined thereby missing potential opportunities to shed light on some of society's pressing problems. This chapter proposes and demonstrates data mining in general and an iterative attribute-elimination process in particular as important analytical tools to exploit more fully these important data from the social sciences. We use an iterative domain-expert and data mining process to identify attributes that are useful for addressing distinct and nontrivial research issues in social science—presidential vote choice and living arrangement outcomes for maltreated children—using the American National Election Studies (ANES) from political science and the National Survey on Child and Adolescent*

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*Well-Being (NSCAW) from social work. We conclude that data mining is useful for more fully exploiting important but under-evaluated data collections for the purpose of addressing some important questions in the social sciences.*

## Data Mining: Ethical, Theoretical, and Practical Motivations

*[D]ata are expensive in terms of time, effort, money, and other resources… If the research was worth doing, the data are worth a thorough analysis, being held up to the light in many different ways so that our research participants, our funding agencies, our science, and society will all get their time and their money's worth.* (Rosenthal, 1994, p. 130)

More than $2.4 billion was spent on social science research and development at colleges and universities alone in 2003 (Jankowski, 2005). This investment represents not only a financial expenditure, but also the expenditure of countless hours, days, weeks, and months of researcher, participant, and administrator time and effort and other exertions. The results of this immense investment are often embodied in extensive data sets such as the general social survey, which has collected data on American society and attitudes approximately every two years since 1975 (NORC, 2006), the Uniform Crime Reporting Program, which has gathered information on crime levels and law enforcement administration, operation, and management in the U.S. since 1930 (Federal Bureau of Investigation, 2004), and the Baccalaureate and Beyond Longitudinal Study, which has followed about 11,000 students who completed their baccalaureate degree in 1992-93 to assess their education and work experience (Wine, Cominole, Wheeless, Dudley, & Franklin, 2005).

Minimally, the desire for basic efficiency demands reasonable output from this substantial input into research and development. But the challenge accepted by social investigators is greater than simple efficiency. For example, the mission statement of the National Science Foundation indicates its objective is "to promote the progress of science; to advance the national health, prosperity, and welfare; [and] to secure the national defense…" (National Science Foundation, 2005). The challenge, then, is to resolve important issues, often calling for high risk, high payoff research. In these endeavors, large quantities of data often are collected, some of which may contain the keys to resolving important social issues but which remain inaccessible due to the inability to evaluate comprehensively and efficiently the mass of data. Given the resources invested and potential for important findings, then, there is an ethical imperative to exploit these data to their fullest extent.

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In addition to the ethical motivations, there are a number of theoretical and practical reasons to data mine social science data. The objective of secondary analysis is to test the robustness of previous findings and to generate new hypotheses for further examination in an effort to advance practice and the discipline. If valid, findings from previous research should withstand scrutiny using different analytical tools in a different analytical context. That is, relationships that are significant in bivariate and limited multivariate analyses should also be significant in the large-scale multivariate analyses performed in data mining. Further, broadening the scope of the analysis, which data mining accomplishes by including larger sets of attributes, may result in new hypotheses that can inform practice and advance the discipline.

Finally, large quantities of data present problems in and of themselves. Scientific research is motivated by the desire to find patterns in data, and advances in data collection and processing have made the creation of massive data sets viable. While modern scientists are fortunate to have access to these vast stores of data, the data sets are often so immense that comprehensive analysis is all but impossible using conventional techniques. For example, the American National Election Studies Cumulative File presents the researcher more than 900 attributes and 47,000 records, while the National Survey on Child and Adolescent Well-Being presents more than 20,000 attributes on 5500 children and their caregivers collected over a 36-month time period. Moreover, while conventional techniques typically constrain multivariate analysis to a handful of carefully selected and, therefore, a potentially biased set of attributes (Shapiro, 2002), data mining techniques, including the iterative attribute-elimination process presented below, effectively assess the significance of dozens of seemingly related and unrelated attributes. Not only do data mining and this iterative attribute elimination more efficiently exploit data, they also reduce bias inherent in conventional model specification.

Following these motivations, this chapter proposes and demonstrates data mining in general and our iterative attribute elimination process in particular as important analytical tools to exploit more fully data from the social sciences. We use a data mining process to identify attributes that are useful for addressing nontrivial research issues in social science—presidential vote choice and living arrangement outcomes for maltreated children—using the theoretically and structurally distinct American National Election Studies (ANES) from political science and National Survey on Child and Adolescent Well-Being (NSCAW) from social work. In the end, we conclude that data mining can be useful for more fully exploiting important but under-evaluated data collections for the purpose of addressing some important questions in the social sciences.

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## Data Mining Across Disciplines

Data mining is the exploration and analysis of large quantities of data using mathematical algorithms and computer learning techniques to detect hypothesized and previously unknown relationships and patterns and to predict behaviors and outcomes. It is a process that allows researchers to evaluate data in-depth to discern complex patterns, generate rules, uncover previously unknown relationships, spark new ideas, refocus questions, and make predictions. Data mining focuses data in that it takes advantage of large quantities of data and allows those data to suggest which relationships are and are not worthy of attention. It is a multivariate method that acknowledges that previously ignored attributes often have predictive and, therefore, applied value when combined with other attributes. Data mining employs a number of techniques such as classification, cluster analysis, and association analysis to identify the complicated relationships and to predict behaviors and outcomes.

Data mining has been applied to and produced noteworthy results in a number of disciplines including political science, library science, criminal justice, education, business, health care, and environmental sciences. For example:

- Scime and Murray (forthcoming) show how data mining can be used to reduce and to focus the information used to predict vote choice in presidential elections to a small and manageable number of predictor attributes.
- Nicholson (2006) describes how librarians have used data mining to detect user patterns that can be used to allocate resources.
- Whiting (2006) describes how a city police department uses data mining to determine the probability that a particular type of crime—armed robbery, auto theft, and murder—will occur in a specific area at a given time.
- Romero and Ventura (2006) demonstrate how data mining can be used to better inform education administrators of student enrollment and to allow them to make institutional changes.
- Huang, Chen, and Cordes (2004) demonstrate how data mining can be used to identify causes of fatal automobile accidents.
- Spangler (2003) describes how personal video recorder companies are able to exploit collected data to show advertisers that their commercials are reaching certain segments of the population by connecting specific television shows with specific family members.
- McCarthy (1997) describes how a pharmaceutical company used data mining to find less expensive but equally efficacious drug treatments for certain categories of patients.

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- Mitchell (1997) describes how data mining has been used to predict women at high risk of needing an emergency C-section.
- Fayyad, Piatetsky-Shapiro, and Smyth (1996) describe the use of data mining in earthquake detection, sky image analysis, and the location of geological features on other planets.

This brief review suggests that data mining is an important technique for exploiting large data sets and uncovering important relationships within them. It comprises a wide and flexible set of tools designed to perform a broad array of analyses for predicting outcomes and behavior. Moreover, data mining is a technique that has proven useful in a variety of disciplines from political science to health care and planetary geography.

## Iterative Attribute Elimination for the Social Sciences

The data mining process, defined as a data mining life cycle (DMLC), involves both human and software resources (Hofmann & Tierney, 2003). In this life cycle, experts in the domain and the data interact with the data source to identify the purpose of the data mining project. Data preparation involves selecting data and metadata with the objective of selecting an appropriate data set. This prepared data set is then data mined to create a model of the data. The domain experts convert the resulting data mining model into rules (Anand, Bell, & Hughes, 1995).

There has been much work done in the selection and reduction of classification rules and association rules (Deshpande & Karypis, 2002; Freitas, 2000; Jaroszewicz & Simovici, 2004; Padmanabhan & Tuzhilin, 2000). Classification algorithms construct decision trees by looking at the past performance in records with respect to a goal attribute (dependent variable). Classification rules are constructed from these decision trees. These rules are then used to predict the behavior in future cases, classifying the case into one of the values of the goal class.

The large number of rules leads to a need to mine the discovered patterns (rules) to find the most useful and interesting. There are two main approaches: (1) sorting the rules on some measure of interestingness or (2) pruning followed by deleting redundant or similar rules (Deshpande et al., 2002; Jaroszewicz et al., 2004; Padmanabhan et al., 2000). While rule reduction is necessary, it is possible to reduce the number of attributes used in decision tree construction and, hence, simplify the rule-reduction problem. Furthermore, reducing the number of attributes reduces the need for data collection in predicting future behavior.

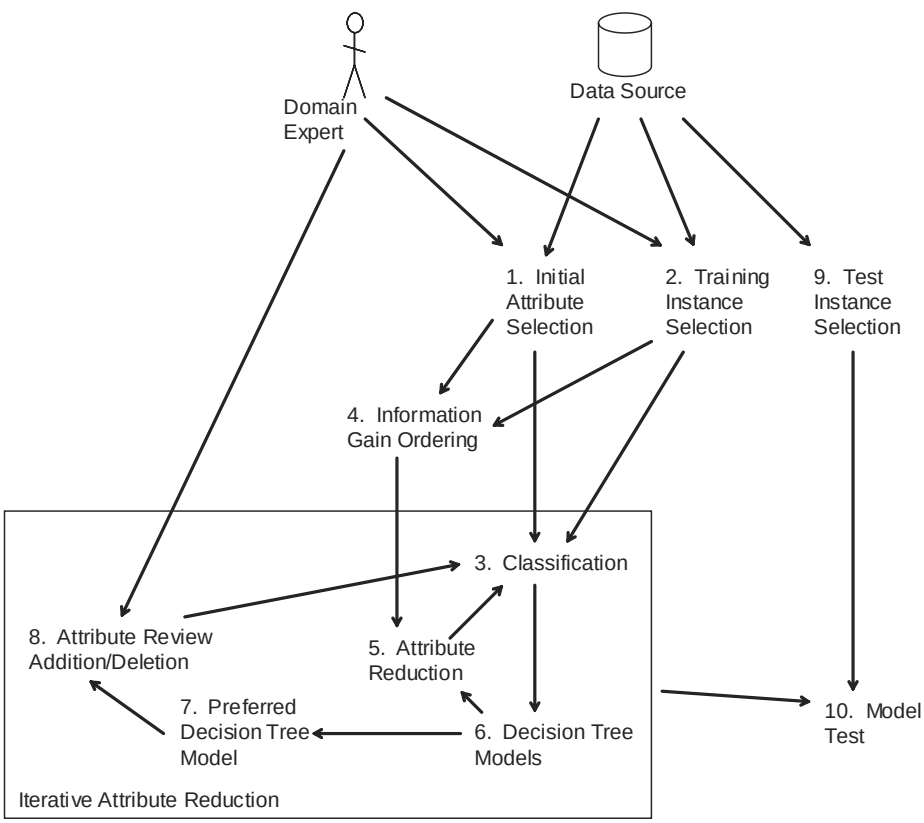
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Iterative attribute elimination is a method for data dimensionality reduction (DDR). DDR is used to simplify the data without reducing the classification capabilities of the resultant model. It has been shown to improve classification by neural networks and to produce concise and accurate rules (Fu & Wang, 2005).

The intent of iterative attribute elimination is to use the domain expert's knowledge to refine the problem domain and select the appropriate research problem. This is done by both iterative attribute elimination (Scime & Murray, forthcoming) and record selection (Scholz, 2005). As represented in Figure 1, the process of iterative attribute elimination using domain knowledge to produce a classification decision model is as follows:

1. The domain expert selects attributes that are likely to produce interesting and useful results.

*Figure 1. Iterative domain knowledge and attribute elimination*




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2. The domain expert selects records that are likely to produce interesting and useful results; alternatively these records can be selected randomly.
3. A classification algorithm produces a decision tree model from the data.
4. Information gain for each attribute with respect to the goal attribute is calculated. The attributes are ordered by information gain.
5. Progressively removing attributes from the data, beginning with the lowest information-gain-ranked attribute, the classification algorithm is repetitively executed.
6. A model for each collection of attributes is produced. The resulting decision trees are compared for predictive accuracy.
7. The preferred model is selected.
8. The expert reviews the preferred model and uses domain knowledge for further attribute additions and deletions. The expert may add attributes that are likely to increase the model's accuracy and/or usefulness. Attributes may be deleted when the domain expert expects that a combination of attributes is redundant or overlapping.
9. An independent set of data from the original data source is used for testing the final model.
10. The final model is tested with the test records.

If the test is deemed acceptable, the decision tree can be used to classify new data and can be converted into rules. The rules can then be analyzed to gain greater understanding of the particular data set and the problem domain as a whole.

## Data Mining Example 1: Presidential Vote Choice

The American National Election Studies (ANES) is an ongoing, long-term series of public opinion surveys intended to produce research-quality data for researchers who study the theoretical and empirical bases of American national election outcomes using voting behavior, public attitudes, and measures of political participation. Given this objective, the ANES collects data on items such as voter registration and choice, social and political values, social background and structure, partisanship, candidate and group evaluations, opinions about public policy, ideological support for the political system, mass media consumption, and egalitarianism. The ANES has conducted pre- and post-election, face-to-face interviews of a nationally representative sample of adults every presidential and midterm election year since 1948,

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except for the midterm election of 1950. The most common study design has been a cross-sectional, equal-probability sample.

The 1948-2004 ANES cumulative data file is a single file composed of the merged records and attributes from each of the studies conducted since 1948 (47,438 records). The file includes most, but not all, of the questions that have been asked in three or more ANES surveys conducted during the multi-decade time period. It is composed, therefore, of more than 900 attributes, which, for comparability, have been coded in a consistent manner from year to year. The consistent coding of the data avoids a common problem in data mining. That is, when data are collected by different methods, it is possible that attribute values representing the same data characteristic may not be coded in the same way (Jeong & Lee, 2005). This can happen based on simple changes in the coding method (e.g., representing education as "advanced degrees" or "graduate degrees") or by a using both generalizations and specializations in attribute values (such as coding education as just "graduate degree," "master degree," or "doctorate degree"). Because the data set is prepared for analysis, all the attribute values are coded numerically with predefined meanings.

The data set includes both discrete and continuous attributes. Discrete-value attributes contain a single defined value such as party registration, which is indicated as Democrat, Republican, or other. Continuous-value attributes measure affect toward a specified target such as 0-100-scale "feeling thermometers" and subtractive scales that indicate the number of "likes" minus the number of "dislikes" mentioned about a target.

## Classification Mining the Voting Data

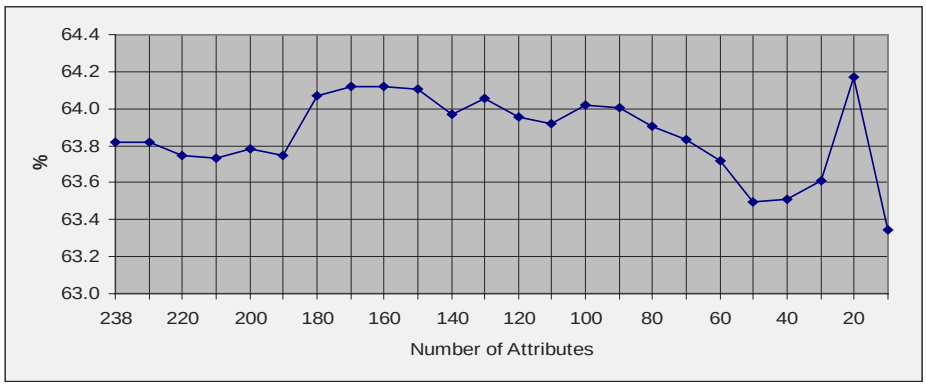
With the objective of finding attributes that together are indicative of a respondent's presidential vote choice, we reviewed the 1948-2004 ANES cumulative data file. The domain expert selected a total of 237 attributes in addition to the goal attribute, which together are intended to predict a respondent's presidential vote choice for the Republican, Democrat, minor-party candidate, or the decision not to vote.

The data were initially limited to records of surveys from presidential election years, thereby excluding midterm election years since 1948. This selection is based on the recognition that more extensive surveys were administered during presidential years as well as the increased attention and interest paid by citizens during these high-profile elections. From this data set of more than 26,000 presidential-year records, a subset of 6678 records was randomly selected as the training set.

The ANES survey questions have varied from year to year. This creates the possibility that the records may be sparsely populated (i.e., that is, they may contain large numbers of NULL values). Within the randomly selected data, this was found not to be the case. Data are said to be sparse if only a small fraction (no more than

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*Figure 2. Success rate of presidential vote classification as attributes are reduced*




20%) of the attributes are non-NULL for given record. Sparse data occur when there are many more attributes that could possibly have values than actually do. For example, in market basket problems there might be 10,000 products in a store, and the average size of the collection of distinct items that a customer purchases in a typical transaction is 50 or fewer products (Taft et al., 2003). In the initial data set, only 3.25% of the records could be considered sparse.

In addition to selecting the attributes most likely to represent the overall records, it is necessary with ANES data to clean and prepare the data for data mining. The original ANES data are represented numerically in the cumulative file. The "feeling thermometer" attributes represent a continuous spectrum of possible feelings. The affect-value attribute values represent a point on a continuum of the differences between "likes" and "dislikes." Both were left as continuous attributes. The discrete-value attributes have a number value representing possible responses to the attribute question. These values need to remain discrete and not be considered as continuous. To discretize the numeric discrete-value attribute values the character "a" was pre-pended to each value during data preparation.

A decision tree was constructed using the WEKA implementation of the classification decision tree C4.5 algorithm with three-fold cross-validation (Witten & Frank, 2005). The information gain of the 237 attributes against the goal attribute was calculated, and then the attributes were ranked based on information gain from high to low. The attributes with the lowest information gain were systematically removed from the data and the C4.5 algorithm re-applied. The results, which are presented in Figure 2, show the greatest success of predicting vote choice with 20 attributes, which are the goal attribute and the19 attributes with the most information gain.

Domain expertise is again used to evaluate the reasonableness or interestingness of the attribute selection. Among the 19 attributes (20 including the goal attribute) that best predict presidential vote choice is a question asking for whom they voted

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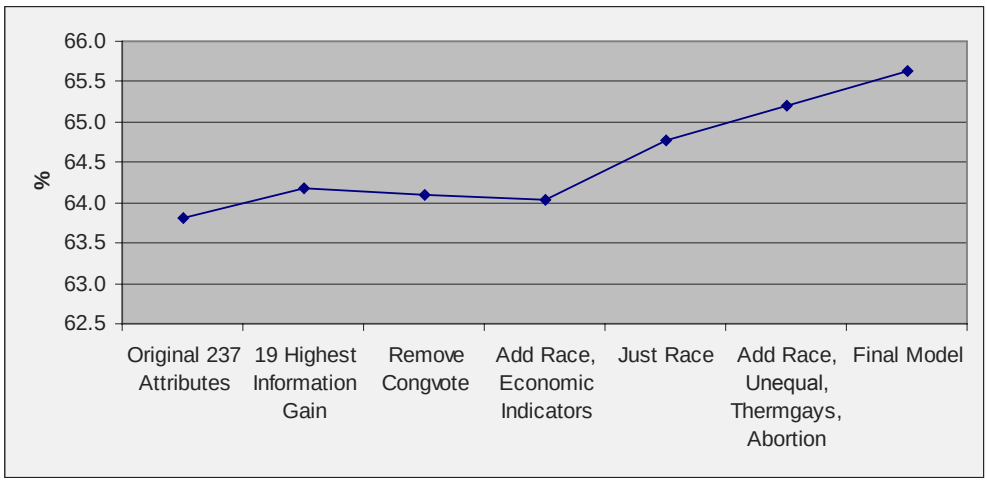
in the corresponding congressional election. This is a post-election question and the intent of our process is to predict voter behavior prior to the election. The removal of this attribute results in a slight decline in the success rate from 64.1730% to 64.1052%.

The domain expert's first review of the decision tree also led to the decision to include race and aggregate indicators of economic assessments to improve the performance of the model. Race is considered fundamental to voting behavior (e.g., Burton & Shea, 2003) and was forced back into the attribute list for evaluation. There is also a close relationship between economic performance and vote choice (Abramson, Aldrich, & Rohde, 2003). Following Nadeau and Lewis-Beck (2001), we enhanced the data set with measures of aggregate evaluations of retrospective (National Business Index [NBI]) and prospective (Economic Future Index [EFI]) economic performance.

Adding the attributes for race and economic assessment to the 19 attributes identified by the information-gain selection processes reduces the success rate to 64.0373%. Adding these attributes in pairs and individually shows that the economic indicators (NBI and EFI) decrease the ability of the resulting decision trees to predict voter behavior. However, adding the race attribute with the 19 information-gain-selected attributes provides better results than the 19 information-gain-selected attributes alone, 64.7668%.

The race attribute was kept as part of the model and additional attributes were also added. The attributes tested by the domain expert were those that would be expected

*Figure 3. Success improvement of presidential vote classification with domain knowledge*




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to improve the model's accuracy and usefulness. These include attributes pertaining to beliefs about traditional values and religion; feeling thermometers about immigrants, the economy, blacks, gays, welfare, and abortion; residence in the South; voting in the primary election; and tolerance of different moral standards.

The review of the resulting models found that the tree that includes the 19 information-gain-identified attributes plus the expert-identified attributes for race, the assessment of whether all people receive an equal chance and feelings toward gays and abortion provided the best success rate at 65.2078%. The expert then deleted attributes from this model and tested the effect of each deletion. The attributes that improved the success rate when deleted were concern for war, the actual closeness of the election, the political party most likely to do a better job, feelings toward the Democratic vice presidential candidate, and feelings toward conservatives. This provided the final model, which was satisfactory to the domain expert and yielded 65.6319% success (Figure 3). See Appendix A for details about the survey items used in the final model. This success rate may be compared to Lacy and Burden (1999) who produced a theoretically motivated multinomial probit model that included abstention as a vote choice and correctly classified 50.6% of respondents.

This final model contains 223 branches in the decision tree, which can be converted into 223 rules. These rules are actually composed of only 13 of the 17 attributes. Pruning of the tree by the C4.5 algorithm using the default confidence (25%) during tree construction obviated the need for four attributes (attitudes about abortion and unequal treatment and feelings toward liberals and gays) while increasing accuracy. This indicates that these four attributes are needed to build the more accurate decision tree, but are not needed for future prediction.

Finally, in order to test the robustness of the model, the final decision tree is used to classify test data. These records are formatted with the same attributes as the training data, but are records that were not used to build the model. The remaining presidential-year data were used as the test set (20,033 records). The final model proved to be correct for 64.3588% of these records (vs. 65.6319% for the training set), which indicates that the decision tree model is robust.

## Resulting Rules

The final decision tree can be converted into 223 rules. For example, Rule 12 indicates that respondents with the following characteristics will abstain 66.4% of the time:

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*Rule 12: IF affect toward Republican presidential candidate is neutral or slightly negative (0 or -1) AND affect toward Democratic presidential candidate is neutral or negative (≤0) AND party identification is weak Democrat AND interest in campaign is "not much" AND feeling for Democratic Party is positive (>55) AND affect toward the Republican Party is slightly negative to positive (>-3)*

*THEN individuals ABSTAIN.*

Importantly, though, Rule 11 indicates that respondents with these same characteristics except for greater negative affect toward the Republican candidate (<-2 vs. 0 or -1) will vote for the Democratic presidential candidate 49.4% of the time. This suggests, among other things, the potential to convert non-voters to Democratic voters through strategic considerations related to campaign message and tone such as negative or comparative advertising.

As one would expect, some rules are more useful than others. For example, Rule 203 is the rule that applies to the largest proportion of the electorate. While it is consistent with what most knowledgeable observers would expect and while it supports the validity of the data mining process, it is not strategically useful or informative. It states:

*Rule 203: IF affect toward Republican presidential candidate is positive (>0) AND party identification is weak or leaning Republican AND race is white*

*THEN individuals vote REPUBLICAN.*

At the same time, a number of rules, such as Rule 12 above, hold theoretical and strategic interest. For another example, Rule 200 addresses a segment of independent identifiers:

*Rule 200: IF affect toward Republican presidential candidate is positive (>0) AND affect toward Democratic presidential candidate is positive to negative (<3) AND party identification is independent AND interest in the campaign is "somewhat" AND affect toward the Democratic Party is negative to positive (>-3) AND affect toward the Republican Party is negative to positive (>-3)*

*THEN individuals vote REPUBLICAN.*

Importantly, though, Rule 200 implies a means to convert these independents from voting Republican to voting Democratic. When evaluated in conjunction with Rule 199, Rule 200 suggests that independent respondents with these same characteristics except for greater negative affect toward the Republican Party (≤-3 vs. >-3) will vote for the Democratic presidential candidate 47.3% of the time. Rules 199 and

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200 taken together suggest, of course, the potential to convert a Republican voter to a Democratic voter through strategic campaign efforts such as increasing negative affect toward the Republican Party.

In a similar vein, Rule 9 addresses a segment of non-voters. It states:

*Rule 9: IF affect toward Republican presidential candidate is neutral (0) AND affect toward Democratic presidential candidate is neutral (0) AND party identification is weak Democrat AND feeling toward the Democratic Party is slightly warm to cold (≤55)*

*THEN individuals ABSTAIN.*

Importantly, though, Rule 8 indicates that respondents with these same characteristics except for greater negative affect toward the Democratic candidate (<-1 vs. 0) will vote for the Republican presidential candidate 48.1% of the time. This suggests the potential to convert non-voters to Republican voters through strategic considerations related to campaign message and tone such as increasing negative affect toward the Democratic candidate.

In all, this brief review of a few of the rules suggests that campaign message and tone may be adjusted to influence certain groups of voters. Importantly, decision tree analysis identifies characteristic by characteristic who these voters are. This review, then, demonstrates some of the political science analysis that is possible with decision tree classification following data mining and the iterative attribute-elimination process.

## Data Mining Example 2: Living Arrangements of Maltreated Children

The National Survey on Child and Adolescent Well-Being (NSCAW) is a rich collection of data designed to represent children and families who enter the child welfare system. The NSCAW data set contains descriptive information on children, families, parents and caregivers, community environmental factors, and caseworker/intervention factors that may shed light on ways to improve outcomes for children who enter the child welfare system. NSCAW tracks service interventions over time to assess the effects of support on parenting in families of origin or in foster families. This two-stage stratified random sample was drawn from cases investigated or assessed by 97 child welfare agencies nationwide between October 1999 and December 2000. It includes data on children being served in and out of their homes that were collected in four waves, at entry of the children into the child welfare system and at

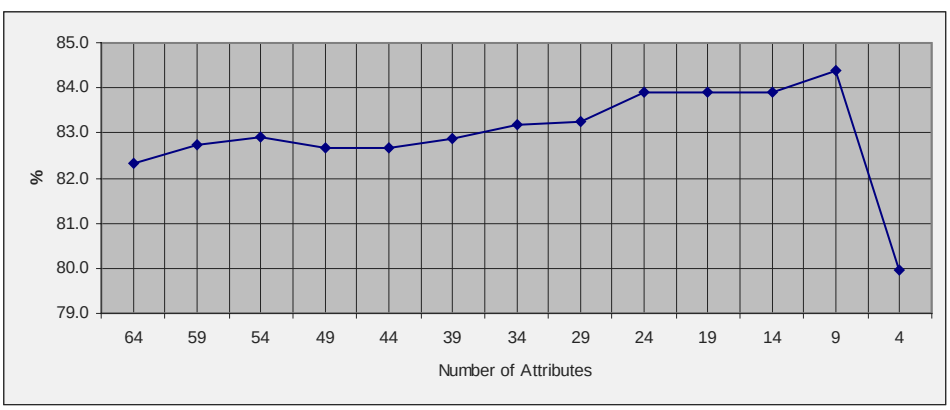
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12, 18, and 36 months post-entry. The data were collected in a manner designed to allow in-depth evaluations of subgroups of particular interest (e.g., young children or adolescents in foster care) while providing estimates for the nationwide population of children and families entering the system. These data are appropriate for analysis of child welfare outcomes such as the safety, permanence of care, and well-being of children, which are major child welfare outcomes of the state reviews required by federal law (U.S. Department of Health and Human Services, 2001). The data set includes approximately 5,500 children and more than 20,000 attributes. This data set provides an opportunity to employ the process on data that are theoretically (i.e., different disciplines) and structurally (i.e., different record and attribute structures) distinct from the ANES. Like the ANES data set, the NSCAW data set includes both discrete and continuous attributes. These data were cleaned and prepared in a process similar to that of the ANES data.

## Classification Mining the NSCAW Data

With the objective of finding attributes that together are indicative of a subject child's living arrangements after entering the child welfare system, we reviewed the attributes of the NSCAW data set. The domain expert selected a total of 63 attributes in addition to the goal attribute, which indicates child living arrangements (i.e., foster care, kin care, group home, other out-of-home placement, or in home with parents). The data were limited to records in the first wave of the four-wave panel; that is, the data collected when the children entered the child welfare system as opposed to post-entry data. This selection reflects our modest goal here to demonstrate the utility of data mining in the social sciences generally and in the field

*Figure 4. Success rate of living-arrangement classification as attributes are reduced*




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of social work more specifically. It also reflects the sparse nature of the NSCAW data. For instance, many of the psychological and behavioral measures are age specific (e.g., depression is not measured before age 7 and exposure to violence is not measured before age 4). In most instances, the sparse attributes are eliminated from the analysis. In cases where the attribute has significant theoretical import, a "not eligible" category is created and classified to be distinguished from records in which data are NULL because the researchers did not collect the data. Further, from the full set of 5,501 records, a subset of 3630 records was randomly selected for the training set with the remainder held in reserve as a test set.

Like the analysis of presidential vote choice, a decision tree was constructed using the WEKA implementation of the classification decision tree C4.5 algorithm with three-fold cross-validation (Witten et al., 2005). The information gain of the 63 attributes against the goal attribute was calculated, and then the attributes were ranked based on information gain from high to low. The attributes with the lowest information gain were systematically removed from the data and the C4.5 algorithm re-applied. The results, which are presented in Figure 4, show the greatest success of predicting living arrangements with nine attributes, which includes the dependent variable and the eight attributes with the most information gain.

The expert's first review of the tree led to the decision to force back into the nine-attribute data set a number of theoretically important attributes that the domain expert expected to improve the model's accuracy. These seven attributes include indicators about the child (overall health, level of depression, race/ethnicity, and living skills), the caregiver (age and employment status), and type of abuse or neglect. More specifically, race seems to play a role regarding which children enter foster care through the child welfare system (Roberts, 2002; Shireman, 2003), but the challenges that families entering the system face are much more complicated and often interrelated. Evidence suggests that adequate income and stable employment are critical to a family's ability to meet the needs of children (Crosson-Tower, 2002). In fact, parental income is a better predictor of a child's removal to out-of-home placement than is the severity of the alleged or substantiated maltreatment (Lindsey, 1994), which, of course, is also a significant predictor of removal (Roberts, 2002). As well, child functioning such as mental health and social and living skills can impact placement decisions, as all are part of child welfare assessments (Samantrai, 2004).

Adding all seven of the previous attributes to the most successful model failed to improve the success rate whether added in groups or individually. The most successful combination included adding child's overall health and living skills, type of abuse, and caregiver's employment status, which together enjoyed a success rate of 84.0771% but failed to surpass the success rate of the nine-attribute tree (84.3802%). The nine-attribute tree, then, serves as the final model. This final model contains 88 branches in the decision tree, which translates into 88 rules. These rules are composed of all eight of the predictor attributes (the goal attribute being the ninth), which are presented in Appendix B.

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Again to test the robustness of the model, the final decision tree is used to classify the test data. These records are formatted with the same attributes as the training data, but are records that were not used to build the model. The remaining living-arrangement data were used as the test set (1,871 records). The final model proved to be correct for 84.5003% of these records (vs. 84.3802% for the training set), which indicates that the decision tree model is robust.

## Resulting Rules

The final decision tree can be converted into 88 rules. For example, Rule 46 indicates that children with the following characteristics will be placed in home with their parent(s) 98.8% of the time:

*Rule 46: IF child is not eligible for inpatient treatment AND child's age is greater than one (> 1)*

*THEN child will be placed IN HOME.*

Rules 1 through 45 address children who are ineligible for inpatient treatment under age 1. These children's living arrangements depend on the age of the caregiver, how long the child has lived with the caregiver, family income, whether child welfare services have been received or not, and the number of children in the household. Put otherwise, children under age 1 are placed in a variety of settings based on a number of conditions. Once the child is over age 1, though, Rule 46 in indicates they are highly likely to be placed in home with their parent(s).

Rules 47 through 51 highlight the importance of income. Rule 47 states that children with the following characteristics will be placed in kin care with other relatives 50.7% of the time:

*Rule 47: IF child has had no inpatient treatments (= 0) AND child has lived with caregiver for 9 years or less (≤ 9) AND child's age is 9 years or less (≤ 9) AND total family income is $0-9,999 (0-9,999)*

*THEN child will be placed in KIN CARE.*

Taken together, this group of rules indicates an important income point at which living arrangements change. Children are placed in kin care with other relatives when family income is less than $30,000 (Rules 47, 49, and 50) 57.6% of the time but in foster care when family income is $30,000 or more (Rules 48 and 51) 63.8% of the time.

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A similar analysis of Rules 67 through 71 identifies an important caregiver age point. Rule 71 indicates:

*Rule 71: IF child has had no inpatient treatments (= 0) AND child has lived with caregiver for more than 9 years (> 9) AND child's age is greater than 9 years (> 9) AND caregiver age is 55 years or greater (≥ 55)*

*THEN child will be placed in KIN CARE.*

This group of rules suggests that children in the care of the oldest caregivers (55 or older) will be placed in kin care with other relatives (Rule 71) 64.2% of the time, while those with younger caregivers (younger than 55) will be placed in home with parents (Rules 67-70) 79.5% of the time.

Like the small set of rules presented from the vote choice decision tree, these rules identify important relationships, in this case regarding the living arrangements of maltreated children. More specifically, the rules suggest important inflection points pertaining to the living arrangements for these children. This brief review of a few of the rules again demonstrates the viability of decision tree classification following data mining and the iterative attribute-elimination process.

## Conclusion

Following a number of ethical, theoretical, and practical motivations to data mine social science data, we proposed and set out to demonstrate data mining in general and an iterative attribute-elimination process in particular as important analytical tools to exploit more fully some of the important data collected in the social sciences. We have demonstrated the iterative attribute-elimination data mining process of using domain knowledge and classification modeling to identify attributes that are useful for addressing nontrivial research issues in social science. By using this process, the respective experts discovered a set of attributes that is sufficiently small to be useful for making behavioral predictions, and, perhaps more importantly, to be useful for shedding light on some important social issues.

We used the American National Election Studies (ANES) and National Survey on Child and Adolescent Well-Being (NSCAW) data sets to identify a small number of attributes that effectively predict, respectively, the presidential vote choice of citizens and the living arrangements of maltreated children. The results suggest that the process is robust against theoretically and structurally distinct data sets: the ANES data set is used primarily in the field of political science and contains a large number of records (more than 47,000) and attributes (more than 900), while the

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NSCAW data set is used in the fields of social work and child welfare and contains many fewer records (5,501) but many more attributes (more than 20,000). In all, we believe the results of these analyses suggest that data mining in general and the iterative attribute-elimination process in particular are useful for more fully exploiting important but under-evaluated data collections and, importantly, for addressing some important questions in the social sciences.

## References

Abramson, P. R., Aldrich, J. H., & Rohde, D. W. (2003). *Change and continuity in the 2000 and 2002 elections*. Washington, DC: Congressional Quarterly Press.

American National Election Studies. (2005). *Center for political studies*. Ann Arbor, MI: University of Michigan.

Anand, S. S., Bell, D. A., & Hughes, J. G. (1995). The role of domain knowledge in data mining. *Proceedings of the 4th International Conference on Information and Knowledge Management* (pp. 37-43). Baltimore, MD.

Burton, M. J., & Shea, D. M. (2003). *Campaign mode: Strategic vision in congressional elections*. New York: Rowman and Littlefield.

Crosson-Tower, C. (2002). *Understanding child abuse and neglect*. Boston: Allyn and Bacon.

Deshpande, M., & Karypis, G. (2002). Using conjunction of attribute values for classification. *Proceedings of the 11th International Conference on Information and Knowledge Management* (pp. 356-364). McLean, VA.

Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). The KDD process for extracting useful knowledge from volumes of data. *Communications of the ACM* 99 (pp. 27-34).

Federal Bureau of Investigation. (2004). *Uniform crime reporting handbook* (Revised ed. 2004). U.S. Department of Justice. Washington, DC: Federal Bureau of Investigation.

Freitas, A. A. (2000). Understanding the crucial differences between classification and discovery of association rules—A position paper. *ACM SIGKDD Explorations Newsletter, 2*(1), 65-69.

Fu, X., & Wang, L. (2005). Data dimensionality reduction with application to improving classification performance and explaining concepts of data sets. *International Journal of Business Intelligence and Data Mining, 1*(1), 65-87.

Hofmann, M., & Tierney, B. (2003). The involvement of human resources in large scale data mining projects. *Proceedings of the 1st International Symposium*

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*on Information and Communication Technologies* (pp. 103-109). Dublin, Ireland.

Huang, W., Chen, H. C., & Cordes, D. (2004). Discovering causes for fatal car accidents: A distance-based data mining approach. *Proceedings of 2004 International Conference in Artificial Neural Network in Engineering (ANNIE)*, St. Louis, MO.

Jankowski, J. E. (2005). *Academic R&D doubled during past decade, reaching $40 billion in FY 2003*. National Science Foundation, Directorate for Social, Behavioral, and Economic Sciences. Washington, DC: National Science Foundation.

Jaroszewicz, S., & Simovici, D. A. (2004). Interestingness of frequent itemsets using bayesian networks as background knowledge. *Proceedings of the 10th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Seattle, WA.

Jeong, M., & Lee, D. (2005). Improving classification accuracy of decision trees for different abstraction levels of data. *International Journal of Data Warehousing and Mining, 1*(3), 1-14.

Lacy, D., & Burden, B. C. (1999). The vote-stealing and turnout effects of Ross Perot in the 1992 U.S. Presidential Election. *American Journal of Political Science, 43*(1), 233-55.

Lindsey, D. (1994). *The welfare of children*. New York: Oxford University Press.

McCarthy, V. (1997). Strike it rich. *Datamation*, *43*(2), 44-50.

Mitchell, T. M. (1997). Does machine learning really work? *AI Magazine*, *18*(3), 11-20.

Nadeau, R., & Lewis-Beck, M. S. (2001). National economic voting in U.S. Presidential Elections. *Journal of Politics*, *63*(1), 159-181.

National Science Foundation. (2005). *National Science Foundation FY 2005 Performance Highlights*. Retrieved April 28, 2006, from http://www.nsf.gov/pubs/2006/nsf0602/nsf0602.jsp

Nicholson, S. (2006). Proof in the pattern. *Library Journal*, *131*, 4-6.

National Survey of Child and Adolescent Well-Being (NSCAW) (1997-2010). U.S. Department of Health and Human Services; Administration for Children and Families; Office of Planning, Research, and Evaluation.

NORC. (2006). *GSS study description*. National Organization for Research at the University of Chicago. Retrieved April 27, 2006, from http://www.norc.uchicago.edu/projects/gensoc1.asp

Padmanabhan, B., & Tuzhilin, A. (2000). Small is beautiful: Discovering the minimal set of unexpected patterns. *Proceedings of the 6th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 54-63). Boston.

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Roberts, D. (2002). *Shattered bonds: The color of child welfare*. New York: Basic Books.

Rosenthal, R. (1994). Science and ethics in conducting, analyzing, and reporting psychological research. *Psychological Science*, *5*(3), 127-34.

Romero, C. , & Ventura, S. (2006). Educational data mining: A survey from 1995 to 2005. *Expert Systems with Applications*, in press.

Samantrai, K. (2004). *Culturally competent public child welfare practice.* Pacific Grove, CA: Brooks/Cole.

Scholz, M. (2005). Sampling-based sequential subgroup mining. *Proceedings of the 11th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Chicago.

Scime, A., & Murray, G. R. (forthcoming). Vote prediction by iterative domain knowledge and attribute elimination. *International Journal of Business Intelligence and Data Mining* (Forthcoming).

Shapiro, I. (2002). Problems, methods, and theories in the study of politics, or what's wrong with political science and what to do about it. *Political Theory*, *30*(4), 596-619.

Shireman, J. (2003). *Critical issues in child welfare.* New York: Columbia University Press.

Spangler, W. E. (2003). Using data mining to profile tv viewers. *Communications of the ACM*, *46*(12), 66-72.

Taft, M., Krishnan, R., Hornick, M., Mukhin, D., Tang, G., & Thomas, S. (2003). *Oracle data mining concepts.* Oracle, Redwood City, CA.

U.S. Department of Health and Human Services, Administration on Children, Youth, and Families. (2001). *Safety, permanence, well-being: Child welfare outcomes 2001 Annual Report*. Washington, National Clearinghouse on Child Abuse and Neglect Information.

Wine, J. S., Cominole, M. B., Wheeless, S., Dudley, K., & Franklin, J. (2005). *1993/03 baccalaureate and beyond longitudinal study (B&B:93/03) methodology report*. (NCES 2006–166). U.S. Department of Education. Washington, DC: National Center for Education Statistics.

Whiting, R. (2006). What's next? *CMPnetAsia*, 31 May 2006.

Witten, I. H., & Frank, E. (2005). *Data mining: Practical machine learning tools and techniques* (2nd ed.). San Francisco, CA: Morgan Kaufmann.

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# Appendix A: ANES Survey Items

Discrete-Valued Questions (attribute names):

1. What is the highest degree that you have earned? (AEDUC)

A1. 8 grades or less
A2. 9-12 grades, no diploma/equivalency
A3. 12 grades, diploma or equivalency
A4. 12 grades, diploma or equivalency plus non-academic training
A5. Some college, no degree; junior/community college level degree (AA degree)
A6. BA level degrees
A7. Advanced degrees including LLB

2. Some people don't pay much attention to political campaigns. How about you, would you say that you have been/were very much interested, somewhat interested, or not much interested in the political campaigns this year? (AINTELECT)

A1. Not much interested
A2. Somewhat interested
A3. Very much interested

3. Some people seem to follow what is going on in government and public affairs most of the time, whether there is an election going on or not. Others aren't that interested. Would you say you follow what is going on in government and public affairs most of the time, some of the time, only now and then, or hardly at all? (AINTPUBAFF)

A1. Hardly at all
A2. Only now and then
A3. Some of the time
A4. Most of the time

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4. How do you identify yourself in terms of political parties? (APID)

A-3. Strong Republican
A-2. Weak or leaning Republican
A0. Independent
A2. Weak or leaning Democrat
A3. Strong Democrat

5. In addition to being American, what do you consider your main ethnic group or nationality group? (ARACE)

A1. White
A2. Black
A3. Asian
A4. Native American
A5. Hispanic
A7. Other

6. Who do you think will be elected President in November? (AWHOELECT)

A1. Democratic candidate
A2. Republican candidate
A7. Other candidate

Continuous-Valued Questions:

Feeling Thermometer Questions: A measure of feelings. Ratings between 50 and 100 degrees mean a favorably and warm feeling; ratings between 0 and 50 degrees mean the respondent does not feel favorably. The 50 degree mark is used if the respondent does not feel particularly warm or cold:

7. Feeling about Democratic presidential candidate. (DEMTHERM)
8. Feeling about Republican presidential candidate. (REPTHERM)
9. Feeling about Republican vice presidential candidate. (REPVPTHERM)

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Affect Questions: The number of "likes" mentioned by the respondent minus the number of "dislikes" mentioned:

10. Affect toward the Democratic Party. (AFFDEM)
11. Affect toward Democratic presidential candidate. (AFFDEMCAND)
12. Affect toward Republican Party. (AFFREP)
13. Affect toward Republican presidential candidate. (AFFREPCAND)

Goal Attribute (discrete valued):

14. Summary variable indicating the respondent's presidential vote choice or abstention. (ADEPVARVOTEWHO)

A1. Democrat
A2. Republican
A3. Major third-party candidate
A4. Other
A7. Did not vote or voted but not for president

## Appendix B: NSCAW Questionnaire Items

Discrete-Valued Items (attribute names):

1. Number of children in household. (AHHDNOCH)

A1. 1 child
A2. 2 children
A3. 3 children
A4. 4 children
A5. $\geq$ 5 children

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2. Number of inpatient treatments of child. (ARCS_INNM)

A1. 0
A2. 1
A3. >1
A99. not eligible for inpatient treatment

3. Child welfare services received. (ASERVC)

A1. Yes
A2. No

4. Tot family income ($) per year. (ARIN2A)

A1. 0-9,999
A2. 10,000-19,999
A3. 20,000-29,000
A4. 30,000-39,000
A5. 40,000 and greater

5. Indicator of Substantiated Maltreatment. (ASUBST)

A0. No
A1. Yes

6. Caregiver age. (ARCGVRAGE)

A1. ≤ 25 years
A2. 26-35 years
A3. 36-45 years
A4. 46-55 years
A5. >55 years

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Continuous-Valued Items:

7. How long child lived with caregiver (in years). (YCH18A)
8. Child age (in years). (CHDAGEY)

Goal Attribute (discrete valued):

9. Derived measure indicating child's living arrangements. (ACHDOOHPL)

A1. Foster home
A2. Kin care setting
A3. Group home/residential program
A4. Other out-of-home arrangement
A5. In home arrangement

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## Chapter XIV

# A Machine Learning Approach for One-Stop Learning

Marco A. Alvarez, Utah State University, USA
SeungJin Lim, Utah State University, USA

## Abstract

*Current search engines impose an overhead to motivated students and Internet users who employ the Web as a valuable resource for education. The user, searching for good educational materials for a technical subject, often spends extra time to filter irrelevant pages or ends up with commercial advertisements. It would be ideal if, given a technical subject by user who is educationally motivated, suitable materials with respect to the given subject are automatically identified by an affordable machine processing of the recommendation set returned by a search engine for the subject. In this scenario, the user can save a significant amount of time in filtering out less useful Web pages, and subsequently the user's learning goal on the subject can be achieved more efficiently without clicking through numerous pages. This type of convenient learning is called one-stop learning (OSL). In this chapter, the contributions made by Lim and Ko (2006) for OSL are redefined and modeled using machine learning algorithms. Four selected supervised learning algorithms:*

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*support vector machine (SVM), AdaBoost, Naive Bayes, and Neural Networks are evaluated using the same data used in Lim et al. (2006). The results presented in this chapter are promising, where the highest precision (98.9%) and overall accuracy (96.7%) obtained by using SVM is superior to the results presented by Lim et al. (2006). Furthermore, the machine learning approach presented here demonstrates that the small set of features used to represent each Web page yields a good solution for the OSL problem.*

# A Machine Learning Approach for One-Stop Learning

Using the Web, a global repository of information, for educational purposes requires more accurate and automated tools than general-purpose search engines. Innovative tools should support the learning experience and focus the attention of the learner on his or her desired target subject. A typical learner would be interested in going directly to the point and learn without spending time with useless or non-informative pages. In this context, harvesting the Web using current search engines and technologies, however, looking for concepts, subjects, or general information usually imposes significant overhead that is denoted when the user spends time in filtering irrelevant pages or when he or she is simply distracted with advertisements, latest news, or attractive but not suitable Web sites for learning.

Before the advent of the Web, students and occasional learners studied new subjects by reading books or well-known articles in which they could find all the required information. Certainly, these primary sources of information can be considered adequate and sufficient for learning the subject when the learner satisfies his or her aspirations with them. In most cases, there is no need to look for additional resources for the same subject. This conventional learning strategy is called one-stop learning (OSL) in Lim and Ko (2005). On the other hand, when considering the Web as a repository for learning, the learners very often rely on available general-purpose search engines like Google, Yahoo, or Microsoft Live in finding suitable materials for OSL. Here, it must be emphasized that these search engines were not designed with the specific goal of assisting educational activities. The use of such engines for one-stop learning needs to be revisited in order to optimize the time that learners spend searching for self-contained sources of knowledge/information. One clear advantage of existing search engines is the fact that they maintain billions of updated pages already indexed for fast search and retrieval. Previously proposed strategies for OSL using the Web take advantage of the results returned by search engines (Lim et al., 2005, 2006). The major motivation in this chapter is to present a machine learning approach for the OSL problem making use of existing search

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engines. This approach can be considered as an extension of the previous work presented in Lim et al. (2005, 2006).

There are numerous advantages when using the Web for learning, which allows fast, ubiquitous, and controlled-by-user learning experiences. Paradoxically, the high degree of freedom that online hypermedia environments offer to the learners can turn these advantages into inconveniences, for example, distractions from less informative pages at hand or time spent on clicking and eventually navigating through such pages. With the intention of improving the learning experience of Web users, this chapter evaluates the application of selected machine learning algorithms for automatic Web page classification into suitable pages or unsuitable pages, according to their suitability for OSL. The authors believe that the OSL problem can be modeled as a supervised learning problem where a classifier can be trained to discriminate suitable pages, suitable for learning a given subject. Furthermore, classifiers with probabilistic output can naturally be used for ranking purposes, which is essential for showing the results to the learner. If the classifier being used only provides hard decisions, binary output in this case, then it would be necessary to incorporate a ranking formula among the true positives.

The main challenges in proposing tools for OSL purposes rely on the necessity for accurate responses in real time. In fact, notice that due to the subjective boundary between suitable and unsuitable pages for OSL, which varies from person to person, it is accepted to sacrifice minimal amounts of effectiveness in exchange for efficiency. Usually, classification tasks require a pre-processing stage where discriminant features are extracted from the example cases, hypermedia documents in the context of this chapter. To meet the real time requirement, this process must be as efficient as possible, however, extracting good features often involves more sophisticated processing. The approach presented here considers the concepts and the low cost formulas introduced in Lim et al. (2006), using them as Web page features. In this chapter, it is shown that a subset of such formulas, despite their simplicity, is enough to train an acceptable model for the set of suitable pages. Moreover, the trained model is independent of the subject chosen by the user, enabling its use in Web systems for OSL.

A similar problem is the automatic Web page categorization (Fang, Mikroyannidis, & Theodoulidis, 2006), which has been actively studied in recent years. Roughly speaking, automatic Web page categorization is an instance of a more abstract problem, text categorization, which is used in many contexts, "ranging from automatic document indexing based on a controlled vocabulary, to document filtering, automated metadata generation, word sense disambiguation, population of hierarchical catalogues of Web resources, and in general any application requiring document organization or selective and adaptive document dispatching" (Sebastiani, 2002). In general, approaches for automatic Web page categorization consider either structural patterns or content patterns. For example, the problem of discriminating "call for papers" and "personal home pages" (Glover et al., 2001) can be solved looking for

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features that describe structural patterns inside the Web pages, like links or tags. On the other hand, a classifier trained to label Web pages automatically according to their pertinence to one of a predefined set of thematic categories needs features extracted from the Web page contents. Making a parallel with the OSL problem, using the machine learning approach, one is interested in classifiers that are capable of making accurate predictions with any subject. This deliberately implies that structural features seem to be enough for building an accurate classifier.

Another issue is related with the scope of each class. In general, machine learning approaches for Web page categorization assume a multi-class framework where each class is represented and well-defined by a sufficient number of examples. Here, generally it is assumed that each category has a well-defined scope and respective invariant features can be identified. Conversely, the one-class binary classification problem introduces a different scenario, because the classifier must be able to discriminate one class from all the others, one against the world. In Yu, Han, and Chang (2004), the reader can refer to a recent proposal for this problem. The machine learning approach for OSL is a one-class binary classification problem, where the main focus is on learning to discriminate the suitable Web pages, suitable for OSL, from all the others. This task requires special attention when collecting positive and negative examples. For a given subject, one needs to collect suitable Web pages (positive training examples) and a set of unsuitable Web pages (negative training examples). The authors consider this process difficult because of the small number of positive examples encountered among all the Web pages returned by a search engine after a query on a desired subject.

Considering the context described so far, the focus of this chapter is on the challenge of achieving as much classification effectiveness as possible whilst maintaining a fast Web page pre-processing, enabling real time response. In summary, the contributions of this chapter are as follows:

- A novel machine learning modeling of the one-stop learning problem. Where a small number of features (6) are enough to distinguish suitable Web resources for a given subject, in contrast to the two stage process described in Lim et al. (2006) that involves more calculations;
- An improved effectiveness when compared with the previous proposed method for OSL (Lim et al., 2006). Using supervised learning algorithms it is possible to achieve higher precision (98.9%) and overall accuracy (96.7%) in automatically finding good Web resources for one stop learning. The proposed framework using classifiers makes use of a subset of the simple and efficient properties and formulas proposed in (Lim et al., 2006), to make the prior training and testing steps faster in a machine learning approach.

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The rest of this chapter is organized as follows: The related work section discusses related work for Web page categorization and one-stop learning. The complete description of the proposed approach comes in the learning suitable pages section. The experimental settings and results obtained are presented and discussed in the empirical evaluation section. Finally, conclusions and directions for future work are given in the conclusion and future work section.

## Related Work

An overview of related work is presented in this section, focusing on proposed approaches for post-processing of results returned by Web search engines and also reviewing proposals for automatic Web page categorization/classification and previous work on OSL. It is not the purpose of this section to make an extensive review of the state-of-the-art in clustering or classification of Web pages, instead, a synopsis of similar work is presented.

Various approaches have been proposed to mine useful knowledge from search results. Cluster analysis (Kummamuru, Lotlikar, Roy, Singal, & Krishnapuram, 2004; Wang & Kitsuregawa, 2002; Zhang & Liu, 2004), refinement (Zhou, Lou, Yuan, Ng, Wang, & Shi, 2003), and concept hierarchies (Wu, Shankar, & Chen, 2003) of search results are a few examples and somewhat related to the problem of automatically finding the most suitable pages for one-stop learning. However, to the best of the authors' knowledge, until now the OSL problem was only addressed in Lim et al. (2005, 2006).

In Haveliwala (2002), an approach was presented that attempts to yield more accurate, topic-specific search results from generic PageRank (Brin & Page, 1998) queries by biasing the PageRank computation. The computation is done making use of a small number of representative basic topics taken from the open directory[1]. This approach, however, may not be helpful for topics not present within the set of pre-compiled basic topics. In contrast, the goal of the approach presented in this chapter is to make a post-processing of search results such that no prior information on the subject of interest is required.

In addition, there exist a number of approaches to automatically cluster search results. Clustering approaches could be categorized into two wide categories: term-based clustering and link-based clustering. The bottom-line of clustering-based approaches is the attempt to cluster search results into semantically related groups. Wang et al. (2002) presented a clustering algorithm for Web pages. Their solution is based on a combination of term and link features, in other words, Web pages are processed in order to identify common terms shared among them, moreover, the co-citation and coupling analysis is performed by observing the out-links (from

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the Web page) and in-links (to the Web page). Later Zhang et al. (2004) presented an incremental clustering algorithm that clusters the results from a search engine into subgroups and assigns to each group a short number of keywords. Users are allowed to choose particular groups to further examine the Web pages contained in it, instead of considering the whole number of search results.

Constructing concept or topic hierarchies for search results facilitates navigation and browsing of Web search results. The work presented in Wu et al. (2003) and Kummamuru et al. (2004) are two examples of automatic construction of topic hierarchies. The former uses a co-occurrence based classification technique while the latter approach is based on a hierarchical monothetic clustering algorithm. The word monothetic refers to the fact that a document is assigned to a cluster based on a single feature, while the others are polythetic. Hierarchies can be useful in facilitating navigation and browsing tasks as well as in building the path to finding authoritative Web pages even when they are low-ranked.

However, notice that despite the advantages of current clustering algorithms incarnated in either mere grouping or more sophisticated hierarchical organization of Web pages, these algorithms cannot be applied in a straight sense to solve the OSL problem. The straight use of these approaches for OSL is not appropriate because mere grouping of Web pages still requires the user further examine a subset of Web pages manually to find the most suitable ones. Furthermore, hierarchical organization suffers the same limitation, implying in an overhead to the user forced to browse through the concept/topic hierarchy. Nonetheless, it is worth to consider that previous clustering approaches propose some direction on the relevant features to be considered in the design of a machine learning approach for OSL. The reader can refer to Crabtree, Andreae, and Gao (2006) for a recent proposal and a kind review of related work in clustering of Web pages.

On the other hand, Web page categorization/classification is also very related to the machine learning approach for OSL. Generally speaking, categorization refers to the automatic assignment of category labels to Web pages, and classification usually refers to the same problem. In the context of this chapter, the term classification is adopted because one is interested in discriminating the suitable pages (one class) from all the others, instead of categorizing Web pages into pre-defined categories. In any case, the design of Web page classification systems has common characteristics one another regardless of the final purpose of the classifiers.

Bear in mind that hypertext introduces new research challenges for text classification. Hyperlinks, HTML tags, and metadata all provide important information for classification purposes. According to Yang, Slattery, and Ghani (2002), "how to appropriately represent that information and automatically learn statistical patterns for solving hypertext classification problems is an open question." In fact, Yang et al. (2002) proposes five hypertext regularities that can hold (or not) in a particular application domain. As stated in Fang et al. (2006), existing Web page classification

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approaches can be grouped into three categories according to the type of features used: the hypertext approach, link analysis approach, and neighborhood category approach. In the hypertext approach, the features considered are extracted from content (text) and context (e.g., anchor text). The link analysis approach is based on text components and linkage among them and the neighborhood category approach exploits the category assignment of already classified neighboring pages. For a more comprehensive explanation of the three approaches and a survey of the proposed algorithms for each approach, the reader can refer to Fang et al. (2006). For a broader sense in machine learning algorithms for text categorization/classification, the reader can refer to Sebastiani (2002).

Clustering and classification are established proposals for post-processing the results returned by Web search engines. This trend has motivated the use of a machine learning approach for the OSL problem. Previous work on OSL was presented in Lim et al. (2006), where they proposed the use of a two-stage algorithm. The first stage is devoted to the pruning of unlikely good resources for OSL contained in a recommendation set (results from a search engine) for a given query subject. The pruning is done by using an efficient statistical algorithm based on universal characteristics of Web pages, such as number of words or number of tags. The second stage considers the scoring and subsequent ranking of the remaining pages. Three different approaches are proposed for the second stage: (1) a term distribution (OSL-TD) based algorithm, (2) a term frequency (OSL-TF) based algorithm, and (3) a term weight (OSL-TW) based algorithm. The highest average precision reported by Lim et al. is 87.1% for OSL-TW using the Google search engine. In the present chapter, a subset of the formulas used by Lim et al. is selected to be evaluated using machine learning algorithms. The main difference between the proposed approach and the Lim et al.'s method relies on the use of a one stage solution using a trained classifier to identify the suitable pages for OSL from a recommendation set.

## Learning Suitable Pages

Considering the assumption that search results returned by conventional search engines like Google or Yahoo often include useless or distracting pages that are not suitable for OSL, one is interested in designing a method that can be used to discriminate which are the suitable pages among all the Web pages returned by a simple query posted to a search engine. This method has the main purpose of improving the learning experience of users when searching the Web for arbitrary target topics.

Machine learning techniques during the last 20 years have been successfully used to solve significant real-world applications (Mitchell, 2006). In this chapter, the authors

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propose the use of machine learning for solving the OSL problem. Discovering highly suitable resources for OSL can be nicely modeled as a supervised learning problem, where a classifier is trained to learn the mapping function between a set of features describing a particular Web page and their suitability for OSL. This is the central point on the approach proposed here. In addition, as a posterior step, with the intention of optimizing the results presented to the user, the set of Web pages labeled as suitable by a classifier can be ranked in two different ways: (1) using the probability $p(C|D)$, that is the probability that a given document $D$ belongs to a given class $C$, usually available with classifiers with probabilistic output, and (2) ranking the suitable results given by a classifier with hard decisions (-1 or +1) using the ranking formula introduced in Lim et al. (2006). Recall that the main goal of the experiments conducted here is to maximize the performance of the classifier rather than find the most suitable ranking formula.

The goal of this section is to present how the OSL problem can be modeled as a supervised learning problem. For this purpose, initially the basic definitions and terminology are presented, followed by an overview of the proposed approach together with a detailed explanation of every single stage defined for training and testing purposes.

## Terminology and Basic Concepts

The proposed approach relies on search results returned by existing search engines, Google and Yahoo in particular. In fact, only the top $N$ items from the recommendation set, defined below, were considered for the experiments. Once the recommendation set is known, a previously trained classifier can be used to identify the suitable pages. The following definitions and concepts are significant for the OSL problem.

**Definition.** Let $S$ be a subject and $E$ a search engine. A recommendation set $R$ is the set of pages returned by $E$ after a query for the subject $S$.

**Definition.** A Web page is called *suitable* if it provides a relatively comprehensive discussion on the requested subject.

The identification of suitable Web pages is critical in OSL, because an OSL system is expected to help focused learners during their learning experiences by presenting to the user a minimal set of suitable pages. Focused learners are characterized as the users looking for technical subjects for an educational purpose. Bring to mind that the main motivation for OSL is to enable the fast and efficient access to Web resources for educational purposes. Technical subjects are more frequent than general ones when the bottom line is online learning. An example for a general subject is "travel" whereas "wavelets" is an example for a technical subject.

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Regarding the supervised learning terminology, the following definitions will be considered in the context of OSL. A labeled instance is a pair $(\bar{x}, y)$ where $\bar{x}$ is a vector in the *d*-dimensional space *X*. The vector $\bar{x}$ represents the feature vector with *d* attributes extracted from a given Web page and $y \in \{false, true\}$ is the class label associated with $\bar{x}$ for a given instance, where *true* stands for suitable pages and *false* for all the others. A classifier is a mapping function from *X* to *{false, true}*. The classifier is induced through a training process from an input dataset, which contains a number *n* of labeled examples $(\bar{x}, y)$ for $1 \le i \le n$.

In order to achieve significant results, machine learning techniques, in particular supervised learning techniques, were chosen to automatically identify suitable Web pages in *R*. Naturally, Web pages are central components for the problem, consequently the first issue to address is how to identify the invariants for suitable pages. The approach proposed here is based on observations made previously in Lim et al. (2005, 2006), where practical measures were defined by solely analyzing the internal structure for a given Web page.

When applying machine learning algorithms raises the question of how to select the right learning algorithm to use. In this chapter, four different algorithms are used and their empirical results are compared according to their effectiveness on classifying Web pages as suitable or not. Two of them give probability estimations for the membership of each test case (here the ranking is intrinsic) and the other two classifiers provide hard decisions (suitable page or not) which needs a posterior calculation for ranking.

In deciding which type of information should be used during the learning process, there are several features that can contribute to determine the membership of a page: number of words, number of tags, number of links, number of words in links, etc. These features can be easily extracted during a pre-processing of the page, which precedes the learning phase.

## Feature Vectors

In order to build a classifier, it is needed to determine the set of useful features for deciding the membership of a given page. One important fact is that the set of features must be generic enough to be used with Web pages from totally different domains. Also, the set of features must be enough to easily identify (filter) what are the unsuitable pages, which has been shown achievable with good results in Lim et al. (2006). Surely, all the features proposed here are somewhat borrowed from Lim et al. (2006). The set of features considered *a priori* consists of nine features described next:

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1. **Number of words (NW):** A good candidate should have a considerable amount of text. Here the total number of visible words is measured including stop words like "a," "the," "to," etc., which gives a notion for the size. The reader should be advised that this number is different (larger) from the number of distinct words;
2. **Property 1 (P1):** This feature captures the ratio between the number of words and the number of tags in a page. Given a page *p*, property 1 stands for $N_w/N_t$, where $N_w$ and $N_t$ are respectively the total number of words and the total number of HTML tags. It is expected that suitable pages exhibit high values for property 1. Assuming that *p* is not a suitable page, then most probably it is a hub page or a commercial page containing many links, scripts, meta information, and advertisement with a large number of HTML tags;
3. **Property 2 (P2):** The meaning of property 2 is given by the observation that a suitable page has a relatively small number of links compared with the text size. Given a page *p*, the value of property 2 is $N_w/(N_{in} + N_{out})$ where $N_{in}$ is the number of words occurring in text between the start and end tags of links referencing to resources residing in the same domain, and $N_{out}$ follows a similar definition for links referencing resources at different domains. Notice that these measures include stop words. It is expected to have high values for suitable pages. If *p* contains many links to outside resources, then can be assumed that *p* is recommending the learner to refer to other pages to learn about the subject;
4. **Property 3 (P3):** Here, it is measured the ratio between the number of links referencing the same domain and those referencing outside domains. Given a page *p*, the value of property 3 is $N_{in}/N_{out}$. The rationale is that suitable pages discourage the dense use of links referencing other domains. Here it is noticed that links to the same domain are likely used for descriptive purposes, referencing to arguments or illustrations in the same page or even in the same Website. On the other hand, links referencing outside domains are more likely used for navigational purposes;
5. **Distinct Descriptive Words (DDW):** It is possible to categorize words according to their roles into descriptive or navigational words. If the word *w* occurs in page *p* and *w* is not present in the text of links to outside domains, it is assumed that the intention of *p* is to describe *S* using *w*, thus *w* is called *descriptive word*. This type of words strengthen the suitability of *p* with respect to *S*. The value of this feature is the number of distinct descriptive words;
6. **Distinct Navigational Words (DNW):** On the other hand, a word *w* is called *navigational* if *w* occurs as text in links to outside domains. The intention of *p* with respect to *w* is to confer authoritativeness to other pages motivating the learner to visit the linked pages. The value of this feature is the number of distinct navigational words;

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7. **Distribution 1 (D1):** This feature represent the distribution of descriptive words in a page *p*. It is defined as the number of exclusively descriptive words in *p*, which is the set of words that are descriptive and not navigational. This feature can be defined using:

$$dist1(p_i) = \sum_{j=1}^{|d_i|} \Pr(d_{i,j} \mid V)$$

where $\Pr(d_{i,j} \mid V)$ is 1 if the exclusively descriptive word $d_{i,j} \in p_i$ occurs in *V*, the global (domain) vocabulary, and 0 otherwise. Note that $d_i$ is the set of exclusive descriptive words in $p_i$. For suitable pages it is expected to have the highest values in this feature, since it is a measure of how descriptive is the page using the global vocabulary;

8. **Distribution 2 (D2):** In a similar manner it is possible to define $dist2(p_i)$, which represents the distribution of exclusively descriptive words that are used as navigational words in all the pages in *R*. This feature is defined by:

$$dist2(p_i) = \sum_{j=1}^{|d_i|} \Pr(d_{i,j} \mid N)$$

where $\Pr(d_{i,j} \mid N)$ is 1 if the exclusively descriptive word $d_{i,j} \in p_i$ occurs in *N*, the global set of exclusively navigational words, and 0 otherwise;

9. **Distribution 3 (D3):** The last feature represents the distribution of exclusively navigational words. It is defined by:

$$dist3(p_i) = \sum_{j=1}^{|n_i|} \Pr(n_{i,j} \mid N)$$

where $\Pr(n_{i,j} \mid N)$ is 1 if the exclusively navigational word $n_{i,j} \in p_i$ occurs in *N*, the global set of navigational words, and 0 otherwise. Note that $n_i$ is the set of exclusive navigational words.

## Training Examples

Considering all the features described previously, the training examples are defined in this section. Each training example represents a Web page *p* from the recommendation set *R* for a given subject *S* and is defined as a row vector $\overline{x}$ composed by nine

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*Table 1. Sample training pages for the subject "data mining" returned by Google*

| NW | P1 | P2 | P3 | DNW | DDW | D1 | D2 | D3 | Class |
|---|---|---|---|---|---|---|---|---|---|
| 173 | 0.84 | 1.97 | 16.60 | 41 | 4 | 41 | 33 | 4 | False |
| 342 | 1.21 | 14.25 | 23 | 151 | 0 | 151 | 113 | 0 | False |
| 3993 | 12.10 | 399.30 | 9 | 858 | 0 | 858 | 599 | 0 | True |
| 260 | 2.03 | 3.88 | 0.38 | 87 | 34 | 79 | 58 | 26 | False |

features and a class label. The class label is either *true* or *false*, indicating whether an example is suitable (positive example) or not (negative example) respectively. The features are extracted processing the HTML code of *p*, where NW, P1, P2, P3, DNW, DDW are entirely measured from *p* and the remaining D1, D2, and D3 are calculated using additional vocabulary measures (refer to the feature description) from the whole recommendation set of pages. In Table 1 it is possible to observe some of the training examples about the topic "data mining" returned by Google. After training a classifier to learn the training data, the test examples also can be created directly from the recommendation set returned by a search engine using the same approach.

## Classifiers

Having a considerable amount of training examples, it is desired to train a classifier to learn the mapping function between the set of features and the suitability of a given page. A machine learning approach for this problem, involves the selection of a supervised learning algorithm, with the intention of maximizing the effectiveness and efficiency simultaneously when possible. There are numerous available algorithms for supervised learning, from which four were selected to validate their performance in solving the OSL problem. These are briefly described in the following lines.

## Support Vector Machine

Support vector machines (SVM) have become one of the most popular classification algorithms. SVMs are classifiers based on the maximum margin between classes. By maximizing the separation of classes in the feature space, it is expected to improve the generalization capability of the classifier. Conceived as linear classifiers, SVMs can also work with non-linearly separable datasets by mapping the input feature space into higher dimensions expecting that the same data set become linearly separable in the higher space. Due to the unquestionable success of the SVM classifier

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in the academic community and because SVMs are naturally designed for binary classification problems like the OSL problem, the authors were motivated to select this algorithm for testing its performance on the OSL problem. However, the reader should notice the "hard-decision" nature of the classifier's output, -1 or +1, which does not allow the automatic ranking of pages. This ranking might be performed in a posterior step using the formula given in (Lim et al., 2006) or, if desired, using a different one. More details about SVMs can be found in Burges, 1998).

## AdaBoost

boosting is a general way to improve the accuracy of any given learning algorithm. The basic idea behind boosting refers to a general method of producing very accurate predictions by combining moderately inaccurate (weak) classifiers. AdaBoost is an algorithm that calls a given weak learning algorithm repeatedly, where at each step the weights of incorrectly classified examples are increased in order to force the weak learner to focus on the hard examples. The reader can refer to Freund and Schapire (1999) for a detailed description of AdaBoost. The main motivation to the use of a meta-classifier as AdaBoost is given by the fact that many previous papers have shown stellar performance of AdaBoost with several datasets (Bauer & Kohavi, 1999). In fact, Bauer and Kohavi (1999) show a more realistic view of the performance improvement one can expect. Regarding the weak learner, several algorithms were empirically tested with AdaBoost using the OSL dataset, the highest and most stable performance was achieved by the J48 Decision Tree (DT) algorithm (the Java implementation of C4.5 integrated in Weka[1]). Beyond the good performance of J48 with AdaBoost, the motivation for decision trees is driven by the following characteristics: (1) DT are easy to understand and convert into production rules, allowing fast evaluation of test examples, (2) There are no *a priori* assumptions about the nature of the data. This algorithm can be used as a binary classifier with hard decisions (-1 or +1), again, it will be necessary to rank the true positives.

## Naive Bayes

The Naive Bayes (NB) classifier is a simple but effective text classification algorithm (Lewis, 1998). NB computes the posteriori probabilities of classes, using estimations from the labeled training data. Here, it is assumed that the features are independent of each other. The motivation for testing the NB performance is driven by the fast calculation of the posterior probability $\Pr(y \mid \bar{x})$ using the Bayes rule. NB classifiers hold good popularity and perform surprisingly well, even when the independent assumption does not hold. Once the output of NB is probabilistic, it

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also can be directly used to rank the true positives according to their suitability. The user would obtain a fast ranking of the most suitable Web pages for learning a given topic of interest. For more details about the NB algorithm, the reader can refer to McCallum and Nigam (1998).

## Backpropagation

Artificial neural networks (ANN) have been successfully used to solve classification problems in several domains, specifically the Backpropagation algorithm is very often the favorite to train feedforward neural networks. An ANN is composed by a number of interconnected artificial neurons that process information using a computational model inspired in the human brain behavior. The Backpropagation algorithm, in particular, adaptively changes the internal network free parameters based on external stimulus (input examples from the dataset). After trained, a neural network can make predictions about the membership of every test example. Feedforward networks trained with the Backpropagation algorithm suffers from the high number of parameters that need to be tuned, like learning rate, number of neurons, momentum rate, etc. However, the motivations to select this algorithm arise after observing that they have been used to solve problems in different domains, moreover, the output can be directly used for ranking purposes. An extensive description of ANNs can be found in Abdi (1994).

# Empirical Evaluation

Having in mind the goal of evaluating and comparing the results obtained by the four chosen classifiers, and furthermore, contrasting the results obtained by machine learning algorithms with the previous approach (Lim et al., 2006), this section describes the details about the experiments and their respective results.

*Table 2. Ten technical subjects chosen for training and testing purposes*

| | |
|---|---|
| Data Mining | Global Warming |
| Gulf War Syndrome | Neural Network |
| Neuroscience | Organic Farming |
| SARS | Stem Cell |
| Taoism | Wavelet |

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## Dataset

For educational purposes, which are the main motivation for this proposal, one is interested in technical subjects rather than general ones. An example for general is "travel" whereas "wavelets" is a more technical subject.

The dataset used for experimentation has been built from the data collected in Lim et al. (2006). The dataset was collected using 10 different technical subjects. Each of these subjects was posted against Google and Yahoo search engines for a recommendation set. Upon receiving a recommendation set *R* from the search engine *S*, the top 100 pages were carefully evaluated and tagged as suitable (positive example) or not (negative example) for OSL purposes. The technical subjects, listed in Table 2 correspond to scientific topics reflecting the current research issues at the time that the dataset was collected.

Therefore, the total number of pages is 2000 (2 x 10 x 100). For each of these Web pages, a feature vector $\overline{x}$ was calculated and stored into the dataset. At the same time, all the training examples were already labeled either as *true* or *false*, depending on their suitability for one-stop learning or not, respectively. Therefore, the training set is composed by 2000 $(\overline{x}, \{true, false\})$=pairs. It should be noted that the dataset is highly unbalanced due to the majority of *false* examples, where 1846 are negative examples and just 154 are positive examples. This characteristic introduces a challenge for the learning algorithms, which need to accurately identify the discriminant features for the positive examples.

## Experimental Settings

The experiments were conducted using the latest version of Weka software[1] (Witten & Frank, 2005) and the LibSVM library[4] written by Chang and Lin. The four selected algorithms: support vector machine (SVM), Naive Bayes, AdaBoost with J48, and backpropagation for multilayer perceptrons (MLP) are available in Weka. For each of these algorithms, five different executions were repeated with different seed values. At each execution the seed value was used to perform 5-fold cross-validation, thus, the results can be interpreted/analyzed in a more realistic way.

## Evaluation of Features

Prior to proceeding with the training runnings, all the features extracted in the preprocessing step were analyzed using three different approaches for feature selection, yielding a ranking of the importance of each feature to the class distribution in the dataset. The selected feature selection criteria are: chi-square test, gain ratio, and

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*Table 3. Results from the evaluation of the features using 3 different methods. Notice that D3, DNW, and P3 present the lowest values. For this reason, these features are excluded from the experimentation.*

| Chi-Square | | Gain Ratio | | Information Gain | |
|---|---|---|---|---|---|
| D2 | 647.993 | NW | 0.182 | D2 | 0.164 |
| D1 | 637.973 | D1 | 0.150 | D1 | 0.161 |
| DDW | 609.745 | DDW | 0.146 | D DW | 0.157 |
| NW | 590.159 | D2 | 0.120 | NW | 0.149 |
| P1 | 408.858 | P 1 | 0.072 | P 1 | 0.120 |
| P2 | 284.858 | P 2 | 0.066 | P 2 | 0.097 |
| D3 | 12.929 | D3 | 0.018 | D3 | 0.007 |
| DNW | 12.152 | DNW | 0.017 | DNW | 0.007 |
| P3 | 0.000 | P3 | 0.000 | P3 | 0.000 |

information gain. The ranking is shown in Table 3. Each column corresponds to one of the tests.

After analyzing the feature selection results, the three least important features were removed: D3, DNW, and P3, because these features are not very useful for discriminating {*false*,*true*} classes. Thus, in the remaining of this section when the dataset is referred it implies the reduced dataset with 2000 examples of six features each.

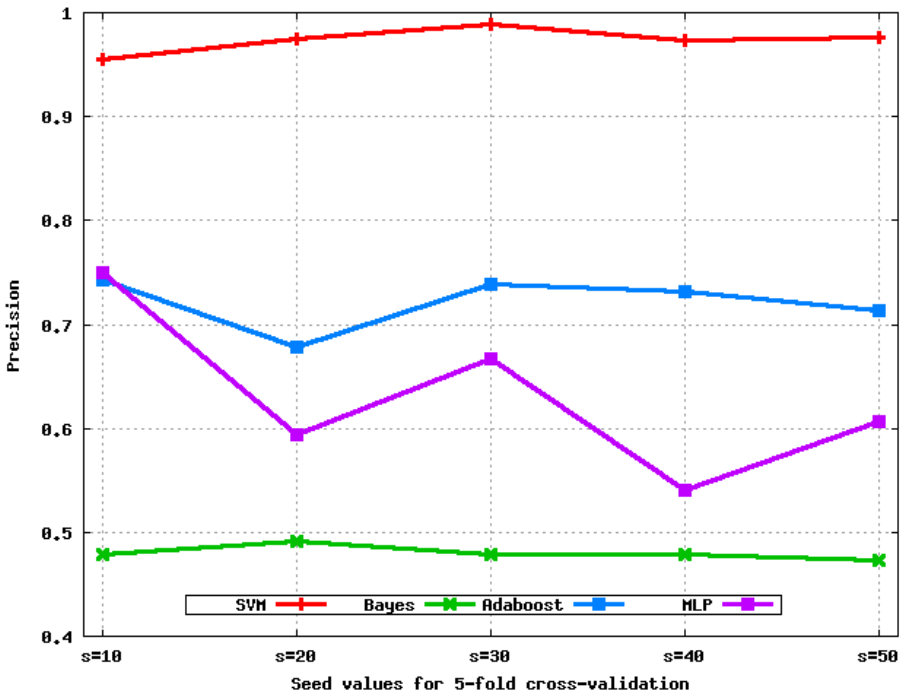
## Evaluation of Supervised Algorithms

In order to discriminate suitable pages for OSL, four different classifiers were applied to the training set. While it is common thought that SVM classifiers perform more accurate than others, bear in mind that, there is no evidence of previous application of supervised learning algorithms to the OSL problem, therefore, the experiments conducted here are basically exploratory, aiming to evaluate the performance of different algorithms. The results reported after the experiments have been compared using several measures including, precision, recall, overall accuracy, MCC, and the ROC space.

For all the runnings, the total number of true positives (*tp*), false positives (*fp*), true negatives (*tn*), and false negatives (*fn*) were counted. Using these indicators it is possible to compare the classifiers by combining them into more sophisticated

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*Figure 1. Precision results using different five-fold cross-validation seeds for all the algorithms. Notice that SVM yields precision values close to 1 (100%) outperforming all the other runnings.*




formulas. The precision obtained by the classifiers is shown in Figure 1, where precision is defined as:

$$P = \frac{tp}{tp + fp}$$

The reader can notice that SVM clearly outperforms all other classifiers. This can be explained by the low number of false positives that SVM algorithm yields for this problem. In fact, it is highly desired to minimize the number of false positives, because an OSL system aims to filter the recommendation set and present to the user only suitable results. Surprisingly the average number of false positives calculated by SVM is just 2.2 out of 154 actual positive examples. SVM was trained with the radial basis function (RBF) kernel and their free parameters were optimized using the traditional grid parameter selection, which yielded the best values for $C = 1.0$, the penalty factor, and $gamma=0.005$, the parameter for the RBF kernel. After the parameter selection, all the runnings for SVMs were done using the same

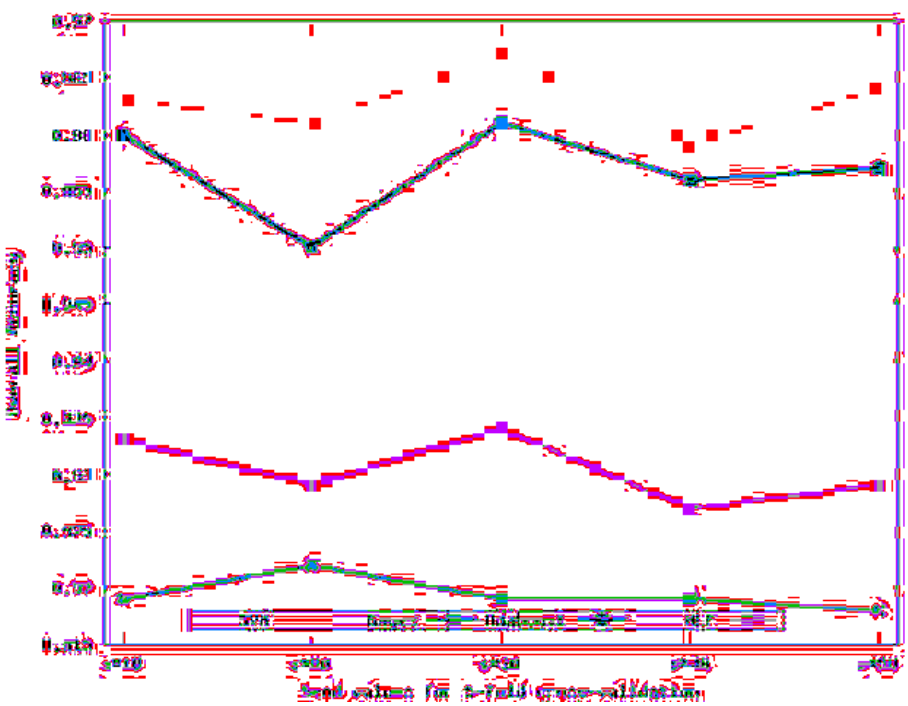
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configuration. It is concluded that SVMs maximize the number of correct positive predictions.

The poorest precision results were obtained by Naive Bayes, because the high number of false positives. The Bayesian approach yielded an average number of 95.2 false positives and only 87.8 true positives. Recall that, there is a total of 154 actual positive examples within the training set. The main motivation for testing the Naive Bayes approach was its efficiency, and simplicity, however, the results obtained show that Naive Bayes is not as effective as desired.

The overall accuracy is another measure commonly used for investigating the quality of classifiers. Accuracy values should be analyzed carefully, because they are not recommended to make decisions about the best classifiers, nonetheless, they can be useful to have intuition about general trends. While precision gives a notion of the proportion of correct predictions out of all the positive predictions, accuracy gives the proportion of correct predictions out of all the examples, either positive or negative. The overall accuracy is calculated using:

*Figure 2. Overall accuracy using different five-fold cross-validation seeds for all the algorithms. SVM and AdaBoost present the highest accuracy values.*



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$$ACC = \frac{tp + tn}{tp + fp + tn + fn}$$

Figure 2 shows the calculated overall accuracy for all the executions. It can be seen that the results are between 97% and 92%, which can be explained because the high number of true negatives. This behavior is very important for OSL purposes. Recall that the approach in (Lim et al., 2006} needs two stages: pruning and ranking. Using machine learning algorithms, both stages are intrinsic to the learned mapping function (model). The high overall accuracy levels shows that at least, the classifiers are very successful in pruning the unsuitable pages.

The Matthews Correlation Coefficient (MCC) is another measure for evaluating the performance of classifiers. MCC measures the correlation between all the actual values and all the predicted values. The formula for calculating the MCC is given next:

$$MCC = \frac{(tp \bullet tn) - (fp \bullet fn)}{\sqrt{(tp + fp) \bullet (tp + fn) \bullet (tn + fp) \bullet (tn + fn)}}$$

Figure 3 shows the MCC values for all the algorithms. According to this correlation measure, the performance of SVM and AdaBoost is similar. Note that perfect prediction is achieved when the correlation value is 1. Values close to 0.75 are justified by the high number of false negatives that appears in all the algorithms. The average number of false negatives are: SVM (72.8), Naive Bayes (66.2), AdaBoost (43.6) and MLP (114.2). False negatives are the suitable pages that were neglected by the classifiers. Certainly these numbers influence the overall performance, but, for the specific purposes of OSL, some false negatives could be tolerated as long as the number of false positives can be reduced and the number of true positives can be maximized.

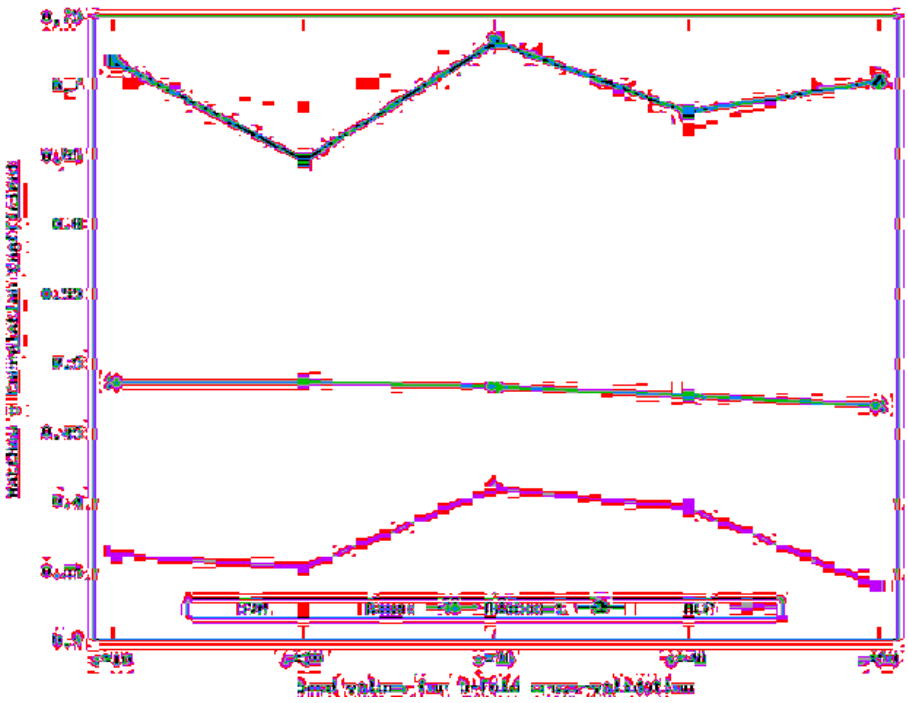
The confusion matrix, which gives the total number of *fp*, *tp*, *tn*, and *fn* for each run can be used to calculate the True Positive Rate (TPR) and the False Positive Rate (FPR). The TPR, often called recall or sensitivity is defined by:

$$TPR = \frac{tp}{tp + fn}$$

TPR represents the proportion of the number of correctly predicted suitable documents (*tp*) out of all the actual suitable documents. On the other hand, the FPR follows

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*Figure 3. Matthews Correlation Coefficient using different five-fold cross validation seeds for all the algorithms, MLP shows the poorest performance far from the acceptable results presented by AdaBoost and SVM.*



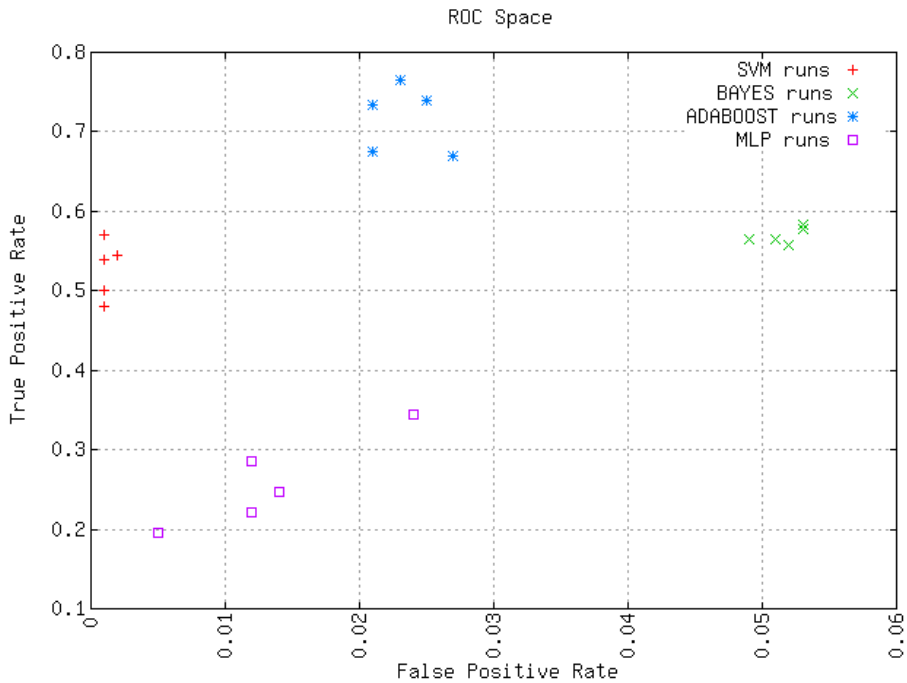
the same calculation for the negative class, in other words, FPR is the proportion of the number of incorrectly predicted unsuitable documents (*fp*) out of all the actual unsuitable documents. FPR is calculated using the following formula:

$$FPR = \frac{fp}{fp + tn}$$

With TPR and FPR values for every execution, it is possible to visualize the classifiers performance in the receiver operating characteristic (ROC) space. The ROC space is a 2-dimensional space where the *x* and *y* axes correspond to the FPR and TPR values respectively. Each execution represents a point in the ROC space. The best classifiers are expected to appear close to the upper left corner at position (0,1), which, to be precise, represents a situation where all the true positives are found and no false positives exist at all. Thus, the (0,1) point denotes perfect classification. Figure 4 depicts the ROC space for all the runnings performed for the OSL problem.

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*Figure 4. The ROC space for all the runnings. Notice that the axes were scaled to facilitate the visualization. Normally the ROC space is presented ranging from 0 to 1 in both axes.*




Note that the results are, to certain extent, clustered according to the algorithm being used. The MLP runnings are more disperse than others because the backpropagation algorithm is very sensitive to initial conditions like the random assignment of weights. SVM and AdaBoost seems to be the best classifiers according to TPR and FPR observations. While the SVM reduces dramatically the number of false positives the AdaBoost algorithm perform better than others identifying the highest number of true positives. Certainly, for an OSL system one is interested in maximizing the TPR and minimizing the FPR. It is important to notice that SVMs are preferred because the minimization of false positives. The user of an OSL system can afford some misclassified suitable documents (false negatives) but it is not desired to have misclassified unsuitable documents (false positives).

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## Analysis

Analyzing the results presented in the previous section, one can infer that supervised learning algorithms are very suitable for solving the OSL problem. Specifically, two of the selected algorithms in this chapter have shown expectant results using 5-fold cross validation. SVM and AdaBoost are the best algorithms, thus, their use is recommended. However, it must be pointed out that SVM minimizes dramatically the number of false positives, but, at the same time, exhibits a high number of false negatives, neglecting a considerable amount of suitable pages. Conversely, the precision of AdaBoost is less aggressive but it shows the highest recall. AdaBoost is characterized by having the lowest number of false negatives. Both algorithms perform better than the previous results reported in Lim et al. (2006). In fact, additional advantages of the machine learning approach over the Lim et al.'s statistical measures are (1) there is no necessity of performing stemming in the vocabulary of a given Web page, and (2) less number of calculations needed for extracting the features of a given Web page. Test examples can be evaluated by extracting just six features and posting the feature vector into the trained classifier. At this point, AdaBoost has the advantage of producing faster example evaluation, after proper training, because the less number of operations. While in AdaBoost it depends on the number of decision trees and the height of each decision tree, SVM requires time proportional to the number of support vectors, usually higher.

Naive Bayes and backpropagation, despite the high overall accuracy number, show the lowest MCC and precision. These two algorithms perform well rejecting unsuitable pages, that is, identifying a high number of true negatives, but still show deficiencies in discriminating properly the positive class.

On the other hand, by using only structural and link-based features, this approach can be extended to process Web pages written in any language. The minimal difference would be imposed by the use of stop words, that are language dependent. Nevertheless, this can be addressed by expanding the set of stop words to include the stop words of the desired language.

## Conclusion and Future Work

The authors believe that there is plenty of room for improvement in boosting the learning experience of educationally motivated Web users. The proposed one-stop learning paradigm based on machine learning demonstrated its effectiveness in achieving a learning goal on the Web while taking advantage of general-purpose search engines. This was done by extending the work presented in Lim et al. (2006) by adopting a subset of the proposed statistical measures in their work.

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The natural next step is a large scale testing of this proposal. In addition, a meta-search system designed for OSL can leverage the recollection of more precise and realistic opinions about which pages are more suitable for OSL. Another future direction is the investigation of what new features can help in reducing the number of false negatives, which is observed in the proposed machine learning-based approach. Reduced false negatives will increase the recall and improve the ROC curve, making the classifier perform better.

The running time for prediction was not fully investigated in this work although the performance of the proposed approach is outstanding. It is justified by the fact that building the feature vector for each page can be done in one pass, and, once the feature vector is ready, the evaluation cost of a new page by the classifier depends on the actual supervised algorithm being used. For example, SVMs require time proportional to the number of support vectors and AdaBoost with J48 depends on the number of base decision trees and the height of each decision tree.

## References

Abdi, H. (1994). A neural network primer. *Journal of Biological Systems, 2*(3), 247-283.

Bauer, E., & Kohavi, R. (1999). An empirical comparison of voting classification algorithms: Bagging, boosting, and variants. *Machine Learning, 36*(1-2), 105-139.

Brin, S., & Page, L. (1998). The anatomy of a large-scale hypertextual Web search engine. *Computer Networks and ISDN Systems, 30*(1-7), 107-117.

Burges, C. J. C. (1998). A tutorial on support vector machines for pattern recognition. *Data Mining and Knowledge Discovery, 2*(2), 121-167.

Crabtree, D., Andreae, P., & Gao, X. (2006). Query directed Web page clustering. *In Proceedings of the 2006 IEEE/WIC/ACM International Conference on Web Intelligence* (pp. 202-210). IEEE Computer Society.

Fang, R., Mikroyannidis, A., & Theodoulidis, B. (2006). A voting method for the classification of Web pages. *In Proceedings of the 2006 IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology* (pp. 610-613). IEEE Computer Society.

Freund, Y., & Schapire, R. E. (1999). A short introduction to boosting. *Journal of Japanese Society for Artificial Intelligence, 14*(5), 771-780.

Glover, E. J., Flake, G. W., Lawrence, S., Kruger, A., Pennock, D. M., Birmingham, W. P., & Giles, C. L. (2001). Improving category specific Web search

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by learning query modifications. *In Proceedings of the 2001 Symposium on Applications and the Internet*, IEEE Computer Society, 23.

Haveliwala, T. H. (2002). Topic-sensitive pagerank. *In Proceedings of the 11th International Conference on World Wide Web* (pp. 517-526). ACM Press.

Kummamuru, K., Lotlikar, R., Roy, S., Singal, K., & Krishnapuram, R. (2004). A hierarchical monothetic document clustering algorithm for summarization and browsing 19 search results. *In Proceedings of the 13th International Conference on World Wide Web* (pp. 658-665). ACM Press.

Lewis, D. D. (1998). Naive (bayes) at forty: The independence assumption in information retrieval. *In Proceedings of the 10th European Conference on Machine Learning* (pp. 4-15). Springer-Verlag.

Lim, S., & Ko, Y. (2006). A comparative study of Web resource mining algorithms for one-stop learning. *International Journal of Web Information Systems*, *2*(2), 77-84.

Lim, S., & Ko, Y. (2005). Mining highly authoritative Web resources for one-stop learning. *In Proceedings of the 2005 International Conference on Web Intelligence* (pp. 289-292). IEEE Computer Society.

McCallum, A., & Nigam, K. (1998). A comparison of event models for naive bayes text classification. *In Proceedings of the Workshop on Learning for Text Categorization* (p. 41-48). AAAI Press.

Mitchell, T. M. (2006). The discipline of machine learning. *Tech. Rep. CMU-ML-06-108*, Carnegie Mellon University - ML Department.

Sebastiani, F. (2002). Machine learning in automated text categorization. *ACM Computing Surveys*, *34*(1), 1-47.

Wang, Y., & Kitsuregawa, M. (2002). Evaluating contents-link coupled Web page clustering for Web search results. *In Proceedings of the 11th International Conference on Information and Knowledge Management* (pp. 499-506). ACM Press.

Witten, I. H., & Frank, E. (2005). *Data mining: Practical machine learning tools and techniques* (2nd ed.). San Francisco, CA: Morgan Kaufmann.

Wu, Y. F. B., Shankar, L., & Chen, X. (2003). Finding more useful information faster from Web search results. *In Proceedings of the 12th International Conference on Information and Knowledge Management* (pp. 568-571). ACM Press.

Yang, Y., Slattery, S., & Ghani, R. (2002). A study of approaches to hypertext categorization. *Journal of Intelligent Information Systems*, *18*(2-3), 219-241.

Yu, H., Han, J., & Chang, K. C. C. (2004). Pebl: Web page classification without negative examples. *IEEE Transactions on Knowledge and Data Engineering, 16*(1), 70-81.

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Zhang, Y. J., & Liu, Z. Q. (2004). Refining Web search engine results using incremental clustering. *International Journal of Intelligent Systems, 19*(1-2), 191-199.

Zhou, H., Lou, Y., Yuan, Q., Ng, W., Wang, W., & Shi, B. (2003). Refining Web authoritative resource by frequent structures. *In Proceedings of the 7th International Database Engineering and Applications Symposium*, IEEE Computer Society, 250.

## Endnotes

1 http://www.dmoz.org

2 http://www.cs.waikato.ac.nz/ml/weka

3 http://www.cs.waikato.ac.nz/ml/weka/

4 http://www.csie.ntu.edu.tw/_cjlin/libsvm/

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# About the Contributors

**David Taniar** received bachelor's, master's, and PhD degrees in computer science, specializing in databases. Since completing his PhD in 1997 from Victoria University, Australia, he has been at Monash University, Australia. He also held a lecturing position at RMIT University in 1999-2000 and was a visiting professor at the National University of Singapore for 6 months in 2006. His research primarily focuses on query processing, covering object-relational query processing, XML query processing, mobile query processing, and parallel query processing. He has published more than 50 journal papers in these areas. He has published a book on Object-Oriented Oracle (IGI Global, 2006) and another book on high performance parallel query processing and Grid databases that will soon be released by John Wiley & Sons. He is a editor-in-chief of the *International Journal of Data Warehousing and Mining* (IGI Global, USA).

* * *

**ABM Shawkat Ali** is a lecturer in the School of Computing Sciences at Central Queensland University, Australia. He holds a BSc (Hons.) and an MSc in applied physics and electronics engineering, an MPhil in computer science and engineering from University of Rajshahi, Bangladesh, and a PhD in information technology from the Monash University, Australia. Ali has published a good number of refereed journal and international conference papers in the areas of support vector machine, data mining, and telecommunication. Recently he published a text book, *Data Mining: Method, Technique and Demo*.

**Marco A. Alvarez** received a BSc in computer science from the Department of Computing and Statistics at Federal University of Mato Grosso do Sul, Brazil (1997).

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He also received an MSc in computer science after working in the Computational Intelligence Laboratory at the University of São Paulo (São Carlos), Brazil (1999). Alvarez has been working as a professor in the Computer Engineering Program at the Dom Bosco Catholic University (Brazil) (1999-2004) and as the head of the same undergraduate program (2002-2004). Alvarez has been serving as vice-president for the Peruvian Computer Society. He currently is a PhD student in the Department of Computer Science at Utah State University, under the supervision of Dr. SeungJin Lim. His main research interests include data mining, machine learning, information retrieval, and computer vision.

**Raju S. Bapi** received a BE in electrical engineering from Osmania University, India, and an MS and PhD from University of Texas at Arlington. He worked for three years as a research fellow in University of Plymouth, UK, and two years in the Kawato Dynamic Brain Project, ATR Labs, Kyoto, Japan. Since 1999, he has been working as a reader in the Department of Computer and Information Sciences in University of Hyderabad, India. His research interest is in various areas of computatonal intelligence, including machine learning and applications, neural networks and applications, neural and cognitive modeling, computational neuroscience, brain imaging and bioinformatics.

**Omar Boussaid** is an associate professor qualified to supervise research in computer science at the School of Economics and Management, University of Lyon 2, France. He received a PhD in computer science from the University of Lyon 1, France (1988). Since 1995, he has been in charge of the master's degree "computer science engineering for decision and economic evaluation" at the University of Lyon 2. He is a member of the decision support databases research group within the ERIC Laboratory. His main research subjects are data warehousing, multi-dimensional databases and OLAP. His current research concerns complex data warehousing, XML warehousing, data mining-based multidimensional modelling, OLAP and data mining combining and mining metadata in RDF form.

**Carol Brownstein-Evans** is associate professor of social work at Nazareth College and program director for the GRC MSW Program of Nazareth College and State University of New York College at Brockport. She received her PhD in social science from Syracuse University. Her research is in the intersection of maternal substance abuse and child welfare issues. Her publications and presentations are in maternal child health, mothers and addictions, and child welfare professionalization. She is a coauthor and project director to several collaborative child welfare grants in the Rochester, NY, arena.

**Longbing Cao**, as an IEEE senior member, has been heavily involved in research, commerce, and leadership related to business intelligence and data mining. He has

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served as PC members and program co-chairs in several international conferences, as editorial board members of international journals. He was chief technical officer, chief investigators, team or program leader business, and academic projects in Australia and China. He has published over 50 refereed papers in data mining and multi-agent systems. He has demonstrated knowledge, experience, and leadership in several large business intelligence and multi-agent research and commercial grants and projects, which amount to over RMB50 millions and AU$1.6m. He has delivered business intelligence and data mining services to industry areas such as capital markets, telecom industries, and governmental services in Australia and China.

**Mingwei Dai** earned a BS from the Math Department of Xi'an Jiaotong University (2003) and an MS from the Computer Science Department of Xi'an Jiaotong University (2007). Now he is a research assistant at the Chinese University of HongKong. His research interests include statistical learning, data mining, and pattern recognition.

**Qin Ding** is an assistant professor in Department of Computer Science at East Carolina University. Prior to that, she was with the Computer Science Department at The Pennsylvania State University at Harrisburg. She received a PhD in computer science from North Dakota State University (USA), an MS and BS in computer science from Nanjing University (China). Her research interests include data mining and database. She is a member of Association for Computing Machinery (ACM).

**Tu Bao Ho** is a professor at the School of Knowledge Science, Japan Advanced Institute of Science and Technology. He received a BTech degree in applied mathematics from Hanoi University of Technology (1978), and MS and PhD degrees from Marie and Pierre Curie University (1984 and 1987). His research interest include knowledge-based systems, machine learning, data mining, medical informatics, and bioinformatics.

**Xiaohua (Tony) Hu** is currently an assistant professor at the College of Information Science and Technology, Drexel University, Philadelphia. He received a PhD in computer science from University of Regina, Canada (1995) and an MSc (computer science) from Simon Fraser University, Canada (1992). His current research interests are biomedical literature data mining, bioinformatics, text mining, rough sets, and information extraction ad information retrieval. He has published more than 100 peer-reviewed research papers in the previous areas. He is the founding editor-in-chief of the *International Journal of Data Mining and Bioinformatics*.

**Wan Huang** is an assistant professor in the Department of Computer Science at State University of New York College at Brockport. She received a PhD in computer science from the University of Alabama (2004). Her current research focuses on wireless security and privacy, e-commerce, and data mining.

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**Yun Sing Koh** is currently a lecturer at Auckland University of Technology, New Zealand. Her main research interest is in association rule mining with particular interest in generating sporadic association rules and interestingness measures. She holds a BSc (Hons.) in computer science and a master's degree in software engineering, both from the University of Malaya, Malaysia. She recently completed her PhD in data mining from the University of Otago, New Zealand.

**Pradeep Kumar** is a PhD student of Department of Computer and Information Sciences, University of Hyderabad, India. He holds an MTech in computer science and a BSc (Engg) in computer science and engineering. For his PhD work, he receives his fellowship grant from Institute for Development and Research in Banking Technology (IDRBT), India. Currently, he is working as a JRA with SET Labs, Infosys Technologies Limited, India. His research interest includes data mining, soft computing, and network security.

**P. Radha Krishna** received a PhD from the Osmania University (1996), and an MTech in computer science from Jawaharlal Nehru Technological University, both in Hyderabad, India. Currently, he is working as associate professor at IDRBT. Prior to joining IDRBT, he was a scientist at National Informatics Centre (NIC), India. He is involved in various research and developmental projects, including implementation of data warehouse in banks, and standards and protocols for e-check clearing and settlement. He has to his credit two books and quite a few research papers in referred journals and conferences. His research interests include data mining, data warehousing, electronic contracts, and fuzzy computing.

**Yue-Shi Lee** received a PhD in computer science and information engineering from National Taiwan University, Taipei (1997). He is currently an associate professor in Department of Computer Science and Information Engineering, Ming Chuan University, Taoyuan, Taiwan. His initial research interests were computational linguistics and Chinese language processing, and over time he evolved toward data warehousing, data mining, information retrieval and extraction, and Internet technology. In the past, he cooperated with CCL (Computer and Communications Research Labs), Taishin International Bank, Union Bank of Taiwan, First Bank, Hsinchu International Bank, AniShin Card Services Company Ltd., Metropolitan Insurance & Annuity Company, Chia-Yi Christian Hospital, Storm Consulting Inc. (for Helena Rubinstein, HOLA, and China Eastern Air), Wunderman Taiwan (for Ford), and Microsoft in data mining and database marketing. He also served as a program committee member and a reviewer in many conferences and journals. He is a member of IEEE (The Institute of Electrical & Electronics Engineers). He has published more than 170 papers in his research areas. Hhe is also a leader of several projects from NSC (National Science Council) and MOE (Ministry Of Education) in Taiwan.

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**SeungJin Lim** received a BS in computer science from the University of Utah (1993) and an MS and PhD in computer science from Brigham Young University (1995 and 2001). In 2003, he joined the faculty of Utah State University, where he is currently an assistant professor in Department of Computer Science. His research interests include mainly data mining.

**Jun Meng** earned both a BS and MS in electrical engineering from Hefei University of Technology, P.R. China (1987 and 1990, respectively) and a PhD in control theory and engineering from Zhejiang University, P.R. China (2000). He has been an associate professor with the College of Electrical Engineering, Zhejiang University since 2001. His main research areas are in fuzzy logic, data mining/data analysis and fuzzy clustering of non-linear systems, and intelligent control of complex systems. He has received several research grants from government and funding agencies to direct his many research projects. He has authored more than 30 research papers published in conference proceedings and journals.

**Riadh Ben Messaoud** received a PhD in the decision support databases research group of the ERIC Laboratory in the School of Economics and Management of the University of Lyon 2, France (2006). Since 2002, he has been an engineer on statistics and data analysis from the School of Computer Sciences of Tunis, Tunisia. He received a research master's degree on knowledge discovery in databases from the University of Lyon 2, France (2003). His research interests are data warehousing, OLAP, complex data, and data mining. Since January 2004, he has actively published his work on several national and international conferences and journals.

**Rokia Missaoui** has been a full professor in the Department of Computer Science and Engineering at UQO (Université du Québec en Outaouais) since August 2002. Before joining UQO, she was a professor at UQAM (Université du Québec à Montréal) between 1987 and 2002. She obtained her Bachelor (1971) and Master of Engineering (1978) in applied mathematics from INSEA (Morocco), and her PhD (1988) in computer science from Université de Montréal. Her research interests include knowledge discovery from databases, formal concept analysis, integration of data mining and data warehousing technologies, as well as content-based image retrieval and mining.

**Gregg R. Murray** is an assistant professor in the Department of Political Science and International Studies at State University of New York College at Brockport. Prior to completing his PhD in political science at the University of Houston in 2003, he worked as a political consultant providing management and public opinion services to candidates for public office and their campaigns. His current research focuses on political behavior, including public opinion and political participation, as well as the application of data mining in the social and behavioral sciences.

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**Thanh Phuong Nguyen** is a PhD candidate in the Japan Advanced Institute of Science and Technology and a lecturer at Hanoi University of Technology. She earned an MS in computer science at Hanoi University of Technology (2005). She pursues her research on data mining, bioinfomatics, and formal methods.

**Richard A. O'Keefe** holds a BSc (Hons.) in mathematics and physics, majoring in statistics, and an MSc in physics (underwater acoustics), both obtained from the University of Auckland, New Zealand. He received a PhD in artificial intelligence from the University of Edinburgh. He is the author of *The Craft of Prolog* (MIT Press). O'Keefe is now a lecturer at the University of Otago, New Zealand. His computing interests include declarative programming languages, especially Prolog and Erlang; statistical applications, including data mining and information retrieval; and applications of logic. He is also a member of the editorial board of theory and practice of logic programming.

**T. M. Padmaja** received an MTech in computer science from Tezpur University, India (2004). She is currently a research scholar with University of Hyderabad, India. She receives her fellowship grant from Institute for Development and Research in Banking Technology, Hyderabad, India. Her main research interest includes data mining, pattern recognition, and machine learning.

**Dilip Kumar Pratihar** received a BE (Hons.) and MTech in mechanical engineering from the National Institute of Technology, Durgapur, India (1988 and 1994, respectively). He was awarded the University Gold Medal for securing the highest marks in the University. He received his PhD in mechanical engineering from Indian Institute of Technology, Kanpur, India (2000). He visited Kyushu Institute of Design, Fukuoka, Japan (2000) and Darmstadt University of Technology, Germany (2001) (under the Alexander von Humboldt Fellowship Program) for his post-doctoral study. He is presently working as an associate professor in the Department of Mechanical Engineering, Indian Institute of Technology, Kharagpur, India. His research interests include robotics, manufacturing science, and soft computing. He has published around 80 technical papers.

**Sabine Loudcher Rabaséda** is an associate professor in computer science at the Department of Statistics and Computer Science of University of Lyon 2, France. She received a PhD in computer science from the University of Lyon 1, France (1996). Since 2000, she has been a member of the decision support databases research group within the ERIC Laboratory. Her main research subjects are data mining, multi-dimensional databases, OLAP, and complex data. Since 2003, she has been the assistant director of the ERIC Laboratory.

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**Nathan Rountree** has been a faculty member of the Department of Computer Science, University of Otago, Dunedin since 1999. He holds a degree in music, a postgraduate diploma in computer science, and a PhD, all from the University of Otago. His research interests are in the fields of data mining, artificial neural networks, and computer science education. Rountree is also a consulting software engineer for Profiler Corporation, a Dunedin-based company specializing in data mining and knowledge discovery.

**Anthony Scime** is a 1997 graduate of George Mason University with an interdisciplinary doctorate in information systems and education. Currently he is an associate professor of computer science at the State University of New York College at Brockport. Prior to joining academia, he spent more than 20 years in industry and government applying information systems to solve large-scale problems. His research interests include the World Wide Web as an information system and database, information retrieval, knowledge creation and management, decision making from information, data mining, and computing education.

**Shibendu Shekhar Roy** received a BE and MTech in mechanical engineering from R.E. College, Durgapur-713209, India (Presently NIT, Durgapur). He worked as a scientist at Central Mechanical Engineering Research Institute, Durgapur-9, India from March 2001 to December 2006. Since January 2007, he has been working as a lecturer in the Department of Mechanical Engineering, National Institute of Technology, Durgapur-713209, India. He has a number of research papers in journals and conferences and filed a number of patents in product development. His research interests include expert systems and application of computational intelligence techniques in manufacturing process.

**Gnanasekaran Sundarraj** received a BE in electrical and electronics from Madurai Kamaraj University (1992) and an MS in computer science from Pennsylvania State University (2005). His main research interests include computational complexity, graph theory, algorithms, and databases. He is currently working as a software engineer.

**Tuan Nam Tran** received a BS in computer science from the University of Electro-Communication, Japan (1998) and a master's degree in computer science from Tokyo Institute of Technology (2000). He earned a PhD in computer science at the Tokyo Institute of Technology (2003). His research interests include machine learning, data mining, and bioinformatics. He is currently the chief technology officer of NCS Corporation, an IT company based in Hanoi, Vietnam.

**Tushar** is an undergraduate student in the Department of Mechanical Engineering, Indian Institute of Technology, Kharagpur, India. He is pursuing dual degree course

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(a five year course) with bachelors of technology in mechanical engineering and masters of technology in manufacturing systems and engineering. He is currently working in the Soft Computing Laboratory, Mechanical Engineering Department, IIT Kharagpur. His research interests include applications of soft computing techniques in the areas of artificial intelligence, robotics, and data mining.

**John Wang** is a full professor at Montclair State University. Having received a scholarship award, he came to the USA and completed a PhD in operations research from Temple University (1990). He has published more than 100 refereed papers and four books. He has also developed several computer software programs based on his research findings. He is the editor of the *Encyclopedia of Data Warehousing and Mining* (1st and 2nd eds.). He is also editor-in-chief of the I*nternational Journal of Information Systems and Supply Chain Management* and the *International Journal of Information Systems in the Service Sector*.

**Can Yang** earned a BS from the EEE Department of Zhejiang University, P.R. China (2003). He is now a graduate student with the EEE Department, Zhejiang University. His research interests include fuzzy systems and data mining. He has two projects supported by the China NNS (National Natural Science) Foundation, and has published five research papers in international conferences and journals.

**Show-Jane Yen** received an MS and PhD in computer science from National Tsing Hua University, Hsinchu, Taiwan (1993 and 1997, respectively). Since August 1997, she has been an assistant professor in the Department of Information Management in Ming Chuan University, and now she is an associate professor in the Department of Computer Science and Information Engineering, Ming Chuan University. Her research interests include database management systems, data mining, Web mining, and data warehousing.

**Justin Zhan** is at the Heinz School Faculty, Carnegie Mellon University. His research interests include privacy and security aspects of data mining, privacy and security issues in bioinformatics, privacy-preserving scientific computing, privacy-preserving electronic business, artificial intelligence applied in the information security domain, data mining approaches for privacy management, and security technologies associated with compliance and security intelligence. He has served as an editor/advisory/editorial board member for 10+ international journals and a committee chair/member for 40+ international conferences. He is the chair of the Information Security Technical Committee Task Force and the chair of Graduates of Last Decade (GOLD), Computational Intelligence Society of the Institute of Electrical and Electronic Engineers (IEEE).

**Chengqi Zhang** has been a research professor in the Faculty of Information Technology, University of Technology, Sydney (UTS) since 2001. He received a bachelor's

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degree from Fudan University, a master's degree from Jilin University, a PhD from the University of Queensland, Brisbane, and a Doctor of Science (higher doctorate) from Deakin University, Australia, all in computer science. His research interests include business intelligence and multi-agent systems. He has published more than 200 refereed papers and three monographs, including a dozen of high quality papers in renowned international journals, such as AI, information systems, IEEE transactions, and ACM transactions. He has been invited to present six keynote speeches in four international conferences. He has been elected as the chairman of ACS National Committee for Artificial Intelligence from 2006. He was also elected as the chairman of the Steering Committee of KSEM (International Conference on Knowledge Science, Engineering, and Management) in August 2006. He is a member of steering committees of PRICAI (Pacific Rim International Conference on Artificial Intelligence), PAKDD (Pacific-Asia Conference on Knowledge Discovery and Data Mining), and ADMA (Advanced Data Mining and Applications), serving as general chair, PC chair, or organizing chair for six international conferences and a member of program committees for many international or national conferences. He is an associate editor for three international journals including *IEEE Transactions on Knowledge and Data Engineering*. He is a senior member of the IEEE Computer Society.

**Dan Zhu** is an associate professor at the Iowa State University. She obtained her PhD from Carnegie Mellon University. Zhu's research has been published in the *Proceedings of National Academy of Sciences*, *Information System Research*, *Naval Research Logistics*, *Annals of Statistics*, *Annals of Operations Research*, etc. Her current research focuses on developing and applying intelligent and learning technologies to business and management.

**Shanan Zhu** earned both a BS and MS in electrical engineering from Zhejiang Univerisity, P.R. China (1982 and 1984, respectively) and a PhD in mechanical engineering from Zhejiang University, P.R. China (1987). He performed his postdoctoral research at Oxford University between 1990 and 1992. He worked for Argonne National Laboratory (USA), Utah University (USA), and National University of Singapore between 1992 and 1998. He has been a professor in the College of Electrical Engineering, Zhejiang University, since 1998. His main research areas are in system identifications, predictive control and its industrial applications, PID selftuning for SISO and MIMO systems, and intelligent control. He has received several research grants from Chinese government and funding agencies, and has published more than 70 research papers in journals and conferences.

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